

Europe's future – a model for assessing and increasing digital sovereignty

Dissertation

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Brno 2023

For my parents.

For Michel, Emma and Julia. For your future.

And for my beloved wife Anne.

I express my gratitude to assoc. prof. doc. Ing. Oldřich Trenz, Ph.D. and JUDr. Ing. Libor Kyncl, Ph.D. for their valuable assistance and support during my dissertation. Their recommendations and aid played a significant role in enabling me to complete my work to a high standard. Their expertise, guidance, and constructive feedback helped me to refine my research questions, reinforce my arguments, and identify pertinent literature sources. Additionally, they assisted me in integrating my research findings effectively and enhancing the overall quality of my work. I would also like to extend my appreciation to assoc. prof. doc. Ing. Pavel Žufan, Ph.D. and assoc. prof. doc. Mgr. David Hampel, Ph.D. for their immense advice and guidance in supporting my research. For their friendly and collegial support in administrative matters and mental challenges I would like to thank Marty-Jörn Klein, Bernhard Stellner and Prof. Dr. Jochem Müller. But the biggest thanks go to my wife Anne for her patience and love.

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In Brno on 1st July, 2023



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1. Introduction

Technological innovation has played a relevant role in the political and social upheavals of the past centuries. Social developments and political action have been influenced by inventions and their scaling, for example, regarding industrial production methods. The interactions between politics, industry and society continue to increase. Particularly in the new millennium, information and communication technology (ICT) is of growing and outstanding social, economic and political importance, which can be observed in, for instance, the power-politically motivated efforts of autocratic states such as China and Russia (Kukkola, et al., 2019). This has given rise to a new, interdisciplinary field of research that examines the interactions between politics, society and technology: Science and Technology Studies (STS) (DeNardis, et al., 2020, 87ff). ICT technologies interact with various areas of industry, civil society and politics. Dependency on ICT has become so great that any restriction leads to significant limitations in these areas. In this respect, industry, civil society and politics have a high interest in ensuring that commonly used ICT technologies are available. The actors want to have sovereign control over these technologies regarding freedom of choice and decision-making.

The National Security Agency (NSA) scandal, centered around Edward Snowden, along with the rise of platform capitalism that sees US companies increasingly encroaching upon state or sovereign domains, and the escalating reliance on ICT, have collectively highlighted the concept of digital sovereignty for political entities within the EU in recent times. In contrast to the techno-positivist capitalist model of the US and the regulated, non-liberal system in China, the search for a value-based digital policy has broadened in the EU states, requiring a digital sovereignty model in the context of European values.

Despite intensive and increasingly relevant discourse, understanding of digital sovereignty is highly differentiated, and there is no uniform model that describes or measures it. In the political sphere, such questions are particularly relevant, as insufficient digital sovereignty could have an impact on state sovereignty, thus calling into question the concept of state that emerged in Europe in the Middle Ages. State sovereignty is limited to a state's territory; does this apply to digital sovereignty influencing state sovereignty? In addition to approaches already realised by autocratic states that digitally colonise themselves and strive for autarky, are there ways for a state to remain sovereign from other states with the use foreign technology? How much digital sovereignty is required to maintain diplomatic negotiating capacity? Is the geopolitical order affected by changes in digital sovereignty?

These questions apply to all countries. However, the answers are particularly relevant for Europe. Europe's industrial supremacy in the manufacturing sector has thus far been maintained, but in the ICT sector, Europe has fallen behind. In the 1980s, the use of digital technologies increased so rapidly that critical infrastructure industries, government

organisations, ministries and authorities in Europe began to use technologies from non-European countries in such a way that dependent relationships developed. Today, hardware and chip components are mainly procured from Asia, while software technology is, with a few exceptions, innovated in the USA. In Europe, with notable exceptions such as Software AG, SAP, Ericsson, Nokia and smaller niche suppliers, there are hardly any relevant manufacturers from whom hardware and software can be procured. The Covid-19 pandemic has made it clear how important digital technologies are for the maintenance of state actions. With a few exceptions, however, technologies from non-European suppliers are used. What would happen if these companies used their power to influence government action by corrupting their systems? What does this mean for the independence of a state and its ability to resist or defend itself?

The dependence of the Western world on Russia's gas supplies, triggered by the war of aggression against Ukraine, has clearly shown how a lack of independence can influence economic and political action. In this respect, the influence of technological dependence on state sovereignty is a highly topical issue.

This dissertation aims to make the term *digital sovereignty* more accessible from the perspective of the European value system, as well as to concretise the term, to indicate how states can be classified, and to create room for improvement. To this end, an etymological derivation of the terms *sovereignty* and *digitalisation*, as well as their classification within a historical context, is first provided. The industrial revolutionary stages and social, technological, industrial, and political parallels are analysed. Through synthesis, a composite conceptual definition of *digital sovereignty* is proposed. The definition is then structured with a qualitative component model and quantified with secondary data. With the development of a composite comparative index of digital sovereignty, states are quantitatively assessed, and recommendations are given to strengthen the digital sovereignty of states, with a focus on the EU.

2. Literature research

2.1. Sovereign digitalisation, or digital sovereignty of the state

Since the concept of sovereignty and digital sovereignty is interpreted very differently in certain countries, a literature review will be carried out first. This is necessary in order to classify the concept of digital sovereignty in various political and economic concepts.

In the state-context, the term *digital* has only been spoken of for around 15 years, before which information was considered a 'means to an end'; this changed with the introduction of the internet. Information became the 'end', and the focus shifted to how information secrecy and the sovereignty of internet users could be preserved. The link between digitalisation and sovereignty was established in an analysis by Brinkerhoff in 2009, the focus of which was the use of ICT by ethnic minority groups to organise themselves into

a diaspora¹. Using Tibet as an example, the author discussed how a state that is not considered sovereign can nevertheless make use of the national identity of its citizens in the government-in-exile and create a sovereign state through digitalisation (Brinkerhoff, 2009, 26ff). Therefore, the term *digital sovereignty*, which is used in the EU at the political level, will be used in the present dissertation (Madiaga, 2020b).

2.1.1. Sovereignty as the basis of state action

In order to be able to conceptually analyse the digital sovereignty of states, the concept of state sovereignty must be considered. It is important in this dissertation, as a normative, conceptual definition of digital sovereignty will be one of the results and the concept of sovereignty in the digital context is critically discussed in literature.

Etymologically, the term *sovereignty* derives from the Latin *supremus*², the Middle Latin *superanus*³ and the Middle French *souveraineté*⁴ (Quaritsch, 1986, p. 14) (Kurz, 1970). In antiquity, concepts of state sovereignty were considered without their explicit conceptual use (Carlsnaes, et al., 2013, pp. 246–247). The term *sovereignty* was mentioned for the first time in the year 1120 (Quaritsch, 1996, p. 14) and was used until the late Middle Ages, mainly in a confessional context or regarding the highest-ranking ruler of the fief pyramid, whose rule was limited to a certain area—a territorial partial sovereignty (Mayer-Tasch, 2007, p. 58). The prevailing state systems of the Middle Ages were characterised by Machiavellian *raison d'état*—the absolute priority of state interests and the legitimisation of all instruments of power to achieve these interests.

The idea of a modern state first found mention in 1576 in Bodin's⁵ *Les Six Livres de la République*⁶ (Nicholls, 2019, pp. 47–66). Bodin is thus considered the founder of the concept of sovereignty from which the modern European state developed. Man as an individual came to the fore, and a social contract emerged as the foundation of the state system. This created an essential prerequisite for the emergence of economic progress. Bodin added to the existing theories that a community should be governed '*summa potestate ac ratione moderata*'—by a supreme power. This power should be undivided. The term *sovereignty* thus stands for absoluteness that cannot be derived from a higher power and was also associated with a direct encroachment on citizens and their 'households' (Mayer-Tasch, 2009, p. 23). Thomas Mun's and Jean-Baptiste Colbert's economic theories on mercantilism focused on strengthening the state through trade for the state's own benefit fell into this era (cf. 2.2.1).

The weaknesses of Bodin's views on sovereignty can be seen in the absence of the 'legitimation of absolute rule', the exclusive linking of 'sovereignty and monarchy' and the

¹ 'Modern diasporas are ethnic minority groups of migrant origins residing and acting in host countries but maintaining strong sentimental and material links with their countries of origin – their homelands—their homelands.' (Sheffer 1986, p. 3)

² 'the highest, supreme'

³ 'above', also 'superior'

⁴ 'supreme', 'most' or 'excellent'

⁵ Jean Bodin (1530–1596)

⁶ *The Six Books of the Republic*, here in the sense of a state.

blurred separation of sovereignty from the concept of right (Heintzen, 2005, pp. 3–4). In the following centuries, this led to monarchical absolutism through Hobbes’⁷ state-theoretical explanations regarding ‘sovereignty under state law and the sovereignty of a sovereign’ becoming one (Kriele, 2003, 56ff).

Bodin’s teachings marked Europe’s transition from the Middle Ages to modernity⁸ and are still influential today. The so-called Westphalian model⁹ was primarily characterised by a territorial principle, in addition to the sovereignty and legal principles. State borders delineated the political, legislative and executive boundaries of the sovereign (Zimmer, 2008, p. 55). Territorially delineated, autonomous and functionally similar states are essential characteristics of a modern state system (Ruggie, 1993, p. 151). The territorialisation of the concept of the state went hand-in-hand with Smith’s¹⁰ and Say’s¹¹ economic theories of classicism, in which state regulation ceded to liberal, self-regulating mechanisms (cf. 2.2.1).

2.1.2. Sovereignty in the modern constitutional state

Although the principles have essentially remained the same, the model of one sovereign is a counter-model to today’s understanding of the state in democratic states, which is characterised by the separation of powers (i.e. shared state sovereignty), valid law for all and popular sovereignty. In the course of democratisation, the monarchical sovereign developed into a popular sovereignty, which is anchored in many European constitutions today. For example, Article 20 of the German Basic Law states, ‘All state power shall emanate from the people’¹². The current interpretation of state sovereignty is found in Zandonella, whose formulation is also used by the Federal Agency for Civic Education¹³: ‘Sovereign states can freely and independently determine the type of government, legal system and social order within their territory (internal sovereignty). International law postulates the independence and equality of all states in international relations (external sovereignty)’ (Zandonella, Oktober 2007).

In this context, it is particularly worthwhile to classify state sovereignty in an intranational context and to consider it in terms of international law. The United Nations, an association of 193 states, aims at maintaining world peace and international security and promoting interstate relations while respecting peoples’ self-determination. The objective was derived from the fundamental principle of equal sovereignty among all members, and it was incorporated into the United Nations Charter in 1945 (United Nations, 1971). A single

⁷ Thomas Hobbes (1588–1679), in whose major work *Leviathan* (1651) the foundation of the state is discussed as a social contract of citizens in favour of a sovereign monarch (the sovereign).

⁸ The existence of states before the use of a concept of sovereignty (Werner, Wilde (2001, p. 289)).

⁹ After the Peace of Westphalia (1648)

¹⁰ Adam Smith (1723–1790)

¹¹ Jean-Baptiste Say (1767–1832)

¹² Similar formulations can be found in other constitutions of European states, e.g. the Czech Republic, Austria and Latvia.

¹³ A German authority affiliated with the Ministry of the Interior working at the interface between society, politics and science (www.bpb.de).

sovereign does not exist in a system with separation of powers (Hillgruber, 2002, 1073ff); it is a system of 'checks and balances', meaning that tasks, functions and spheres of power are distributed and control one another in the best possible ways. Sovereignty is understood through states' relationships with one another as defined by international law. External sovereignty corresponds to 'independence of command from other states' with simultaneous 'binding to existing rules of international law' (Doehring, 2004, p. 70). States are thus externally independent and sovereign when no other state can directly or indirectly interfere with or challenge their internal sovereignty.

Acts of sovereignty regarding internal affairs remain reserved to the states themselves provided no other states are affected. The maintenance of internal sovereignty requires full and ultimate decision-making power over the persons and things within a state's territory, as well as their recognition under international law. This includes the choice of economic, political and social systems. In addition, the states remain self-determining regarding the implementation of their own concerns or of decisions of the United Nations or international organisations on their territory. In this context, independence from the instructions of third parties applies. Internal sovereignty thus represents the highest decision-making power, which must be free from the influence of other states. Legal withdrawal from the United Nations Charter of international law is not possible, which is why states cannot be independent of this international legal order but are legally independent of the will of other states (Doehring, 2004, p. 197). This means that freedom of choice applies within the limits of international law. This elementary value also applied to Bodin's considerations of the binding of the sovereign to the principle of justice. It was only in the following centuries that a concept of absolute, unrestrained and unrestricted sovereignty developed in Europe, which led to the cruel systems of the early 20th century. With regard to an overriding world sovereignty alongside the retention of individual state sovereignty, the Charter of the United Nations led to a new degree of maturity of state expression and international coexistence.

In a modern constitutional state, however, state sovereignty must be interpreted in a special way to be classified in the digital era.

2.1.3. Modern government thinking and action

The works of Bodin and Hobbes were and are foundations of European thought, and they are stable and absolute in their definitions of power. Especially recently, the literature has been advocating the teachings of Michel Foucault regarding international relations and their impact on political analysis. Foucault changed the focus from the state system and the associated principles of sovereignty and introduced the concept of *governmentality* (Vasilache, 2014, pp. 4–7). He further rejected the principle of state sovereignty and the unambiguous definition of power, instead formulating a sovereign administrative action with diverse power influences, which is paradoxically a reference to sovereignty and law (Biebricher, 2014, p. 21).

Foucault was an advocate of 'cybernetic network thinking' and structuralism, focusing on the interplay of individuals and institutions rather than their individual natures (Moebius,

Peter, 2014, p. 20) (August, 2021, pp. 222–226). His approaches to nation–state sovereignty have been described as *biopolitics*, meaning a productive, nurturing ecosystem (Biebricher, 2014, p. 27).

Foucault considered power to be the self-regulation of individual objects: ‘directed against the claim of universal truths, power is conceived as a historical form of multiple power relations and as a complex strategic situation’ (Bublitz, 2014, p. 273). He manifested this in his works ‘Analytics of Power’ and ‘Microphysics of Power’, and in his reflections on ‘governmentality’ he proposed the following categories: ‘The form of power is government; the object of government is the population; and the means of government is the political economy’ (Mohabbat Kar, et al., 2018, p. 187). Essential in Foucault’s network thinking was that the individual—the citizen—is also part of the exercise of power and part of governing. He spoke of the ‘change of speech from power as possession to power as a general structuring element of the social’ (Saar, 2007, p. 32). For him, the emergence of today’s (neo-)liberal state was the result of science and historical developments. As a result, he clarified that the state governs the population and directs its actions to influence independent, free individuals in such a way that an ideal situation is created for the people of a state. Foucault thus sought ‘power as enabler’ rather than ‘power as limitation’. Boyle described how Foucault’s ideas, while not explicitly referring to the internet or 20th century digital technologies, nevertheless influence the discourse of territorial state power and sovereignty on the internet and in cyberspace (Boyle, 2017). Therefore, in this dissertation, Foucault’s approach to state sovereignty is applied. His investigations and argumentations mostly revolved around European ways of thinking and the associated value orientations (Vasilache, 2014, p. 171). He built on the values of Europe and diversified the concept of sovereign power to include administrative action and the people as integral to the exercise of power and governance. States that are sovereign but do not align with the approaches of Bodin, Hobbes and Foucault, such as Russia and China (i.e. autocratic great powers), may not correspond to this understanding of sovereignty in the digital context.

2.1.4. Sovereignty as an instrument for economic self-determination

Sovereignty can be understood as an instrument for the digital positioning and self-determination of states. Strategic autonomy plays an essential role here, as it is an indicator of how sovereign, i.e. self-determined, a state can decide on the selection of technologies to be used (Cottey, et al., 2022) (Madiaga, 2020b). A high degree of strategic autonomy means a high degree of freedom of choice, a low degree means a high degree of dependence (Pohle, Thiel, 2020, p. 10). The focus is on dependency with regard to procuring technologies from other states, which is necessary in a globalised world, but too much dependency can have undesirable effects (van Veen, 2017, p. 15). Russia’s war of aggression against Ukraine has shown the effects that an overdependence on resources, especially gas, can have, significantly influencing political, economic and especially financial systems, as well as competitiveness, within a short period of time (Adekoya, et al., 2022).

Different approaches and strategies can be observed in relation to dependencies. The EU has so far accepted (forced) high dependencies for raw materials and digital technologies and used state sovereignty to weigh mutual dependencies. The US, which has low to medium dependencies (depending on resource), uses its sovereignty to manage competition (van Veen, 2017, p. 22). They dominate technological development in many areas. For example, 15 of the top 50 digital companies in the world come from the US¹⁴, while the other 35 come from countries closely linked to the US: Japan, Western Europe, Taiwan and South Korea. In addition, the US pursues economic protectionism against countries that deviate from its political system. In addition, the US pursues economic protectionism against countries that deviate from its political system. Technologies from certain suppliers are banned (e.g. Huawei/ZTE ban), and corporate policies of major US companies prohibit the use of foreign technology from specific states (Calcara, 2022).

China uses its position of power to control competition and further reduce their dependence. This leads to using digitalisation as a basis for geopolitical power expansion. For an autocratic state like China, the dissemination of ideas via the internet poses a relevant threat. Since 2000, China has therefore been advocating 'cyber sovereignty'; the digital information space is controlled territorially (Pohle, 2020, p. 9). In addition, software products from the US are banned, as are US IPOs of Chinese technology companies. Digital sovereignty is therefore understood in China as an instrument of isolation to protect national interests and enforce political programmes.

China's approaches have been copied by other states, such as Russia. Russia has used its sovereignty to overcome technological dependencies, a practice that has intensified since the Western states stopped exports in response to Russia's war against Ukraine, leading to a strengthening of its relationships with China. Thus, digitalisation and technological strength are becoming platforms of action for the global economy and expressions of political digital sovereignty with divergent understandings of political and economic systems.

2.1.5. Measurability of state sovereignty; secondary data

Qualitative elements of state sovereignty have been described in previous sections. Since this dissertation requires secondary data to construct an index, this Section describes possible data for measuring state sovereignty. The specific selection of data is described in Section 5.4.1.1. For the quantitative assessment of state sovereignty, metric methods can be used. An overview of existing measurement methods is provided below, see also (Goertz, 2006, p. 112). The question of whether a state is fundamentally sovereign can be answered in binary terms, provided that one limits oneself to the legal recognition of a state. Here, one speaks of *de jure* sovereignty (Dahlmann, 2009, p. 31). If a state is a recognised member of the United Nations, it can, in principle, be described as sovereign because it is only through the recognition of others that a state or a government can attain sovereignty.

¹⁴ Source: www.fxssi.com (2022)

For the measurability of state sovereignty and modern governance according to Western values, *de jure* sovereignty is certainly a necessary criterion, but it is not sufficient. Questions arise: How sovereign is a state? How are sovereign characteristics of a state, government or administration measured? What influence does governance have on sovereignty? There are numerous elaborations on how to assess 'good governance', for example by the World Bank (The World Bank, 1996). Not only purely economic aspects are considered but also questions of freedom of speech, press, assembly and information, as well as the participation of the population in political decision-making processes or the fight against corruption. By promoting good governance in borrower countries, the World Bank supports the original interest of strengthening economic development in recipient countries and international economic relations. In principle, this follows the approach of good governance.

However, there is no recognised model for qualitative or quantitative assessments, in part due to the multiple dimensions of state sovereignty. A model that has quantitatively described sovereignty from 100 (high sovereignty) to 0 (low sovereignty) for several years and comprises a relatively comprehensive criteria of governance is the World Governance Indicator (WGI), which, since 1996, has evaluated more than 200 governments with the help of 31 cross-national individual indicators (Kaufmann, Kraay, 2020). The index consists of six composite assessment dimensions of governance of almost all states: 'participation and accountability, political stability, government effectiveness, regulatory quality, rule of law and control of corruption'. Methodologically, world governance is based on governance that is 'the traditions and institutions by which authority is exercised in a country' (Kaufmann, et al., 2010, p. 4). This includes the ability of a 'government to effectively formulate and implement sound policies', as well as the 'respect of citizens and the state for the institutions that govern the economic and social interactions' between them. Thus, essential dimensions describing the manifestations of state sovereignty are as follows (Kaufmann, et al., 2010, p. 6) (Kaufmann, Kraay, 2020):

- **Government effectiveness:** the quality of public service and the degree to which an administration is independent of political pressure; 'the quality of policy formulation and implementation; and the credibility of the government's commitment to such policies',
- **Quality of regulation:** 'the ability of a government to formulate and implement sound policies and regulations that enable and promote private sector development',
- **Rule of law:** the extent to which all actors in a state have 'confidence in and abide by the rules of society', in particular the enforcement of the rights and duties of citizens and state authority (e.g. executive),
- **Control of corruption:** 'the extent to which power is exercised' by state actors for private, unfair gain (including corruption) and the prevalence of elites to the detriment of the economy and freedom,
- **Political stability:** the 'likelihood of political or' socially motivated violence or terrorism or the instability of the state, and

- **Co-determination and freedom of expression:** measuring freedom of expression, free elections and freedom of the media.

Although it can be critically discussed whether the measurement of the WGI is reliable and can be considered a valid indicator of ‘good governance’ in all its individual forms, after more than 25 years, the WGI is a stable and long-standing indicator that has institutional value and allows comparisons to be made over time and between countries.

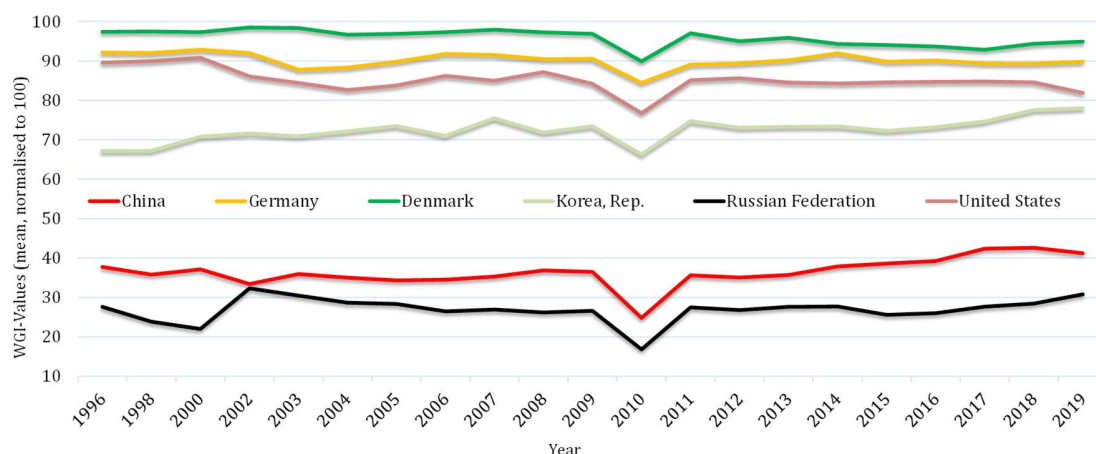


Figure 1 - WGI¹⁵ (mean, normalized), select countries (own representation based secondary data)

Western countries are at almost constant high levels, although their WGI values decline slightly (by 0.1–0.25 points) over the observation period. Russia (stagnating) and China (slightly rising) remain at low levels. Korea, an Asian country with a strong ICT industry (e.g. Samsung), is behind the EU countries but still at a high level. New Zealand, Luxembourg and Norway are ranked the highest, while Georgia, Kosovo and others stand out as particularly positive when comparing the largest deviations within more than 20 years. Badly placed are Venezuela, Bolivia and Syria (not shown).

Based on the WGI, Barnett developed a state sovereignty index by summing the *de jure* index, the six WGI indicators and the World Happiness Index (Barnett, 2017). It is led by countries such as Finland, Denmark, Iceland, Switzerland and the Netherlands. Germany is ranked 14th, the US 16th and the Czech Republic 18th. China is in the middle, ranked 72nd, and the last places are occupied by Zimbabwe, Lebanon and Afghanistan.

¹⁵ Source: Worldbank (2022) (WGI), change metric in 2010

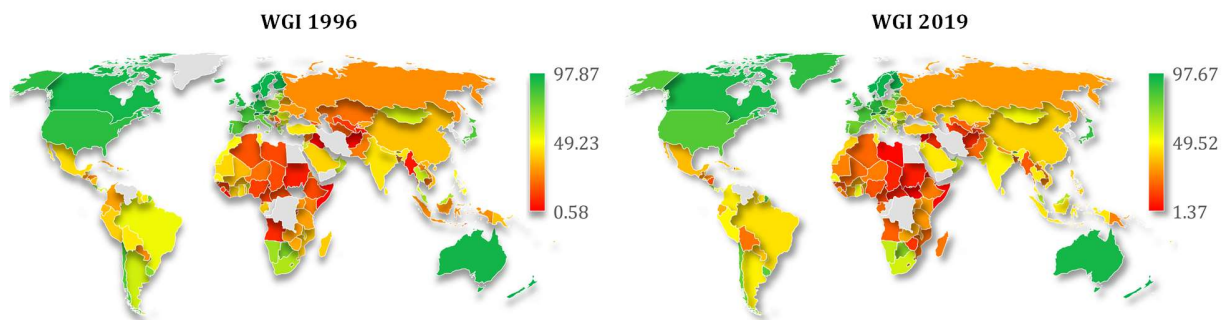


Figure 2 - Development of the WGI¹⁶ (index values) from 1996 to 2019
(own representation based on secondary data)

The latter, here λ^{17} , objectifies the subjective values *happiness*, *well-being* and *satisfaction* from different (e.g. psychological or economic) perspectives to make them measurable (Helliwell, et al., 2022). The Sovereignty Index is calculated using the following formula (Barnett, 2017, p. 21):

$$\text{Sovereignty} = \theta + \varepsilon, \theta = \eta, \varepsilon = \beta_1 + \beta_2 + \beta_3 + \beta_4 + \beta_5 + \beta_6 + \lambda \quad .(1)$$

and presents itself in a global overview, as shown in Figure 3.

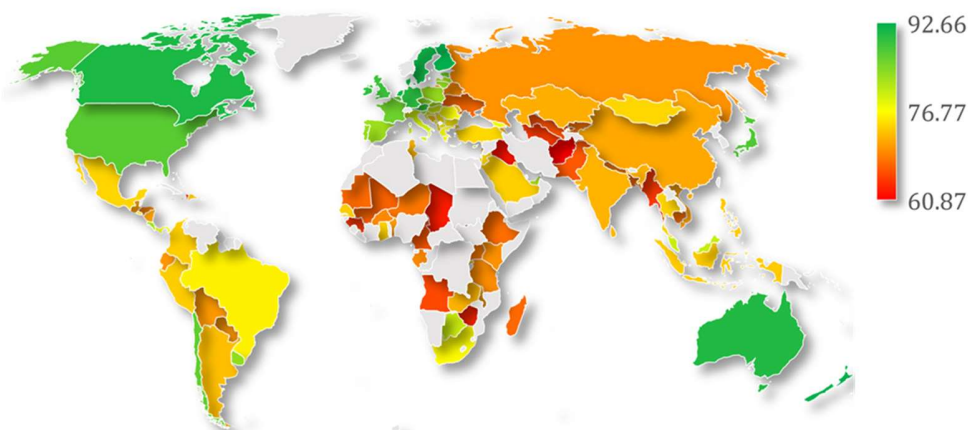


Figure 3- Barnett state sovereignty¹⁸ (index values)
(own representation based on secondary data)

2.1.6. Summary of literature review; the aspects of state sovereignty

The concept of sovereignty can be derived historically, and there are key figures that describe sovereignty comparatively on the basis of select criteria and are documented in time series. The key figures have been subjected to scientific discourse and represent a usable basis for this work. The measurement of sovereignty, according to Barnett, is

¹⁶ Source: Worldbank (2022) (WGI)

¹⁷ Source: Helliwell, et al. (2022)

¹⁸ Source: Barnett (2017, p. 61)

fundamentally suitable for the concept of the sovereignty of ‘good governance’ according to Foucault that is used in this work. The main aspects of the previous analysis of state sovereignty in the digital age can be summarised as shown in Figure 4.

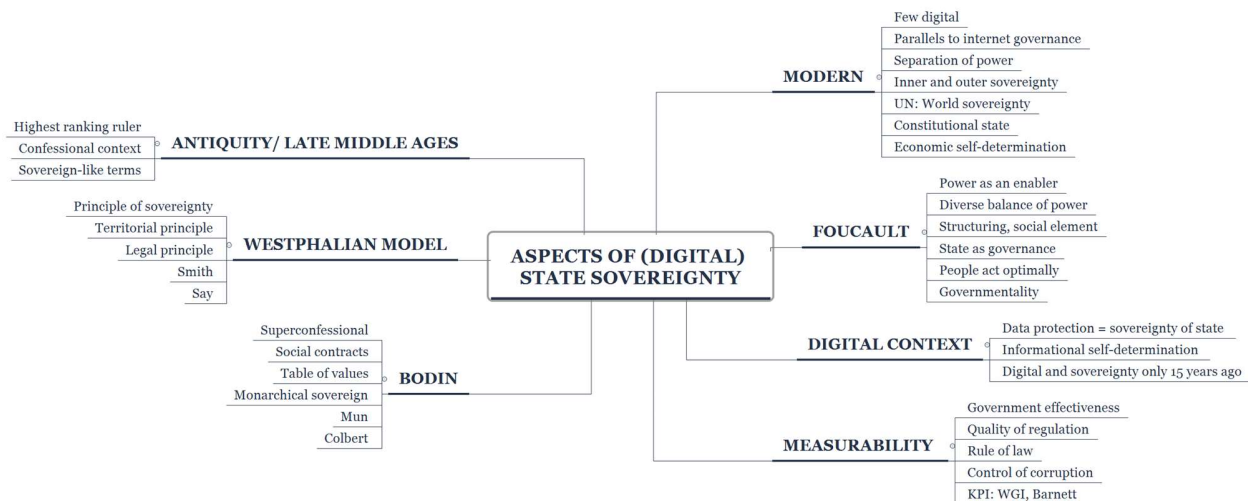


Figure 4 - Select analysed attributes of state sovereignty
(own representation based on literature review)

2.2. Technologisation and the role of the state

In the following sections, select aspects of the Industrial Revolution are examined in chronological order. The focus is on the events that were fundamental to the current digital revolution, as well as on the associated political and social upheavals. The role of the state—like the role of the EU states today—in the digital revolution is of particular interest. It is clear from the literature review that self-regulation of the economy alone has not proved to be suitable. Therefore, state interventions as a consequence of relevant events are elaborated and the connection with the relevant economic theories is indicated.

2.2.1. Analogue revolution on the eve of digitalisation

Digitalisation is not a random phenomenon. It has been classified as the fourth stage of the Industrial Revolution in Europe. By taking a historical look at the industrialisation of the past 250 years, it becomes clear that digitalisation as it is understood today is the result of three preceding, technologically-induced revolutionary stages. This is also the origin of the term *Industry 4.0*, which was coined in the German-speaking world. Here, the intention is not to focus digitalisation only to industry-sector. In mentioning the fourth stage of the Industrial Revolution, *revolution* is meant in the sense of a structural and lasting change in technological, social, political and economic systems and is therefore a suitable term.

Starting in Britain, predominantly in England, at the beginning in the mid-18th century (cf. Figure 5, exact date disputed) with the invention of the steam engine and the spinning machine, the first production plants dominated by machine power became possible, for

example in the textile industry. Already at the beginning of the 18th century, the British economy was more advanced than in the rest of Europe (Tomory, 2016). The management of money and goods and communication and transport routes were well developed (Mohajan, 2019, p. 383).

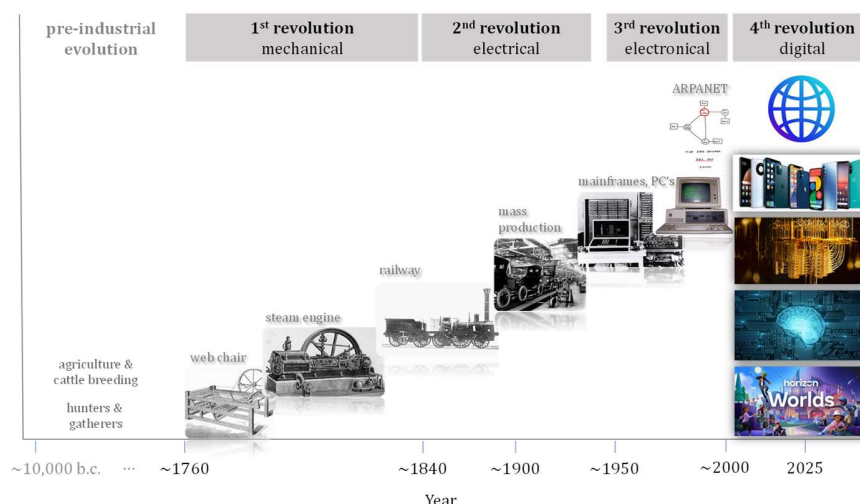


Figure 5 - Chronological classification of industrial revolutionary stages
(own representation based on internet¹⁹ search)

Through naval power and newly acquired colonies, sales territories were significantly expanded beyond the domestic market. The policy of expansion, initially mainly in the cotton economy, promoted technological innovations, especially in transport possibilities such as railways and shipping. This was preceded by the era of mercantilism, in which feudal structures in Western Europe disintegrated and capitalist principles were introduced, including the idea that a state government should control the economy with the aim of increasing the people's wealth and strengthening the state's power (Ormrod, 2008). Influential were, for example, Mun²⁰, who had the idea that trade, and especially foreign trade with a positive foreign trade balance, contributed to the wealth of a state, and Colbert²¹, who is considered the founder of French mercantilist economic and colonial policy and evidence-based action (Karstensson, 2018, p. 9) (Mukerji, 2020, p. 310). State macro-economic interventions and regulations promoted distinctive trade patterns, even with annexed overseas territories. The foundations of the British trading system were laid and had an impact well into the 19th century. With the growth of international trade, which was concentrated in the colonial and thus monopolised markets, the domestic economy also grew: Britain's industrial products were traded with raw materials from overseas (Hobsbawm, 1989).

¹⁹ Source: www.alamy.com/, <https://althistory.fandom.com/>, www.imtest.de/, www.ibm.com, <https://stock.adobe.com/>

²⁰ Thomas Mun (1571–1641)

²¹ Jean Baptiste Colbert (1619–1683)

The innovations of this period were mechanical in nature. One example is Hargreaves'²² spinning machine, which increased productivity at least fourfold in 1767 via the ability to spin with 100 spindles simultaneously. At the same time, the first attempts at industrial automation were made, in which human power was replaced by waterpower. Around 1770, Arkwright²³ succeeded in increasing the productivity of spinning machines sixfold with water mills (Juhász, et al., 2020, p. 41). Smith published²⁴ *The Wealth of Nations* in the same period, in which he, as one of the early advocates of free foreign trade, defined a further building block for the prosperity of a country (Smith, 2005): absolute advantage. As a proponent of neoclassical economics and the 'invisible hand' theory²⁵, Smith is credited with the idea of absolute advantage, in which a country focuses on the production and trade of the goods it can produce the most inexpensively and creates a largely static pattern of trade. Deviating from the neoclassical interpretation of Smith's work, some examples in the literature interpret a dynamic trade system as having 'increasing returns to scale' and intra-industrial rather than inter-industrial specialisation. Another source interpreted Smith's theories as an 'uneven development' between (trading) countries, in which less developed, more agriculture-focused countries develop at a disproportionately lower rate than industrialised countries (Schumacher, 2020, p. 22). The era of the then-new neoclassical economy saw inventions such as the steam engine by Watt²⁶ (1769), the steamship by Fulton²⁷ (1806) and the locomotive by Stephenson²⁸ (1814). The first railway line was built between Nuremberg and Fürth in 1823 (Emrich, 1957, p. 12). These innovations and advances had one thing in common: physical labour was replaced with machines. On the one hand, this led to an economic boom and disproportionate population growth. On the other hand, it led to social changes, as a two-class system was created (Vogel-Heuser, et al., 2017, pp. 1–3): 'The basic continuity of the pre-industrial world was interrupted between 1750 and 1850' (Pfahlmann, 1985, p. 1). Although there are no statistically reliable surveys of population trends for this period, it is undisputed that there was significant population growth (Rivinius, 1976, p. 275). The associated negative effects of the Industrial Revolution triggered controversial discussions and new economic theories. One example is the theory of underconsumption from economist Malthus²⁹ (Brooke, 2020, p. 27), who is considered a pessimist of classical national economics. Similarly, Say, is considered a pioneer of supply theory (Say's theorem) and thus of neoclassicism.

The question as to why the Industrial Revolution began in Europe, and in particular in Great Britain, can be answered as follows (Butschek, 2002, 268ff): the era of mercantilism, which prevailed in Europe until the second half of the 18th century, was characterised by an

²² Edward Hammond Hargraves (1816–1891)

²³ James Arkwright (1732–1792)

²⁴ Adam Smith (1723–1790)

²⁵ Self-organisation of the markets, only minimal state regulation

²⁶ James Watt (1736–1819)

²⁷ Robert Fulton (1765–1815)

²⁸ George Stephenson (1781–1848)

²⁹ Robert Malthus (1766–1834)

economic policy that aimed to maximise the profits of businesses, the taxes on which could be used to finance government spending, namely the military (Gömmel, 1998, p. 41). The sovereignty of the states, characterised by decrees and laws, was the basic prerequisite (Vasilache, 2014, p. 28). Therefore, state action was essential in making economic activities attractive. The 'extension of central administration' to enforce identical principles of action for uniform economic activity to the entire territory was just as fundamental as 'the standardisation of the legal system' (Butschek, 2002, p. 269). Courts ensured that no one was above the law and that the law could be enforced in principle. The levying of taxes and economic activities were linked for the first time since, as was common in the past and in other cultures, taxes did not serve solely to enrich the ruler and exploit the population; rather, they created a win-win situation: higher economic power generated higher tax revenues, and higher tax revenues meant that more could be invested in the state and the community, which promoted even higher economic power. Thus, increasing the social product became a maxim of state action (Butschek, 2002). Bodin had already recognised the relevance of a value-oriented polity, the ideology of reason and the need for a social contract. The recognition of the individual created a basis for economic growth. This open, fundamentally new attitude of the Enlightenment made it possible to promote a revolutionary, scientific way of thinking. States gained new sovereignty, as research was no longer reserved for an elite circle. The opening of scientific discussions to lay people was institutionalised. Here, the example of learned societies for the cultivation of science should be highlighted, first and foremost the British Royal Society, which was founded in 1662. The purpose of the society was to recognise, encourage and support outstanding scientific achievements and to promote the development and use of science for the benefit of humanity (Birch, 1968). Notable inventors emerged from the Royal Society, for instance Hargreaves.

Compared to France, which generated significantly more technological innovations at the time, in Great Britain, technical-economic connections were better recognised and civil liberties were more valued at the state level, which led to the economic flourishing described above (Butschek, 2002, p. 273). This was politically flanked by patent legislation that made sustainable investments worthwhile for businesses. The beginnings of a classical political economy in Europe and the associated gradual abandonment of mercantilism led to the first Industrial Revolution (Butschek, 2002, pp. 268–271), as did the following success factors resultant of Europe's cultural and state development:

- the emergence of the sovereign rule of law, in which everyone is identically subject to the law; this legal certainty created the basis for sustainable action by reducing transaction costs,
- the political participation of the individual, because the initiative and self-confidence of a European citizen require the assumption of responsibility and
- access for large portions of the population to seek scientific knowledge and translate research into practice.

Interestingly, scholarly consensus suggests that the circumstances in 14th century China resembled those in 18th century Europe, raising the intriguing question of why the initial Industrial Revolution did not occur in China four centuries earlier. This phenomenon is also called the Needham Problem (Needham, 1969, p. 109).

Elements of a market economy had already developed in China at an early stage (Schmidt-Glintzer, 2008, p. 45). Scientifically, China was among the world leaders, and the essential innovations that catapulted Europe out of the Dark Ages and into the Modern Age were created there: gunpowder, paper, printing technology and the magnetic compass. What was missing for China to take the next step into industrialisation? There are various theses on this, including the following:

- **The equilibrium trap:** China's population growth at this time was very large. Shaped by the social desire to produce as many progenitors as possible, to enable earlier marriages and to have children as early as possible, the ratio of the amount of work to the number of available workers became unbalanced. This led to economic resources, such as means of production and land, becoming expensive, while labour became cheap (Elvin, 1973). The demand for technical innovation and the desire for automation thus declined. However, this theory criticised on the basis of historical key figures (e.g. ratio of land quantity to number of population) (Schmidt-Glintzer, 2008) (Lin, 1995, p. 275).
- **Experience-based vs. scientific innovation:** China's innovation thrusts have been experience-based for centuries. Assuming that a technical innovation would produce a technical solution that had higher production efficiency, innovation in China was based on experience-based action: 'experimentation' or 'tinkering'. If an innovation was achieved, a new innovative technology was created; if not, it was discarded. The following parameters determine the success of this method:
 - number of experiments, which increases as more people (population size) engage in experiments,
 - time, because with more time the number of experiments increases and
 - effectiveness of experiments.

Unlike in China, the effectiveness of experimentation in Europe was supported by a scientific-experimental approach, which was promoted by the collaboration of science and lay people. Scientific knowledge was used to systematically select experiments in such a way that the probability of success was increased. This increased the efficiency of innovation in such a way that even with a smaller population and less time, more experiments were successful (Lin, 1995, p. 278).

- **Bureaucratic ideological system:** China's centrally controlled empire was run with a pronounced bureaucracy. This served to control the size of the country, was conducive to and important for a highly developed infrastructure and made it possible for a distinct scientific system to form early on. China was doing well economically at the time, so there was no economic pressure to innovate. While in Europe, due to the number of different states, there was economic and political competition between the states themselves and between the states and the church, this competitive pressure

was absent in China. The ideological approach and the absolute power of the political system suppressed any kind of public discourse that stood in the way of innovation and a necessary increase in innovation efficiency (Qian, 1985, p. 91).

A connection between the prevailing political system, the cultural achievements and the need for technological innovation sparked the first Industrial Revolution in the 18th century in Europe. Mechanics and the replacement of human labour with machines were the innovation foci of the first industrial stage. Economic stagnation and pessimism followed in the 1870s and 1880s (Cardwell, 2013, 215ff). It was not until the invention of electricity that a new level of maturity was achieved. This laid the foundation for the development of electricity-producing generators. Steam engines could be replaced by generators that converted mechanical energy into electrical energy. The first generator was developed by Werner von Siemens: 'We can therefore regard 17 January 1866 as the birthday of the electricity industry that now spans the globe' (Emrich, 1957, p. 13).

This was the start of the second—the electrical—phase of the Industrial Revolution. The dominant feature of this phase was that factories were electrified, which led to greater automation. In addition to electric drives, the Otto engine³⁰ was developed between 1860 and 1870, and the diesel engine³¹ was invented in 1897, both of which were made ready for series production and used industrially.

The European states of the time were regionally distinct with regard to industrial progress (Becker, et al., 2011). While Great Britain was responsible for driving industrialisation in the mid-18th century, in Germany³², this process started much later. It was not until the 1830s and 1840s that the industrial sectors, which focused on mining and mechanical engineering, grew disproportionately, followed by railway construction in the 1850s (Hahn, 2011, p. 28). Germany had to catch up with Great Britain in terms of development (Landes, 1983, p. 211). Germany's technological and industrial developments were accompanied by a discussion on industrial policy. A policy that promoted as much industry as possible was not uniformly pursued in the German states, and there were repeated setbacks (Wehler, 1974, pp. 289–325). Thus, the first industrial society did not emerge in Germany until about 50 years after those of Great Britain.

Until the mid-1860s, the railway sector dominated German economic growth (Fremdling, 1985). Germany began flowering during this period and became the forerunner of technological developments in the Second Industrial Revolution. Scientific knowledge was no longer imported but exported. Innovation came about through a high scientific content, and it was often academics who became industrialists. Applied research became the focus of industrial success.

Characteristic of this second stage in the US was the development and introduction of standardised and increasingly automated production processes in various industrial sectors

³⁰ Nicolaus August Otto (1832–1891)

³¹ Rudolf Diesel (1858–1913)

³² German Confederation (1815–1871)

(Mokyr, Strotz, 1998, p. 10). Due to increased urbanisation, there was a high demand for the industrialisation of food production. Assembly line technology was introduced, and it was further developed to perfection in automobile production. This industry was and still is considered a pioneer for the highest standardisation of and most efficient production methods (Ford, 1923, p. 36). This was later systematically introduced in the manufacturing industries through Taylorism³³ in the mid-1880s. Beginning in 1914, the introduction of automatic assembly lines and standards drastically reduced production costs.

As a critique of the classical model of economics that characterised the first Industrial Revolution, a social class system—the bourgeoisie and the workers—emerged during this period of transition to neoclassicism.: bourgeoisie and workers. The classes not only differed economically, but were sharply demarcated from one another in all circumstances of life: culturally, educationally and socially (Wegner, 2022). Questions about the social, economic and political upheavals led to scientific and sometimes dogmatic discussions among economists. The development cycles of the economy, the question of the relevance of state intervention in the economy and the effects of industrialisation on society influenced Schumpeter³⁴, who distinguished overlapping types of business cycles with different durations: the approximately 40-month observed bank clearing cycle (according to Josef Kitchin), the approximately 7–10-year empirically observed price/interest rate series (according to Clement Juglar) and the 50–60 year technology innovation and investment cycle (according to Kondratjev³⁵). In this theory, Schumpeter assumed that technological innovations initiated by companies and ‘dynamic entrepreneurs’ were the determinants of economic cycles (e.g. the invention of the steam engine, electrification and railways) and were responsible for economic upswings and crises (Frambach, 2018, 12 ff.). One of these crises was the global economic crisis triggered by the stock market crash in the US in 1929, which resulted in a worldwide recession, a decline in production and long-lasting unemployment.

Without addressing the dark phase and influences of the First and Second World Wars in this work, reference should be made to the founder of modern macroeconomics, the British national economist Keynes³⁶. Still under the influence of the economic crisis, he wrote *The General Theory of Employment, Interest and Money* (Pigou, 1936). The foundation of Keynesianism can be described as ‘anti-Say’ and contradicts the approach that every supply creates demand (Béraud, Numa, 2019, 228 ff.). From this, Keynes developed a demand-oriented economic policy and ideas in which state instruments are necessary to generate demand and full employment and thus promote business cycles through state intervention. For the first time, approaches were developed in which politics did not have to submit to ‘natural’ business cycles to reduce the powerlessness of governments in the context of the Great Depression. The basis of these ideas was that Keynes discovered structural and long-

³³ Frederick Winslow Taylor (1856–1915)

³⁴ Joseph Schumpeter (1883–1850)

³⁵ Nikolai Kondratiev (1892–1938)

³⁶ John Maynard Keynes (1883–1946)

term weaknesses in demand as well as an increasing growth-productivity weakness (Bontrup, 2006). Keynes therefore recommended optimal microeconomic ('individual economic rationality') and macroeconomic perspectives ('macroeconomic rationality'). Even though the oil crisis in the mid-1970s once again gave rise to a neoliberal countermovement, with his new economic approach, Keynes showed that the widely demanded principle of competition was not sufficient to achieve an optimal economic market and full employment. Especially in crisis situations, state intervention is necessary. Current examples of this are the Covid-19 pandemic and Russia's war of aggression on Ukraine.

The third Industrial Revolution, which began in the 1960s in Europe and the US, is associated with the development and spread of computer technology. In part, an alternative definition can be found in the literature, wherein the third Industrial Revolution is discussed as replacing the resource-intensive, growth-oriented model of the first (coal) and second (oil) stages. The focus was on shaping a sustainable industrial environment using only renewable, climate-friendly resources to reduce the impact on the environment as much as possible (Voss, 2010, p. 46). This interpretation is explicitly not considered in this thesis.

Here, the third revolutionary stage is understood as the preliminary stage of digitalisation, which is why it is important to discuss which effects describe this phase and continue to have an impact today (Schwab, 2018).

2.2.2. Electronic revolution through central technology innovation

The foundations for the technological development of the calculators that accompany our lives today were laid over centuries. As early as 1100 BC in the Indo-Chinese region, the abacus, the first calculating machine³⁷, was developed. Many such early developments were rediscovered in medieval Europe. Germany and England created innovative and scientific foundations for today's digitalisation, but US companies were the first to implement and achieve economic success with it.

Among the first achievements were a mechanical calculating machine mastering the four basic arithmetic operations developed by Leibnitz³⁸ in 1673 and the rediscovery³⁹ of the dual number system, which is the basis of digitalisation. In the following decades, calculating machines were steadily perfected. The first mechanical calculating machine, which formed the essential basis for the computer technology of today, was Babbage's⁴⁰ difference machine (Campbell-Kelly, et al., 2018, pp. 23–27). Based on this design, numerous innovations in mechanical calculating machines and input and storage methods were realised in the following decades (Richard, 2017). Particularly noteworthy is the Hollerith⁴¹ machine, which used punched cards for programming and data storage. Mechanical

³⁷ Discovered off the Greek island of Antikythera around 1900

³⁸ Gottfried Wilhelm Leibniz (1646–1716)

³⁹ Indicative approaches to a dual number system already existed in India in the third century BC, in China in the 11th century and in Polynesia.

⁴⁰ Charles Babbage (1791–1871)

⁴¹ Hermann Hollerith (1860–1929)

calculating machines and punched cards were in use until the 1970s. Modern computers in binary technology were first developed by Zuse⁴² around 1935 (Strawn, 2019).

IBM⁴³, which grew out of the Tabulating Machine Company founded by Hollerith in 1896, was established in 1911 and was instrumental in shaping the mass phenomenon and scaling of computer technology. IBM developed computers in the 1950s and 1960s, initially with a focus on scientific applications. These found their way into research and teaching, as they could be used for programming exercises in mathematics studies. This laid the foundation for education in computer science content, another driver of the mass phenomenon of digitalisation (Bashe, 1986). IBM technology moved into industrial companies, for instance, with the first mainframe computers (O'Regan, 2019, pp. 243–248). In 1970, another evolutionary stage developed: the small computer with Datapoint 2200 from the company CTC⁴⁴. It was quickly recognised that it was suitable for payroll accounting and other applications required by industry. Scaling and user-friendliness were considered early on as success factors for digitalisation, with a focus on increasing efficiency by digitising analogue processes.

With the development of semiconductor technology, the first integrated circuits were designed in 1958 and industrially manufactured beginning in the 1960s (O'Regan, 2019, pp. 155–156). Drivers were requirements for miniaturisation, essentially by the space and defence industries. CTC also initiated a leap into a new generation of processors—the microprocessor. In 1969, circuit designs were given to Texas Instruments, an early integrated circuit manufacturer, and Intel to produce a new type of processor architecture. Texas Instruments was unable to deliver, and Intel was unable to meet deadlines. This led to amended contracts between CTC and Intel, which secured Intel the rights to the new processor architecture and set the stage for Intel to dominate the computer market today (Wood, 2012, pp. 113–120).

As early as 1965, Moore⁴⁵ predicted that the invention of integrated circuits would rapidly increase technology innovation: 'Integrated circuits will lead to such wonders as home computers—or at least terminals connected to a central computer-automatic controls for automobiles, and personal portable communications equipment. The electronic wristwatch needs only a display to be feasible today' (Moore, 1965). Moore predicted, with a focus on presumed areas of application for processor technology, in which rhythms processor development would progress (Moore, 2003). His approaches changed several times over the decades. Moore's Law, which is more like a regularity and has no scientific basis, was and is usually interpreted as follows: the processor power for computers (i.e. the number of transistors housed in a chip of the same size) doubles every 18 months. In the literature, it is often discussed whether Moore's statements followed an empirical, extrapolable observation or whether the technological development in the chip industries

⁴² Konrad Zuse (1910–1995)

⁴³ International Business Machine Corporation

⁴⁴ Computer Terminal Corporation

⁴⁵ Gordon E. Moore (1929–2023)

matched Moore's forecasts—whether they were a self-fulfilling prophecy (Kish, 2002, Lundstrom, 2003, Waldrop, 2016). It can be said that Moore's Law is still in use today.

Initially, in the 1960s and 1970s, economically motivated and industrially centrally controlled chip innovation led to an increase in integration density—the intelligent placement of as many components as possible on a chip's surface. This development was driven by the development in semiconductor technology and the miniaturisation of components in order to fit as many logic gates as possible on one board. In 2007, Moore himself predicted the end of his law in 10–15 years, as no further leaps in efficiency could be achieved with the technologies available. The extraordinary technological development of processors, which was predicted by Moore, led to the success of the chip and thus the PC industry, and it was the basis for mass digitalisation. Computing power could be increased, and costs could be reduced. This made mainframes and PCs affordable for companies as well as private environments, and they became widespread beginning in the 1970s. This was the basis for and at the same time a feature of the third industrial, electronic revolutionary stage.

2.2.3. The internet as a basis for decentralisation

In addition to innovation in computer technology, another phenomenon was relevant for mass digitalisation: even before computers became accessible to the general public, predecessor structures of the internet emerged (Abbate, 2001, pp. 168–172). At the beginning of the 1960s, the US Department of Defense (DoD) was looking for a way to network scientific institutions involved in military research. On the one hand, it was a matter of exchanging data; on the other hand, it was about being able to use scarce computing resources across the board. In 1966, the ICT division of the DoD, as part of the Defense Advanced Research Projects Agency (DARPA), initiated activities for a new kind of networking of computers under the leadership of Taylor⁴⁶, the then-director. As early as 1962, Licklider⁴⁷ noted ideas for creating symbiosis between people and machines by means of a large computer network (Schmitt, 2016, p. 125). He is quoted as having envisaged a network 'imagined as an electronic commons open to all, "at the main and essential medium of informational interaction for governments, institutions, corporations, and individuals."' (Garreau, 2005, p. 22). Together with the chief engineer of DARPA, Roberts⁴⁸, the development of the ARPANET began. The basic idea was that computers should be able to access each other's data, regardless of location, computer type or operating system. The data packets were to be transmitted by Interface Message Processors (IMP), which, as small computers, exclusively took over the data communication and translation tasks of the computers integrated in the network (Abbate, 2001, pp. 150–152). This created an essential, new element of system design that moved away from centrally planned, centrally controlled systems to decentralised and smaller networked systems. It was the basis for the growth of self-organised systems as part of a larger system community with only a few superordinate

⁴⁶ Robert Taylor (1932–2017), member of the Intergalactic Computer Network Lickliders

⁴⁷ Joseph Carl Robnett Licklider (1915–1990), Taylor's predecessor as director

⁴⁸ Lawrence Roberts (1937–2018)

rules. Linked to this was the idea that the decentralised systems would develop successfully on their own and thus contribute to the robustness of the overall system.

In his work *Small is Beautiful*, economist Schumacher⁴⁹ contradicts the production principles that are still valid today with a striving for maximum output and yield without including non-replicable goods (raw materials) in the calculation. He places a 'medium technology' at the core of his consideration; on the one hand, this is favourable, but on the other hand, to leave room for degrees of freedom, it is only suitable for application on a smaller scale: 'But people can be themselves only government effect in small comprehensible groups. Therefore we mustn't learn to think in terms of an articulated structure that can cope with a multiplicity of small-scale units. If economic thinking cannot grasp this it is useless' (Schumacher, 1973, p. 57). In this way, Schumacher propagated a model of decentralised, small-scale production (in comparison to classical economics) and described a kind of community in technology development. He also contradicted the plan to establish high-tech production methods in emerging countries that had not developed corresponding degrees of economic or social maturity. Although these countries misinterpreted Schumacher's concept 60 years ago, the triumphant advance of the internet, also—or especially—in second and third world countries, has proven how forward-looking and realistic his ideas were. However, the ideas of Schumacher, Licklider and Roberts had to be translated into functioning technological systems.

Beginning in 1969, the ARPANET was technically realised. The individual distributed locations enabled access to centrally and de-centrally held data and computing power. Telephone lines with 56 kbit/s were used. In the event of an attack, the robust and resilient design would enable the preservation of parts of the network. The network structures implemented up to then had a 'single point of failure'. The ARPANET became a reality through DARPA and MIT⁵⁰. Initially, a purely military network of computer centres was created between US universities that conducted research for the DoD. On 30 August 1969, the first IMP was installed at the University of California in Los Angeles, connecting with the one at the Stanford Research Institute, then at the University of California in Santa Barbara and the University of Utah (Schmitt, 2016, p. 139).

Electronic mail based on the 'file transfer protocol' was used from 1971 onwards and was more of a by-product, but its utility was immediately high. In Europe, there were also developments of computer networks in France, Norway and Great Britain. These were not interoperable but, like ARPANET, focused exclusively on the networking of computers with each other rather than on the comprehensive networking of networks. Together, a network protocol was developed that was able to mediate between networks via gateways. The result was the TCP/IP⁵¹, and it was agreed that all previous protocols would be converted to TCP/IP on 1 January 1983. This date is therefore considered the birthday of the internet (Ronda Hauben, 1999). The original ARPANET, which was used exclusively for military research

⁴⁹ Ernst Friedrich Schumacher (1911–1977)

⁵⁰ Massachusetts Institute of Technology

⁵¹ Transmission Control Protocol/Internet Protocol

purposes and was still experimental at that time, was sealed off and expanded in a MILNet to become the Defence Data Network and later the NIPRNet⁵², which provided the operational data network for non-confidential information of the US DoD on the basis of TCP/IP (Defence Communications Agency, 1985, p. 11).

Until the mid-1980s, the number of internet users increased linearly, at which point it increased exponentially (cf. 2.2.6.1). The high demand for international, cross-border communication infrastructure and the further establishment of computers in industrial companies accelerated the scaling of computer technology in industry. This laid the foundation for questions about the digital sovereignty of a territorially limited state.

Computers not only found their way into companies' production processes; beginning in the 1980s at the latest, computer technology became affordable for private individuals. In 1993, there were 21 million PCs in private households worldwide; this number grew to 30 million in 1994 and 45 million in 1995 (Tapscott, 1996, p. 33). The number of internet users grew at a similarly remarkable rate. In 2018, the number of internet users worldwide rose to around 3.9 billion, and in 2022 it rose to approximately 5 billion⁵³. In Germany, more than half of the 40 million private households already had PCs after the turn of the millennium. In 2022, this figure was 92.3%⁵⁴. Today, most households have access to computers and usually also the internet. Digitalisation has thus permeated society, and industry and private households have been computerised. Strong networking, the availability of computer technology and, above all, the mobilisation of computer technology can be seen as the triggers of the fourth revolutionary stage. It is difficult to date the beginning of this stage, and there is scientific consensus. An interval around the turn of the millennium can serve as a benchmark.

2.2.4. Decentralisation and mobilisation

In principle, electronification and digitalisation can be seen as one coherent stage, but there are good reasons to differentiate these into the following two stages of the Industrial Revolution: the third stage, involving electronification and computerisation and the associated evolutionary expansion of technology into certain areas of life, and the fourth stage, which revolutionarily affects almost all areas of the lives of citizens, customers and users. The third stage was largely centrally controlled, as was the case in the first two stages of the Industrial Revolution: technology was invented and established in business and industry by political or industrial decision-makers. The effects of this were politically flanked and partly counteracted. A new, much faster type of innovation has created mechanisms that have enabled digitalisation everywhere and, notably, in a decentralised manner. From an economic perspective, consumers are 'moving more and more to the centre' (Schwab, 2016), and technology is becoming life-determining in essential areas. The following elements can be named as decisive for decentralisation and mobilisation.

⁵² Non-classified Internet Protocol Router Network

⁵³ Source: Statista (2023)

⁵⁴ Ibid.

First is the establishment of mobile radio technologies beginning at the end of the 1950s. Initially, access to mobile technologies was limited to a few subscribers. Only with the development and expansion of digital technologies could exponential growth be achieved. The strongest growth emerged at the end of the 1990s and again from 2016 with the introduction of fast transmission technologies.

With the introduction of faster data technologies through the UMTS network⁵⁵, the basic infrastructure was in place to transport data sensibly and use ICT applications in addition to telephony. Currently, following the most widespread technology LTE (LTE+ with up to 1 mBit/s and latency of approximately 20 milliseconds), 5G networks are being established that enable speeds of up to 20 gBit/s and latency times of less than 20 milliseconds. This makes real-time applications, which are needed, for example, in the contexts of industrial production or autonomous driving, realistic. 5G is geared more towards industrial automation than towards digital society and thus enables a further step into the digital, fourth revolution.

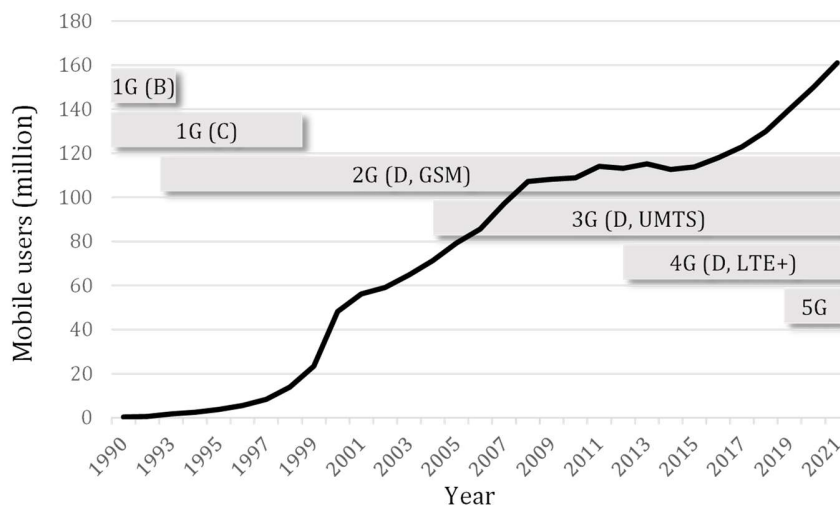


Figure 6 - Number of mobile phone users from⁵⁶ 1990–2020 in Germany and the generations of mobile phone technologies⁵⁷ (own representation based on secondary data)

The second element is the introduction of mobile terminals with data access at the turn of the millennium, initially the Blackberry with a fixed keyboard and screen. The company RIM⁵⁸ initially led the mobile terminal market with the device. With the introduction of the iPhone by Apple in 2007 and touch screens including a soft keyboard, the market was

⁵⁵ 7.2 mBit/s up to 42 Mbit/s, previously 52.6 kBit/s and with EDGE technology up to 220 kBit/s

⁵⁶ Source: Bundesnetzagentur (2022)

⁵⁷ G for generation of mobile radio technologies, additional designations in Germany such as A-network (1958–1977, analogue network with 10,500 subscribers), B-network (1972–1995, analogue network with 27,000 subscribers) and C-network (1986–2000, analogue network with 850,000 subscribers)

⁵⁸ Research in Motion, later Blackberry

revolutionised. Already in 2010, Apple had a market share of 33% with the iPhone (Blackberry 16%). As unit costs decreased, mobile devices were able to scale. In May 2012, mobile devices had a global market share of only 10% compared to desktop computers; in November 2014, this had increased to 30%, and in June 2019 to 50%. Today, Asia is well above 60% (60–65%), North America is in line with the global average and Africa fluctuates between 58–62%. The European average is below 44%⁵⁹: Germany (<45%), Czech Republic (<35%), Austria (<46%). Worldwide, there were 2.5 billion mobile devices in 2016; there were 2.9 billion in 2018 and 3.8 billion in 2021. This means that today, almost half of the world's population owns a mobile device⁶⁰. As of 2016, more than half of the internet was accessed via mobile devices. This shows that the mobility of knowledge, data management and networking, in both society and the economy, is unstoppable.

The final and perhaps most important element for decentralisation and mobilisation is the penetration of the internet in society. There are hardly any areas of life in which the use of internet-based platforms is unthinkable. Commerce, mobility, education, sports, media, medicine and communication are dominated by internet-based tools, applications and databases. Of the 50 most visited⁶¹ websites, 18 are based in China or Hong Kong and 21 in the US. Only two are based in Europe. In Germany, internet usage has increased by 50% in the last 20 years. In 2019, 86% of the population of Germany used internet content at least occasionally, and 74% also had mobile access.

In summary, the decentralisation of and increase in computing power, knowledge and networking created a new precondition for innovation and progress compared to the previous stages of the Industrial Revolution. The fourth stage can thus be clearly differentiated from the previous three. Users' mobile devices, the sensors in many electronic devices and the digitalisation of production and logistics chains have enabled a high degree of digital networking, which lays the foundation for decentralised self-organisation and reduces the relevance of higher-level control. Whereas previously overarching production control systems were relevant in industrial companies, and production steps and goods were passive components of the production and logistics process, with the help of today's technological possibilities, production goods, raw materials and semi-finished or finished products actively control the production and logistics process through self-identification and status reporting. In the past, the production process reported the completion of a production good for forwarding to the logistics process; today, it is possible for the production good to 'do it on its own'. The past delivery processes were designed, optimised and handled as a central, static specification; today, the goods involved in the delivery process can localise and optimise themselves independently (Wagner, et al., 2017, pp. 128–131). Cyber-physical systems have emerged in which the 'emerging product thus controls the production process itself, monitors the relevant environmental parameters via embedded sensor technology and triggers appropriate countermeasures in the event of malfunctions - it becomes both an

⁵⁹ Source: Statcounter (2022)

⁶⁰ Source: Statista (2023)

⁶¹ Determined on the basis of time of use, number of page visits and amount of data used.

observer and an actor' (Kagermann, et al., 2011). The availability of high computing power, the increase of mobile devices and real-time communication are creating entirely new markets where supply and demand converge on sharing platforms and threaten old industrial assets. Digital assets are on par with or even displacing physical ones.

The term *Industry 4.0* was first presented to the German government on 1 April 2011 by the Promoter Group Communication of the Research Union Industry – Science⁶². The aim was to make the German economy more competitive as a manufacturing location by means of networking and internet technology: 'Now we need to take the next step towards the Internet of Things in the industrial environment so that Germany becomes the leading provider in this new market by 2020' (Kagermann, et al., 2011). Elements of networking the production chain, from demand to the interaction (Broeders, van den Berg, 2020) of distributed production steps in the 'smart factory' to the delivery of a product, are essential drivers in the new, cyber-physical system (Bauernhansl, et al., 2014, pp. 14–30). Smart production control means that systemic management theories, such as constraint theory, must be fundamentally rethought (Hartmann, Halecker, 2015, p. 237). On this theoretical basis, performance improvements in production processes can be achieved through buffer management, and effects can be observed in automation technology, process industry, logistics, human-machine communication and ICT security (Kuo, et al., 2021).

Schwab summarised the reasons and effects that justify the term *fourth revolution*. The speed of change is no longer linear but exponential; almost every industry in every country is being disrupted, and the entire industrial system is changing in terms of leadership, production mechanisms and management systems (Schwab, 2018). He also cited the high mobile availability of computing power, access to knowledge and the convergence of the biological, physical and digital worlds as key features of this change. Within this context, interactions with governments, ministries and authorities can also be found. While these still act in parallel to the second stage of the Industrial Revolution (i.e. mechanically and processually sequentially driven), citizens (i.e. civil society) and industry are developing quickly. On the one hand, civil society is gaining more access to information and can contact an administration directly or organise itself via platforms; on the other hand, technological progress is giving the state ever greater access to citizens' data. Centralised control is taking a back seat to the decentralisation of knowledge and power. The decisive factor for government systems will be how flexibly they can adapt to this new, volatile, complex and ambiguous world. Administrative processes are still characterised by centralised control, concentration of knowledge and power, and the appropriation of information relevant to decision-making in order to guarantee reliable administrative action (Rölle, 2020, p. 144). This leads to slower processes than required by industry and civil society in the fourth Industrial Revolution. Future administrations will only be successful if they adapt to the faster environment by maintaining independent action and learning to deal agilely with the

⁶² Among others, Professor Kagermann, acatech – German Academy of Science and Engineering e. V.

uncertainty and volatility that has arisen. Only through this ability to evolve can governments ensure a necessary innovative capacity in the future.

Today, citizens are largely digitised. They use internet platforms and online services, share (sometimes very private) data on social networks and use cloud services and assistants to manage their private lives. In Europe, 82%⁶³ of the population use the internet. In Germany, 60%⁶⁴ of the population use online banking, 54%⁶⁵ use social networks and 56%⁶⁶ search online for information about public authorities.

Such user behaviour is becoming an attitude of entitlement in all areas of life, including the contact between citizens and the state and administration. In this respect, the term *State 4.0* will have even greater relevance in the future for establishing connectivity to the fourth Industrial Revolution and the associated challenges regarding functionality, sovereignty and ICT security (Deckert, 2020). Thus, the implementation of a digitalisation strategy at the state level makes sense. Management methods⁶⁷ for implementation can be applied, and influence can be exerted on the implementation of digitalisation in companies, as has already been established in other policy areas, such as transport, environment and energy (Budzinski, Eckert, 2018).

2.2.5. Industrial revolutions and quantitative characteristics; secondary data

In the following, select key figures that can be explained and evaluated with at least one stage of the Industrial Revolution are described. These can be the basis for quantifying digital sovereignty, but they may not be. So, there is 'no golden rule'. The specific selection of secondary data is described in Sections 5.4.2 and 5.4.3. The following consideration does not claim to be exhaustive. The countries of the EU, China, the US, Korea and Japan, as well as other relevant countries, are compared.

2.2.5.1. Population

The Industrial Revolution enabled the cheaper production of food and clothing, an improvement in working conditions and the avoidance of structural famines. This had an impact on population growth. Global per capita GDP grew roughly identically with population (Brynjolfsson, et al., 2018, pp. 17–20). Population growth was the strongest in the country of origin of the first Industrial Revolution: 'While the total population of Europe had grown to less than one and a half times its original size in the period 1800 to 1850, the population of England had doubled in the same period (from 8.9 to 17.9 mill.)' (Kulischer, 1988, p. 27).

⁶³ Source: Destatis (2022), max 99% in Iceland

⁶⁴ Source: Destatis (2022)

⁶⁵ Ibid.

⁶⁶ Ibid.

⁶⁷ E.g. Mintzberg's 5P's: plan, pattern, position, perspective, ploy

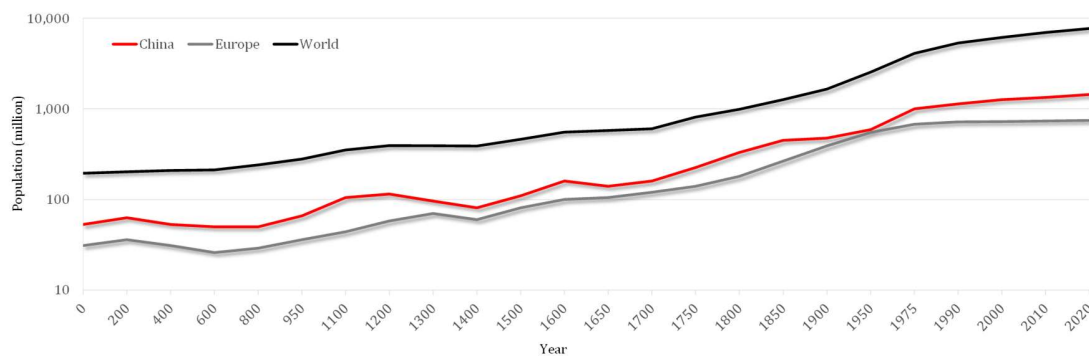


Figure 7 - Population development in China, Europe and worldwide⁶⁸ from year 0–2020, logarithmic y-axis (own representation based on secondary data)

The second stage of the revolution led to greater prosperity through the emergence of mechanical engineering, chemical and electrical industries and automobile production and was thus the basis for further population growth (Bauernhansl, 2020, pp. 11–13).

Growth in developing countries, such as China, continued to be exponential, while population growth in Europe began to stagnate during the third stage of the revolution.

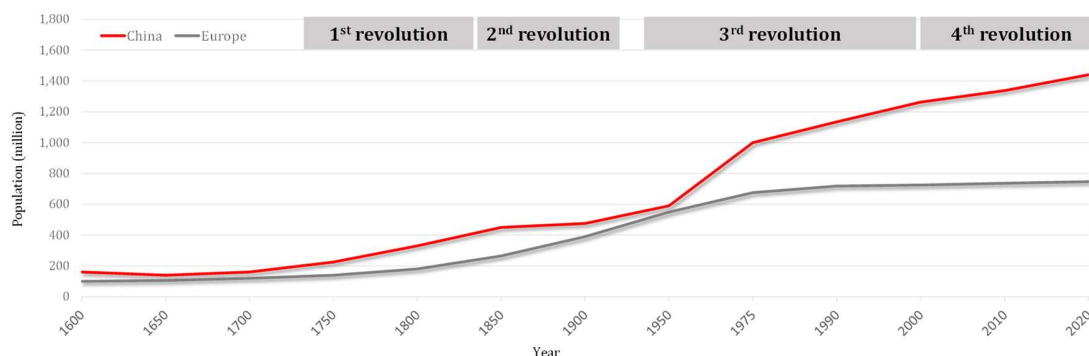


Figure 8 - Population development in China and Europe⁶⁹ during the four stages of the Industrial Revolution from 1600–2020 (own representation based on secondary data)

The concentration of industrial production in cities intensified the rural exodus to cities (Bauernhansl, 2020, p. 19).

2.2.5.2. Urbanisation

Since an large proportion of the population lives in rural areas, especially in a highly technologised country like China, it is important to look at rates of urbanisation. The first Industrial Revolution led to a decreasing importance of agriculture and motivated the population to move to cities with the prospect of better paid work. This effect continues today, and, by 2050, the proportion of urban dwellers is expected to rise to around 70%. (Bauernhansl, 2020, p. 19). A good analysis and description of population development and

⁶⁸ Source: Our world in data (2022) (world-population-growth)

⁶⁹ Ibid.

urbanisation from 10,000 BC onwards can be found in Klein Goldewijk, et al. (Klein Goldewijk, et al., 2010, 567ff).

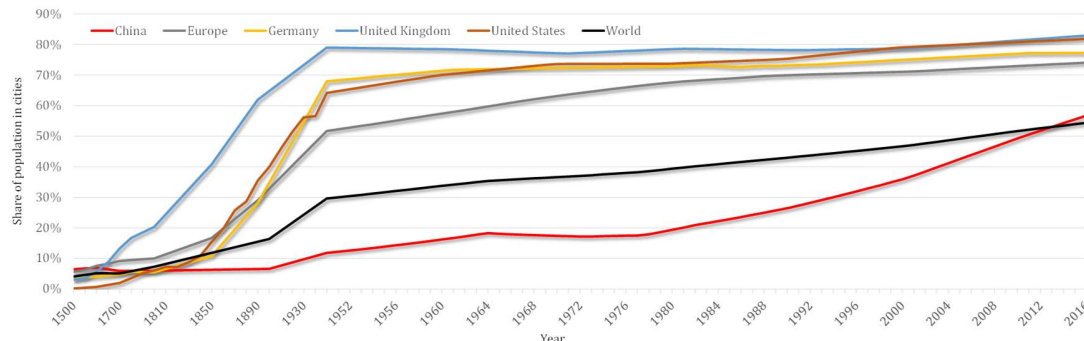


Figure 9 - Urbanisation⁷⁰ of selected regions 1500–2016
(own representation based on secondary data)

A detailed look at Prussia and Germany shows a shift in employment structures and the resulting urbanisation during the second stage of the revolution. Employment structures, including economic power, shifted significantly from agriculture, forestry and fishing towards trade, services and public services beginning at the end of the 18th century (Aubin, 1976, p. 528). This affected export volumes and goods as well as rural exodus; in 1871, only 36.1% of Germany's population lived in cities, but by 1910 this figure had risen to 60% (Reulecke, 1992, p. 202).

2.2.5.3. Unemployment

The question of whether technological or process innovations reduce the demand for human labour—so that not only individual occupational profiles but entire economies are affected by technological upheavals in such a way that unemployment rises—is the subject of much discussion (Buhr, et al., 2017, p. 112). The term *technological unemployment* was introduced by Keynes: 'We are being afflicted with a new disease of which some readers may not yet have heard the name, but of which they will hear a great deal in the years to come—namely, technological unemployment. This means unemployment due to our discovery of means of economising the use of labour outrunning the pace at which we can find new uses for labour.' (Keynes, 2010, p. 324). He saw technological development as both a risk and an opportunity and described this type of unemployment as temporary (Möller, 2016). If labour substitution through technology leads to inelastic demand (i.e. if the demand for goods does not increase due to falling prices, for example in saturated markets), technological unemployment follows (Appelbaum, Schettkat, 1995). If, however, relative price reduction leads to a disproportionate and thus elastic demand, the effect of labour substitution is overcompensated and, in the opposite direction, leads to lower unemployment (Möller, 2016, 51ff). This theory promoted the emergence of the Phillips curve, which establishes a dependence between unemployment and inflation. It is undisputed in the scientific

⁷⁰ Source: Our world in data (2022) (Urbanization last 500 years)

literature that technological development has an impact on the labour market, but the existence of structurally based unemployment and the systematic, lasting effects of technological development on unemployment are disputed. In this regard, reference is made to various discussions on the relevance of, for example, the introduction of agricultural machinery, computers and other new technologies on the destruction of labour (Rifkin, 1998) (Brynjolfsson, et al., 2018) (McGaughey, 2018).

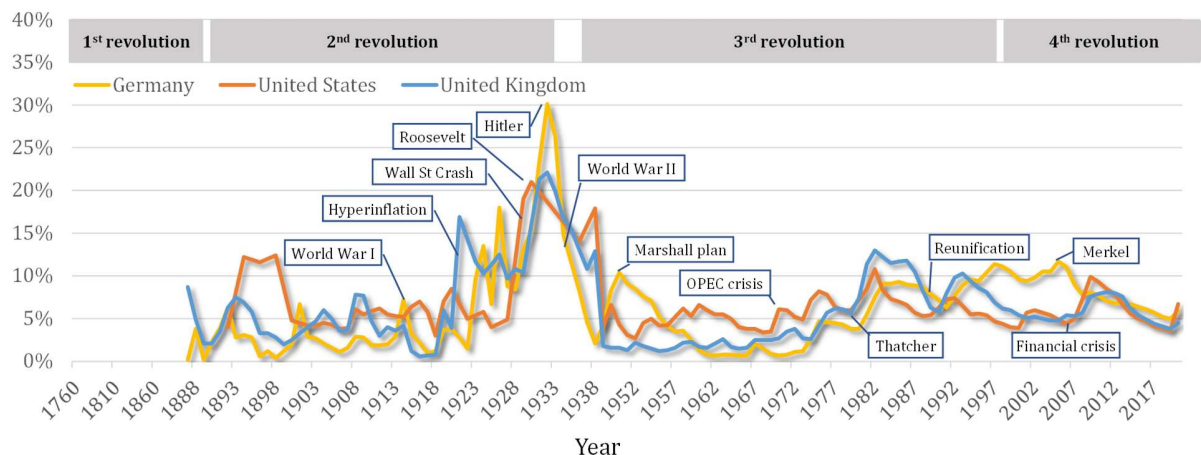


Figure 10 - Unemployment rates⁷¹ in the US, the UK and Germany 1887–2020, including individual events (own representation based on secondary data and McGaughey, 2018)

Systematically recorded unemployment figures have been available since the end of the 19th century, but their development over 140 years shows no direct, systematic, long-term correlations between technological innovation and unemployment. Triggering moments relevant to the unemployment rate were cyclical (e.g. recession) or political (monetary policy, wage bargaining systems) effects or wars. Nevertheless, it is worth examining possible countermeasures regarding technological unemployment, as this has a direct influence on cost increases in a national economy. The concept of the NAIRU⁷² provides an explanation by defining an inflation-stable unemployment rate: the ratio that may not be exceeded in order to not impact inflation. The basic concept is that when inflation rises, prices rise, real wages and thus labour costs fall, and companies hire more staff. The effects are opposite when inflation falls. If nominal wages, which tend to be contractually bound for longer periods, were adjusted to real wages in a timely manner, this effect would not occur.

This value can be an indicator for economic policy measures, for example, in the sense of a stabilisation policy, to determine whether unemployment is subject to cyclical fluctuations or extraordinary events (e.g. financial market crisis, the Covid-19 pandemic or the war against Ukraine) or is of systematic origin (e.g. technological innovation). Economic policy instruments can therefore be adjusted. In his post-Keynesian model, Bontrup

⁷¹ Sources: Destatis (2022), Mitchell (1975, 166 ff)

⁷² Non Accelerating Inflation Rate of Unemployment, a further development of the (Keynesian) Philips curve that explains the relationship between inflation and unemployment, with the NAIRU as the intercept $y = 0$, i.e. the unemployment rate not accelerating inflation

proposed regulated, centrally organised ‘wage bargaining systems’ to stabilise wage costs and inflation (Bontrup, 2006, p. 6). However, it is controversial to regard the NAIRU as the sole indicator of the reduction of structural unemployment (Heimberger, et al., 2017). Based on economic theory and empirical evidence, technological development only temporarily leads to structural unemployment. However, it remains unclear whether the (unobservable) effects of the unemployment rates in the previous three stages of the Industrial Revolution apply to the fourth stage, or whether the use of digital and disruptive technologies simultaneously and rapidly affect so many industries that many activities that previously required human labour are replaced, or if the use of these technologies is even leading to a ‘positive job balance’ (Brynjolfsson, et al., 2018, 199ff) (Fend, Hofmann, 2022, pp. 305–331). The founder of the World Economic Forum has predicted that the establishment of cyber-physical systems will not lead to the fundamental destruction of labour, but it will have a disruptive effect on the labour market through substitution (Schwab, 2018). However, interdependence analyses of unemployment rates and digitalisation, and especially digital sovereignty, cannot be found in the literature. This also applies to edge technologies such as Generative Pre-trained Transformer and Artificial Intelligence (AI) (Eloundou, et al., 2023). However, interdependence analyses of unemployment rates and digitalisation, and especially digital sovereignty, cannot be found in the literature. A forward-looking labour market policy will continue to be an important economic policy success factor to counteract possible effects of the digital stage of the industrial revolution.

2.2.5.4. *Gross domestic product*

Influenced by relevant economists⁷³, gross domestic product⁷⁴ (GDP) has been considered a measure of a state’s economic performance since the 1940s. The Maddison Project examines economic statistics over periods of the Industrial Revolution, including the wealth of states in terms of per capita GDP (Maddison, 1983). Here, and also by Lorberg, it was empirically proven that Western Europe ceded leadership to the US in the second stage (Lorberg, 2018b, p. 146) and they kept it for a long time, while the countries of Asia (with the exception of Japan) played a subordinate role. Productivity in the US increased tenfold between 1870 and 1973 (Maddison, 2003, p. 379). The prosperity of Germany, the US and the rest of the world developed largely identically in terms of per capita GDP between 1913 and 1972, although there were deviations in the catch-up phases after the First and Second World Wars.

⁷³ Keynes, among others

⁷⁴ Alternatively, the state quota as a parameter for the state’s capacity to act, source: Our world in data (2022)

Table 1 - Growth factor per capita GDP, (GK\$⁷⁵) (Lorberg, 2018a, p. 150)

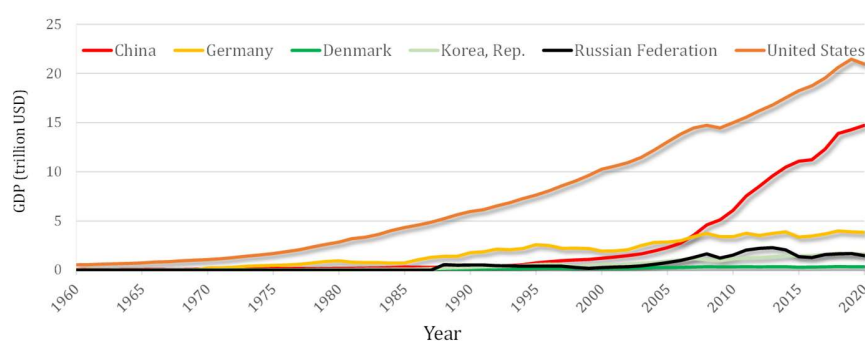
Growth factor	Germany	Western Europe	US	World
1913–1973	3.28	3.22	3.15	2.64
1973–1998	1.51	1.58	1.61	1.40
1913–1998	4.94	5.07	5.07	3.71
1913–2010	5.66	5.82	5.75	5.06
1913–1950	1.06	1.32	1.80	1.36
1950–1973	3.08	2.44	1.75	1.94

The deviations in absolute GDP values show in the high differences in population growth; while the population roughly doubled in Europe between 1870–1998, it grew sixfold in the US.

Table 2 - Population growth factor 1870–2010 (Lorberg, 2018a, p. 149)

Population growth	Germany	West Europe	US	World
1870–1913	1.66	1.40	2.43	1.41
1870–1950	1.74	1.58	3.78	1.99
1870–1973	2.01	1.85	5.27	3.08
1870–1998	2.09	1.93	6.72	4.65
1913–1973	1.21	1.32	2.17	2.19
1973–1998	1.04	1.04	1.28	1.51
1913–1998	1.26	1.37	2.77	3.30

National GDP values have been systematically collected since the 1960s. It is striking that the growth of global GDP in the second half of the 20th century was driven by industrialised countries, such as the US, Germany and Japan (Lorberg, 2018a). The GDPs of select countries are presented as examples.



*Figure 11 – GDP of China, the US, Germany, Denmark, Korea and Russia⁷⁶
(own representation based on secondary data)*

⁷⁵ 'Geary-Khamis dollar', a fictitious currency unit with a value equal to the purchase value that the US dollar had on a certain date (here 1990)

⁷⁶ Source: Worldbank (2022) (GDP)

While European countries, Russia and Korea show only moderate growth rates from the 1990s onwards, GDP has grown steadily in the US and disproportionately in China since 2000—during the third and fourth stages of the Industrial Revolution.

Furthermore there is a direct correlation between degree of digitalisation and per capita GDP: the E-Government Development Index (EGDI) was compiled by the UN and describes the degree of digitalisation of a country on the basis of the Online Service Index, the Telecommunications Service Index and the Human Capital Index. The top countries are Denmark, Finland, the Republic of Korea, New Zealand, Iceland, Sweden, Australia, Estonia, the Netherlands and the US (Yera, et al., 2020) (United Nations, 2022). For the EU countries (with the exception of outliers Luxembourg and Ireland), the higher the per capita GDP, the higher the EGDI rank. However, Estonia has a very high EGDI rank despite having a relatively low per capita GDP (Peters, 2020).

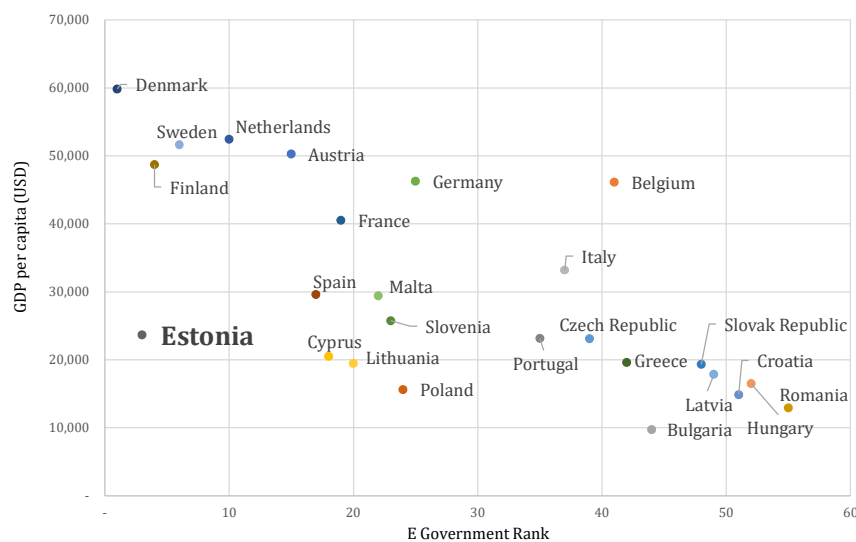


Figure 12 - EGDI rank and per capita GDP (2020, in USD), adapted from (Peters, 2020) (own illustration)

Peters found the reasons for Estonia's special positioning in a consistent, nationwide provision of fast internet (decided in 2009, implemented by 2015), the early and consistent positioning of digital relevance for the state since 1992 and the consistent exploitation of the 'comparative advantage of poor infrastructure', as the country was able to invest in a fundamentally new infrastructure with innovative technologies without the existence of established technologies. Essential for Estonia's reorientation was the 'exogenous shock' caused by the collapse of the Soviet Union and the state's endeavour to digitise all relevant processes of the country, including government activities, to allow Estonia to continue to exist, even without territory, in the event of a crisis. This included the establishment of a digital embassy in Luxembourg⁷⁷.

⁷⁷ Source: personal conversations of the author with, among others, the CTO of the Estonian Ministry of Economy in Tallinn on 11 August 2022

2.2.5.5. Foreign direct investment

One indicator of multinational trade is foreign direct investment (FDI), which is the ability or willingness of private or state investors to invest in business enterprises or state structures of other nations⁷⁸ in order to exert influence on the performance of the investment subject. They are an important indicator of globalisation. Net FDI inflows are direct investments made domestically by non-resident investors in an economy. Net FDI outflows are the value of direct investment abroad by the citizens or institutions of the reporting economy. They can serve to open new sales markets, take advantage of optimal production locations and conditions, diversify an investment portfolio and avoid exchange rate risks. Negative values for foreign direct investment arise when outflows exceed inflows, for example when liabilities are repaid or through disinvestment or reinvestment (Kogut, 1983). Within 20 years, the FDI volume grew tenfold worldwide (1980: approximately USD 700 billion; 2002: approximately USD 7,122 billion) (Krüger, Ahlfeld, 2005).

The level of FDI shows a correlation with the use of digital technologies. Empirical evidence does not show that the use of ICT to reduce transaction and production costs increases FDI inflows (Narula, Driffield, 2012). Furthermore a correlation between investments in ICT and FDI in 23 countries between 1976–1999 was examined; it could not be proven that higher FDI leads to an increase in GDP (Garcia-Murillo, Vinod, 2005). However, ICT infrastructures have to be fortified to enable FDI, which in turn leads to an increase in investment in ICT (Gholami, et al., 2005) (Kogut, Chang, 1991).

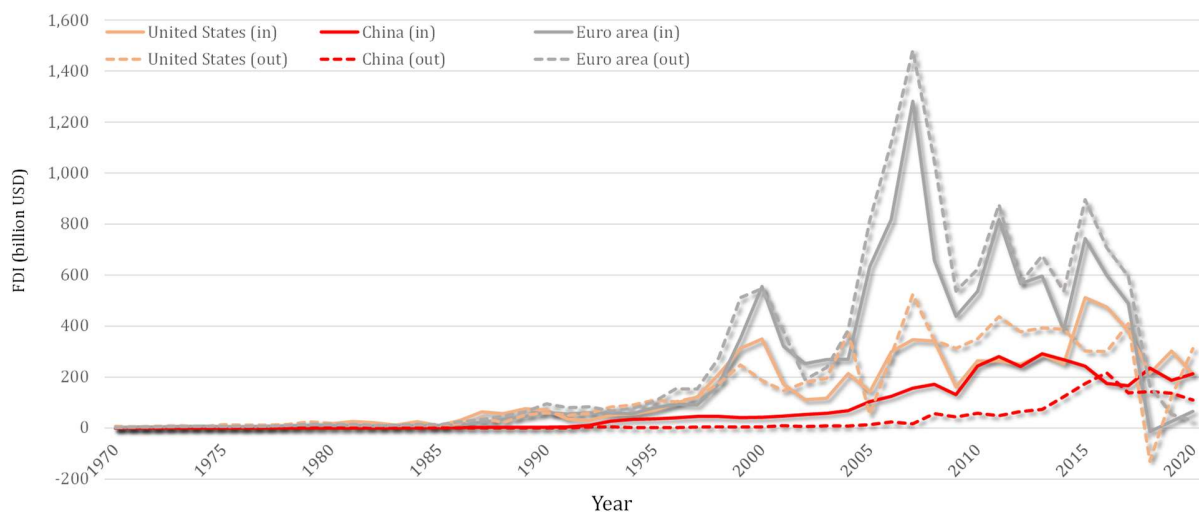


Figure 13 - Foreign Direct Investment⁷⁹ (own representation based on secondary data)

Anand and Kogut, in their correlation analysis of technology and FDI, identified two types of characteristics for countries that attract FDI. The first is the resident technology of the recipient country; the availability of superior technology in the recipient country motivates investors. The second is the market attractiveness of the recipient country, especially if there

⁷⁸ Minimum participation 10%

⁷⁹ Source: Worldbank (2022) (net inflows, net outflows)

is not a high density of competition there (Anand, Kogut, 1997). In an earlier analysis focusing on Japan and the US, Kogut and Chang observed pull effects only in joint ventures (Kogut, Chang, 1991). This observation also applies to the fourth stage of the revolution. Adedoyin et al. added that economies that exhibit a high degree of ICT maturity and are open to exchange are more likely to attract direct investment than closed economies (Adedoyin, et al., 2020, p. 4). With regard to Europe, FDI is predominantly in digital projects.

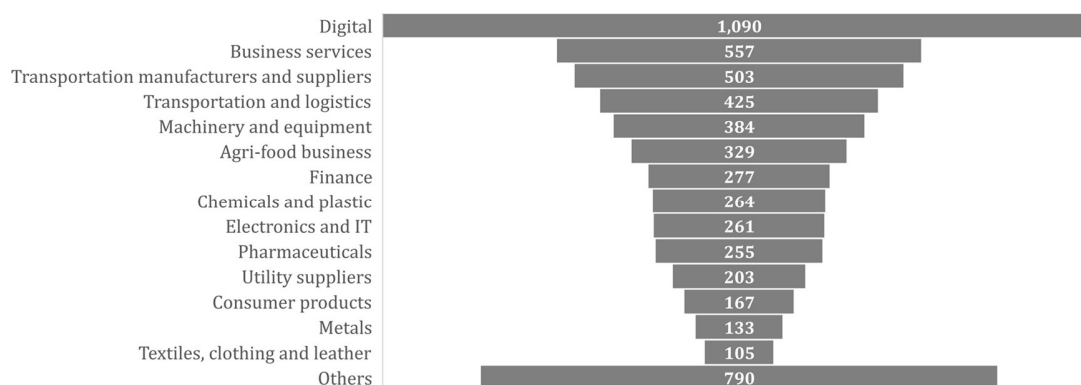


Figure 14 - Number of FDI projects⁸⁰ in Europe (own representation based on secondary data)

With a strong digital economy, imports, exports and direct investments are a feature of sovereignty and a high capacity to act. It may seem contradictory that high imports of ICT goods can be a characteristic of technological independence. At this point, it is referred to the discussion on the Digital Dependency Index (cf. 5.15.3): countries with high volumes of ICT-goods imports also have high degrees of technological independence. With a weak digital economy, a strong surplus of imports or FDI inflows would likely be problematic. However, since the focus of FDI in Europe is on digital projects, high FDI values are interpreted as a sign of high sovereignty.

2.2.5.6. Bureaucratisation

With regard to the relevance of bureaucratisation in the Industrial Revolution (cf. 2.2.1), it can be an important factor in measuring digital sovereignty. Measuring bureaucratisation can be done in different ways. For example, the World Governance Index used the Ease of Doing Business Index (EDBI) to measure how (un)complicated it is to start a business in 181 countries (World Bank, 2008). The EDBI used 41 input parameters to compare countries' Start-up environments; it is often used for regulatory purposes. However, the consistency and validity of the EDBI have also been criticised, and the World Bank's de facto regulatory power with the index is controversial. The index measures the extent to which business Start-ups that can contribute to the creation of new technologies or services, including digital ones, contribute to a country's digitalisation capacity—the higher the score, the easier it is to start a business. The index was collected by the World Bank until 2021, but The EDBI was

⁸⁰ Source: Ernst & Young (2022, p. 24)

discontinued due to allegations of manipulation⁸¹. The Economic Freedom Index can also be used as an alternative to the EDBI⁸².

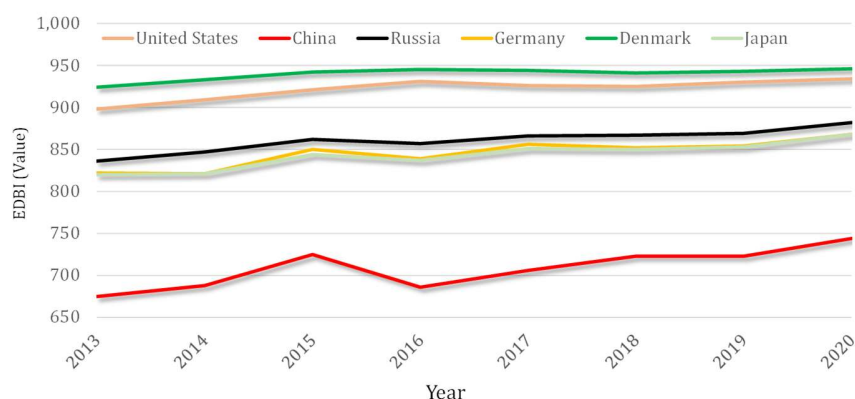


Figure 15 - EDBI⁸³ for select states⁸⁴ (own representation based on secondary data)

Another criterion to show bureaucratic barriers at the state level is the Services Trade Restrictiveness Index (STRI), which the OECD measures in the categories of logistics, accounting, law, air transport, maritime transport, road freight transport and rail freight transport, among others (OECD, 2020c) (Geloso Grosso, Massimo et al., 2015). The STRI describes which elements promote or restrict globalised trade and serves as a stand-alone metric to measure trade restrictions (cf. 2.2.6.3).

2.2.6. Digitalisation in numbers; secondary data

The previous chapter focused on parameters that can be assigned to the stages of the Industrial Revolution. The third (digital) stage gave rise to further indicators that describe the degree of digitalisation of states and societies as well as of digital effects. Select data that can serve as a basis for measuring digital sovereignty are presented below. It is important to note that when considering an index, the selection of secondary data itself is always going to be discussed in a critical way. In the later part of the dissertation it becomes even more important to verify and validate the index to be constructed. The specific data selected for the index are described in Sections 5.4.2 and 5.4.3.

2.2.6.1. Households with a PC and/or internet

The participation of private households in ICT developments can be an indicator of ICT affinity; however, the explicit ICT literacy level of the individual cannot be seen as such. The indicator describes the population's access to information and the basis for using digital

⁸¹ Investigation of Data Irregularities in Doing Business 2018 and Doing Business 2020, Worldbank (2022)

⁸² Fraeser Institute (2022)

⁸³ Source: Worldbank (2022)

⁸⁴ Not available for Korea, therefore replaced by Japan as a comparative value in this case

services. In the 1980s, PC use in households was unique to the United States, with other countries following suit in the early 1990s. In the Western industrialised countries, private internet use followed beginning in the mid-1990s; countries such as China and the Baltic States followed after a five-year gap but caught up quickly. It is striking that, in China, internet use differs greatly between urban and rural populations; in 2020, 46.2% of the rural population and 76.5% of the urban population used internet services⁸⁵.

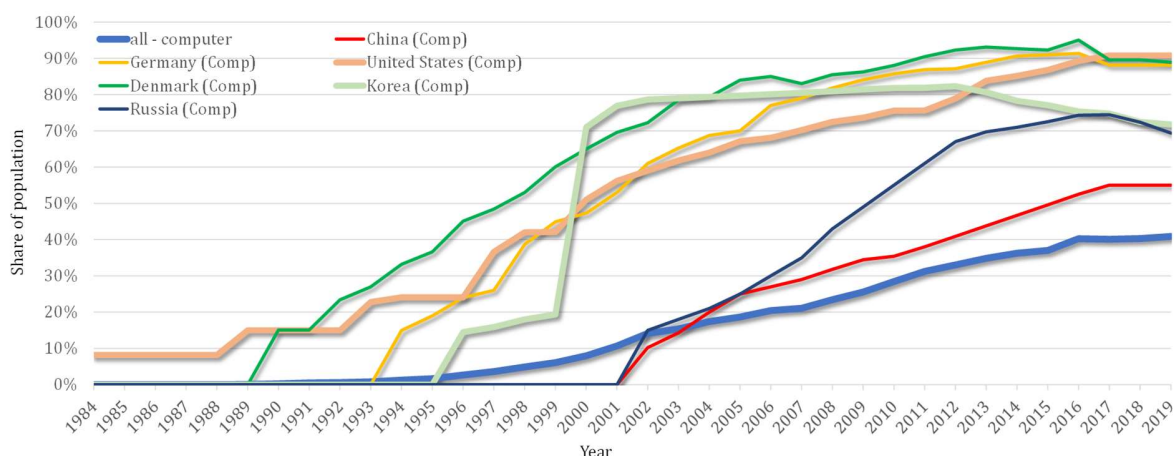


Figure 16 - Computer use of private households⁸⁶ (own representation based on secondary data)

In the industrialised countries, the availability of PCs and internet services in private households is nowadays over 85%.

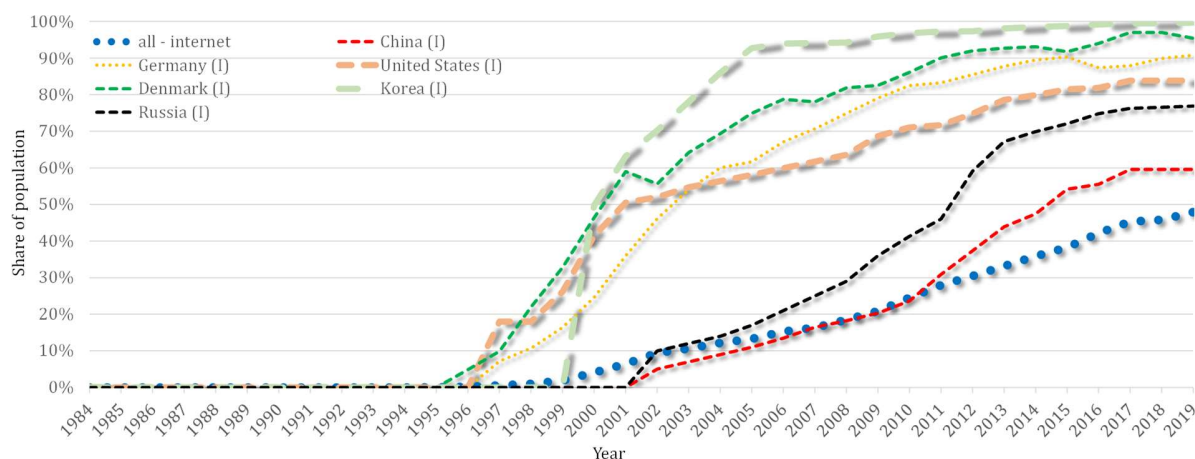


Figure 17 - Internet usage of private households⁸⁷ in China, Germany, the US, Denmark, Korea and Russia (own representation based on secondary data)

⁸⁵ Source: Statista (2023)

⁸⁶ Source: ITU (2022) (households with a computer)

⁸⁷ Source: ITU (2022) (households with internet access at home)

2.2.6.2. Trade in ICT goods

With the onset of the Industrial Revolution, inter-national trade took on significant importance (cf. 2.2.1). Importing cheaper (raw) materials was and is a success factor in a globalised world, even if discussions about de-globalisation are generally on the rise due to a destabilising geopolitical situation (Jordaan, 2022). However, what happens when goods that are relevant to critical infrastructure cannot be produced independently in sufficient quantities? A recent example is the availability of face masks at the beginning of the Covid-19 pandemic. In 2019 many states were unable to equip their populations with sufficient protection as this previously unimportant commodity quickly became systemically relevant. Another example is the high dependence of Western states on gas from Russia. However, what about more complex technology or ICT services? What is the relevance of dependence on the production goods of other states? Can this be measured? Biryukova and Matiukhina examined the role of the export of ICT services for Brazil, Russia, India, China and South Africa. They concluded that the use of imported services led to higher dependence and fewer development opportunities for these countries (Biryukova, Matiukhina, 2019). Using China as an example, they showed that by focusing policy on the digital industry, the growth of this sector significantly exceeded that of the GDP. Hardware products accounted for the largest share of profits (approximately 80%) and represented 75% of the value added in the ICT sector, while ICT services accounted for 25%⁸⁸. India, as another example, had a restrictive import and export policy until 1984 when trade restrictions were eased, and since the turn of the millennium, the state has intensively promoted the ICT hardware and software industry. A key component was the allowance of FDI. For both countries, the availability of telecommunications services, in the broadband fixed-network as well as the mobile sector, was a growth factor for the successful ICT service sector. The presentation of select countries for the export of ICT services shows that the volume increased significantly at the turn of the millennium.

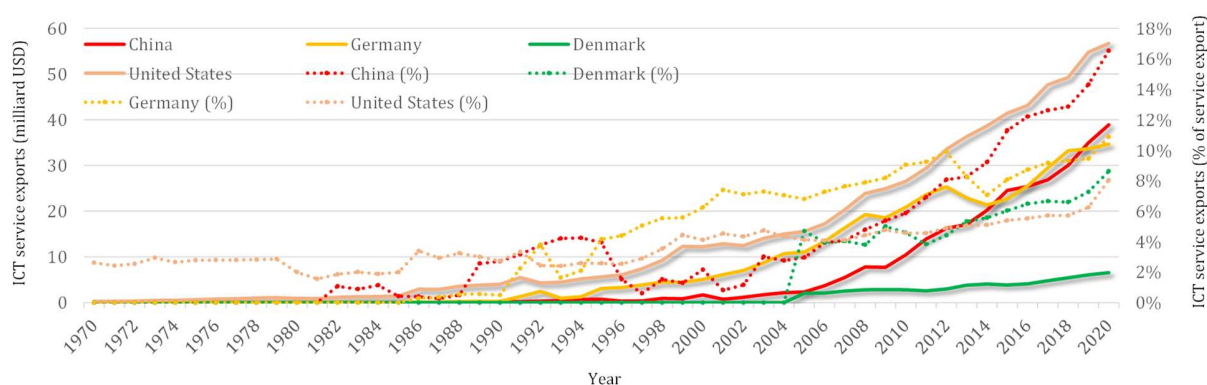


Figure 18 - ICT-service exports⁸⁹ (own representation based on secondary data)

⁸⁸ Source: Worldbank (2022)

⁸⁹ Source: Worldbank (2022) (ICT service exports, BoP, current US\$)

2.2.6.3. Freedom of trade for digital services

Complementary to the STRI trade barrier measurement, digital aspects of trade barriers are measured with the digital STRI (OECD, 2020a), which ‘identifies, catalogues and quantifies barriers to trade in digitally enabled services’. This information is collected for 46 countries in the sub-indices of ‘infrastructure and connectivity, electronic transactions, payment systems, intellectual property rights and other barriers’ that impede ‘trade in digitally enabled services’ (e.g. restrictions on online advertising). The index can be used to identify regulatory bottlenecks and assess the competitiveness of markets for digital trade (Ferencz, 2019). An index value of 0 means fully open; an index value of 1 indicates fully restrictive digital trade.

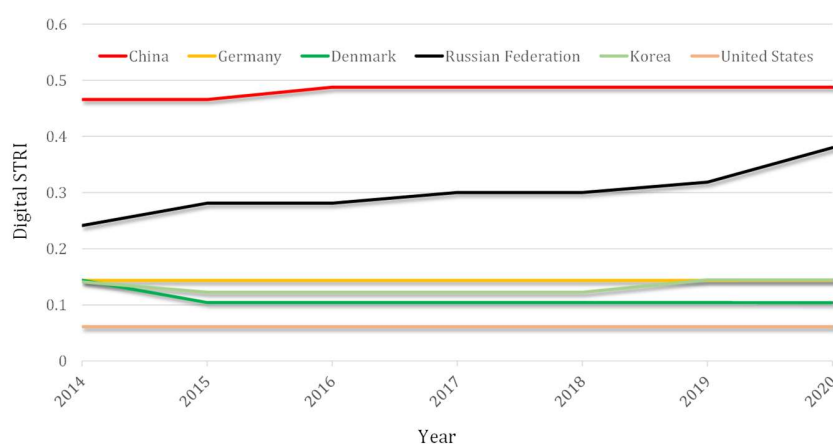


Figure 19 - Digital STRI⁹⁰ (own representation based on secondary data)

Based on select countries, it can be seen that China has very high trade barriers for digital services, while Russia and India have high barriers. This indicates a strong protectionism of these states to tightly control inflows and outflows of digital trade. The US and the European states have low barriers (i.e. they are more open to digital trade); thus, they are more liberal and use fewer regulatory options to differentiate themselves digitally⁹¹.

2.2.6.4. (Education) Training

In the context of digitalisation, the question arises of the education and training needed to meet the challenges of the fourth revolutionary stage.

With regard to the first phase of the Industrial Revolution, virtually no evidence can be found that education played a relevant role in Britain’s leadership: ‘education [was] not a significant factor in England’s economic growth during the Industrial Revolution’ (Mitch, 1993, p. 307). Technological developments, however, create a need for education, and the new technologies introduced exogenously during the second stage of the revolution required a basic standard of education. Using Prussia as an example, Becker et al. demonstrated

⁹⁰ Source: OECD (2022b) (Digital Services Trade Restrictiveness Index)

⁹¹ The US restriction on Chinese technologies is likely to have an impact on the US digital STRI in the future.

quantitatively that regions were able to adapt particularly well to (also disruptive) technologies and participate in economic growth when there was basic education for citizens. An additional correlation has been demonstrated regarding regions that were particularly close to the capitals of England and Prussia. Formal education was not the most important factor during the early phase of the Industrial Revolution, but it increased in relevance.

Becker et al. suggested that formal 'education had a significant effect on aggregate industrialization in both phases of the Industrial Revolution in Prussia'. In the textile industry, which was less disruptive and mainly characterised by manual labour, the effect cannot be demonstrated. However, for the industries that changed significantly or involved the emergence of new technology, the effect was significant. Regions with educated populations were more aware of the potential of new technologies coming to Prussia from England and were better able to adapt to the associated changes. The high variation in education levels in the counties of Prussia was decisive for the success of the Industrial Revolution in this region. The higher the degree of change associated with a technology, the more schooling influenced the economic upswing. Education can therefore be a factor in catch-up development (Becker, et al., 2011, p. 120).

Peters and Jandrić discussed a form of education that needs to be changed, arguing that the (non-digitally oriented) *homo oeconomicus*, who was educated with a focus on competition and individualised performance, is being replaced by the (digitally oriented) *homo collaborans*, who lives in a shared economy and focusing on 'radical openness', 'collective intelligence' and the development of common ideas in education (Peters, Jandrić, 2019, p. 14).

Basic education no longer plays an essential role in qualifications in the fourth (digital) revolution. Instead, the vocational ICT qualification must be considered. Suitable indicators include the number of employees who have acquired ICT training or indicators that quantify the number of employees in the ICT sector. Secondary data in the same form are not available for the EU, China and the US. Eurostat provides comparable data for the EU and a few non-EU countries, while alternative data sources are used for the US and China.

Globally, most skilled ICT people live in China, followed by India and the US. Although the EU countries have caught up in the last 10 years with a growth of 52% and are comparable to the US with almost 3 million qualified ICT persons, China still dominates, with over 9 million qualified ICT persons.

Within the EU countries, Germany, France, Spain and Poland lead, followed by Italy and the Netherlands. The remaining EU countries have less than 100,000 ICT workers.

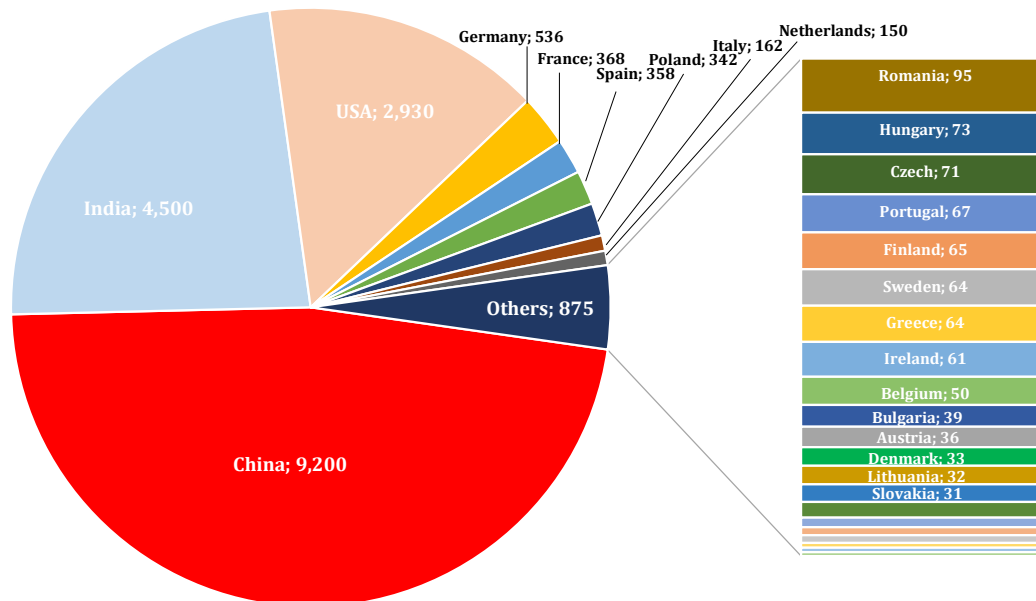


Figure 20 - Qualified employees⁹² in 1,000 in ICT (own representation based on secondary data)

Table 3 - Number of employed persons (in thousands) with ICT education, EU-27,
Source: Eurostat and CEIC (2022)

Year	EU-27 ⁹³	China ⁹⁴ (est.)
2010	1,815	n/a
2011	1,913	n/a
2012	1,869	8,200
2013	1,971	8,500
2014	2,117	8,800
2015	2,159	8,300
2016	2,339	9,100
2017	2,367	8,300
2018	2,528	9,150
2019	2,625	8,700
2020	2,703	7,300
2021	2,792	9,200

⁹² Sources: Eurostat (2022) (persons with ICT education by labour status), Statista (2023) (number of US IT workers)

⁹³ Source: Eurostat (2022) (persons with ICT education by labour status)

⁹⁴ Source: CEIC (2022) (number of employees in the computer, communication, other electronic equipment industries in China)

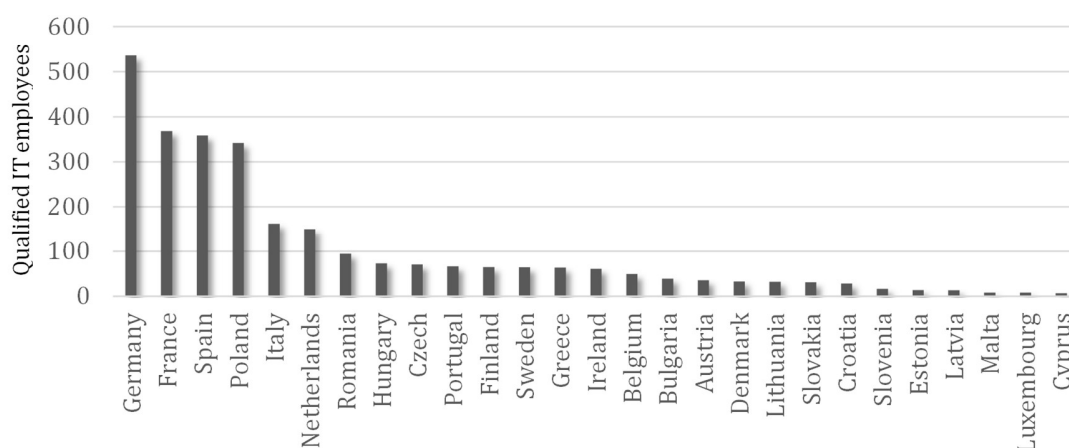


Figure 21 - Number of employed persons (in 1,000) with ICT education⁹⁵ in 2021, EU-27 (own representation based on secondary data)

Digital skills are a basic prerequisite for the digital capability of a state, and the EU states have some catching up to do. This also applies to digital sovereignty, as the citizens of a state must be digitally sovereign in order to shape a digitally sovereign state (Goldacker, 2017, p. 20).

Other indicators of the level of education of a country's citizens that can be significant in the context of digitalisation include the Trends in International Mathematics and Science Study (TIMSS), an international comparative school performance survey that has been conducted regularly since 1995; the Human Capital Index; the Assessment in Reading, Mathematics, and Science Index; the QS university ranking (average score of the top three universities) and the Expenditures on Education of the World Bank's WGI.

In addition to pure (university) education, it is also relevant how the transfer of research results or innovations to industry occurs (Nsanzumuhire, Groot, 2020). A study can be used to assess countries in terms of how private co-financing of public expenditure on research and development as a percentage of GDP or public-private co-publications per million inhabitants are implemented in the countries (Morisson, Pattinson, 2020). Another regional study focused on European states, measuring their potential for university-industry collaboration, and found an east-west divide in Europe. A quantitative criterion for cooperation between universities and industry can be the number of joint publications. If these are set in relation to the total number of publications and restricted to mathematically and information-technically relevant publications, a key figure for the degree of cooperation in the ICT environment can be determined. This secondary data can be found in the Leiden Index (van Eck, 2022).

⁹⁵ Source: Eurostat (2022) (persons with ICT education by labour status)

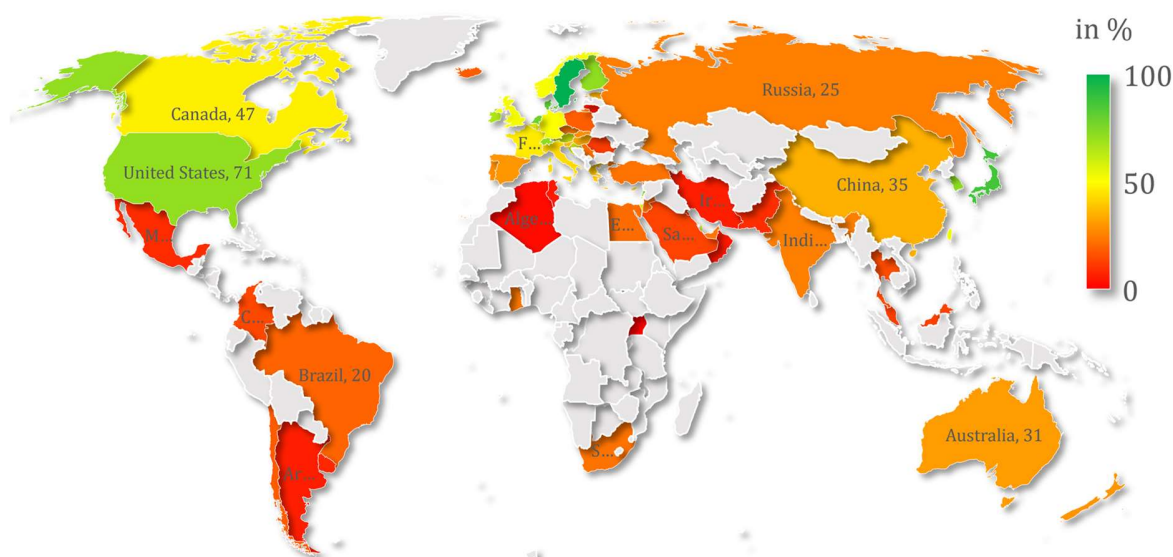


Figure 22 - Proportion of cooperative publications compared to the total number of publications⁹⁶ (ICT and mathematics, 2017–2020), normalised to 100 (own representation based on secondary data)

The list of the number of publications (mathematics and computer science) is headed by China (approximately 212,000), followed by the US (approximately 123,000) and Great Britain (approximately 38,000). The number of publications cooperating with industry is led by the US (approximately 10,000), China (approximately 8,500) and South Korea (approximately 2,500). The ranking of the quotient is led by Sweden (11.5%), Japan (10%) and Denmark (9.5%).

2.2.6.5. Degree of digitalisation of the administration

Discussion on state digital sovereignty raises the question of the degree of digitalisation of the state and its administrations. The UN's governments online services measures how digital the interaction between citizens and the administration is on four levels. The lowest level is that of simple online presence, in which authorities present themselves with simple, static internet representations; the highest level is that at which ministries and authorities interact with each other and with citizens in a bi-directionally networked manner, up to and including participatory decision-making. The model is widely discussed in the literature (Rorissa, et al., 2011) (Yera, et al., 2020).

The EGDI can be used to measure the digital maturity of a state and its administrations. It measures the online capability of the authorities, the status of the telecommunications infrastructure and the existing digital skills (cf. 2.2.5.4) (United Nations, 2022) (Yera, et al., 2020). At this point, reference should be made to Ma and Zheng's correlation analysis between citizens' satisfaction with a state's digital services and digital ranking (Ma, Zheng, 2019).

⁹⁶ Source: van Eck (2022)

The Digital Economy and Society Index⁹⁷ (DESI) for European countries can also be used as a measure. It consists of the following categories: ‘connectivity’ (‘fixed broadband’ usage, ‘fixed broadband coverage’, ‘mobile broadband’ and ‘broadband prices’), ‘human capital’ (‘internet user skills and advanced skills’), ‘internet usage’ (citizens’ use of internet services and online transactions), ‘digital technology’ integration (‘business digitalisation and e-commerce’) and ‘digital public services’ (e-government). It describes the digital performance and the development of digital competitiveness of EU member states (European Commission, 2022) (Anderton, et al., 2020).

2.2.6.6. Investments in Start-ups

The willingness to invest in Start-ups, the respective investment volume and the number of Start-ups in relevant key enabling technologies (KETs) can be an indication of how capable a country is or will be to participate in technology and innovation in the future (Ma, Downs, 2020). The US and China dominate in terms of number (445 and 117, respectively) and investment volume in Start-ups, with investment volumes of more than one billion USD⁹⁸.

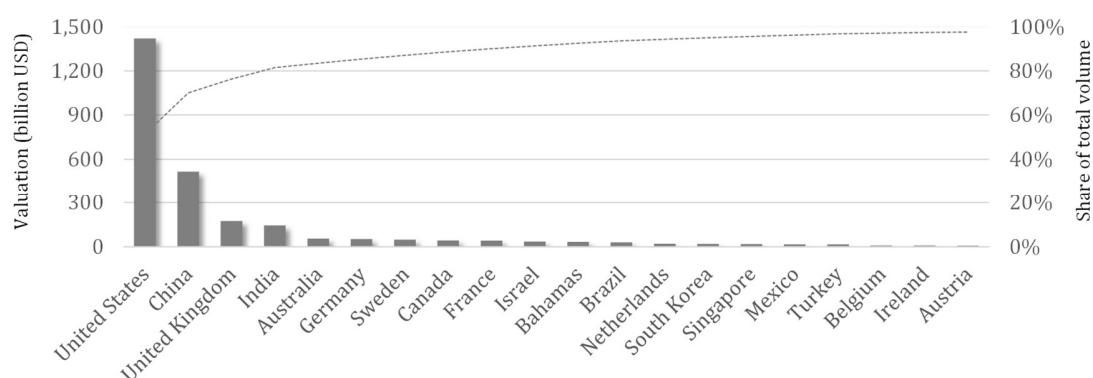


Figure 23 – Start-up Investments, Unicorns, ICT and Tech⁹⁹, 2014–2021, (own representation based on secondary data)

Germany and Sweden top the list of EU countries, with a total investment volume of about USD 50 billion (2014–2021).

While the EU states have been able to increase the annual investment volume from USD 55 million to USD 8.2 billion in the last 15 years, the US has increased the volume from USD 29 billion to USD 135 billion. In China, an investment volume of USD 130 billion is estimated¹⁰⁰.

A focus on Start-ups that produce KETs can be particularly profitable but must be in sync with a state’s capability profile. Räsänen used the key enabling technology quantum computing in Europe to examine how Start-ups and investments are dealt with. High investments in Start-ups without corresponding levels of qualification of skilled workers and

⁹⁷ Source: European Commission (2022)

⁹⁸ Source: CB Insights (2022)

⁹⁹ Ibid.

¹⁰⁰ Source: bloomberg.com (2021)

an existing manufacturing industry are not sufficient for successful establishment (Räsänen, Mäkynen, Möttönen, Goetz, 2021).

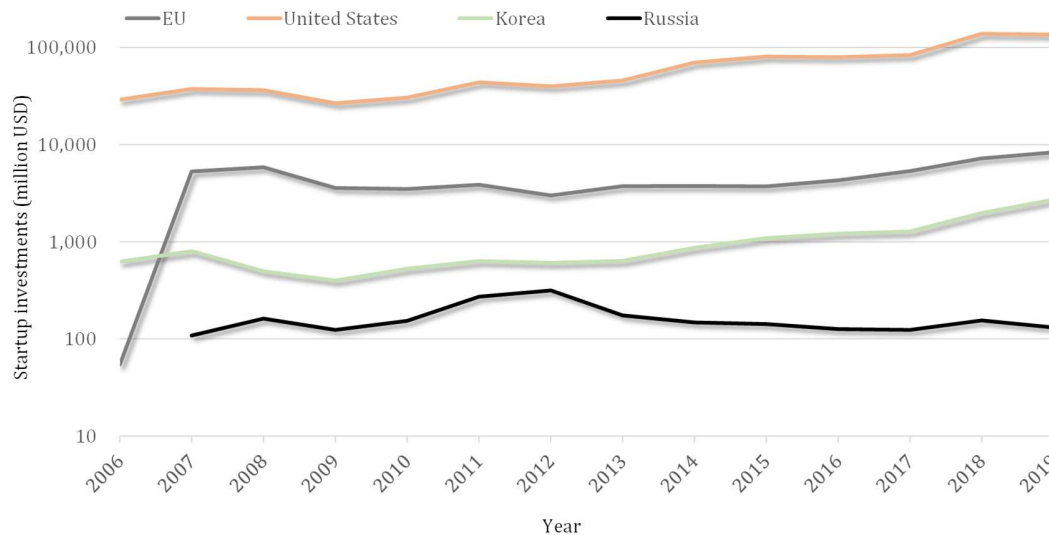


Figure 24 - Investments in Start-ups¹⁰¹ in the EU, the US, Korea and Russia; logarithmic y-axis (own representation based on secondary data)

In terms of technology Start-ups in general, China and the US invest in much higher numbers with corresponding volumes. The states of the EU remain far behind.

2.2.6.7. Digital competitiveness

The Institute for Management Development (IMD), as an independent academic institution with a focus on education and research, uses the Digital Competitiveness Index (WDC) ranking to assess the ability of countries to research and use digital technologies—that is, to provide transfer services from research to application. This evaluates the capabilities of countries to implement digital technology in order to strengthen industry, government operations and society a whole.

The primary assumption of the WDC ranking is that the transformation takes place at the state or private company level. The three main categories of knowledge, technology and sustainability are assessed, each of which is subdivided into three equally weighted subcategories that consist of a total of 50 individual criteria. Nineteen of the criteria are derived from survey-based primary data (e.g. agility of companies) or secondary data (e.g. internet bandwidth) and are shared with the World Competitiveness Ranking (WCR), which measures competitiveness in general (IMD business school, 2022, p. 30).

¹⁰¹ Source: OECD (2022b) (venture capital investments)

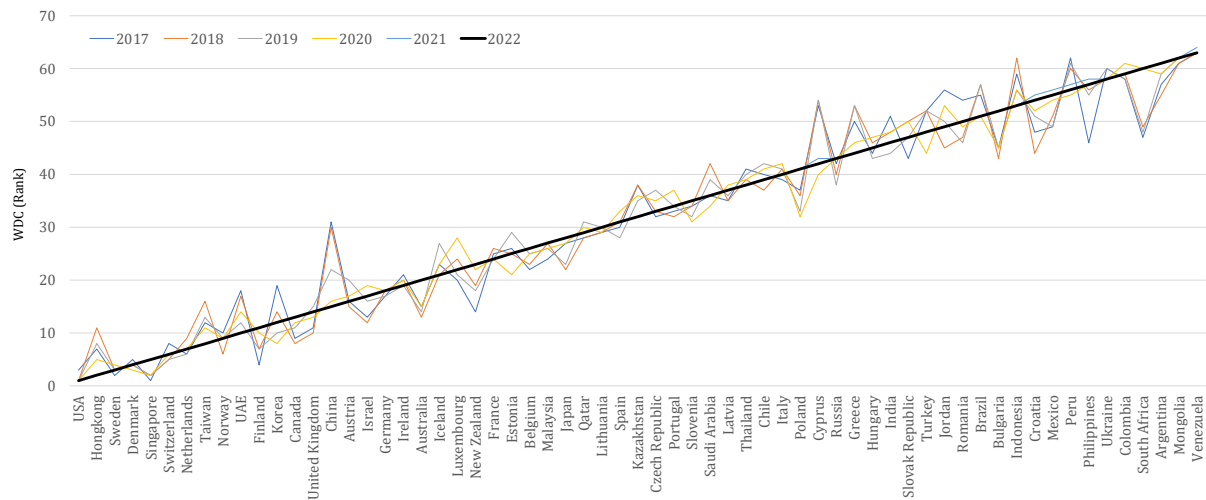


Figure 25 - WDC¹⁰² - Rank 2017–2022, states sorted by rank in 2022
(own representation based on secondary data)

The aggregation of the results of the individual criteria and subcategories leads to the overall ranking, which is shown in Figure 25. The states are sorted according to the ranking of the reference year 2022; the remaining annual rankings thus represent deviations from the year 2022. For a later index calculation, the WDC index value of the year 2022 is used.

2.2.6.8. Key enabling technologies

KETs are those that have an ‘above-average broad impact’ and are important as a ‘contribution to pending transformations of the economy and society’ (Kroll, et al., 2022, p. 12). KETs play a role regarding the future of digital sovereignty, as their mastery will be essential for states’ strategic autonomy (cf. 2.3). The US and China are making long-term investments in the research and development of KETs and state participation. The states of the EU have fallen behind in this respect (Kroll, et al., 2022, p. 17). To illustrate this, KETs’ quantum technology, AI and blockchain are considered below.

The market for quantum technology will grow to USD 850 billion¹⁰³ in the next decades. Since quantum technology not only enables higher computing power for specific use in general but also represents disruptive technology for today’s cryptography methods due to its effectiveness for prime factorisation algorithms, mastering this technology is essential. China recognised quantum technology as a KET early on and is investing around USD 2.5 billion annually. The sum of Chinese investments in quantum technology from the mid-1980s to the present is sometimes given as USD 25 billion, with an additional USD 15 billion supporting the National Laboratory for Quantum Information Science¹⁰⁴. Other countries

¹⁰² Source: IMD business school (2022, p. 30)

¹⁰³ Source: www.bcg.com (2022)

¹⁰⁴ Source: CQI (2022)

follow only at a great distance. Among them is Germany, with a total investment volume of USD 2.7 billion.

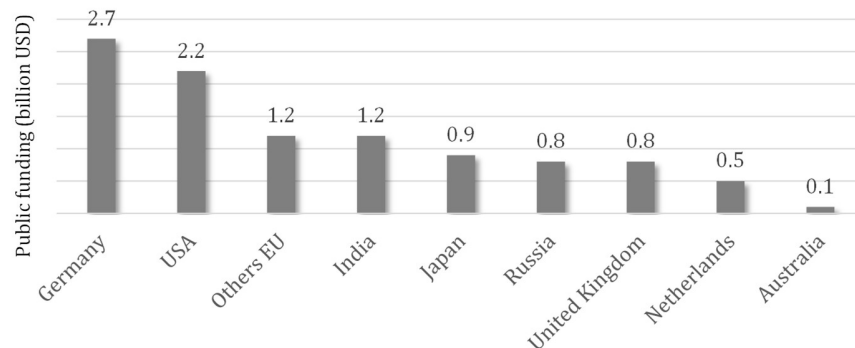


Figure 26 - Public funding for quantum computing¹⁰⁵ in select countries until 2020
(own representation based on secondary data)

China's dominance can also be seen in the number of previous quantum patents, followed by the US.

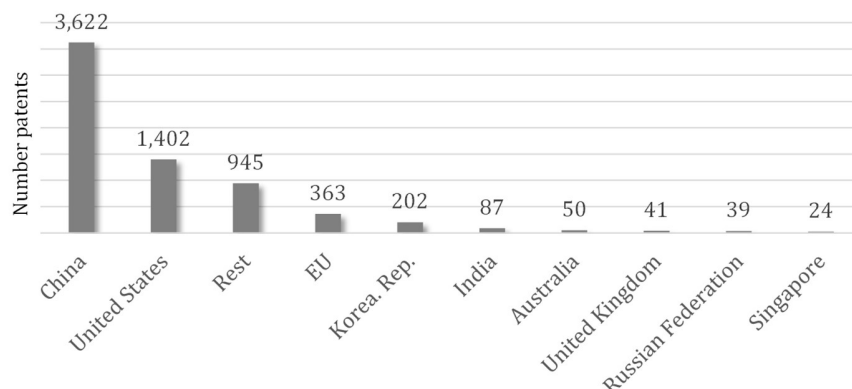


Figure 27 - Quantum patents¹⁰⁶ in select countries by 2020
(own representation based on secondary data)

The EU countries are in the process of catching up and are particularly strong in the research sector¹⁰⁷, but they are still far behind the US and China.

The patent situation for AI looks similar. The countries of the EU are largely behind the curve. The development of intelligent systems, from initially rule-based expert systems to today's neural networks and deep learning to heuristic models in the future, is taking place primarily in the USA and China. The need to deal with large amounts of data and machine learning was recognised early in these two states, which is reflected in the dominance of the provider landscape and the AI models used.

¹⁰⁵ Source: McKinsey Global Institute (2020, p. 5)

¹⁰⁶ Source: World Intellectual Property Organization (2021)

¹⁰⁷ Source: McKinsey Global Institute (2020)

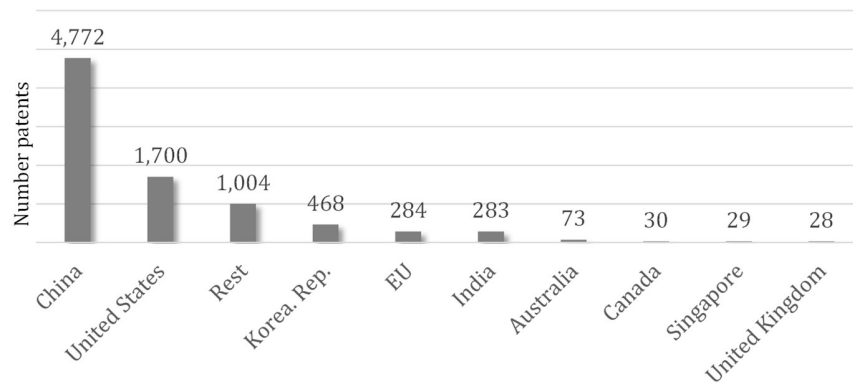


Figure 28 - AI patents¹⁰⁸ in select countries by 2020
(own representation based on secondary data)

The country positioning is also reflected in the Cambrian Index for AI, which examines the positioning of countries in terms of their ability to develop AI technology on the basis of 26 individual criteria, such as the availability of data, training in AI, expenditure on research and development, investor capital and number of patents.

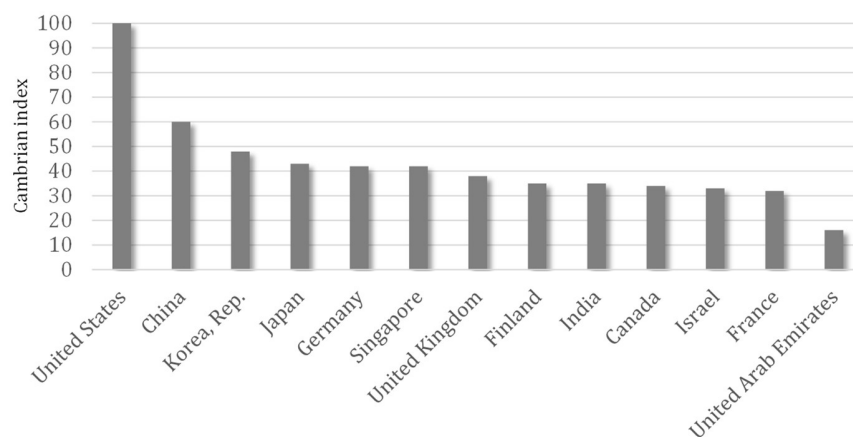


Figure 29 - Cambrian AI Index¹⁰⁹ (own representation based on secondary data)

Finally¹¹⁰, the KET blockchain-technology is considered. In addition to technological relevance, blockchain technologies are increasingly coming into political focus, especially in the context of electronic digital identities. For example, 'hotel check-in will be digital in the future—and secure thanks to blockchain' (Geuter, 2021). The introduction of such technology could be society-changing, as the 'process could rapidly change even the basic principles (...) and challenge the traditional role of the state and centralised institutions.

¹⁰⁸ Source: World Intellectual Property Organization (2021)

¹⁰⁹ Source: Cambrian Institute (2020)

¹¹⁰ It should be noted that the question of which technology is identified as a KET can be individual; therefore, the technologies discussed here should be understood as a selection.

Indeed, many blockchain proponents claim that civil society could organise itself' (Atzori, 2017, p. 48).

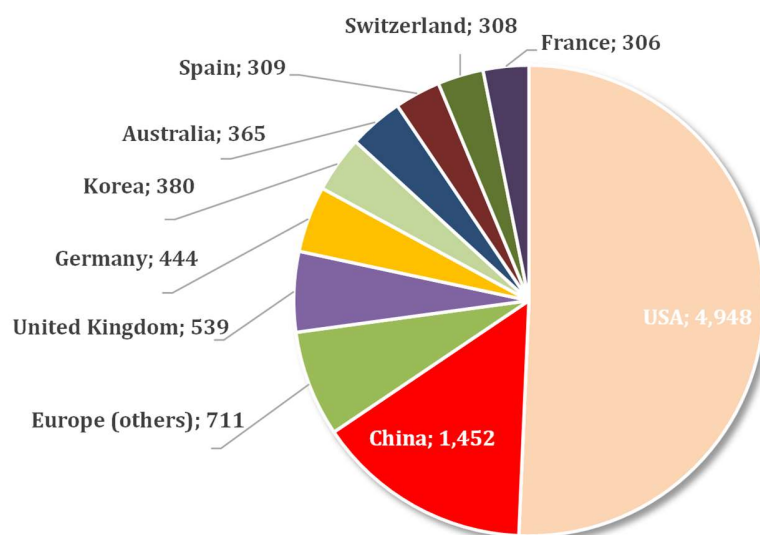


Figure 30 - Blockchain patent applications¹¹¹ (2008-2019)
(own representation based on secondary data)

The US dominates this sector. However, China is catching up; the country's focus on blockchain was confirmed by President Xi Jinping in 2019. Bitcoin has also been approved as a currency since 2020. This is reflected in the number of patents per state. Companies with a blockchain focus and numbers of patents have been rising steadily over the past 15 years.

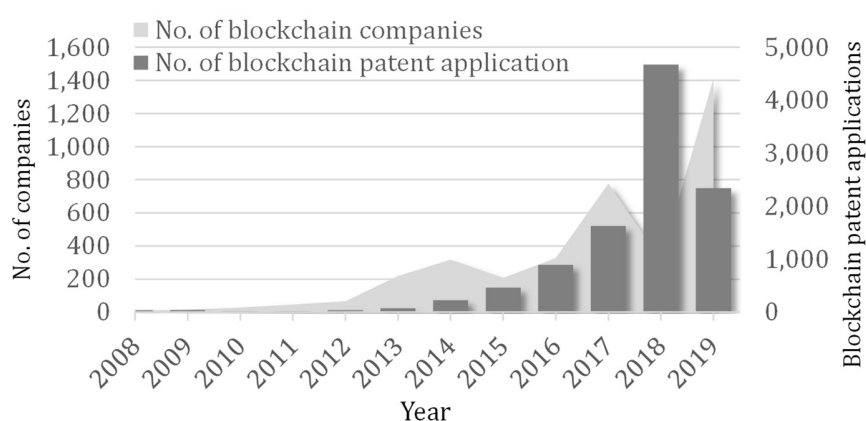


Figure 31 - Blockchain Start-ups and the total number of blockchain patent applications¹¹²
(own representation based on secondary data)

¹¹¹ Source: IP Lytics (2019, pp. 5-8)

¹¹² Ibid.

2.2.6.9. Data protection level

The EU uses select criteria to assess whether a state has a level of data protection equivalent to that of the EU. If a state is classified accordingly, then the same conditions apply to data traffic as in an EU state. The classification takes place within the framework of an adequacy decision in the following steps, which are published on the EU's website: proposal by the European Commission, opinion of the European Data Protection Board, agreement by the representatives of the EU countries and adoption of the decision by the European Commission (Duque de Carvalho, Sara Leonor, 2019). The following non-EU countries were accredited by an adequacy decision: Andorra, Argentina, Canada, Switzerland, Cyprus, the UK, Ireland, Iceland, Israel, Japan, South Korea, Liechtenstein, New Zealand, Uruguay and the Jersey Channel Islands.

In addition, there are parameters from country-specific data protection studies. The most comprehensive from the author's point of view is the study from the research company Comparitech¹¹³, which surveys the status of privacy protection and the state of monitoring for 47 countries in 14 individual criteria. From this, a measurable overview of states' compliance with data protection concerns can be derived. The methodology quantitatively assesses various sub-categories, including privacy under the constitution, laws, enforceability, data sharing, government oversight and access to data. The results are categorised into seven levels from 'endemic surveillance' (1.1–1.5 points) to 'consistent compliance with data protection standards' (4.2–5.0 points). The scores are weighted and range from 1.8 (China) to 3.2 (Ireland). An overview can be seen in Figure 32.

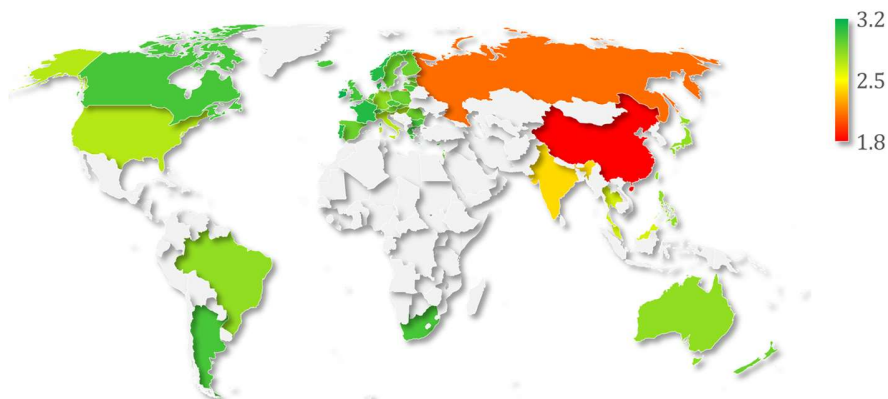


Figure 32 - Classification of data protection¹¹⁴ (own representation based on secondary data)

By appropriately adding the data protection assessment methods outlined above, a consolidated metric can be used to assess the implementation of data protection concerns which will be explained in the later part of the dissertation.

¹¹³ Source: Bischoff (2019)

¹¹⁴ Ibid.

2.2.6.10. Cyber security level

Cyber security is a relevant component of digital sovereignty, as increasing cyber attacks can threaten the sovereignty of states. Numerous examples of governmental and industrial organisations, especially critical infrastructure such as banks or universities and hospitals, being either blackmailed or compromised by cyber-attacks show the impact of cyber-attacks on the global economy (Balbaa, et al., 2022). The number of cyber attacks has increased, especially since Russia's war of aggression against Ukraine. For this reason, cybersecurity has a direct and increasing impact on the sovereignty of states and is conceptually linked to it (Madiaga, 2020a). There are various indices that measure the level of cyberresilience and cybersecurity at the state level, such as the Global Cybersecurity Index, the Index of Cybersecurity (ICS), or the National Cybersecurity Index (NCSI), with ICS and NCSI being assessed as the most realistic. Of particular interest is the NCSI, which measures the effectiveness of 173 countries to defend themselves against cyber attacks such as denial of service attacks or data integrity breaches using 46 indicators in largely administrative areas. According to the NCSI definition, state-level capabilities include, for example, the ability to regulate cyber attacks, cooperation, security policy and crisis management. Since 2018, the index has been compiled by the e-Governance Academy in Estonia (Kravets, 2019). The graphical analysis of the ranking in Figure 33 shows that many American, Asian and European states, as well as Australia, have a relatively high cybersecurity rank. Many of these states are also autocratic states, for which cyber defence is also an essential attribute of digital sovereignty and protection of autocratic structures. It can therefore be said that cyber resilience is an essential attribute of digital sovereignty for democratic states, including all EU member states, but also for non-democratic states. A relevant example is Saudi Arabia, which is ranked 14th in the NCSI, but 117th in the Barnett Index. For this reason, cybersecurity is an essential aspect for states to assess digital sovereignty.

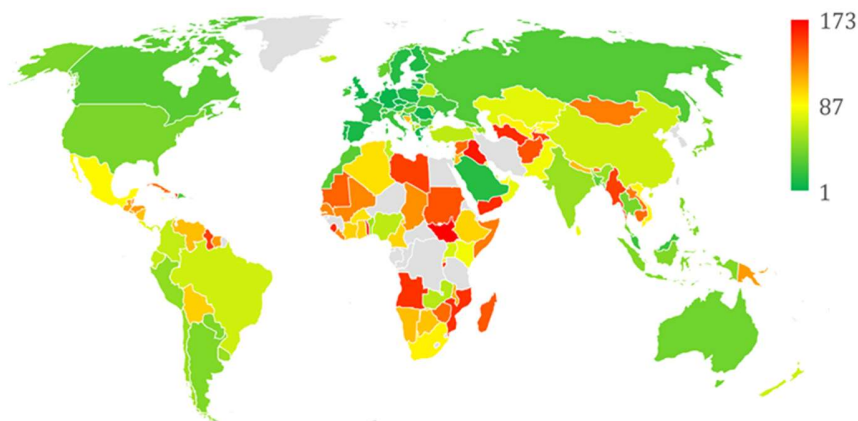


Figure 33 - NCSI Rank¹¹⁵ (own representation based on secondary data)

¹¹⁵ Source: e-Governance Academy (2023)

However, since this is also true for autocratic states, a cybersecurity index such as the NCSI should not be included in the modelling of digital sovereignty for European or other democratic states, but can be used as a comparative index.

2.2.7. Summary of literature review; the aspects of digitalisation

The current phase of digitalisation can be understood as the fourth stage of the Industrial Revolution.

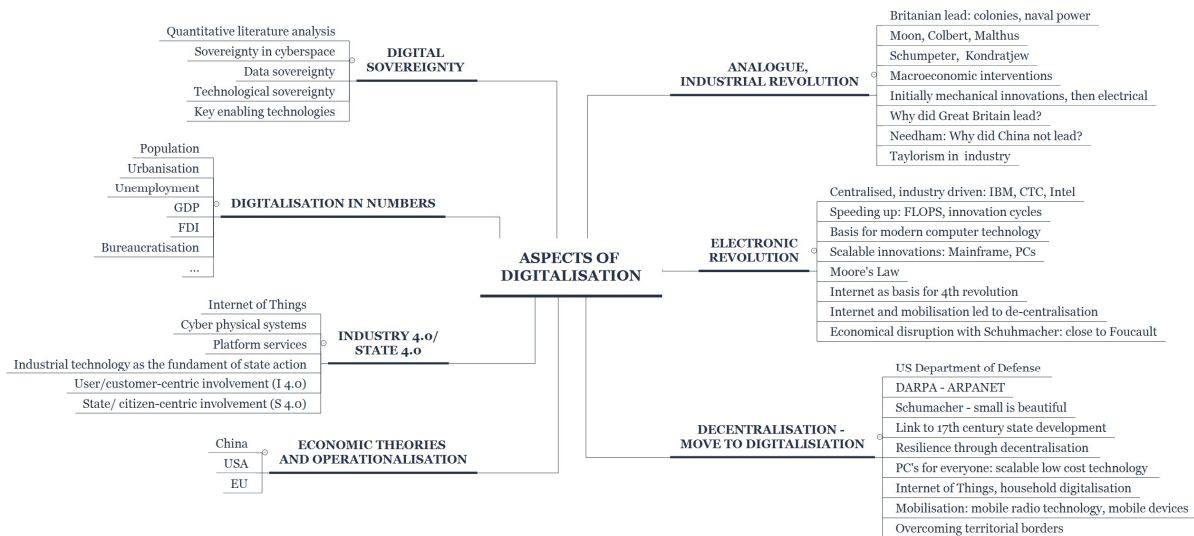


Figure 34 - Literature research; aspects of digitalisation
(own representation based on literature review)

The digitalisation of industry and civil society also impacts politics and the state. Key figures that can be used to measure the stages of the Industrial Revolution are widely available and can be used as a basis for quantifying states' digital sovereignty. The main aspects considered in the previous analysis of digitalisation have been summarised in Figure 34.

2.3. Digital Sovereignty ≠ Digitalisation + Sovereignty

Following the analysis of the individual terms *sovereignty* and *digitalisation*, the following analysis is devoted to the composite relationship of the two terms. Sovereignty in the digital context has various connotations in the literature: technological sovereignty, data sovereignty or cybersovereignty. Other combinations, such as information sovereignty, network sovereignty and computer sovereignty, will not be discussed.

Table 4 - Mentions of 'sovereignty' in a digital context¹¹⁶ structural based on (Couture, Toupin, 2019) (own analysis)

	Data sovereignty		Technological sovereignty		Digital sovereignty	
	academic	other	academic	other	academic	other
before 2011	2	24	6	62	0	0
2011–2015	168	1,165	14	144	14	144
2016–2020	294	5,187	75	1,021	69	984
2021–2022	289	3,224	48	904	73	1,505

The analysis of the frequency of use of the term *digital sovereignty* in online academic articles and books, trade magazines and newspapers shows a growing but not yet frequent mentioning (Couture, Toupin, 2019, p. 2308). In academic articles, the term was used 69 times between 2016–2020 and 73 times since 2021. If we examine the context of the term *digital* and expand the search to include *data* and *technology*, we see a higher usage of the term: approximately 850 mentions between 2016–2022. Use of the individual terms *digital* (1.3 million) and *sovereignty* (140,000) is clustered in scientific articles between 2000–2022. The approaches to defining digital sovereignty are discussed in many different ways in the literature; these are divided into three categories for a better overview (Kaloudis, 2021b, pp. 6–8).

2.3.1. Digital sovereignty in the sense of sovereignty in cyberspace

The term *digital sovereignty* originated in the sense of sovereignty in cyberspace. In the first growth phase of the internet, when only 30 million people (1% of today's users) used the internet, the idea of a state-independent sovereign space arose, which was to be created without territorial borders and with its own rules for users—the people. This was intended as an opposition to traditional claims to sovereignty. In 1996, the idea of net activism was supported by the US Telecommunication Act, with which the regulation of internet content in the US was decided in a separate chapter titled 'Decency in Communications Technology'. With the state's encroachment on the internet, internet activists sought a dedicated space—cyberspace—that would be free of state regulation and state sovereignty (Turner, 2008, p. 13). This approach risked shaking the nation-state, a territorially limited and sovereign state system that had grown over 250 years. The discourse of the approach can be found early in a manifesto by Barlow¹¹⁷, which he presented at the World Economic Forum in 1996. One of his most famous essays, 'A Declaration of the Independence of Cyberspace', he argued for the end of colonial, freedom-restricting rules in cyberspace and the reduction of physical intervention options for regulation. He propagated a new cyberworld that was equally accessible to all, without privileges related to origin or economic or military strength. The first lines of the manifesto show the non-acceptance of traditional nation-state sovereignty: 'Governments of the industrial world, you tired giants of flesh and steel, I come from cyberspace, the new home of the mind. In the name of the

¹¹⁶ Source: ProQuest Central (2022)

¹¹⁷ John Perry Barlow (1947–2018)

future, I ask you from the past to leave us alone. You are not welcome here. You have no sovereignty where we gather.’ (Barlow, 1996).

Although these approaches did not catch on, ideas related to the sovereignty of cyberspace are still present almost 30 years later. The idea of ‘popular sovereignty in cyberspace’, in which the state plays a subordinate role and the main actors are the people who act in cyberspace is currently still under discussion (Couture, Toupin, 2019, p. 7) (Mueller, 2020, 780 ff.). Mueller equated popular sovereignty with multistakeholder participation, thus questioning state sovereignty over the internet. Even if a separate cybersovereignty space were not established, a certain degree of self-organising legal freedom is evident on the internet. Virtual worlds, in which the boundaries between virtual and physical reality merge, are the metaverse of the future and will be centrally organised either by companies (e.g. Meta) or by decentralised autonomous organisations (e.g. KONG Land) (Nabben, 2021).

2.3.2. Concept of data sovereignty

After Snowden’s¹¹⁸ publication of secret documents from intelligence services, a debate arose about the storage and provision of data. The question of how to deal with data ownership concerns citizens as much as it does companies and state institutions. The concept of data sovereignty is understood differently in the literature; most articles refer to *data protection*, discussing the ICT architectures required for this and providing legal explanations of data sovereignty (Hummel, et al., 2021). In companies, the purpose of data sovereignty is related to aspects of digital law and data protection for employees and customers. At the state level, the focus is also on maintaining national security. In addition, the protection of citizens’ data is relevant, since in a democratic nation–state, citizens demand that the state protect their rights, such as the right to ‘forget data’ or ‘take data with them’. The question is whether the user—whether a private individual, a company or a public organisation—can decide independently and securely which data is stored where, for how long it is stored, if it is used within the framework of data analyses and how it is used. Absolute data sovereignty means being able to (re)decide on each of these aspects at any time. The German Ethics Council distinguishes data sovereignty from data protection and defines it in the context of the use of big data in healthcare as informational self-determination, which is ethically and constitutionally founded (Deutscher Ethikrat, 2017, p. 40). The protection of data should be ‘multidimensional’, ‘with a view to different groups of actors’ and ‘reflect contexts of action’. This means that protection classes are not assessed on the basis of statically collected data, as has been the case, but that the information that arises in collection and processing, in the sense of a correlation, plays an essential role. Ultimately, drawing conclusions about the data provider is of particular interest; it is they who should be protected and whose sovereignty must be safeguarded. This is intended to represent an ‘informational freedom shaping of the individual within a just and solidary

¹¹⁸ Edward Snowden (*1983)

society' (Deutscher Ethikrat, 2017, pp. 253–261). What is common to almost all views of sovereignty is that they have 'supreme control' (Goldsmith, Jack Landman, 1998, 475ff) over a territory and are 'independent of other sovereigns' (Bodin, 1994) (Woods, 2018). In this respect, the question arises whether and how a state can administer data in a sovereign nation–state. The rights of citizens are defined in the Charter of Fundamental Rights of the European Union, which expresses the 'character of the Union as a community of values' (Schmitz, 2004, p. 692), as follows:

'(1) Everyone has the right to the protection of personal data concerning him or her. (2) Such data may be processed only fairly and lawfully for specified purposes and with the consent of the data subject or on any other legitimate basis laid down by law. Any person shall have the right of access to data collected concerning him or her and the right to have such data rectified. (3) Compliance with these provisions shall be monitored by an independent body' (Article 8, European Parliament, 2012).

Here, the aspect of monitoring plays an essential role, as it guarantees the data sovereignty of citizens and institutions and ensures informational self-determination. This can be ensured through transparent ICT services that offer users the opportunity to decide for themselves which data is collected, stored, deleted and processed by whom. This decision should be made independently of the ICT provider and the technical solution used and, above all, without (state) control (Beyerer, et al., 2018, p. 278). Data sovereignty can also be defined as a subset of cyber sovereignty, in which states seek to subject cyberspace data to national jurisdiction. Approaches to protect national security or implement state interests correspond to 'strong data sovereignty', while 'weak data sovereignty' describes the privately oriented, liberally motivated implementation of digital legal aspects to protect users' data (Polatin-Reuben, Wright, 2014, p. 6). China, Russia and other autocratic states have the strongest interpretations of data sovereignty, as they limit data on a nation–state basis and cite digital sovereignty as an argument for implementing their political agendas to protect their own regimes. The states of the EU rely on weak data sovereignty, which is modelled on the data protection requirements for industrial companies and essentially serves to protect citizens' data.

2.3.3. Technological sovereignty and key enabling technologies

The issue of sovereignty specifically regarding the use of KETs is the subject of political and technological debate. The reason for this is the lack of 'free technology' (Haché, 2014, pp. 165–173). From a sociological perspective, sovereignty is defined as that which 'deals with technologies developed by and for civil society, and the initiatives that form it attempt to create alternatives to commercial and/or military technologies' (Haché, 2014, p. 168). The focus is on civil society, wherein 'individual and collective actions are not motivated primarily by profit, but rather by attempt to meet desires and needs while promoting social and political transformation' (Haché, 2014, p. 166). Haché opposes a profit-maximising, neoliberal attitude and calls for alternatives to commercial hardware and software with a focus on the use of the open-source technologies that form the basis of technological

sovereignty. Betr  n built on this concept and defined technological sovereignty in a political context as ‘an anti-establishment practice in the field of ICT, in order to integrate it in other anti-imperialist struggles’ (Beltr  n, 2016, p. 18) (Couture, Toupin, 2019). In doing so, she proposed to fundamentally question the ‘imperialist approaches’ of big ICT companies and to think of alternatives to reappropriate digital sovereignty (Couture, Toupin, 2019).

Other considerations focus on an exclusively technological (i.e. hardware and software-oriented) definition. The construction of dedicated submarine cables (European Union, Finland), territorially localised data routing (Germany, France), national encrypted mail systems (Germany), territorial local data storage, including for cloud services (Germany, Poland, Switzerland, France), and own encryption technologies (Germany, Great Britain) are politically motivated and technologically realisable approaches to achieve higher strategic autonomy and thus sovereignty (Maurer, et al., 2015).

The design of nation–states is flanked by political recommendations on how to manage the internet. For example, since 2014, OECD countries have wanted to help preserve the fundamental openness of the internet and simultaneously achieve certain public policy goals. To this end, recommendations for policymakers have been formulated to, among other reasons, ‘promote and protect the free flow of information globally’, ‘ensure transparency, fair procedures and good governance’, ‘promote and enable cross-border service delivery’ and ‘promote cooperation to foster cybersecurity’. However, these recommendations do not always correspond to nation–state efforts to increase technological sovereignty (Maurer, et al., 2015, pp. 13–14).

Technological sovereignty and data sovereignty control current discussions in politics and business (Cottey, et al., 2022, pp. 19–27). The question must be clarified: is there is a need for all technologies to be equally sovereign or can economically appropriate prioritisation via KETs regulated by individual states be found? In addition, how a solid assessment can be made when a technology becomes a KET must be examined. The time periods in which a technology develops into a KET is illustrated by the example of AI: how can computers be trained to learn independently and solve certain problems more effectively than humans? The term *Artificial Intelligence* has existed since 1955, and research has been conducted on AI since then; however, it was not until the early 2000s that processors and storage media began to be able to process the necessary amounts of data and algorithms for functional AI. Scientifically, a study of AI was first proposed in 1956 at a conference at Dartmouth. The hypothesis was that ‘that any aspect of learning or any other feature of intelligence can in principle be described so accurately that a machine can be made to simulate it.’ (McCarthy, 1955, Krotz, 2022).

When is the right time to identify the technologies that are becoming relevant as KETs and ensure technological sovereignty in the future?

There are many definitions of *KET* in the literature. For instance, Wolan indicated that these technologies should not be considered separately from one another, and AI, as a super KET, will dominate and be integrated into them. He defined KETs as technologies that grow

exponentially or become cheaper, thereby becoming more widespread in industry (Wolan, 2019, p. 170).

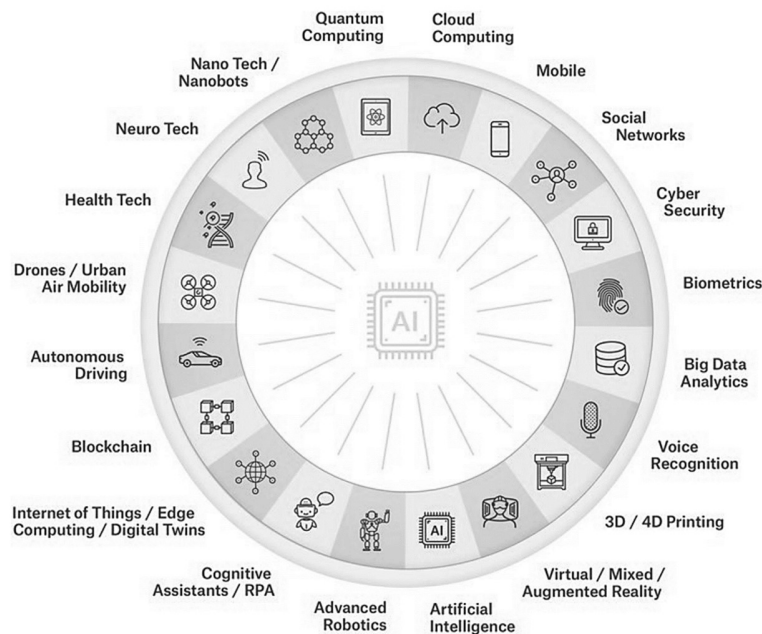


Figure 35 -Technology radar (Wolan, 2019, p. 169)

The VDE¹¹⁹ distinguishes digital from technological sovereignty, considering aspects of digital sovereignty such as trust, ICT and data security, access to raw materials, access to components and access to knowledge as sub-areas of an overarching technological sovereignty (Illgner, 2020).

Technological sovereignty is understood as independent action and decision-making—the ‘ultimate decision-making power’. It is not a matter of ‘technologies for their own sake’, but of ‘being able to use technologies in a sovereign manner because of their economic, social and political relevance’. Technology fields in which the basic requirements do not change and KETs that serve to implement their basic requirements technologically and can change over time are features of the VDE’s consideration.

¹¹⁹ Association for Electrical, Electronic and Information Technologies

Table 5 - KETs according to the VDE¹²⁰

Technology field	Key enabling technology
Information and communication technology	5G Swarm communication Optical communication Quantum computer Microcomputer (IoT) High-performance computer Automatic safety analyses Encryption Blockchain
Microelectronics	Mixed signal integration Energy minimisation Generative Adversarial Networks Processor architectures TPU/GPU architectures
Software	Automatic verification Virtualisation Vulnerability analysis Digital twin Speech/face/gesture recognition Mental control
AI	Deep learning Cognitive systems Training procedures Data aggregation and storage
Digital platforms	Business models Identity management E-payment Authentication procedure Software as a service Micro services
Energy	Micro power plants Bio-power plants Smart networks Battery technology Gas conversion
Automation	Digital twin Human-robot interaction Cognitive recognition Sensors
Medical technology	Metrology OP technology Bio-sensors/-actuators Micro-robot (bloodstream)

¹²⁰ Source: VDE, Illgner (2020, p. 8)

The VDE emphasises that their selection of KETs can only be exemplary, and controllability in the relevant stages of the value chain is more important than the technology itself. The VDE assesses the relevance of the technology fields on the basis of criteria such as economic benefit, future viability, social acceptance, safety policy requirements and sustainability.

The European Commission understands KETs as technologies that are at the core of European industrial policy for economic strength across all sectors. They 'include the Internet of Things, the use of industrial data, advanced manufacturing, robotics, 3D printing, blockchain technology and AI'. These KETs drive innovation across the economy, ensuring Europe's future leadership, and are therefore particularly worthy of support and part of the European industrial strategy¹²¹.

The BDI¹²² differentiates in the definition Europe's 'security technological sovereignty'; on the one hand, European industries are leaders in the development and production of KETs. On the other hand, the state and industry must be able to assess the products and services of other providers with regard to 'performance, integrity, confidentiality and availability' and use them if they are suitable. In the view of the BDI, this includes a fundamental systemic understanding of KETs and thus 'the ability of Germany as an industrialised country to be able to make self-determined decisions about the production, further development and application of KETs, services and platforms' (BDI, 2015, p. 4).

The ICT industry association Bitkom e.V.¹²³ considers AI, security technologies, digital identities, blockchain, digital platforms, communication systems and networks, data centres and cloud infrastructure, high-performance computing and quantum computing and micro- and nanoelectronics as KETs from the 'foreign, security and European policy interests' (Bitkom e.V., 2019, p. 21).

For the defence and security industry, the German Federal Government has defined KETs with the aim of strengthening, protecting and promoting research and development. The strengthening is done with the aim of 'ensuring cyber security' and the 'sovereignty of Germany and Europe'. In addition to security and military technologies, the KETs defined here include ICT/communications hardware, security-relevant ICT and communications technologies, networked operations management/crypto and AI (Bundesrepublik Deutschland, 2020). It is emphasised that technologies are to be 'procured from trustworthy manufacturers' in the long term, and dependence on non-EU states is to be reduced.

2.4. Assignment to economic theories

The interrelations between the economic and social effects of the different stages of the Industrial Revolution and state sovereignty could be countered with instruments from mercantilism, the political economy, neo-classical economics and (neo-)Keynesianism. Economic theories have been incorporated into the historical reflections on the clarification

¹²¹ https://ec.europa.eu/growth/industry/policy/advanced-technologies_en

¹²² Federal Association of German Industry

¹²³ German Association for Information Technology, Telecommunications and New Media e.V.

of the concept of digital sovereignty (cf. 2.1 and 2.2). Today, (economic) policy instruments remain relevant for the current phase of the digital revolution through appropriate digital policies. In addition to education, economic stimulus and growth policy, competition and regulatory policy are relevant policies, regardless of whether they are viewed from a political science or an economic science perspective (Budzinski, Eckert, 2018). Policies should not be limited to one economic sector—technology—as digital policy comprises a broad spectrum of interdependencies, and essential state instruments and institutions can be influenced by their actions. It is therefore important to promote the right digital (economic) policy: an approach that can be classified between ideas of the ‘invisible hand’ (i.e. the liberal approach of self-regulation of the markets) and over-regulated institutionalism.

China’s territorially delimited institutionalism can be seen, for example, in the model of the ‘Great Firewall’, which produces digital technologies largely independently through strong regulation and isolation, following the desire for cyber sovereignty. This is an expression of the Chinese Communist Party’s concept to exercise state control over private-sector industries and technologies and to enforce the interests of the nation-state’s cybersecurity strategy, which is in stark contrast to multi-stakeholder management (Creemers, 2020, pp. 107–131). China is pursuing a state-institutionalised, autarky-oriented, digital-policy concept to strengthen their sovereignty. Since 1972, technology transfer from abroad has been systematically ensured: regulation, state control and a five-year plan in the 30 most important policy fields (including KETs) have been the focus of government action; 12 possibilities of legal technology transfer, 12 of extra-legal technology transfer and 8 of illegal technology transfer have been described (Hannas, Tatlow, 2021), indicating that China will further strengthen its digital dominance and independence (Holtmann, 2018, p. 169).

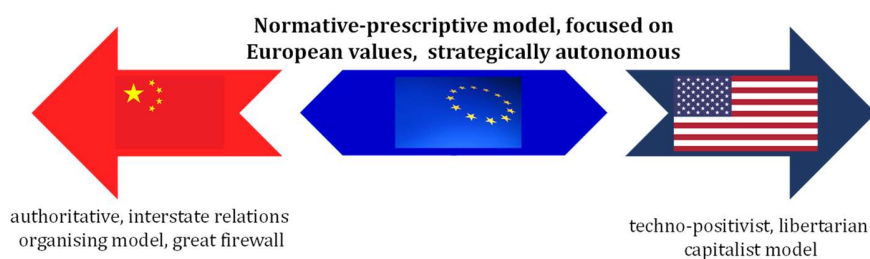


Figure 36 - Model of digital sovereignty based on (Pohle, Thiel, 2021)

In contrast, the ‘invisible hand’ concept can be seen in the successes of digitalisation in the US, which are characterised by a platform capitalism in which private companies have grown into quasi-governmental agencies responsibilities, which was made possible because the US follows a techno-positivist, capitalist and libertarian basic model. The US has always relied on pronounced economic liberalism, which is the basis for the continuing liberal economic policy and for the dominance of the values of freedom and equality (Holtmann, 2018, p. 78). There was almost no government intervention in the economy until the financial crisis of 2007. In the ICT market, as in other industrial fields, the US relies on a liberalised, deregulated market. Competition laws have only recently been considered to limit the

market dominance of the largest ICT companies, such as Alphabet, Amazon, Meta, Apple and Microsoft. The Biden Administration is pursuing greater independence by focusing on KETs, such as rare earths and semiconductors, to make supply chains digitally sovereign and by regulating the use of Chinese products.

Both concepts have led to dominance of China and the US in the digital economy, solidified by a strong policy focus. However, the notion of digital sovereignty is not discussed intensively in either country (Werthner, et al., 2022, pp. 227–231). As consequences of dominance, the ‘rupture of the global market economy’ and a ‘new phase of international mercantilism’ are predicted (Bendiek, Neyer, 2020). In this respect, both models could be seen as successful in reducing dependencies. This is also confirmed by the Digital Dependence Index (DDI), in which the US, China and South Korea received relatively high independence scores, while the other 20 countries considered were rated as having low autonomy due to high use of ‘digital technologies under foreign control’ (Mayer, Lu, 2022, p. 5) (cf. 5.15.3).

The EU states could (economically) politically adapt one or the other model to achieve greater autonomy. However, the model of the ‘middle’, a European model, may combine the best of both. The values of Europe as a stabilising foundation should be incorporated alongside a moderate economic interventionism that is characterised more by the promotion of positive, digital, value-based developments than restrictive regulation and, above all, focuses actively on strategic autonomy. Here, *strategic autonomy* means neither setting oneself apart technologically (in the sense of autarky) nor falling into absolute dependence, but becoming aware of global dependencies and actively shaping them in the sense of freedom of choice—‘being able to perform and compete in a self-determined manner’ (Weber, 2022, p. 155).

It is obvious, however, that the EU will not adopt too much regulation; the (low) salience of economic liberal approaches in recent decades have not led to EU states developing a strong digital economy (Kreutzer, Vogelsang, 2022). Instead, the dependency has become so great that a consistently focused and implementation-oriented digital policy is needed. The values of Europe are at the centre of the discourse on digital sovereignty. These are understood to be liberal, democratic values, as underpinned, for example, by the EU states’ focus on data protection. Only data that natural or legal persons want to make available will be published. Approaches such as state intervention through regulation (China) or the US’s Cloud Act¹²⁴ are not compatible with the European value system. However, Europe will not be able to catch up with the digital leaps of the US and China in the coming decades; thus, an international imbalance will remain. The companies of the EU states’ digital economy must make decisions in uncertainty, which is similar to post-Keynesianism. The task of the state is to prevent this uncertainty from becoming a competitive disadvantage. The instruments go far beyond the protection of mercantilism and require intelligent and interconnected fields of action (FoA). State intervention aimed at fostering the digital economy can be observed through various measures such as the regulation of crucial technologies, the

¹²⁴ Clarifying Lawful Overseas Use of Data Act

enhancement of the labour market through digital skills training and adaptation, the integration of public administration systems, economic policies that endorse key enabling technologies (KETs); and innovations in Industry 4.0. These approaches can also be found in public administration (Teixeira, Stefani, 2020).

The aim of such interventions is to strengthen the digital economic fields relevant for state sovereignty in such a way that dependence on states such as the US, China and Russia does not become so great that state sovereignty is weakened. The aim is to be achieved on the basis of EU values and with a market potential based on 450 million EU customers. In the EU, there are expressions of willingness to act, but no quick, effective concept is in place. Long-term strategic global autonomy is the goal, and medium-term self-sufficiency in certain KETs can be a means to an end (Metakides, 2022).

2.5. Research gap

The literature shows that the understanding of digital sovereignty is not globally uniform but diametrically contradictory. Autonomy and autarky describe the extremes of understanding. In the EU states, strategic autonomy and freedom of choice are formulated as basic attitudes, but there is no uniform understanding of digital sovereignty. The EU states and local industry associations have different, somewhat contradictory understandings that are shaped by specific interests. From an academic perspective, there are numerous sociological, political and technological articles that use the digitally connoted concept of sovereignty, but they are not based on a uniform definition. There is a lack of qualitative elements describing the term and a resulting ambiguity. A clear demarcation and acceptance of the interdependencies of the digital sovereignty of states, citizens and industry do not exist; thus, there is also no interdependency analysis. Therefore, the first research gap and research question are as follows: There is no uniform definition of digital sovereignty; this is especially the case for states in Europe. *How can digital sovereignty of the EU states be defined scientifically?*

The literature review showed that the digital sovereignty of states is not measured. There are neither qualitative nor quantitative criteria or classifications. Metrics that measure state sovereignty are available, as are assessments of diverse digital aspects at the state level. However, there is no consolidated qualitative and quantitative assessment of state digital sovereignty. Therefore, the second research gap and research question are as follows: Digital sovereignty has not yet been measured. *How can the digital sovereignty of a state be measured quantitatively?*

There are numerous FoA that can be taken to strengthen digital sovereignty. For example, the EU is working on an action plan to increase digital sovereignty. However, the effectiveness of the measures cannot be evaluated. Therefore, these gaps need to be addressed, and an analysis of how FoA affect digital sovereignty is needed. In particular, target values must be derived as to how strongly FoA must affect individually measured variables in order to strengthen the digital sovereignty of a state. Thus, the third and final research gap and research question are as follows: The effectiveness of FoA to increase

digital sovereignty cannot be evaluated so far due to a lack of a qualitative and quantifiable definition. *What is the quantitative impact of (economic) policy FoA on the digital sovereignty of a state?*

By closing these research gaps, a contribution can be made to strengthening the EU states' digital sovereignty, as a decision-making support system for decision-makers in politics and industry can be derived.

3. Objectives

The objective of this dissertation is to close the identified and above defined research gaps by answering the research questions and to develop recommendations on the basis of a scientifically derived definition of the compound concept of digital sovereignty in order to strengthen digital sovereignty, especially for the EU states. The recommended FoA can be implemented by political or economic decision-makers as legislative programmes or used as a basis for planning corporate strategies. The objective is hereby defined as modelling digital sovereignty and providing recommendations for strengthening it. In order to achieve this objective, four sub-objectives have been formulated, which are addressed sequentially, resulting in a hierarchy of objectives (Figure 37).

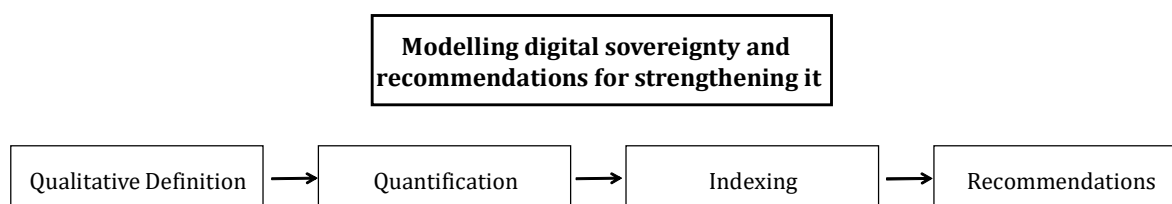


Figure 37 - Objective hierarchy of this dissertation

Qualitative definition: A proposal will be developed for the EU states on how the term *digital sovereignty* can be defined qualitatively, in the sense of a normative description. Historical consideration of the individual terms *digitalisation* and *sovereignty* forms the basis for their synthesis into a new term. The objective is to create a foundation for a normative, uniform understanding and for further analyses. In this way, a uniform, hitherto non-existent basis is proposed so that policy-makers and decision-makers in industry can achieve better certainty of action in order to be able to strengthen the EU states' digital sovereignty.

Quantification: From the previously proposed qualitative (i.e. normatively descriptive) definition, structured components are derived that specify digital sovereignty in greater depth and thus form a basis for an index that can be metrically measured by existing secondary data. This leads to a transition from a normative, qualitative description of digital sovereignty to a quantitative description. For the first time, the basis for a metric classification of the term *digital sovereignty* is generated. The classification takes place via existing key figures, so that decision-makers can assess the evaluation of digital sovereignty

by influencing these figures, thus creating a metric foundation for the development of proposals to strengthen digital sovereignty.

Index building: Another sub-objective is to develop a country-oriented, composite digital sovereignty index (DSI) based on the quantified components that metrically assesses the digital sovereignty of and ranks the states. Indexing and ranking will make it possible to measure the absolute and relative distances between the components and between the states as a whole. A strengthening (or weakening) of digital sovereignty can then be tracked as a repositioning in the index.

Recommendations: The sub-objective is to develop FoA and recommendations from the quantified components and the overall ranking to strengthen the EU states' digital sovereignty. The impact on the DSI is described qualitatively and quantitatively, giving decision-makers the opportunity to adopt the suggestions for strengthening digital sovereignty.

If the four sub-objectives are fulfilled, the objective of this dissertation is met, and the above-mentioned research gaps are closed. Thus, the term *digital sovereignty*, which has previously been used contradictorily, is defined as a uniform proposal for use by the EU states that has been structured into components and measured. For the first time, (economic) political developments can be evaluated in terms of their effectiveness regarding digital sovereignty and discussed and decided on in a more focused manner.

4. Methodology

4.1. Research design

In order to answer the research questions and achieve the objectives, select methods are used, which are presented below (Table 6).

Table 6 - Methods used to achieve the objectives

	Analysis	Synthesis	Induction	Comparison	Deduction	Descriptive approach	Normative approach
Objectives	Qualitative Definition	x					x
	Quantification		x		(x)	x	
	Indexing		x		(x)	x	
	Recommendations	x		x			x

Following is a discussion of the methods used to address the sub-objectives.

4.1.1. Qualitative definition of the term digital sovereignty

The terms *digitalisation* and *sovereignty* are analysed individually in terms of meaning and derivation, including historical consideration regarding the political and economic contexts. The basis for the analysis is the literature review. Essential factors are separated from non-essential factors to arrive at a focused definition. The composite term is defined by synthesis. The insights gained from the analysis of the individual concepts of sovereignty and digitalisation are transformed into a composite concept that has a normative character for the EU states. Findings from analysis and synthesis form the basis for the second sub-objective.

4.1.2. Quantifying digital sovereignty

The relevant information obtained from the analysis, which describes sovereignty and digitalisation in abstract terms, is transferred into a structured, hierarchical component model. This is an inductive procedure, as aggregated statements on the digital sovereignty of a state are drawn from details of the analysis and on the basis of the definition through synthesis. Through the (quantitative) analysis of the component model, statements about regularities or predictions can be derived. The procedures used in induction are based on quantitative methods and statistical procedures for data evaluation, such as linear regression, imputation and correlation analyses. Even if the focus is on inductive procedures, existing theories can be examined by means of deductive data evaluation.

4.1.3. Index formation

A composite index, the DSI, is inductively formed from a quantitatively assessable component model; it quantifies the digital sovereignty of a state with values from secondary data and allows a ranking to be formed. Quantitative methods are used for its creation, closely following the OECD's recommendations for index formation. The index of a state describes, by means of a descriptive approach, which characteristics of the components of digital sovereignty exist in a state. This information can be verified with the help of quantitative methods and used in a validated way by comparing it with existing country-specific indices. This proves the factual correctness of the index formation as well as the correct representation of reality. It is thus suitable for testing existing theories, which is why deductive methods can be used for this analysis.

4.1.4. Recommendations for strengthening digital sovereignty

After the index has been created, EU countries in particular are analysed and compared with regard to their ranking in the DSI. The criteria that led to a specific ranking are identified. From this, measures are derived as FoA that can help select EU states to strengthen their digital sovereignty. The proposals have a normative, macroeconomic character for a political economy, as they are intended to ensure a stronger positioning of the EU states in the future.

4.1.5. Summary of the method selection

The methodological focus of the present work is on the following:

- **Synthesis** with preceding (historical) analysis,
- **Induction** with approaches to deduction and
- **Normative approach** based on descriptive analysis and comparison.

The research design is supported by a qualitative analysis in which the elements of synthesis, induction and normative approach are formed through personally guided surveys. In the following chapters, the specific procedures of the outlined methods will be discussed.

4.2. Structured interviews; qualitative research

In order to substantiate the research design, structured interviews were conducted with five experts. States' differentiated approaches to the implementation of digital sovereignty should thus be structured and qualitatively evaluated with regard to its effect. In addition, the selection of accurate parameters for the DSI should be identified and their weighting verified. The results of the interviews are to be understood as verification evidence and will be integrated into the modelling and research design. The results of the model are cross-checked with the interview results. The questionnaire presented in Table 7 was used.

Table 7 – Questionnaire: Expert interviews

Introduction

The countries of the EU play a rather subordinate role in benchmarks on digitalisation; even wealthy states are only occasionally to be found at the top. In the top digital rankings (e.g. WDC, eGovernment Rank, government online services), the following countries can be found in the top 10: Sweden, Denmark, Korea, USA, Finland, Estonia, Singapore, New Zealand, Switzerland, Iceland, Netherlands, United Kingdom, Australia, Austria and Canada. In addition, dependence on states such as China for hardware and the USA for software is in some cases so strong that the state sovereignty of the EU states depends on it; thus, completely independent action and decision-making is no longer possible, even in state institutions.

The following questions serve to identify the essential FoA of the EU states must take in order to derive recommendations for political decision-makers.

Questions

1. What (in your opinion) is the reason why European countries do not play in the digitalisation 'top league' like China and the USA?
2. What is the main reason why European neighbouring countries, such as Finland, Denmark, Estonia and Sweden, lead the digitalisation rankings in Europe?
3. Which factors in particular should the EU states promote in order to strengthen digital sovereignty in Europe?

-
4. How relevant is the differentiation of political and economic systems (e.g. the US, China and the EU) for success in digitalisation?
 5. What are the three most important KETs on which EU countries are particularly dependent and in which they need to invest heavily?
 6. What are the three most important success factors for bringing Europe to a higher level of digitalisation?
 7. Where should Europe be in 10 years' time in terms of digital sovereignty?
 8. Where will Europe be in 10 years' time in terms of digital sovereignty?
 9. What is your recommendation to Europe to strengthen digital sovereignty?
 10. Please evaluate the AHP relevance table¹²⁵ for the parameters of a digital sovereignty index.
-

4.3. Index development

Indices that are suitable for comparing countries are discussed in many ways in the literature, with the OECD serving as a frequent source. Although the use of indices is criticised, their numbers are steadily increasing (Bjerre, et al., 2019, pp. 649–662). For the development of a composite index—an index that consists of already existing indices or parts thereof or other existing parameters (i.e. secondary data)—the OECD's Statistics and Science, Technology and Industry Division has developed a guide together with the European Commission's Research Centre. The statistical methods outlined therein are used for the considerations of this dissertation (OECD, 2020d, p. 13). In order to ensure that an index reflects reality, it is recommended to not only verify the correctness of the index formation but also to validate it.

Bandura listed almost 180 composite indices in a review, two with digital content (complementary to indices such as DESI or EGDI, described above): the ICT Opportunity Index and the E-Business Readiness Index (Bandura, 2008). In SCOPUS, up to 600 composite indices can be found up to the year 2016 (Greco, et al., 2019, p. 65). Their construction requires high levels of attention regarding the verification of all steps, as inaccuracies can lead to misconstructions and misinterpretations. The verification and validation of an index is of great importance, especially to increase its general acceptance.

The aforementioned construction method is suitable for the representation of a state's digital sovereignty since sovereignty and digital parameters for states already exist. Advantages of a composite index are that it summarises multi-layered information and is easier to interpret. Moreover, it can be measured repeatedly for multi-year comparisons and developments. The challenge in its construction is that the parameters must be carefully selected, and the determination of weights must be validly justified. In the context of political interpretation, the selection of the individual indices or the weighting itself can become a political issue. Focus must be placed on the weighting, as this massively influences the index formation within the framework of the mathematical model calculation. This also leads to academic discourse. In his overview of the frameworks used for the formation of macro-indices, Sharpe focused on different weighting variants (Sharpe, 2004). He assessed the

¹²⁵ cf. 5.9.3

OECD's index formation as extremely positive, reinforcing the procedure for the following implementation and forming the basis of a verifiable and validatable index. The OECD recommends the following (OECD, 2020d, 20ff):

Theoretical framework (cf. 4.4): A theoretical framework is required for index formation. The parameters of the input variables for the primary or secondary data of the index must be justified, above all, appropriate; the parameters' relevance of regarding the informative value of the overall index must be clearly described.

Data selection (cf. 4.5): The selection of each parameter should be evaluated with regard to the criteria of the evidence and the acceptability of the existing data, including its measurability and availability. 'Parameters should be selected based on their analytical soundness, measurability, country coverage, relevance to the phenomenon being measured and relationship to each other' (Srakar, et al., 2018). If data for a country is limited, the use of surrogate data may be helpful.

Missing data (cf. 4.6): Missing data may lead to the exclusion of a state in the index, or data records can be replaced in the case of data scarcity via data imputation procedures. There are various mathematical approaches for determining missing data, the impact of which (on the index) can be massive.

Statistical analysis (cf. 4.6): The parameters should be analysed regarding their statistical dependence on each other using multivariate methods. The results of this analysis can impact the relevance and weighting of the parameters.

Normalisation (cf. 4.7): Since differently scaled data are available, they must be made comparable (i.e. subject to a common normalisation). In this context, outliers and distortions require special attention.

Weighting and aggregation (cf. 4.8 and 4.9): Finally, the parameters are added or multiplied, and each is multiplied or exponentiated by the corresponding weights or otherwise suitably aggregated. The correct selection of the weighting and aggregation procedure is of great importance.

Assuming that the composite index consists of three essential sub-indices yet to be constructed, Δ_1 , Δ_2 and Δ_3 , the result is a composite index with a function (F) to be defined for a state (j) in the following form:

$$DSI_j = F(\Delta_{1j}, \Delta_{2j}, \Delta_{3j}) \quad (1)$$

4.4. Quantitative analysis: model bases

The parameters serving as the basis for the index must be examined for correlation to ensure that they are as uninfluenced and free from overlap as possible. A model is needed that can explain pairwise dependence in order to determine whether a functional correlation exists.

Here, m parameters are used for n countries¹²⁶, describing sovereignty or digitalisation aspects. The pairwise variables $X=X_i=\{x_{i1}, x_{i2}, \dots, x_{ij}\}$ and $Y=Y_i=\{y_{i1}, y_{i2}, \dots, y_{ij}\}$ with $x_{ij}, y_{ij} \in \mathbb{R}$ are parameters for the model to be developed. X and Y are each vectors with $n, m \in \mathbb{N}$.

$$X = X_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ij} \end{pmatrix}; Y = Y_i = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{ij} \end{pmatrix}, x_{ij}, y_{ij} \in \mathbb{R}; n, m \in \mathbb{N} \quad (2)$$

For a statistical analysis, independent and dependent variables are formed. In a pairwise comparison, X represents an independent variable and Y a dependent variable. For the pairwise comparison, an independent variable is examined with $n-1$ dependent variables for possible correlations.

The coefficient of determination is used as a quality criterion for measuring the relationship. R^2 is used as a quality criterion for measuring the correlation, which is determined within the framework of regression analyses (Wolf, Best, 2010) (Fahrmeir, et al., 2009, 98ff.). The aim of the analysis is to determine how large the proportion of measured values is that can be explained by a linear function. In addition to linear regression, other methods are also applicable. Possible outliers can then be examined in order to draw conclusions about special features. A linear forward regression was chosen as the forecasting model, which builds a forecasting model step by step, adds the best variable in each case and optimises the model after adjusting the least squares method. This is 'a model with a (...) regressor (X) that has a relationship with a' regressand (Y) and can be represented by a linear function (Montgomery, et al., 2021, p. 12). This simple linear regression determines a function described by regression coefficients, 'where the intercept (β_0) and the slope (β_1) are unknown constants and ε is a random error component':

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon \quad (3)$$

The deviations (i.e. errors) of the determined values from the linear function determined by regression are denoted by e_i .

$$y_i = \hat{y}_i + e_i \quad (4)$$

It is assumed that 'errors have a mean of zero and an unknown variance σ^2 . In addition, it is usually assumed that errors are uncorrelated. This means that the value of one error does not depend on the value of another error' (Montgomery, et al., 2021, p. 38). For simplicity, it is also assumed that X is a dependent variable and Y is an independent (i.e. random) variable.

¹²⁶ In the implementation in this work, $m = 30$ parameters were collected for $n = 212$ countries. The 212 countries (see Appendix) included all 193 UN states, corresponding to a subset of the World Bank's state database (e.g. additionally Taiwan or island groups).

‘There is a probability distribution for Y at each possible value for X ’. The expected value of this distribution is

$$E(Y|X) = \beta_0 + \beta_1 X \quad (5)$$

and the variance is

$$Var(Y|X) = Var(\beta_0 + \beta_1 X + \varepsilon) = \sigma^2 \quad (6)$$

Thus, the expected value ‘of Y is a linear function of X , although the variance of Y does not depend on the value of X . Furthermore, the results are uncorrelated because the errors are uncorrelated’ (Montgomery, et al., 2021, p. 13). The coefficient of determination was used as a quality criterion for the analysis. R^2 is used as a quality criterion for the analysis (i.e. the proportion of the scatter of the measured points) insofar as a predictor (X) and a regressor (Y), which can be explained by a regression function (Montgomery, et al., 2021, 35ff). This is defined as follows:

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - y)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{explained variation}}{\text{total variation}}. \quad (7)$$

4.5. Selection of secondary data

The qualitative definition of digital sovereignty was derived from historical analysis and research of the individual terms and synthesised into a new composite term. The new term was decomposed into qualitative sub-indices and sub-components based on the structured derivation. The sub-components were quantitatively assessed using parameters describing around 200 states that were available as secondary data. Secondary data sources underlying the analysis for a new DSI included the following:

- World Bank Open Data database, which provides a wide range of data, including country-specific data used in this analysis (www.worldbank.org),
- OECD (Organisation for Economic Co-operation and Development) statistics, which provides country-specific statistics (<https://stats.oecd.org/>),
- IEA (International Association for the Evaluation of Educational Achievement), which is a network for the improvement of education with data on the educational status of more than 100 countries (www.iea.nl),
- Global Innovation Index, which includes detailed measurements of 131 countries that rank the degree of innovation of states (www.globalinnovationindex.org),
- EU database, a country-oriented database focusing on states of the European Union (www.ec.europa.eu),
- Patentscope, which includes worldwide patent applications (www.patentscope.wipo.int),

- UN database, a system on UN states maintained by the Department of Economic and Social Affairs (www.publicadministration.un.org), and
- selected studies by, among others, management consultancies and scientific institutions on special topics (e.g. McKinsey [www.mckinsey.com], Cambrian Intelligence [www.cambrianintelligence.com], IP Lytics [www.iplytics.com], International Institute for Management Development [www.imd.org], International Telecommunication Union [www.itu.int], Comparitech [www.comparitech.com], and Leidenranking [statistics from over 1,200 universities worldwide; www.leidenranking.com]).

For m states, n parameters describing elements of digital sovereignty of the state were collected. The data for each parameter are stored in a vector.

$$X = X_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ij} \end{pmatrix}; x_{ij} \in \mathbb{R}; i, j \in \mathbb{N} \quad (8)$$

with $j \in [1, \dots, m]$ and $i \in [1, \dots, n]$ and n = number of parameters collected

The vector stores the data for up to m states for one parameter at a time.

After data collection, one obtains a $m \times n$ - matrix with $k \leq m \cdot n$ - coefficients, since not all parameters are completely available for each state.

4.6. Replacement methods for missing data

Complete data are not available for every $X_i = \{x_{i1}, x_{i2}, \dots, x_{ij}\}$, leading to the question of how to deal with states that have missing data in the statistical analysis. There are the following variants (Rässler, 2000, p. 67). Missing data are

- 1) treated as one of the available answers,
- 2) analysed/interpreted in the meaning if a value is not missing by chance,
- 3) excluded from the construction of the index and the analyses,
- 4) replaced by the mean value of the existing data or
- 5) assigned values by statistical procedures (imputation).

Variants 1) and 2) can be excluded here, as they are used in cases of non-response in surveys but are not applicable to the data of this dissertation. Variant 3), the 'complete case analysis' (CC) method, is applicable. With CC, missing data sets are not replaced or estimated, but all data sets (i.e. the states that have missing values) are not considered in the context of quantitative analyses. The prerequisite for the CC method is that the missing data have no explicit significance for the analysis; they are missing completely at random. Such is the case here, as only secondary data that consider certain states are used. However, the CC method leads to a high loss of data, which is why substitute procedures should be considered. The

simplest one is to use the arithmetic mean of the available x_j in each case as a substitute value (variant 4):

$$\bar{x}_i = \frac{\sum x_j}{\text{number } j} \text{ with } x_j \text{ present and } x_i \text{ not present} \quad (9)$$

The replacement of non-existent values with mean imputation can significantly influence the positioning of states either positively or negatively. This applies to all data replacement procedures.

Alternatively, the data are suitably replaced by imputation (variant 5). For this purpose, the 'nearest neighbour classification' (kNN) method can be used. By applying a distance function, within the dataset X with missing data records, a search is made within the entire dataset Y for data records that has no missing data there (Seidl, 2009). There are also more advanced methods (Nordholt, 1998). For the calculation of the missing values of X , the k -nearest Y are used. Here, k is an integer and n is the number of all possible X :

$$(X \times Y)^n \times X \rightarrow Y \quad (10)$$

This method can be influenced by the selection of the nearest neighbours to be included (number k) and the choice of distance method. The Euclidean distance is most frequently used (Elkan, 2009, p. 1591):

$$d(X, Y) = \|X - Y\| = \sqrt{(X - Y)(X - Y)} = [\sum (x_i - y_i)^2]^{1/2} \quad (11)$$

The following example illustrates that the validity of correlation analyses is significantly changed by the use of an imputation method. An advantage of the method is that it can replace all data (i.e. completely fill in missing data). However, the results should be used with caution. In the present case, states obtain values for parameters that do not exist in reality and thus falsify the overall result of the statistical analysis.

The regression analysis of the raw data of the Barnett Sovereignty Index and WDC with 57 paired values yields a coefficient of determination of 0.445 and a p-value of 1.073×10^{-8} . In contrast, after kNN imputation with 212 data sets, a coefficient of determination of 0.348 and a p-value of 1.82×10^{-21} is obtained, representing a worse correlation.

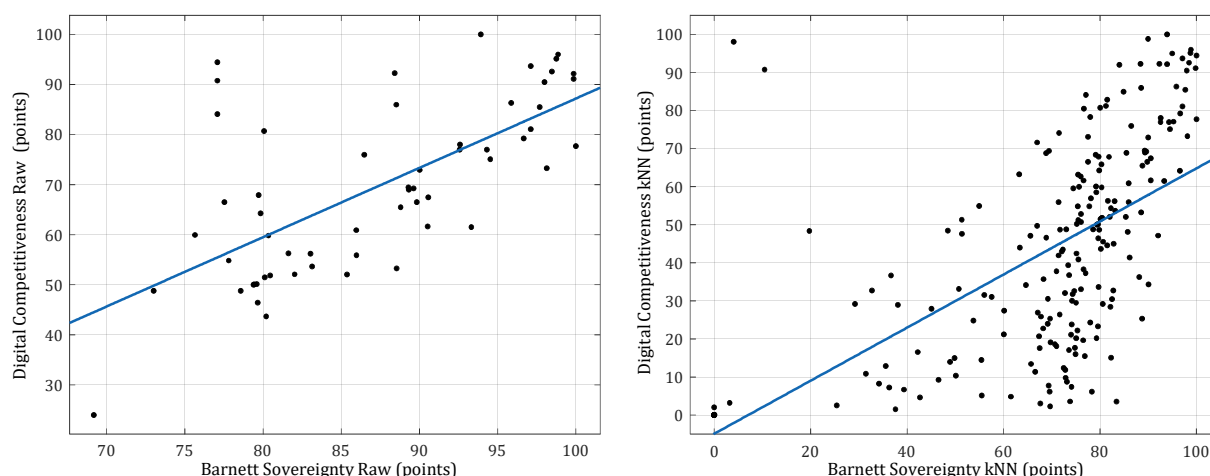


Figure 38 - Comparison of correlation without/with kNN imputation

Therefore, although an imputation procedure can be used to replace missing values, parameters with fewer existing values could also be weighted less in the context of index formation. The coefficient of determination can be used as a quality criterion for the estimation (OECD, 2020d, p. 64).

4.7. Normalisation methods

To aggregate parameters additively or multiplicatively and make the resulting index values comparable, they are normalised. Various procedures are suitable for this (Table 8).

Table 8 - Select normalisation methods (OECD, 2020d, p. 30)

Method	Pros	Cons
Ranking	No impairment due to outliers	No valuation in absolute figures
z-score	Outliers have high influence on index	Strong falsification of the original data
Min-max transformation	Maintaining the distribution and proximity to the original value	Small deviations in components have a stronger effect on index, outliers have high influence on index
Distance to the reference value	For this application: none	Outliers have an extreme effect on index, reference point not clearly definable
Scale transformation	Easy to use, no data loss	None
Scale categorisation	Simple value assignment	Falsification of original data

Scale transformation

The secondary data (X) are in most cases available in different scales. Therefore, it becomes necessary to normalise the data so that the parameters to be formed become comparable. To achieve normalisation to a maximum value of 100, the following function is used.

$$x'_i = 100 \frac{x_i}{\max(x_1, \dots, x_n)} \quad (12)$$

This avoids distortions due to stretching or compressing other normalisation procedures and makes the data comparable across scales (OECD, 2020d, p. 86).

Min-max normalisation

The values calculated without distortions (X) and without bias are then distributed by stretching to values between 0–100. The resulting normalised vector is formed by applying the min-max method (OECD, 2020d, p. 85).

$$x'_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \quad (13)$$

z-Score standardisation

A z-score normalisation normalises data in a scale around a mean value by the number of (standard) deviations from the mean value (OECD, 2020d, p. 30).

$$x'_i = \frac{x_i - \text{mean}(X)}{\sigma(X)} \quad (14)$$

Further normalisation, such as distance calculation to a reference country, scale categories, cyclical indicators or relative annual differences of consecutive years, are not explained further due to non-applicability.

4.8. Appropriate weighting methods

The index can be significantly influenced by the appropriate selection of weights. The models suitable for the identification of appropriate weights (Greco, et al., 2019, 70ff) are presented in Table 9.

Table 9 - Select weighting procedures (OECD, 2020d, pp. 31–32)

Method	Pros	Cons
Correlation analysis	Simple calculation through regression method	If applicable, not statistically significant; correlation does not have to mean causality.
Multiple linear regression	Simple calculation through regression method	Assumes strict linearity, which is usually not given

Method	Pros	Cons
Principal Component Analysis/Factor Analysis (PCA)	Identified commonalities of the components and corrected them via weighting	Not applicable in case of weak correlation of the components
Data envelopment analysis/benefit of the doubt	For this application: none	Suitability for comparative benchmarks only
Unobservable Components Model	Suitability for time series	None
Public opinion	Simple survey through questioning/expert appraisal	Uncertainty in the estimate
Analytic hierarchy process (AHP)	Quantitative support for expert estimators	Expert estimates have uncertainties
Conjoint analysis	Quantitative support for expert estimators	Focus on marketing and market research issues
Equal weighting	Simplicity	No differentiation of the relevance of components

For further analysis, factor analysis, AHP, equal weighting and the number of existing measured values are used to compare weighting options of the index formation.

Equal weighting

Each of the n parameters is given the same weight: $\gamma_i = \frac{1}{n}$.

Factor analysis (OECD, 2020d, 71ff)

The basic idea of a factor analysis is to build the index with sub-indices and sub-components that are as appropriate as possible: the more relevant a factor is for the overall index, the greater its weighting. The factors X partly overlap in terms of content, as they can consist of parameters that are partly similar. In this respect, it is important to work out how the factors are interrelated so that possible overlaps are compensated for by the weighting. A factor analysis is carried out to filter out possible redundant information from the indicators (OECD, 2020d, p. 68). For this purpose, the coefficients of determination R^2 are calculated from the factors X and Y . From the resulting 30×30 matrix, eigenvalues and variances are formed, with which it can be determined how large the proportion of the variance of a variable is within the total variance. The weights can then be formed from the order and the variance proportion (Abdi, Williams, 2010).

Analytical hierarchy process (OECD, 2020d, p. 98)

The AHP developed by Saaty¹²⁷ enables the decomposition of a decision problem into hierarchical levels and a pairwise comparison with each other. This is based on pairwise comparative management methods. For example, parameter importance is compared; in this case, 1=equally important, 3=somewhat more important, 5=clearly more important, 7=much more important and 9=extremely important. Comparative values can also be used as $\frac{1}{3}$, $\frac{1}{5}$, $\frac{1}{7}$ or $\frac{1}{9}$.

The values are determined by expert estimates. Thus, a numerical equation is formed for each hierarchical level, which then leads to a weighting of the components (Saaty, 2008, 86ff). To check the consistency of the decision matrix, Saaty uses a random index calculated for a number of components up to 15. Larger numbers are estimated, for example, for $n = 30$ with 1.6809.

Influence of number of measured values on weightings

Since the number of available measured values per country varies greatly in the present analysis and little data are available in some cases, the number of available measured values should be considered as a further criterion. $|X_i|$ should be taken into account in the weighting. If no measured values are available in the secondary data, an imputation procedure is used. The advantage of this is that missing data can be simulated, but this can falsify the result. Therefore, the countries for which less data is available should be weighted less heavily. The focus here is on the determined weights (γ_i). In general, the less data available for each country, the lower the weighting of all parameters.

4.9. Index aggregation methods

Aggregation functions are additive or multiplicative or use other methods to combine the individual parameters into an index. Depending on the parameters used in terms of compensability, independence and standardisation method, aggregation procedures can be excluded (Bjerre, et al., 2019, OECD, 2020d). An additive aggregation function is most frequently used (i.e. the summation of the parameters weighted with γ_i weighted parameters X) (OECD, 2020d, p. 103) (Goertz, 2006, p. 111):

$$Indexvalue = \sum_i \gamma_i x_i \quad (15)$$

In reality, parameters can cancel each other out (i.e. compensate for each other), and information can thus be lost. The weightings that are considered important can also cancel each other out, which can lead to problems (Greco, et al., 2019, 75ff). The disadvantage of

¹²⁷ Thomas Lorie Saaty (1926–2017)

the compensability of the parameters can lead to undesirable effects. To reduce these, geometric methods can be used:

$$Indexvalue = \prod_i x_i^{\gamma_i} \quad (16)$$

A multiplicative approach can also have compensatory effects, but these are smaller (OECD, 2020d, p. 32). Combinations between additive and geometric models are also possible and are used for the present purpose. The index values determined can then be ranked in the sense of ‘ranking by points’.

To avoid compensatory effects, non-linear ranking procedures can be applied, for example, according to Condorcet’s or Borda’s rules, which perform pairwise comparisons and lead to a ranking (OECD, 2020d, 106ff). States are compared with each other in pairs across all parameters, and the result is stored in an outranking matrix. The parameters do not have to be metric or normalised. The evaluation of the combination of states that results in the highest ranking can be used as a ranking. Based on Condorcet’s and Burda’s ranking methods, various—and very different—ranking methods have been developed, for example by Kemeny, Arrow and Raynaud, and Copeland (Klamler, 2005) (Munda, 2012). These should be used if the weighting of the parameters reflects their importance (this is not the case with equal weighting) or if the assumption of preferential independence cannot be confirmed (this is not the case due to low linear dependencies of the individual parameters) (Munda, 2012, p. 338). Non-linear methods have difficulties, especially when a large number of parameters are involved, due to the intensive computing power required, which is why they are not particularly popular for comparative country indices (Munda, 2012, p. 351) (Greco, et al., 2019, p. 77). For the sake of completeness, non-linear methods are also tested in this paper to verify that linear methods are suitable.

4.10. Methods of index and parameter analysis

To support the sub-objectives and identify FoA, the states are structured according to the DSI values. To analyse the total set of suitable states, they are sorted into categories and subsets of states. A suitable cut point can be found by using the percentiles from descriptive statistics. To distinguish top from average and less good performers, the best quartile can be used as the cut point. When calculating the intersection of two sub-indices, one thus obtains four categories that can be examined.

In each category, the deviations of the index parameters, e.g. to the top performers, can be analysed. This can be done through deviation analyses, graphical interpretation and heatmaps.

4.11. Methods of uncertainty/sensitivity analysis

There is no golden rule for the construction of indices, partially because subjective decisions are made in parts (Bjerre, et al., 2019, 649ff). In the present case, these are the selection of the right procedure for aggregation, for the replacement of missing data, the normalisation and weighting and the selection of the parameters themselves. These elements are sources of uncertainty, even if the selection was justified and partly supported by the expert interviews. Since it is not possible for most readers to assess the extent to which the data used are valid or have been changed or influenced, a sensitivity analysis is necessary (Kennedy, 2010, 366ff). This involves varying the sources of uncertainty and then observing the result (e.g. rank) (Saisana, et al., 2005). A sensitivity analysis is also considered a component of index validation (Janová, et al., 2019, p. 917). Sensitivity and uncertainty analyses are common in management methods and can also be used to examine the robustness of the index and thus the ranking of the countries. In addition to examining the validity of the method, uncertainty analysis helps to focus on the countries that are altered by changes in the above-mentioned factors in the ranking and thus are subject to a certain sensitivity. This helps in discussions about the index and its further development. The uncertainty and sensitivity analyses must be conducted individually depending on the index formation. In the present case, the DSI is formed for all states with all normalisation, imputation and weighting procedures applied.

Table 10 - Overview of normalisation, weighting and imputation procedures

Normalisation	Weights	Imputation
Raw	PCA	without
MinMax	AHP	Mean
z-score MN = 0	equal	kNN
z-score MN = 0, StD = 1		

For each of the aforementioned combinations of procedures, an index value (DSI) can be formed for each state. On the basis of a freely selectable reference index, differences of the ranks are formed for the uncertainty analysis. Mean and variance are then calculated for each combination of procedures.

$$R'_U = \text{mean} (\text{Rank}_{ref}(DSI_n) - \text{Rank} (DSI_n)) =$$

$$\frac{1}{M} \sum_{n=1}^M \text{Rank}_{ref}(DSI_n) - \text{Rank} (DSI_n) \quad (17)$$

$$R''_U = \text{var} (\text{Rank}_{ref}(DSI_n) - \text{Rank}(DSI_n))$$

R'_{ij} and R''_{ij} can be taken as measures of the uncertainty of the chosen combination. The combination with the lowest mean value (variance) is therefore considered the 'best method' in the sense of uncertainty analysis (OECD, 2020d, 118ff). The best combination of imputation, normalisation and weighting is then selected, for which R'_{ij} and R''_{ij} are minimal. Each parameter can be changed in a sensitivity analysis, and the effect on the results (in this case the DSI) can be observed.

4.12. Verification and validation

The selection of parameters, their normalisation and weighting, and aggregation make it possible to calculate an index value. The values of the states can be ranked. To check whether the index value or the resulting ranking was calculated correctly, the results should be verified and validated. To do this, Sargent recommends 'ensuring that the computer programme of the computer-based model and its implementation are correct' and 'demonstrating that a model has a satisfactory range of accuracy within its domain of application that is consistent with the intended application of the model' (Sargent, 2013, p. 157). Only if the model produces correct results can it be used as a decision support system. Finding the right cost/benefit ratio remains a challenge in finding a credible model at an acceptable cost.

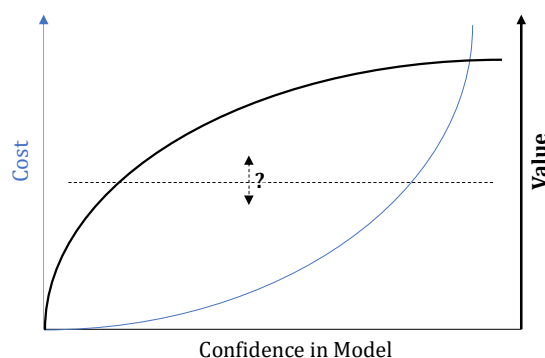


Figure 39 – Cost–benefit relation of a model, based on (Sargent, 2013, p. 158) (own illustration)

Various methods are proposed for verification and validation, including animation, comparison with other models, correctness of data relationships, apparent validity (also based on case studies), validity of events, testing for extreme conditions, operational graphs and analysis of the variability of parameters (Sargent, 2013, pp. 161–162).

As a subfield of operations research, Janová et al. followed Landry et al. in proposing conceptual, experimental and operational procedures for verification in the sense of the formal correctness and validation regarding checking the benefit-oriented effectiveness of the model. After verification and validation, the confidence in a model for decision support increases. For example, the model and its results should be checked by those who are familiar with the model (in this case experts, face validation and convergent expert validation), comparisons with the results of other models (in this case other indices) should

be carried out, and historical comparisons or sensitivity analysis should be applied (Janová, et al., 2019, p. 917) (Landry, et al., 1983, p. 215). For the present purpose, select procedures are used to apply what the author considers to be an appropriate mix of verification and validation procedures that allows the results of the DSI to be used as a basis for (economic) policy decisions.

5. Empirical analysis and results

Using the methods and procedures outlined above, the results of the present dissertation are presented. These support the sub-objectives of qualitative definition, quantification and index formation. The derivation of recommendations as a further sub-objective will only take place after the results have been presented and discussed.

5.1. Qualitative definition of digital sovereignty in the European value context

‘We preserve the capacity for autonomous action’ (Bitkom e.V., 2019, p. 11); this idea could be the basis for digital sovereignty in Europe, and also for the states that conform to non-free basic democratic values. The industry Bitkom does not use the above definition in the sense of digital autarky; autarkic states seal themselves off, do not participate in a globally networked world and develop and operate KETs themselves, even if they are less powerful.

In Section 2, an intensive literature review was conducted regarding the terms *sovereignty* and *digitalisation*. The individual terms were recognised in terms of historical emergence. From the analysis of the individual terms in Sections 2.1–2.4, a definition for *digital sovereignty* in the European understanding of values is synthesised. Part of the definition is based on the understanding of good, modern governance. For this, the understanding of Michel Foucault is used (cf. 2.1.3).

KETs, a clear understanding of which are relevant and the prioritisation of political and economic action are further bases for the definition (cf. 2.3). Ultimately, how technologically sovereign a state is over the KETs it uses and will use in the future is relevant to deal with technology and its related elements, such as data storage, in a self-determined manner (cf. 2.3). It should not be assumed that a state controls each KET individually or produces them independently. However, it should be clear that KETs are understood in principle, that freedom of choice exists and that a state is thus regarded as a negotiating partner vis-à-vis other technologically sovereign states.

In addition, shaping globally oriented markets is an integral part of EU economic policy. This should remain the focus of EU policy. It remains important that the EU countries not only build up their own technological competence in the sense of autarky, especially vis-à-vis their digitally strong partners. The EU states should be perceived as strong negotiating partners who can enforce requirements for imported technology. Therefore, digital sovereignty also means maintaining a seat that is 'eye level at the negotiating table'.

5.2. Results of interviews

Interviews were conducted with five experts to ascertain the basis for the index formation and the classification of the EU states' positioning regarding digital sovereignty. The questionnaire, the description of the interview partners and the results can be found in the Appendix. The main statements from the interviews are summarised as follows.

The currently weak position of the EU states regarding digital sovereignty is based on the previous strength of other economic sectors and the underestimation of the relevance of technological dependence. In addition, there is too little technological foresight and, as a result, no consistent, sustainable political orientation of the EU states. The positionings of the individual states differs significantly, and there is no position of the EU as a whole that can offer the digitally strong partners (the US and China) a counter-position in order to be perceived as a negotiating partner. The states that lead the digital sovereignty position in Europe tend to be smaller states with a long-standing political focus on strategic autonomy, including regarding defence policy, which may be a reason for their front positioning. In order to strengthen their position of the EU states, the state should play a stronger role as a role model and use its purchasing power to promote freedom of choice in technologies and the development of a European technology ecosystem. This includes the consistent training of digital experts and an increase in the attractiveness of the European labour market for European and non-European professionals. In addition, a non-bureaucratic funding ecosystem must be created with strategic farsightedness so that companies and entrepreneurs are incentivised to invest in the EU's locations. Tenders in accordance with EU public procurement law must be designed in such a way that companies that strengthen the EU states' digital sovereignty are given preference. The supposed advantages of other political systems to promote digital sovereignty should at least be understood to strengthen the EU's position, also as a negotiating partner. Technologies on which EU states should focus to strengthen their digital sovereignty are quantum technology, mobile technology, AI with various application areas, digital identities and blockchain.

In the opinion of some interviewees, the EU states should strengthen the KETs described above, use the economic power of the EU to establish a funding ecosystem and strengthen their own EU industry in select areas by strengthening the competence of EU professionals. Regarding where Europe should be in 10 years' time, the interviewees' opinions varied between the strengthening of a digitally self-sufficient EU to the acceptance of the use of technologies from the US and China combined with regulatory measures by the EU that only allow use of technologies from these states if they are trustworthy in terms of

data protection-compliance and cyber security requirements. However, this presupposes strong market power for Europe. The common position was that the development of already existing technologies cannot be caught up with in the short to medium term, and the focus must be on future technologies. The interviewees' predictions of where Europe will be in 10 years were critically reserved, as the timeframe was deemed too short to make up for the failures of the past decades. Without exogenous factors that require greater technology independence or consistent implementation-oriented digitalisation policy at the EU level, the EU states may be further sidelined. Therefore, digital sovereignty is more of a multi-decade project, unless there is an exogenous shock comparable to Russia's gas embargo. To form an index, the interviewees conducted relative weightings of 30 parameters (cf. Appendix) proposed by the author.

5.3. From quality to quantity: theoretical framework

From the proposal of the qualitative descriptive definition of digital sovereignty, a structured hierarchical model is derived below, which will be the basis on which to build an index that describes the reality of digital sovereignty in a measurable way (Kaloudis, 2022).

5.3.1. Sub-indices

The metric classification of the qualitative definition of digital sovereignty requires a theoretical framework (OECD, 2020d). This is developed through a hierarchical model. The sub-indices of the model are based on the definition of digital sovereignty proposed previously (cf. 5.1). A state is digitally sovereign if the three sub-indices are 'fulfilled':

- Sub-index 1: in the sense of the Foucaultian approach of 'good governance',
- Sub-index 2: technology by means of KETs and
- Sub-index 3: technology by the means of technological sovereignty.

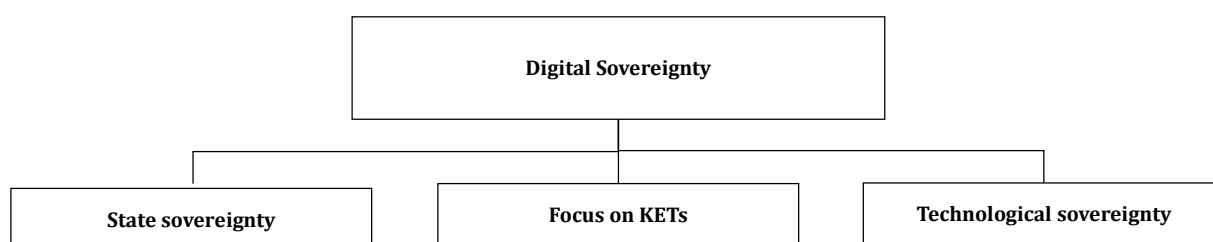


Figure 41 - Digital sovereignty sub-indices

For the quantitative classification of digital sovereignty, various secondary data sources are used to make statements on the sovereignty of states and the handling of digitalisation issues and technologisation.

Sub-index 1: State sovereignty (in the sense of 'good governance')

The first sub-index is the classification of state sovereignty in the value context of the EU. This component is intended to have a significant influence on the DSI, as otherwise an

autocratic state with a high autarkic digitalisation share would receive a high digital sovereignty index. Depending on a state's understanding of sovereignty, this assesses how strong a democratic or (semi-)autocratic state becomes for the quantitative measurement of digital sovereignty.

Sub-index 2: Digitalisation/focus on KETs

KETs are the focus of the technological consideration of digital sovereignty. Therefore, another sub-index is the assessment of a state regarding how KETs are defined, focused on and placed with regard to academic education and economic relevance. Total investments in technologies—and especially in KETs—also play a role here.

Sub-index 3: Digitalisation /technological sovereignty

The third sub-index focuses on technological understanding, plans to increase strategic autonomy and the ability to use technologies from other states and thus increase technology sovereignty.

5.3.2. Sub-components

Each sub-index is divided into sub-components to reinforce them.

Sub-components of sub-index 1: State sovereignty

State sovereignty is guided by elements measuring the quality of governance, such as government effectiveness, quality of regulation, rule of law, control of corruption, political stability and participation and freedom of expression. Furthermore, elements of *de jure* sovereignty and the well-being of a state's citizens, which is an important indicator of political stability, are included. These are complemented by an assessment of the prosperity of a state, which is directly correlated to good state sovereignty. This assesses the concept of good governance and 'good governmentality'. For further details, please refer to Section 2.1.5.

Sub-components of sub-index 2: Digitalisation/focus on KETs

There are many different metrics with which to examine, evaluate and develop how to deal with KETs. These include the ability of a state to promote the technological economy through foreign investments, the training of specialists in schools and universities, the focus on mathematical-technical education and the overall focus on education. It also includes harnessing the potential of citizens in the workforce. With the help of these criteria, governments gain the ability to identify which KETs are significant for a state and how to deal with them. Examples of these KETs include AI, quantum computing and blockchain technology. Another indicator of progress is the extent to which a state translates research results into products, with the number of patent applications playing a key role. However, states do not yet have a uniform understanding of which KETs are relevant (Kaloudis, 2021a, p. 7).

Sub-components of sub-index 3: Digitalisation /technological sovereignty

A good understanding of technology makes it possible to assess which data is sensitive and which measures are necessary to keep third-party dependence at a tolerable level. Various parameters can be chosen for this. The selection is based on the assumption that digital sovereignty in a basic liberal order does not come about through digital autarky in the political model of autocratic states but through an interstate controlled exchange of technology in an ecosystem (in the sense of global sovereignty). The parameters of how technological sovereignty is evaluated in a state include the trade relations with other countries that enable the exchange of digital goods and services. The assessment of bureaucratic obstacles to setting up a business in a state and a country's capacity for innovation also play a role here. In addition, the digitalisation progress of state institutions or society itself can be relevant. All these factors contribute to determining the extent to which a state has developed technologically compared to other countries and whether it remains 'in control' in terms of technology.

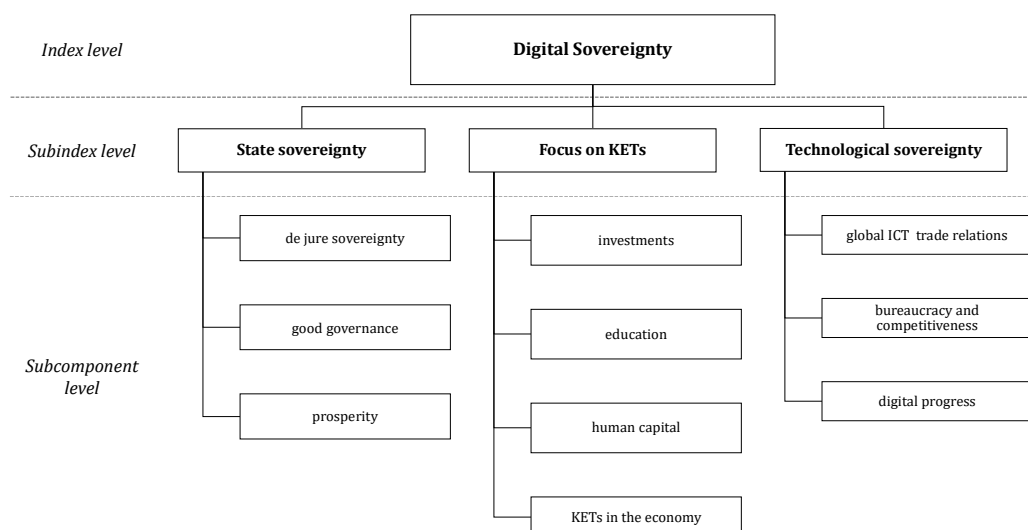


Figure 42 – Index-sub-index-sub-component hierarchy of the digital sovereignty index

For further consideration, the hierarchical model (Figure 42) is proposed as a basis for quantification; it is derived from the qualitative descriptive definition of digital sovereignty and thus from the sub-indices and sub-components.

5.4. Secondary data as a calculation basis for the sub-components

To make the model measurable, secondary data representing the subcomponents are defined in the following sections. The data are publishable, available in freely accessible databases and established. The general usability of the following parameters for quantifying digital sovereignty has been discussed, argued and derived above. Therefore, a reference to Section 2 is made as a reference in the explanation of each parameter. As there is no 'golden rule' regarding index formation, the parameters presented can be supplemented, reduced or

changed. This is described in Section 7 as part of the discussion of this dissertation. Therefore, if the results of the DSI are criticised, these parameters should be seen as a selection that should be further developed in more in-depth research. Therefore, in this dissertation, special emphasis is placed on the validation and verification of the index in Sections 5.14 and 5.15.

5.4.1. State sovereignty

5.4.1.1. *De jure sovereignty*

Whether a state is fundamentally sovereign or not is assessed with *de jure* sovereignty; only states that are recognised in principle are considered. As a basic prerequisite, it is assumed that only if a state is a member of the United Nations is it a state at all. This interpretation is common in the literature (Dahlmann, 2009) (Stepputat, 2015) (Fowler, Bunck, 1996). For the purpose of this dissertation, the degree of recognition is the quotient of the number of states that recognise another in comparison to all UN states, standardised to 100; a value of 100 means that all 193 UN states have recognised this state (Barnett, 2017, 58ff):

$$de\ jure\ -\ sovereignty = \theta \quad (18)$$

5.4.1.2. *Good governance*

As discussed in Section 2.1.5, the WGI has been mapping essential elements of good governance since 1996. Methodologically, it is based on ‘the traditions and institutions through which authority is exercised in a country’ (Kaufmann, et al., 2010). Good governance measures the extent to which a state is able to describe and write the foundations for its own policies and to establish them meaningfully and effectively in politics and society. In addition, the extent to which various political and social institutions interact with citizens is surveyed. This approach corresponds to the quantification of good governance in the Foucaultian sense. Following Barnett, the World Happiness Index is used as a supplement. The state sovereignty index is derived from the sum of the six WGI indicators (β) and the World Happiness Index λ (cf. 2.1.5) (Barnett, 2017, p. 21):

$$good\ governance\ \varepsilon = \beta_1 + \beta_2 + \beta_3 + \beta_4 + \beta_5 + \beta_6 + \lambda \quad (19)$$

For the index calculation, the sum of *de jure* sovereignty and good governance is formed:

$$X_1 = \theta + \varepsilon \quad (20)$$

5.4.1.3. *Prosperity*

The per capita GDP is used to measure the material prosperity of a country (Worldbank, 2022, GDP) (Worldbank, 2022, Population). It can be considered problematic to integrate the prosperity of a state into the index, as this gives relatively prosperous states a particularly high index value. However, good governance and prosperity are relatively strongly correlated, so any outliers produced are manageable. In addition, the economic

strength of a state can reflect how high its willingness to invest in digital technologies is, so this parameter is not omitted.

$$X_2 = GDP_{per\ capita}^{128} \quad (21)$$

5.4.2. Focus on key enabling technologies

5.4.2.1. Investments

Essential, digital infrastructures must be available for foreign direct investments; without these, they could not be effectively operated. In addition, such investment activities are a characteristic of the ability of states or their industries to interact with global markets (cf. 2.2.5.5). A distinction is made between net inflows and outflows, both of which are seen as a positive feature of investment in general, thus also of potential investments in KETs.

$$X_3 = FDI_{inflow}^{129} \quad (22)$$

$$X_4 = FDI_{outflow}^{130} \quad (23)$$

Net inflows can be assessed critically for digital sovereignty (cf. 7.5.5), as this can be a characteristic of dependency. However, since technological autarky is not intended for the EU states, inflows are also assessed positively for this work. The level of investment in Start-ups is used as a key figure (cf. 2.2.6.6) to assess the ability of a state to promote new business models.

$$X_5 = Investment\ Startups^{131} \quad (24)$$

5.4.2.2. (Education) Training

The qualifications of a state's citizens is the basis for (technological) progress (cf. 2.2.6.4). On the one hand, this contributes to the ability of citizens to be digitally sovereign or to participate as professionals in the digitalisation of the state. A country's performance is assessed on the basis of the digital competence its citizens acquire through education. This begins with school education, continues with training and studies and ends with ICT specialists in the labour market. The following key figures are suitable for measuring the educational level.

$$X_6 = TIMMS^{132} \quad (25)$$

$$X_7 = Assessment\ in\ reading,\ mathematics,\ and\ science^{133} \quad (26)$$

¹²⁸ Source: Worldbank (2022)

¹²⁹ Ibid.

¹³⁰ Ibid.

¹³¹ OECD (2022b)

¹³² IEA (2022) (Students reaching the internat. benchmarks of mathematics achievement, grade 4)

¹³³ Source: GII (2022)

$$X_8 = \text{Expenditure on education as a percentage of gross domestic product}^{134} \quad (27)$$

5.4.2.3. Human capital

Human capital—the knowledge-based economic power of a country's workforce—is also assessed. The number of employed ICT specialists per country represents the amount of ICT know-how that can be contributed to the digitalisation of a country. These quantitative data are well recorded for the EU; for other countries, only singular sources can be found (EU Open Data Portal, 2022, Persons with ICT education by labour status).

$$X_9 = \text{Number of employed ICT specialists}^{135} \quad (28)$$

In addition, the number of successfully filed patents with an ICT focus is evaluated to provide an overview of a country's ability to generate innovative ideas. Patent submissions are documented worldwide for patent law reasons and can be regulated (World Intellectual Property Organization, 2021). For this dissertation, published patents in the following categories are evaluated: cloud, AI, quantum, information technology and telecommunication.

$$X_{10} = \text{Number of patents}^{136} \quad (29)$$

The quality of universities is included in the evaluation as a key figure. Here, a university's ranking (QS University Ranking) is used as a parameter, with the average value from the top three universities.

$$X_{11} = \text{QS university ranking, average score TOP 3}^{137} \quad (30)$$

Another evaluation parameter is the extent to which research is transferred to industry. For this purpose, the sub-indicator for the number of publications in mathematics and computer science in cooperation between universities and industry is used from the Leiden Ranking in comparison to all publications with mathematical or information technology content (cf. 2.2.6.4) (van Eck, 2022).

$$X_{12} = \frac{\text{num.of pub. in Maths \& Computer Science in industry-collaboration}}{\text{number of publications in Maths and Computer Science}}^{138} \quad (31)$$

This refers to the research and development management method and the importance of applied research in industry.

¹³⁴ Ibid.

¹³⁵ Source: Eurostat (2022) (Persons with ICT education by labour status)

¹³⁶ Source: World Intellectual Property Organization (2021)

¹³⁷ Source: GII (2022)

¹³⁸ van Eck (2022) (parameter used calculated independently)

The Human Capital Index is used as a complementary factor. The index, collected by the World Bank, measures the extent to which a country actually uses the potential of its citizens (cf. 2.2.5.4, 2.2.6.4 and 2.2.6.5).

$$X_{13} = \text{Human Capital Index}^{139} \quad (32)$$

5.4.2.4. KETs in the economy

As a characteristic of transparency and data availability, an Open Data supporting ecosystem, the index for government support of data use is generated from the characteristics of availability and data accessibility (Lafortune, Ubaldi, 2019).

$$X_{14} = \text{Open Data Index}^{140} \quad (33)$$

To assess how KETs that may be relevant to states are promoted in a state's economy through research and development, the following measure is used:

$$X_{15} = \text{Research Development Expenditure}^{141} \quad (34)$$

With a focus on concrete key enabling technologies¹⁴², investment volumes, existing indices and Start-up activities are evaluated.

$$X_{16} = \text{Public funding for Quantum}^{143} \quad (35)$$

$$X_{17} = \text{Artificial Intelligence Index}^{144} \quad (36)$$

$$X_{18} = \text{Number of blockchain Start-ups}^{145} \quad (37)$$

5.4.3. Technological sovereignty

5.4.3.1. Global ICT trade relations

The measurement of digitised trade is used. To assess the extent to which states trade in digital goods and services, parameters for imports, exports, restrictions and freedoms in the context of trade in digital goods and services are analysed (cf. 2.2.6.2). The selection of individual indices focuses on those that describe the trade relations of ICT services or goods.

$$X_{19} = \text{Amount of exported ICT services}^{146} \quad (38)$$

¹³⁹ Source: Worldbank (2022)

¹⁴⁰ Source: EU Open Data Portal (2022)

¹⁴¹ Source: GII (2022)

¹⁴² To be understood as a selection.

¹⁴³ Source: McKinsey Global Institute (2020)

¹⁴⁴ Source: Cambrian Institute (2020)

¹⁴⁵ Source: IP Lytics (2019)

¹⁴⁶ Source: Worldbank (2022)

$$X_{20} = \text{Share of exported ICT services in total exports}^{147} \quad (39)$$

$$x_{21} = \text{Amount of imported ICT goods}^{148} \quad (40)$$

$$X_{22} = \text{Amount of imported ICT services}^{149} \quad (41)$$

A high volume of imports is considered positive for digital sovereignty, as this is a characteristic of the ability to trade and exploit these services and goods.

The freedom of trade digital goods and services, measured by the digital Services Trade Restrictiveness Index (digital STRI), is also used (cf. 2.2.6.3) (OECD, 2020c).

$$X_{23} = \text{digital STRI}^{150} \quad (42)$$

5.4.3.2. Bureaucracy and competitiveness

To assess the bureaucracy of a state, the index for assessing the promotion of business activities and the index for assessing digital competitiveness are used (cf. 2.2.6.7).

$$X_{24} = \text{EDBI}^{151} \quad (43)$$

$$X_{25} = \text{WDC}^{152} \quad (44)$$

Another criterion for competitiveness in the sense of technological sovereignty is the ability of a state to deal with (weak, according to the definition of digital sovereignty) data protection. The EU's assessment of whether a state has an adequate level of data protection is used to measure this. Based on Article 45 of EU Regulation 2016/679, the EU can define 'whether a country outside the EU offers an adequate level of data protection' (Tarr, et al., 2022). If confirmed, data transfers to the country in question are treated the same as data exchanges within the EU. The EU data evaluation takes place digitally: 1 means a country is classified accordingly, otherwise a 0 is assigned. To assess data protection, the score from the Comparitech study is used and added to the result of the EU data protection assessment (cf. 2.2.6.9).

$$X_{26} = \text{EU Data Protection} + \text{Score Comparitech}^{153} \quad (45)$$

5.4.3.3. Digital progress

The ranking of a state regarding digital capabilities, digital progress and digital openness is formed by individual parameters. On the one hand, the population's access to digital content is assessed.

¹⁴⁷ Source: Worldbank (2022)

¹⁴⁸ Ibid.

¹⁴⁹ Ibid.

¹⁵⁰ Source: OECD (2022b)

¹⁵¹ Source: Worldbank (2022) (www.doingbusiness.org)

¹⁵² Source: IMD (2022)

¹⁵³ Source: Eurostat (2022), Bischoff (2019)

$$X_{27} = \text{Internet Access Households}^{154} \quad (46)$$

$$X_{28} = \text{SIM Cards to Population}^{155} \quad (47)$$

On the other hand, the digitalisation of state organisations, ministries and authorities is used with the following indices (cf. 2.2.5.4):

$$X_{29} = \text{Government's online service}^{156} \quad (48)$$

$$X_{30} = \text{E-Government Rank (EGDI)}^{157} \quad (49)$$

Thus, 30 parameters are defined that can form the basis of the composite digital sovereignty index.

5.5. Summary of the data

The secondary data described above are normalised in such a way that a value of 100 indicates maximum support for the goal of the EU states according to the definition proposed in Section 5.1; a 0 indicates that it does not support this goal. Alternative value interpretations are not pursued¹⁵⁸.

Table 11 - Parameter overview

X_i	description	X_i	description	X_i	description
1	Sovereignty	11	TOP University	21	Import ICT goods total
2	GDP p.c.	12	Leiden Index	22	Import ICT services
3	FDI in	13	Human capital	23	Digital STRI
4	FDI out	14	OpenData Index	24	EDBI
5	Start-up invest	15	R&D expenditure	25	WDC
6	TIMMS	16	Quantum pub. funding	26	GDPR
7	Assesm. reading, etc.	17	AI Index	27	Internet access
8	Expend. education	18	Blockchain Start-ups	28	SIM cards
9	Qualif. ICT employees	19	Exp. ICT services total	29	Governments online
10	Patents	20	Exp. ICT serv./total ¹⁵⁹	30	E-Government rank

¹⁵⁴ Source: ITU (2022)

¹⁵⁵ Ibid.

¹⁵⁶ Source: GII (2022)

¹⁵⁷ Source: United Nations (2022)

¹⁵⁸ Alternatively, other assessments could be made, e.g. 100 digital autarky, 0 digital autocracy and 50 digital strategic autonomy. This approach is explicitly not followed in this work.

¹⁵⁹ X_{20} is the share of exported IT services in total exports vs. X_{19} is the amount of total exported IT services

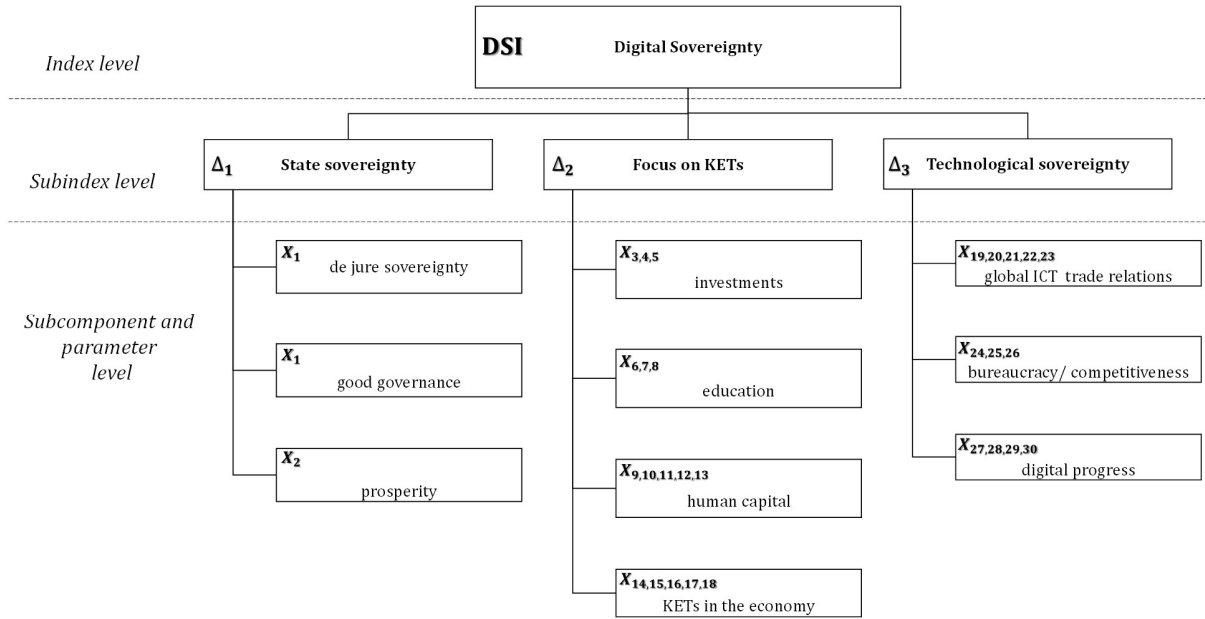


Figure 43 - Classification of parameters in the model¹⁶⁰

The parameters of the subcomponents are aggregated into subindices: Δ_1 , Δ_2 , Δ_3 . The multiplicative or additive aggregation procedure is explained in Section 5.10. The state sovereignty sub-index consists of sovereignty and wealth parameters. Since the parameters can compensate each other, a multiplicative sub-index of Δ_1 is formed, which describes the sub-index 'state sovereignty' with the following:

$$\Delta_1 = \prod_{i=1}^2 X_i^{\gamma_i} \quad (50)$$

The two digitalisation factors in the sub-indices 'KETs' and 'technological sovereignty' are assessed with parameters that may have compensatory effects, but this is not detrimental to the informative value of the index to be formed. The parameters are weighted and added together in sub-indices Δ_2 ('focus on KETs') and Δ_3 ('technological sovereignty') as follows:

$$\Delta_2 = \sum_{i=3}^{18} \gamma_i X_i \quad (51)$$

$$\Delta_3 = \sum_{i=19}^{30} \gamma_i X_i \quad (52)$$

State sovereignty (Δ_1) and the both sub-indices for digitalisation, focus on KETs (Δ_2) and technological sovereignty (Δ_3), are examined for correlations in further analyses.

¹⁶⁰ X_1 represents *de jure* sovereignty and good governance.

5.6. Analysis of the data

5.6.1. Descriptive statistics

The data in Sections 5.4 and 5.5 are first subjected to a scale transformation by transforming the largest value from X to 100 as a reference value. Thus, a value of 100 means that the corresponding country is in the top position for the respective parameter. The scale-transformed secondary data are described below for all X_i with the number of values per country available through secondary data: minimum, maximum, mean and 25th, 50th and 75th percentiles.

Table 12 - Descriptive statistics X_i , raw data without imputation

X_i	Val. no.	Min	Max	Mean	25 th quantiles	50 th quantiles	75 th quantiles
1	121	65.69	100.00	81.23	74.89	79.65	88.53
2	194	.00	100.00	9.50	1.15	3.72	11.11
3	185	.00	100.00	13.82	11.74	11.94	12.86
4	167	.00	100.00	38.35	36.68	36.70	37.14
5	29	.00	100.00	4.70	.03	.20	.44
6	55	41.85	100.00	82.15	76.51	85.88	89.16
7	76	1.00	100.00	54.59	27.28	55.49	83.69
8	127	.00	100.00	42.52	32.29	43.75	55.21
9	37	.00	100.00	6.67	.25	.91	2.64
10	10	.01	100.00	11.25	.09	.24	.96
11	127	24.00	100.00	46.83	24.00	36.00	68.75
12	59	.00	100.00	35.15	15.99	29.73	50.98
13	187	.00	100.00	68.81	55.76	73.96	84.29
14	34	.00	100.00	60.86	52.85	63.57	74.41
15	126	.00	100.00	7.68	.71	2.08	7.76
16	14	3.70	100.00	23.05	3.70	15.93	29.63
17	13	16.00	100.00	42.92	33.75	38.00	44.25
18	9	3.99	100.00	18.73	6.40	8.40	12.29
19	200	.00	100.00	3.39	.00	.08	.90
20	127	.00	100.00	8.88	1.94	6.20	11.63
21	170	.00	100.00	4.12	.14	.53	2.47
22	127	.00	100.00	19.60	8.96	16.42	26.49
23	45	.00	100.00	68.28	50.36	77.23	86.15
24	127	1.00	100.00	50.61	24.85	51.38	75.62
25	62	23.99	100.00	69.73	54.84	69.11	84.11
26	42	56.25	100.00	89.73	87.50	90.63	93.75
27	199	.27	100.00	52.88	23.64	57.51	82.02
28	195	5.96	100.00	32.22	25.53	32.56	38.70
29	127	38.97	100.00	70.13	54.93	69.95	85.68
30	187	1.00	100.00	51.37	26.00	51.78	76.52

For graphical clarification and position determination, the 30 parameters from Table 12 are visualised in a box plot. The boxes in Figure 44 show the minimum, maximum, median, 25th and 75th percentiles, as well as the outliers.

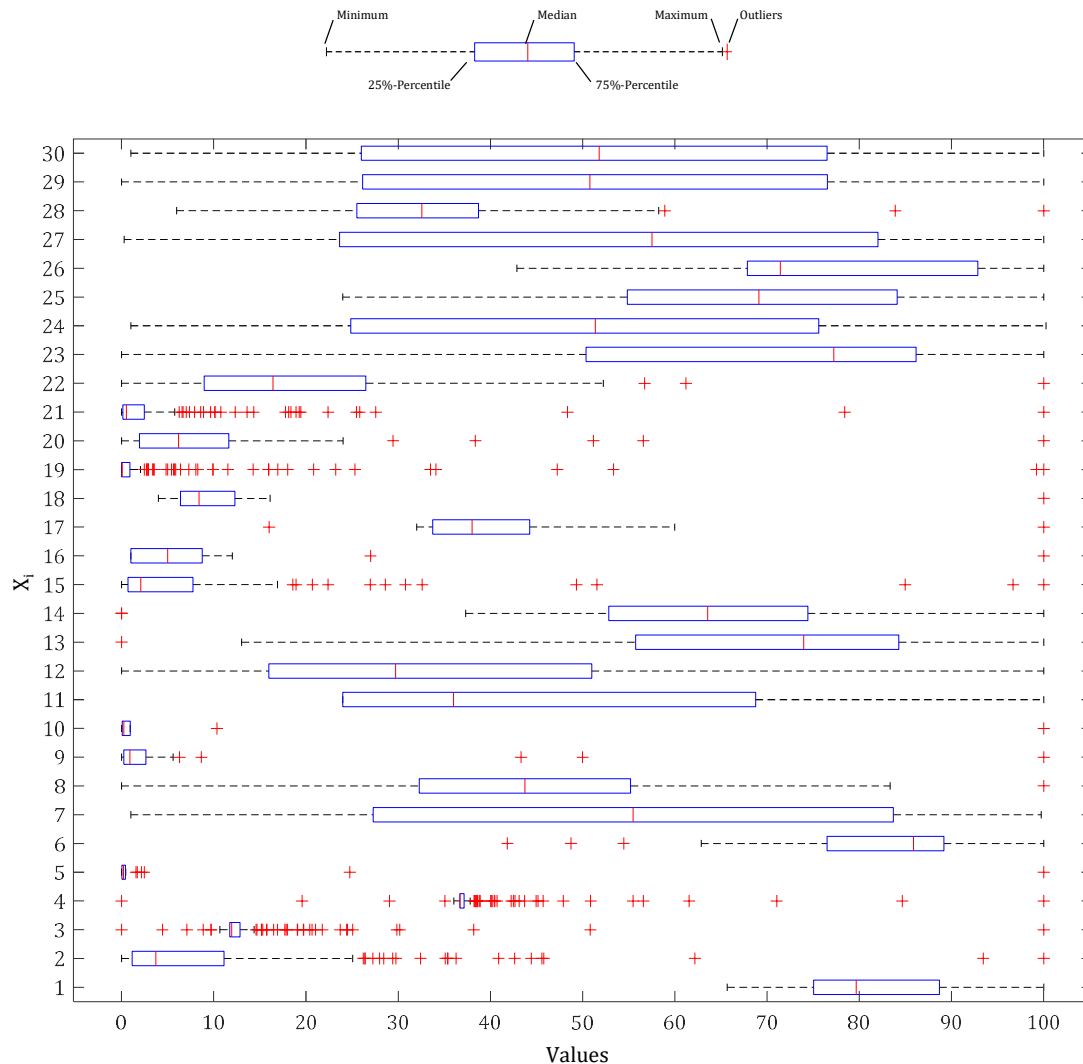


Figure 44 - Boxplot of raw data

5.6.2. Measures of determination and p-values

For the development of indices, the sub-indices or parameters used should be as free of overlap as possible in order to not have secondary data flow into an index calculation more than once, thus reducing double effects. In reality, however, overlaps are often found, which can be compensated for by suitable weighting procedures. First, the respective pairwise coefficients of determination are determined to identify dependencies and correlations between the parameters. An example of this is done below (Figure 45) on the basis of the transformed raw data. Any other data modification by normalisation or imputation can also

be analysed. For visualisation, the scatterplots of all X s, including their histograms, are shown.

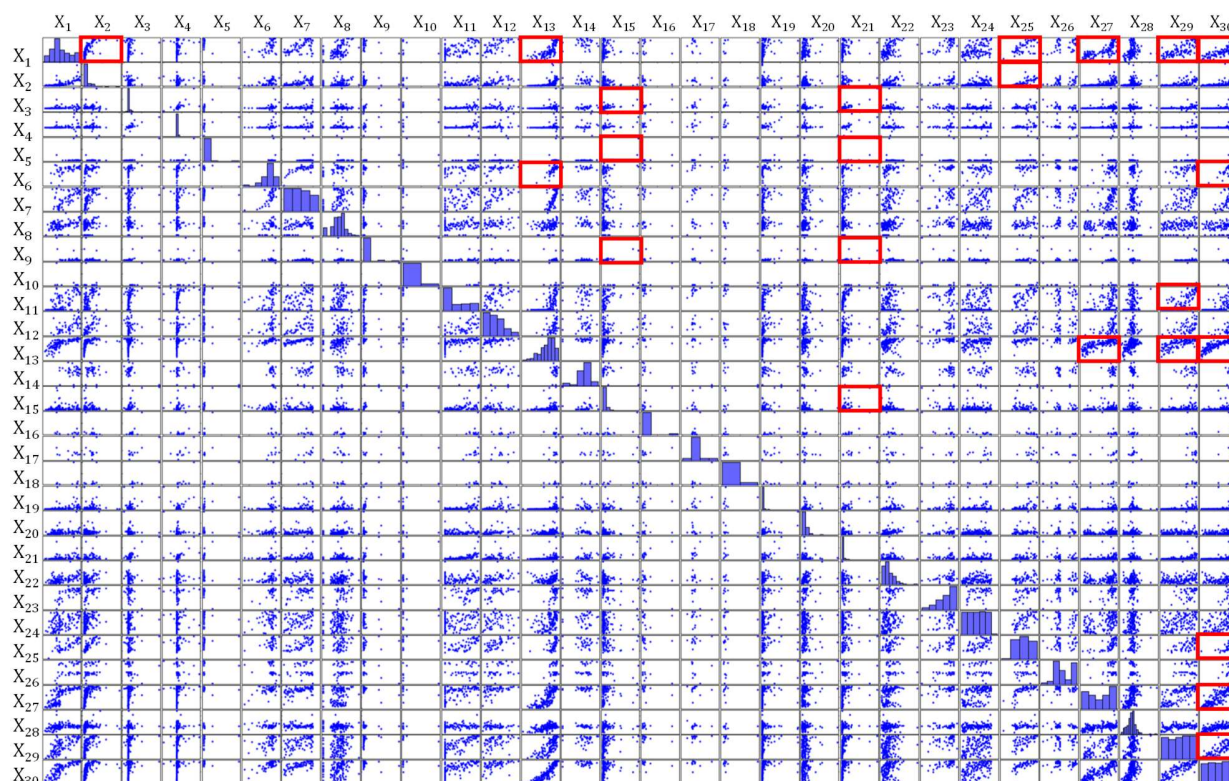


Figure 45 - Scatter plot¹⁶¹ of all variables $X_{i,j}$ with $i, j = 1, \dots, 30$,
diagonal: histograms X_i , outlined in red: $R^2 \geq 50\%$

The pairs $X_{i,j}$ with a coefficient of determination greater than or equal to 50% (i.e. those that show a relatively high correlation¹⁶²) and a significance level greater than 5% are marked. Linear regression analysis is used to determine the proportion of dispersion of the measured points in pairs—a predictor (X_i) and a regressor (X_j)—which can be explained by a regression function.

¹⁶¹ Enlarged graphic in Appendix

¹⁶² Values $>30\%$ can be considered moderate and $>70\%$ strictly positive, cf. Ratner (2009).

Table 13 - Top: Measures of determination X_i , bottom: p-values X_i , both: normal fit

X_i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
1	1.00	.66	.04	.05	.01	.26	.33	.16	.14	.17	.37	.34	.54	.00	.01	.01	.00	.06	.11	.06	.09	.20	.49	.16	.58	.21	.54	.17	.50	.67	
2	.66	1.00	.06	.04	.01	.21	.32	.06	.03	.10	.35	.38	.23	.00	.01	.08	.04	.17	.07	.08	.12	.22	.30	.14	.56	.12	.35	.10	.41	.32	
3	.04	.06	1.00	.46	.87	.07	.03	.01	.43	.10	.20	.04	.04	.18	.56	.10	.84	.84	.16	.01	.78	.00	.01	.02	.10	.45	.04	.02	.13	.07	
4	.05	.04	.46	1.00	.37	.06	.04	.00	.12	.08	.15	.09	.02	.02	.17	.14	.29	.53	.01	.08	.47	.00	.00	.00	.07	.11	.04	.02	.09	.05	
5	.01	.01	.87	.37	1.00	.07	.00	.01	.32	.02	.09	.03	.00	.44	.66	.03	.96	.92	.14	.01	.69	.03	.00	.00	.12	.76	.06	.00	.04	.01	
6	.26	.21	.07	.06	.07	1.00	.49	.10	.09	.02	.15	.25	.50	.02	.02	.06	.14	.00	.07	.01	.08	.09	.13	.08	.14	.02	.30	.01	.17	.54	
7	.33	.32	.03	.04	.00	.49	1.00	.15	.06	.09	.20	.29	.40	.09	.00	.03	.06	.04	.14	.04	.10	.07	.14	.17	.42	.00	.22	.02	.25	.44	
8	.16	.06	.01	.00	.01	.10	.15	1.00	.00	.05	.07	.11	.05	.03	.00	.01	.05	.01	.01	.02	.01	.10	.11	.03	.08	.12	.02	.00	.05	.06	
9	.14	.03	.43	.12	.32	.09	.06	.00	1.00	.77	.14	.01	.22	.25	.81	.05	.21	.00	.23	.00	.57	.07	.50	.00	.02	.78	.36	.01	.04	.05	
10	.17	.10	.10	.08	.02	.02	.09	.05	.77	1.00	.12	.01	.11	.88	.34	.00	.08	.01	.01	.01	.33	.15	.45	.03	.02	.80	.07	.00	.37	.02	
11	.37	.35	.20	.15	.09	.15	.20	.07	.14	.12	1.00	.28	.34	.02	.26	.01	.42	.21	.24	.02	.37	.04	.00	.09	.39	.05	.34	.08	.58	.45	
12	.34	.38	.04	.09	.03	.25	.29	.11	.01	.01	.28	1.00	.20	.00	.00	.00	.16	.48	.12	.10	.08	.29	.25	.05	.43	.04	.19	.01	.15	.21	
13	.54	.23	.04	.02	.00	.50	.40	.05	.22	.11	.34	.20	1.00	.00	.02	.02	.04	.04	.06	.03	.07	.04	.32	.14	.24	.28	.62	.28	.51	.77	
14	.00	.00	.18	.02	.44	.02	.09	.03	.25	.88	.02	.00	.00	1.00	.11	.13	.56	.80	.00	.02	.08	.00	.02	.02	.00	.01	.00	.02	.00	.00	
15	.01	.01	.56	.17	.66	.02	.00	.00	.81	.34	.26	.00	.02	.11	1.00	.09	.44	.30	.32	.00	.63	.00	.18	.00	.00	.58	.01	.00	.14	.05	
16	.01	.08	.10	.14	.03	.06	.03	.01	.05	.00	.01	.00	.02	.13	.09	1.00	.01	.01	.30	.01	.66	.00	.05	.47	.16	.09	.04	.00	.00	.07	
17	.00	.04	.84	.29	.96	.14	.06	.05	.21	.08	.42	.16	.04	.56	.44	.01	1.00	.83	.09	.07	.78	.03	.00	.02	.16	.71	.01	.03	.02	.02	
18	.06	.17	.84	.53	.92	.00	.04	.01	.00	.01	.21	.48	.04	.80	.30	.01	.83	1.00	.05	.06	.51	.02	.11	.01	.24	.01	.02	.01	.14	.07	
19	.11	.07	.16	.01	.14	.07	.14	.01	.23	.01	.24	.12	.06	.00	.32	.30	.09	.05	1.00	.38	.31	.03	.00	.01	.07	.04	.04	.00	.14	.09	
20	.06	.08	.01	.08	.01	.01	.04	.02	.00	.01	.02	.10	.03	.02	.00	.01	.07	.06	.38	1.00	.00	.21	.02	.03	.01	.03	.03	.00	.03	.05	
21	.09	.12	.78	.47	.69	.08	.10	.01	.57	.33	.37	.08	.07	.08	.63	.66	.78	.51	.31	.00	1.00	.00	.01	.02	.17	.44	.08	.02	.20	.13	
22	.20	.22	.00	.00	.03	.09	.07	.10	.07	.15	.04	.29	.04	.00	.00	.00	.03	.02	.03	.21	.00	1.00	.12	.03	.09	.07	.07	.00	.05	.10	
23	.49	.30	.01	.00	.00	.13	.14	.11	.50	.45	.00	.25	.32	.02	.18	.05	.00	.11	.00	.02	.01	.12	1.00	.07	.23	.37	.28	.00	.08	.34	
24	.16	.14	.02	.00	.00	.08	.17	.03	.00	.03	.09	.05	.14	.02	.00	.47	.02	.01	.01	.03	.02	.03	.07	1.00	.40	.01	.18	.07	.18	.20	
25	.58	.56	.10	.07	.12	.14	.42	.08	.02	.02	.39	.43	.24	.00	.00	.16	.16	.24	.07	.01	.17	.09	.23	.40	1.00	.01	.44	.12	.38	.50	
26	.21	.12	.45	.11	.76	.02	.00	.12	.78	.80	.05	.04	.28	.01	.58	.09	.71	.01	.04	.03	.44	.07	.37	.01	.01	1.00	.17	.01	.00	.14	
27	.54	.35	.04	.04	.06	.30	.22	.02	.36	.07	.34	.19	.62	.00	.01	.04	.01	.02	.04	.03	.08	.07	.28	.18	.44	.17	1.00	.29	.46	.76	
28	.17	.10	.02	.02	.00	.01	.02	.00	.01	.00	.08	.01	.28	.02	.00	.00	.03	.01	.00	.00	.02	.00	.00	.00	.07	.12	.01	.29	1.00	.18	.39
29	.50	.41	.13	.09	.04	.17	.25	.05	.04	.37	.58	.15	.51	.00	.14	.00	.02	.14	.14	.03	.20	.05	.08	.18	.38	.00	.46	.18	1.00	.72	
30	.67	.32	.07	.05	.01	.54	.44	.06	.05	.02	.45	.21	.77	.00	.05	.07	.02	.07	.09	.05	.13	.10	.34	.20	.50	.14	.76	.39	.72	1.00	

p	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	.00	.00	.02	.02	.66	.00	.00	.00	.03	.27	.00	.00	.00	.72	.44	.82	.98	.56	.00	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
2	.00	.00	.00	.01	.56	.00	.00	.01	.33	.37	.00	.00	.00	.96	.28	.34	.49	.27	.00	.00	.00	.00	.00	.00	.00	.02	.00	.00	.00	.00
3	.02	.00	.00	.00	.00	.05	.11	.42	.00	.38	.00	.14	.01	.01	.00	.30	.00	.00	.00	.41	.00	.67	.60	.16	.01	.00	.01	.09	.00	.00
4	.02	.01	.00	.00	.00	.08	.10	.69	.04	.43	.00	.02	.05	.46	.00	.20	.06	.03	.16	.00	.00	.48	.73	.72	.04	.03	.01	.11	.00	.00
5	.66	.56	.00	.00	.00	.22	.94	.68	.00	.80	.11	.36	.72	.00	.00	.71	.00	.00	.04	.61	.00	.38	.80	.91	.07	.00	.21	.96	.29	.56
6	.00	.00	.05	.08	.22	.00	.00	.02	.12	.74	.00	.00	.00	.49	.36	.56	.32	.89	.06	.52	.05	.03	.05	.04	.02	.50	.00	.48	.00	.00
7	.00	.00	.11	.10	.94	.00	.00	.00	.14	.44	.00	.00	.00	.10	.66	.62	.44	.63	.00	.10	.01	.02	.02	.00	.00	.93	.00	.29	.00	.00
8	.00	.01	.42	.69	.68	.02	.00	.00	.68	.54	.00	.01	.01	.29	.94	.78	.44	.82	.25	.10	.41	.00	.02	.05	.03	.02	.13	.93	.01	.01
9	.03	.33	.00	.04	.00	.12	.14	.68	.00	.05	.02	.63	.00	.01	.00	.64	.26	.96	.00	.92	.00	.12	.00	.91	.47	.00	.00	.49	.24	.18
10	.27	.37	.38	.43	.80	.74	.44	.54	.05	.00	.33	.78	.36	.02	.08	.98	.54	.82	.83	.76	.08	.27	.05	.65	.70	.01	.46	.91	.06	.71
11	.00	.00	.00	.00	.11	.00	.00	.00	.02	.33	.00	.00	.00	.45	.00	.75	.02	.22	.00	.11	.00	.02	.92	.00	.00	.16	.00	.00	.00	.00
12	.00	.00	.14	.02	.36	.00	.00	.01	.63	.78	.00	.00	.00	.82	.94	.98	.20	.04	.01	.02	.03	.00	.00	.08	.00	.24	.00	.58	.00	.00
13	.00	.00	.01	.05	.72	.00	.00	.01	.00	.36	.00	.00	.00	.76	.11	.62	.51	.62	.00	.04	.00	.03	.00	.00	.00	.00	.00	.00	.00	.00
14	.72	.96	.01	.46	.00	.49	.10	.29	.01	.02	.45	.82	.76	.00	.06	.31	.02	.02	.93	.46	.10	.73	.42	.47	.76	.70	.85	.39	.71	.86
15	.44	.28	.00	.00	.00	.36	.66	.94	.00	.08	.00	.94	.11	.06	.00	.31	.01	.13	.00	.73	.00	.49	.00	.89	.97	.00	.27	.63	.00	.02
16	.82	.34	.30	.20	.71	.56	.62	.78	.64	.98	.75	.98	.62	.31	.31	.00	.77	.80	.05	.80	.00	.86	.51	.01	.16	.39	.48	.94	.95	.40
17	.98	.49	.00	.06	.00	.32	.44	.44	.26	.54	.02	.20	.51	.02	.01	.77	.00	.00	.31	.37	.00	.59	.98	.69	.18	.00	.81	.55	.63	.68
18	.56	.27	.00	.03	.00	.89	.63	.82	.96	.82	.22	.04	.62	.02	.13	.80	.00	.00	.54	.53	.03	.71	.43	.85	.18	.84	.75	.80	.32	.49
19	.00	.00	.00	.16	.04	.06	.00	.25	.00	.83	.00	.01	.00	.93	.00	.05	.31	.54	.00	.00	.00	.07	.81	.21	.04	.20	.00	.60	.00	.00
20	.01	.00	.41	.00	.61	.52	.10	.10	.92	.76	.11	.02	.04	.46	.73	.80	.37	.53	.00	.00	.93	.00	.42	.07	.49	.26	.06	.74	.07	.01
21	.00	.00	.00	.00	.00	.05	.01	.41	.00	.08	.00	.03	.00	.10	.00	.00	.00	.03	.00	.93	.00	.71	.61	.14	.00	.00	.00	.06	.00	.00
22	.00	.00	.67	.48	.38	.03	.02	.00	.12	.27	.02	.00	.03	.73	.49	.86	.59	.71	.07	.00	.71	.00	.02	.04	.02	.08	.00	.98	.01	.00
23	.00	.00	.60	.73	.80	.05	.02	.02	.00	.05	.92	.00	.00	.42	.00	.51	.98	.43	.81	.42	.61	.02	.00	.09	.00	.00	.00	.86	.06	.00
24	.00	.00	.16	.72	.91	.04	.00	.05	.91	.65	.00	.08	.00	.47	.89	.01	.69	.85	.21	.07	.14	.04	.09	.00	.00	.61	.00	.00	.00	.00
25	.00	.00	.03	.04	.07	.02	.03	.03	.47	.70	.00	.00	.00	.76	.97	.16	.18	.18	.04	.49	.00	.02	.00	.00	.00	.59	.00	.01	.00	.00
26	.00	.02	.00	.03	.00	.50	.93	.02	.00	.11	.16	.24	.00	.70	.00	.39	.00	.84	.20	.26	.00	.08	.00	.61	.59	.00	.01	.51	.92	.01
27	.00	.00	.01	.01	.21	.00	.00	.13	.00	.46	.00	.00	.00	.85	.27	.48	.81	.75	.00	.06	.00	.00	.00	.00	.00	.01	.00	.00	.00	.00
28	.00	.00	.09	.11	.96	.48	.29	.93	.49	.91	.00	.58	.00	.39	.63	.94	.55	.80	.60	.74	.06	.98	.86	.00	.01	.51	.00	.00	.00	.00
29	.00	.00	.00	.00	.29	.00	.00	.01	.24	.06	.00	.00	.00	.71	.00	.95	.63	.32	.00	.07	.00	.01	.06	.00	.00	.92	.00	.00	.00	.00
30	.00	.00	.00	.00	.56	.00	.00	.01	.18	.71	.00	.00	.00	.86	.02	.40	.68	.49	.00	.01	.00	.00	.00	.00	.00	.01	.00	.00	.00	.00

For selected $X_{i,j}$, it is shown that other types of regression yield better coefficients of determination than linear regression¹⁶³. The latter is used for the sake of simplicity. The use of predictor and regressor is commutative; therefore, the result is a mirror matrix.

The significance level of p-values above 5% confirms the normal distribution hypothesis; thus, the regression result can be considered usable. The coefficients of determination greater than or equal to 50% that have a p-value of less than 5% are relevant, which is clear from the previous table. In addition, the regression is reduced to the parameters for which at least 50 states have values (approximately 25% of the population), as a small number of input values may lead to uncertainty. This identifies the parameters that have a relatively high linear correlation. This is true for individual combinations¹⁶⁴, but both parameters are dependent on only a few others with high coefficients of determination, so neither should be omitted from the index calculation. This applies to all pairwise high coefficients of determination.

For this reason, the 30 parameters should be used in their entirety for the index calculation. Effects of dependence can be mitigated by suitable weightings, as explained previously.

5.7. Replacement of missing data

As can be seen in Table 12, complete parameter values are not available for all states. In order to replace the missing data sets, the following imputation procedures are applied (following Section 4.6): imputation by kNN procedure, imputation by mean value and omission of certain states with the consequence that no index can be calculated for them. Imputation changes the descriptive statistics, which is illustrated by the comparison of mean imputation versus kNN procedure.

The replacement of missing data changes the results of the descriptive statistics per imputation procedure compared to the raw data as well as the results of the linear regression analysis.

The number of linear correlations with a coefficient of determination greater than or equal to 50% is increased by the application of the kNN method and reduced by replacing missing data with mean values. Since as few linear correlations as possible should be present in the context of index formation, the replacement of missing data by mean values seems to be more suitable.

¹⁶³ Example: human capital is part of the index formation of the eGovernment rank. Linear regression analysis with X_{13} (human capital) and X_{30} (eGovernment rank) results in an $R^2 = 77\%$, an exponential regression results in an $R^2 = 81\%$.

¹⁶⁴ Example: X_9 (qualified ICT employees) and X_{15} (R&D expenditure) with $R^2 = 81\%$

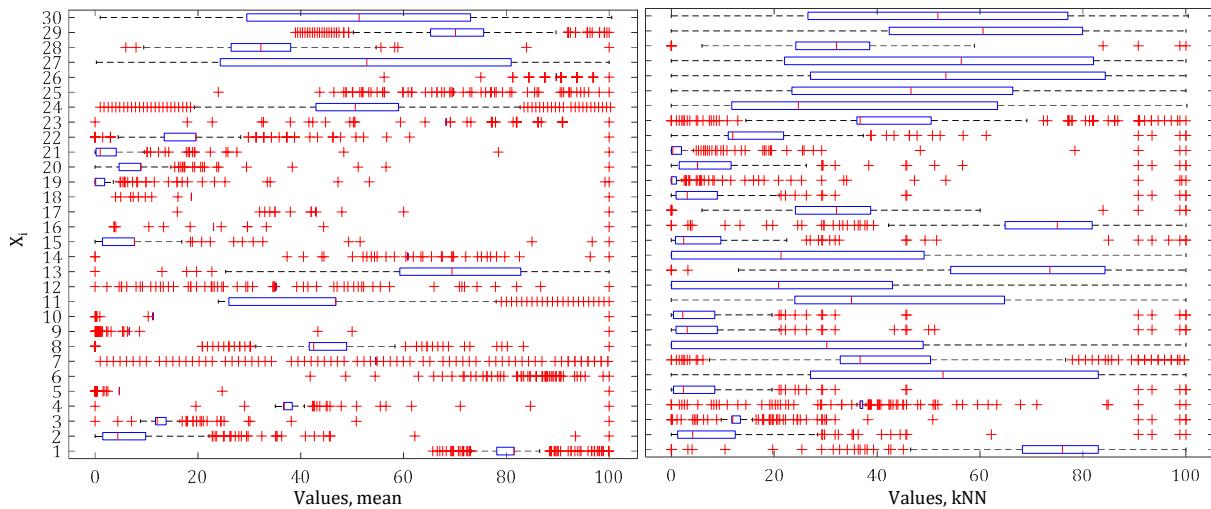


Figure 46 - Descriptive statistics imputation procedure, left: Mean imputation, right: kNN imputation

Table 14 - Measures of determination X_i^{165} , left: Mean imputation, right: kNN imputation

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	X
1	1.00	0.30	0.02	0.03	0.00	0.08	0.10	0.12	0.02	0.01	0.27	0.12	0.32	0.00	0.00	0.00	0.02	0.05	0.13	0.13	0.10	0.15	0.05	0.31	0.05	0.33	0.40				1.00
2	0.30	1.00	0.02	0.01	0.00	0.03	0.10	0.03	0.00	0.00	0.15	0.10	0.19	0.00	0.00	0.01	0.01	0.04	0.02	0.10	0.06	0.06	0.13	0.02	0.27	0.09	0.17	0.27			2.03
3	0.02	0.01	1.00	0.57	0.55	0.06	0.03	0.02	0.25	0.02	0.16	0.01	0.01	0.00	0.69	0.17	0.31	0.26	0.35	0.00	0.72	0.00	0.03	0.06	0.00	0.02	0.00	0.13	0.07		3.01
4	0.03	0.01	0.57	1.00	0.17	0.05	0.04	0.01	0.07	0.12	0.04	0.00	0.14	0.11	0.00	0.66	0.22	0.05	0.54	0.00	0.00	0.01	0.05	0.01	0.02	0.00	0.10	0.05			4.11
5	0.00	0.00	0.06	0.17	1.00	0.00	0.00	0.26	0.04	0.01	0.01	0.00	0.00	0.23	0.04	0.72	0.72	0.02	0.18	0.01	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00			5.00
6	0.08	0.03	0.06	0.05	0.00	0.07	0.05	0.01	0.00	0.06	0.09	0.04	0.00	0.01	0.01	0.01	0.00	0.02	0.01	0.04	0.04	0.02	0.03	0.05	0.02	0.02	0.00	0.04	0.05		6.16
7	0.10	0.10	0.03	0.04	0.00	0.17	0.00	0.01	0.01	0.12	0.15	0.02	0.01	0.00	0.01	0.00	0.03	0.03	0.05	0.04	0.06	0.09	0.23	0.00	0.02	0.00	0.10	0.04			7.16
8	0.12	0.03	0.02	0.01	0.00	0.05	0.10	0.00	0.00	0.07	0.04	0.02	0.01	0.01	0.00	0.00	0.03	0.02	0.02	0.09	0.03	0.04	0.02	0.01	0.01	0.00	0.06	0.04			8.18
9	0.02	0.00	0.25	0.07	0.26	0.01	0.01	0.00	0.00	0.53	0.04	0.00	0.01	0.00	0.34	0.49	0.19	0.02	0.03	0.00	0.13	0.02	0.22	0.00	0.01	0.15	0.01	0.00	0.00		9.04
10	0.01	0.00	0.02	0.01	0.04	0.00	0.01	0.00	0.00	0.53	0.00	0.00	0.00	0.05	0.07	0.01	0.00	0.00	0.02	0.00	0.16	0.00	0.00	0.00	0.07	0.00	0.00	0.00			10.08
11	0.27	0.15	0.16	0.12	0.01	0.06	0.12	0.07	0.04	0.00	0.00	0.11	0.17	0.01	0.19	0.00	0.01	0.00	0.07	0.02	0.15	0.04	0.09	0.13	0.00	0.21	0.03	0.08	0.23		11.28
12	0.12	0.10	0.01	0.04	0.01	0.09	0.15	0.04	0.00	0.00	0.11	0.00	0.03	0.00	0.01	0.02	0.00	0.08	0.00	0.17	0.13	0.02	0.20	0.01	0.02	0.00	0.02	0.02			12.12
13	0.32	0.19	0.01	0.00	0.04	0.02	0.01	0.00	0.00	0.17	0.03	0.00	0.00	0.00	0.01	0.00	0.00	0.02	0.00	0.03	0.01	0.06	0.02	0.00	0.33	0.21	0.23	0.72			13.08
14	0.00	0.00	0.01	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00			14.15
15	0.00	0.00	0.06	0.34	0.23	0.01	0.00	0.01	0.34	0.05	0.19	0.00	0.00	0.00	0.31	0.11	0.06	0.54	0.00	0.72	0.01	0.09	0.01	0.00	0.01	0.00	0.14	0.03			15.05
16	0.01	0.01	0.17	0.11	0.04	0.01	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.16	0.00	0.11	0.01	0.18	0.00	0.06	0.01	0.00	0.00	0.00	0.00	0.00	0.00			16.09
17	0.00	0.00	0.31	0.08	0.77	0.01	0.00	0.19	0.07	0.01	0.02	0.00	0.00	0.00	0.11	0.04	0.00	0.01	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			17.19
18	0.00	0.01	0.26	0.06	0.77	0.00	0.00	0.02	0.01	0.00	0.02	0.00	0.00	0.00	0.06	0.01	0.00	0.01	0.05	0.00	0.03	0.00	0.02	0.02	0.00	0.00	0.00	0.00			18.09
19	0.02	0.01	0.35	0.22	0.02	0.03	0.03	0.03	0.00	0.07	0.00	0.00	0.00	0.01	0.54	0.16	0.00	0.00	0.10	0.74	0.00	0.00	0.02	0.01	0.03	0.00	0.02	0.00			19.09
20	0.05	0.04	0.00	0.05	0.00	0.01	0.03	0.02	0.00	0.00	0.02	0.08	0.02	0.01	0.00	0.00	0.01	0.10	0.00	0.00	0.00	0.01	0.03	0.01	0.03	0.02	0.00	0.03			20.09
21	0.03	0.02	0.72	0.56	0.18	0.04	0.05	0.02	0.13	0.02	0.15	0.00	0.00	0.01	0.72	0.31	0.08	0.05	0.74	0.00	0.00	0.00	0.00	0.03	0.05	0.00	0.01	0.00			21.01
22	0.17	0.10	0.00	0.01	0.04	0.09	0.02	0.00	0.04	0.17	0.03	0.00	0.01	0.01	0.00	0.00	0.00	0.21	0.00	0.03	0.03	0.05	0.00	0.00	0.05	0.00	0.05	0.06			22.03
23	0.13	0.06	0.00	0.00	0.02	0.06	0.03	0.22	0.16	0.00	0.13	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			23.03
24	0.10	0.06	0.03	0.01	0.00	0.03	0.09	0.04	0.00	0.00	0.09	0.02	0.06	0.00	0.00	0.00	0.00	0.02	0.03	0.03	0.02	0.00	0.13	0.00	0.10	0.02	0.19	0.11			24.18
25	0.15	0.13	0.06	0.05	0.03	0.05	0.23	0.02	0.01	0.00	0.13	0.20	0.02	0.00	0.00	0.00	0.01	0.02	0.01	0.05	0.05	0.11	0.13	0.00	0.06	0.02	0.09	0.04			25.29
26	0.01	0.02	0.00	0.01	0.00	0.02	0.00	0.01	0.15	0.07	0.00	0.01	0.00	0.01	0.00	0.00	0.02	0.03	0.03	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	0.00			26.32
27	0.31	0.27	0.02	0.00	0.02	0.02	0.01	0.01	0.00	0.21	0.02	0.03	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.05	0.01	0.10	0.06	0.00	0.00	0.27	0.26	0.66			27.32
28	0.05	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			28.15
29	0.33	0.17	0.10	0.00	0.04	0.10	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			29.42
30	0.40	0.27	0.07	0.05	0.00	0.05	0.04	0.04	0.00	0.00	0.21	0.02	0.72	0.00	0.03	0.00	0.00	0.00	0.04	0.03	0.07	0.06	0.11	0.04	0.00	0.66	0.24	0.30			30.42

When missing data are replaced by imputation, imprecision can occur. This is particularly important for the parameters X_1 and X_2 due to their relevance for the formation of the index. No values are available for the Barnett Index X_1 regarding states that are relevant to the discussion on digital sovereignty. Replacing these with mean or kNN imputation produces incorrect estimates for certain states, which is why the missing data is replaced by the author after manual analysis to determine more realistic DSI values:

- Norway are valued at the same level as Finland,
- Liechtenstein and Switzerland are valued at the level of Germany and
- Hong Kong, Macao and Taiwan are replaced with value from China.

5.8. Normalisation

Applying the normalisation methods (cf. 4.7) and z-score standardisation produces a clear change in position compared to the (scale-transformed) raw data; an example is visualised

¹⁶⁵ Tables enlarged in Appendix

in Figure 47. Raw data and the normalisations MinMax, z-score Mean = 0 and z-score Mean = 0 with standard deviation are selected.

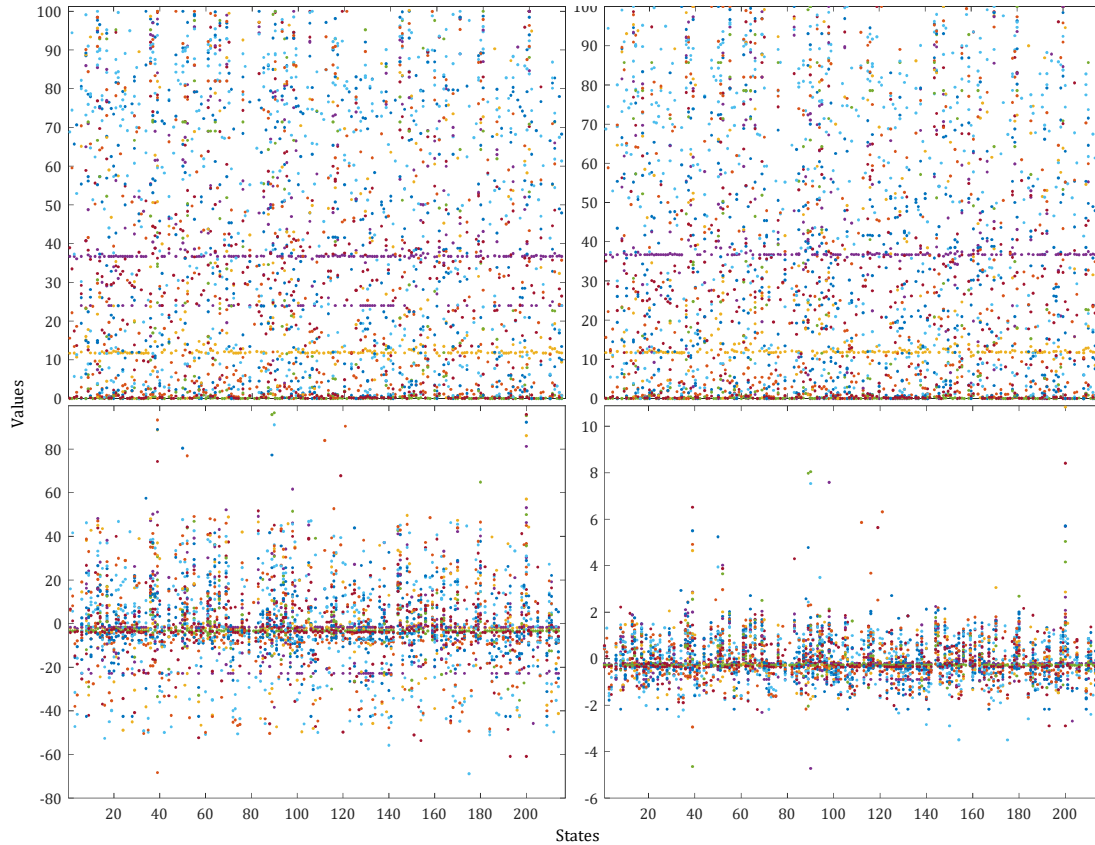


Figure 47 - Visualised comparison top: raw data, MinMax, bottom: z-score Mean = 0, z-score Mean = 0 with standard deviation

Shown here is the normalised raw data without imputation methods applied. For each type of imputation, the normalisation methods are applied for further analysis.

5.9. Weightings

To differentiate the relevance of the parameters (X) by means of weighting factors (γ), different weighting procedures are used. A distinction is made between the weighting of the parameters for Δ_1 and Δ_2 and Δ_3 .

For sub-index 1, the parameters X_1 (Barnett Index) and X_2 (per capita GDP) are equally weighted for later index formation with a focus on the understanding of sovereignty used in this work¹⁶⁶, so the following applies:

¹⁶⁶ This determines that states with a high sovereignty index and high prosperity receive a high index value. The factual content can be critically discussed (cf. Section 7).

$$\gamma_{1,2} = \frac{1}{2}. \quad (53)$$

The high weighting ensures that autocratic states cannot receive a high DSI value. For the weightings of the parameters of sub-indices 2 and 3 (i.e. for X_i , $i > 2$), the weighting methods outlined in Table 9 are applied and evaluated.

5.9.1. Equal weighting

Assuming that all X_i , $i > 2$ are identically relevant for the index formation, these receive an identical weighting, namely

$$\gamma_i = \frac{1}{28} \quad (54)$$

independent of normalisation and imputation procedures.

5.9.2. Principal Component Analysis

In contrast, the determination of weights by means of the component analysis is based on the respective X_i and must therefore be conducted separately depending on the normalisation and imputation procedures. Using two combinations of normalisation and imputation procedures as examples, the weights are calculated this section according to the procedure in Section 4.8 and then determined and presented below for all combinations.

Example 1: Raw data without imputation

First, the loadings for each combination are calculated from the parameters.

Table 15 - Loadings PCA, raw data, no imputation

X_i	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
3	.00	.00	.00	.00	.00	.00	.01	.00	.00	.04	.00	.03	.04	.01	.00	.01	.01	.02	.01	.00	.01	.00	.24	.02	.01	.04	.10	.39	
4	.00	.00	.00	.00	.00	.00	.01	.00	.00	.03	.00	.00	.04	.01	.07	.01	.00	.01	.01	.02	.01	.01	.29	.06	.09	.00	.04	.28	
5	.98	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	
6	.00	.02	.03	.00	.00	.19	.01	.15	.23	.02	.04	.09	.03	.05	.04	.00	.00	.01	.01	.02	.00	.03	.00	.01	.00	.00	.00	.00	
7	.00	.00	.14	.03	.02	.00	.26	.10	.00	.10	.13	.01	.06	.08	.03	.00	.01	.00	.00	.01	.01	.00	.00	.00	.00	.00	.00	.00	
8	.00	.00	.01	.00	.02	.00	.05	.09	.03	.24	.00	.04	.00	.14	.07	.14	.11	.04	.01	.00	.00	.00	.00	.00	.02	.00	.00	.00	
9	.00	.00	.02	.48	.24	.01	.09	.00	.00	.06	.00	.01	.00	.01	.01	.03	.01	.00	.00	.00	.00	.00	.01	.00	.00	.00	.00	.00	
10	.00	.00	.00	.00	.00	.00	.01	.00	.00	.00	.16	.01	.15	.07	.01	.01	.04	.00	.11	.06	.01	.11	.04	.01	.06	.07	.05	.02	
11	.00	.00	.09	.00	.02	.00	.07	.03	.07	.10	.09	.03	.07	.08	.01	.05	.08	.07	.00	.01	.00	.01	.00	.00	.10	.00	.00	.00	
12	.00	.00	.08	.00	.00	.60	.00	.01	.21	.00	.00	.02	.01	.01	.01	.00	.00	.00	.01	.01	.00	.01	.01	.00	.00	.00	.00	.00	
13	.00	.00	.07	.01	.01	.02	.00	.00	.00	.00	.01	.01	.01	.05	.01	.00	.01	.07	.02	.02	.02	.29	.16	.00	.06	.00	.00	.15	.01
14	.01	.94	.01	.00	.01	.00	.00	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	
15	.00	.00	.00	.00	.01	.00	.06	.01	.00	.10	.00	.01	.01	.01	.00	.01	.00	.01	.07	.08	.00	.04	.23	.19	.13	.00	.00	.00	
16	.00	.00	.02	.00	.02	.01	.05	.04	.00	.03	.00	.06	.00	.00	.03	.08	.00	.13	.05	.28	.04	.01	.02	.04	.07	.01	.00	.01	
17	.00	.00	.00	.00	.01	.00	.03	.01	.00	.00	.01	.01	.08	.01	.02	.00	.01	.04	.02	.01	.13	.01	.00	.05	.00	.01	.50	.04	
18	.00	.00	.00	.00	.00	.00	.00	.00	.00	.02	.05	.01	.09	.05	.00	.20	.14	.05	.03	.00	.00	.02	.00	.02	.00	.14	.00	.03	.16
19	.00	.00	.00	.00	.00	.01	.00	.01	.00	.02	.05	.09	.09	.08	.00	.01	.00	.05	.06	.01	.00	.05	.00	.06	.02	.38	.00	.01	
20	.00	.00	.00	.00	.00	.01	.01	.00	.00	.03	.02	.10	.09	.03	.00	.03	.01	.03	.03	.10	.12	.05	.01	.18	.03	.11	.00	.00	
21	.00	.00	.00	.00	.00	.00	.03	.00	.01	.07	.00	.05	.03	.02	.01	.04	.00	.00	.02	.00	.02	.06	.00	.25	.03	.33	.01	.00	
22	.00	.00	.00	.00	.00	.01	.02	.01	.04	.09	.02	.08	.00	.03	.28	.02	.01	.19	.01	.01	.00	.00	.08	.01	.03	.02	.00	.01	
23	.00	.00	.06	.36	.46	.00	.00	.06	.00	.01	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	
24	.00	.00	.05	.04	.00	.01	.09	.22	.00	.00	.36	.02	.06	.00	.02	.01	.05	.00	.00	.00	.01	.02	.01	.00	.01	.00	.01	.00	
25	.00	.00	.04	.01	.01	.05	.02	.10	.29	.00	.00	.18	.00	.06	.04	.02	.07	.02	.00	.04	.01	.01	.00	.00	.00	.00	.00	.00	
26	.00	.00	.00	.05	.05	.00	.02	.02	.01	.01	.05	.02	.04	.01	.13	.27	.04	.02	.06	.04	.12	.00	.00	.02	.01	.00	.00	.01	
27	.00	.00	.16	.00	.06	.01	.11	.07	.06	.00	.01	.01	.07	.03	.07	.01	.01	.01	.01	.16	.00	.06	.01	.05	.02	.00	.00	.01	
28	.00	.00	.01	.00	.01	.03	.00	.04	.00	.01	.01	.00	.01	.00	.00	.02	.15	.09	.39	.00	.04	.12	.02	.01	.02	.00	.00	.03	
29	.00	.00	.05	.00	.01	.00	.03	.00	.01	.02	.00	.00	.03	.15	.01	.20	.06	.00	.06	.00	.06	.23	.00	.00	.04	.00	.02	.00	
30	.00	.00	.15	.00	.05	.01	.00	.01	.01	.00	.00	.09	.02	.02	.06	.02	.03	.08	.00	.08	.09	.01	.00	.00	.16	.00	.08	.00	

The maximum loading per variable X_i is used to determine the weight for the variable.

Table 16 - PCA, maximum loadings and weights for raw data without imputation

X_i	max Loading Weight		X_i	max Loading Weight	
3	.392	.040	17	.497	.051
4	.286	.030	18	.199	.021
5	.976	.101	19	.380	.039
6	.233	.024	20	.179	.019
7	.262	.027	21	.332	.034
8	.239	.025	22	.283	.029
9	.482	.050	23	.455	.047
10	.164	.017	24	.359	.037
11	.103	.011	25	.289	.030
12	.605	.063	26	.271	.028
13	.289	.030	27	.164	.017
14	.942	.097	28	.393	.041
15	.234	.024	29	.227	.023
16	.283	.029	30	.156	.016

The variables yield the extent to which the dispersion can be explained. It can be quantified with the help of the coefficient of determination (i.e. the share of the explained variance in the total variance) (cf. 4.6).

Example 2: Data MinMax normalised with kNN imputation

For the combination of the MinMax normalisation procedure and imputation using the kNN procedure, the determination of the weights is as follows in Table 17.

Table 17 - Loadings PCA, MinMax normalisation, kNN imputation

X_i	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
3	.00	.01	.03	.00	.00	.00	.00	.00	.09	.02	.00	.02	.00	.00	.02	.01	.01	.01	.01	.00	.01	.02	.01	.36	.26	.04	.07	.00
4	.01	.01	.01	.00	.01	.01	.01	.00	.00	.18	.05	.01	.01	.04	.02	.13	.03	.03	.06	.18	.01	.08	.11	.00	.00	.00	.00	.00
5	.00	.05	.06	.02	.01	.00	.00	.01	.01	.03	.01	.00	.01	.01	.00	.00	.06	.04	.02	.01	.01	.00	.03	.01	.00	.02	.00	.56
6	.07	.00	.00	.08	.13	.00	.02	.00	.02	.01	.00	.00	.02	.01	.01	.00	.00	.29	.05	.03	.21	.02	.01	.00	.00	.00	.00	.00
7	.06	.00	.00	.12	.09	.00	.05	.00	.00	.08	.01	.02	.16	.03	.00	.00	.08	.04	.10	.03	.05	.06	.00	.00	.00	.00	.00	.00
8	.04	.14	.04	.03	.04	.03	.00	.03	.00	.02	.01	.01	.09	.02	.17	.21	.00	.01	.02	.05	.01	.00	.02	.00	.00	.00	.00	.00
9	.00	.06	.07	.01	.00	.00	.01	.01	.02	.00	.00	.01	.02	.00	.01	.00	.08	.00	.00	.00	.02	.00	.13	.07	.29	.03	.14	.00
10	.00	.05	.06	.02	.01	.00	.01	.02	.02	.00	.00	.06	.03	.00	.00	.00	.05	.02	.01	.00	.08	.00	.09	.01	.17	.00	.29	.01
11	.06	.12	.01	.08	.27	.01	.01	.00	.01	.05	.16	.04	.06	.02	.00	.04	.01	.01	.01	.00	.01	.00	.00	.00	.00	.00	.00	.00
12	.03	.13	.07	.00	.03	.03	.01	.04	.01	.00	.01	.01	.10	.00	.03	.26	.05	.01	.05	.01	.02	.00	.10	.00	.00	.00	.00	.00
13	.04	.00	.06	.03	.00	.03	.02	.00	.00	.02	.00	.00	.04	.02	.02	.00	.21	.39	.00	.04	.01	.05	.00	.00	.01	.00	.00	.00
14	.04	.18	.04	.00	.05	.01	.03	.05	.07	.01	.01	.00	.00	.00	.26	.00	.03	.04	.08	.02	.00	.01	.06	.00	.01	.00	.00	.00
15	.01	.08	.05	.01	.01	.00	.03	.00	.00	.00	.00	.05	.02	.05	.01	.01	.02	.01	.05	.02	.52	.00	.00	.01	.02	.00	.00	.00
16	.01	.00	.10	.24	.04	.22	.06	.04	.00	.04	.02	.00	.01	.01	.00	.00	.04	.14	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00
17	.01	.00	.03	.00	.00	.03	.18	.01	.13	.01	.01	.04	.00	.02	.03	.00	.00	.00	.01	.02	.01	.01	.01	.25	.02	.11	.05	.00
18	.00	.05	.06	.02	.01	.00	.00	.01	.02	.04	.00	.00	.03	.00	.00	.00	.09	.08	.01	.00	.00	.01	.08	.00	.00	.07	.00	.41
19	.01	.00	.03	.00	.02	.01	.00	.00	.02	.16	.13	.22	.01	.01	.03	.01	.00	.00	.00	.02	.01	.01	.01	.03	.00	.12	.10	.00
20	.01	.03	.04	.01	.01	.00	.04	.00	.16	.11	.06	.00	.01	.00	.02	.00	.00	.00	.03	.01	.00	.00	.05	.04	.00	.32	.03	.00
21	.01	.01	.03	.00	.01	.01	.01	.00	.18	.00	.00	.07	.00	.00	.01	.00	.00	.00	.03	.01	.11	.00	.02	.00	.19	.05	.25	.01
22	.01	.00	.02	.00	.00	.02	.26	.00	.02	.03	.00	.12	.00	.00	.02	.00	.02	.01	.00	.03	.00	.00	.00	.14	.02	.21	.06	.00
23	.06	.00	.00	.07	.08	.00	.00	.00	.02	.03	.03	.01	.17	.05	.04	.14	.00	.00	.01	.06	.00	.07	.11	.00	.00	.00	.00	.00
24	.08	.04	.02	.13	.04	.50	.00	.08	.01	.01	.04	.00	.02	.01	.01	.00	.00	.01	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
25	.05	.00	.00	.05	.04	.00	.00	.00	.01	.03	.00	.00	.09	.00	.27	.00	.14	.05	.15	.04	.01	.01	.05	.01	.00	.00	.00	.00
26	.04	.00	.00	.01	.00	.04	.12	.14	.16	.03	.04	.11	.03	.09	.00	.01	.00	.00	.00	.07	.00	.03	.07	.00	.00	.00	.00	.00
27	.10	.01	.08	.04	.00	.01	.11	.43	.01	.00	.02	.09	.03	.00	.02	.00	.01	.01	.00	.02	.00	.01	.00	.00	.00	.00	.00	.00
28	.01	.00	.00	.02	.00	.00	.00	.01	.01	.05	.18	.06	.00	.47	.00	.06	.02	.02	.00	.00	.05	.01	.00	.02	.00	.00	.00	.00
29	.09	.02	.03	.00	.07	.02	.00	.08	.00	.03	.16	.03	.02	.12	.01	.10	.02	.00	.03	.06	.00	.08	.00	.00	.00	.00	.00	.00
30	.11	.00	.05	.02	.01	.00	.00	.00	.00	.01	.05	.01	.02	.00	.00	.00	.01	.06	.02	.24	.03	.30	.01	.03	.00	.01	.00	.00

The maximum loadings and weights can then be determined as follows in Table 18.

Table 18 - Maximum loadings PCA, weights, MinMax normalisation, kNN imputation

X_i	max Loading	Weight	X_i	max Loading	Weight
3	.358	.042	17	.248	.029
4	.182	.021	18	.411	.049
5	.565	.067	19	.216	.026
6	.286	.034	20	.322	.038
7	.162	.019	21	.245	.029
8	.212	.025	22	.262	.031
9	.291	.034	23	.173	.020
10	.293	.035	24	.498	.059
11	.273	.032	25	.273	.032
12	.258	.031	26	.164	.019
13	.386	.046	27	.433	.051
14	.256	.030	28	.474	.056
15	.519	.061	29	.162	.019
16	.238	.028	30	.304	.036

With all further combinations of normalisation and imputation procedures, the weights can be calculated by means of a component analysis, which is shown in Table 19. The maxima and minima of the weights of X_i deviate significantly from each other (on average by 0.04), which means that, depending on the application of the imputation procedure, deviating weights and thus also deviating DSI values are calculated.

Table 19 - PCA, all normalisation procedures, all imputation procedures: maximum loadings (L) and weights (W)

X_i	no Imputation								kNN Imputation								Mean Imputation							
	Raw		MinMax		MN=0		MN=0, StdD=1		Raw		MinMax		MN=0		MN=0, StdD=1		Raw		MinMax		MN=0		MN=0, StdD=1	
	L	W	L	W	L	W	L	W	L	W	L	W	L	W	L	W	L	W	L	W	L	W	L	W
3	.39	.04	.31	.03	.35	.04	.17	.02	.61	.07	.36	.04	.29	.04	.18	.03	.62	.07	.36	.04	.25	.02	.24	.03
4	.29	.03	.31	.03	.22	.02	.26	.03	.43	.05	.18	.02	.28	.04	.16	.02	.46	.05	.23	.02	.31	.03	.29	.03
5	.98	.10	.97	.11	.37	.04	.94	.11	.56	.06	.56	.07	.49	.07	.28	.04	.38	.04	.54	.06	.52	.05	.30	.03
6	.23	.02	.20	.02	.39	.04	.56	.06	.35	.04	.29	.03	.38	.05	.30	.04	.79	.08	.36	.04	.71	.07	.31	.03
7	.26	.03	.51	.06	.47	.05	.23	.03	.20	.02	.16	.02	.29	.04	.27	.04	.29	.03	.24	.03	.25	.02	.28	.03
8	.24	.02	.28	.03	.29	.03	.28	.03	.26	.03	.21	.03	.16	.02	.29	.04	.22	.02	.38	.04	.37	.04	.61	.07
9	.48	.05	.44	.05	.30	.03	.47	.05	.22	.02	.29	.03	.32	.04	.16	.02	.35	.04	.29	.03	.27	.03	.28	.03
10	.16	.02	.32	.04	.70	.08	.22	.03	.29	.03	.29	.03	.20	.03	.46	.07	.36	.04	.22	.02	.22	.02	.60	.06
11	.10	.01	.19	.02	.19	.02	.13	.02	.25	.03	.27	.03	.15	.02	.26	.04	.20	.02	.24	.03	.21	.02	.12	.01
12	.60	.06	.20	.02	.31	.03	.21	.02	.21	.02	.26	.03	.32	.04	.18	.03	.22	.02	.22	.02	.32	.03	.19	.02
13	.29	.03	.21	.02	.20	.02	.32	.04	.34	.04	.39	.05	.20	.03	.39	.06	.33	.03	.21	.02	.18	.02	.15	.02
14	.94	.10	.47	.05	.49	.06	.89	.10	.22	.02	.26	.03	.18	.02	.25	.04	.20	.02	.35	.04	.31	.03	.35	.04
15	.23	.02	.26	.03	.30	.03	.23	.03	.39	.04	.52	.06	.12	.02	.16	.02	.21	.02	.27	.03	.28	.03	.31	.03
16	.28	.03	.25	.03	.29	.03	.25	.03	.35	.04	.24	.03	.24	.03	.26	.04	.43	.05	.40	.04	.39	.04	.30	.03
17	.50	.05	.19	.02	.57	.06	.20	.02	.23	.03	.25	.03	.21	.03	.15	.02	.32	.03	.47	.05	.61	.06	.54	.06
18	.20	.02	.29	.03	.31	.03	.51	.06	.39	.04	.41	.05	.21	.03	.56	.08	.51	.05	.33	.04	.33	.03	.67	.07
19	.38	.04	.19	.02	.36	.04	.15	.02	.32	.04	.22	.03	.16	.02	.23	.03	.27	.03	.16	.02	.23	.02	.22	.02
20	.18	.02	.23	.03	.23	.03	.20	.02	.26	.03	.32	.04	.27	.04	.18	.03	.24	.03	.14	.02	.24	.02	.31	.03
21	.33	.03	.27	.03	.36	.04	.27	.03	.18	.02	.25	.03	.34	.05	.23	.03	.24	.03	.26	.03	.26	.03	.30	.03
22	.28	.03	.30	.03	.23	.03	.25	.03	.25	.03	.26	.03	.18	.02	.20	.03	.22	.02	.22	.02	.38	.04	.30	.03
23	.46	.05	.38	.04	.25	.03	.43	.05	.18	.02	.17	.02	.17	.02	.22	.03	.21	.02	.31	.03	.27	.03	.34	.04
24	.36	.04	.43	.05	.20	.02	.24	.03	.68	.08	.50	.06	.52	.07	.27	.04	.70	.07	.55	.06	.45	.04	.28	.03
25	.29	.03	.20	.02	.16	.02	.43	.05	.38	.04	.27	.03	.18	.03	.18	.03	.16	.02	.24	.03	.39	.04	.27	.03
26	.27	.03	.49	.05	.19	.02	.26	.03	.21	.02	.16	.02	.40	.06	.18	.03	.23	.02	.53	.06	.44	.04	.19	.02
27	.16	.02	.20	.02	.31	.04	.18	.02	.28	.03	.43	.05	.22	.03	.16	.02	.32	.03	.31	.03	.35	.03	.51	.06
28	.39	.04	.44	.05	.33	.04	.17	.02	.13	.01	.47	.06	.30	.04	.18	.03	.34	.04	.76	.08	.80	.08	.47	.05
29	.23	.02	.37	.04	.21	.02	.18	.02	.19	.02	.16	.02	.17	.02	.25	.04	.24	.03	.49	.05	.36	.04	.14	.02
30	.16	.02	.27	.03	.20	.02	.19	.02	.52	.06	.30	.04	.35	.05	.34	.05	.35	.04	.26	.03	.29	.03	.36	.04

5.9.3. Analytic hierarchy process

Compared to PCA, the determination of weights according to the AHP method is independent of the normalisation and imputation methods, as the parameters of X_i are weighted in relation to each other. The weighting table used for this is shown below. The relevance of

each pair of two parameters, X_i and X_j , is evaluated with the following question: how relevant is the parameter X_i when assessed in comparison to the parameter X_j ?

Table 20 - Weighting matrix AHP

X_i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
2	.00	1.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00	.00
3	.00	.00	1.00	1.00	.33	.33	.33	1.00	1.00	.33	1.00	.33	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	.33	.33
4	.00	.00	1.00	1.00	.33	.33	.33	1.00	1.00	.33	1.00	.33	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	.33	.33
5	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	3.00	3.00	1.00	1.00
6	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	1.00	1.00
7	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	1.00	1.00
8	.00	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	3.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
9	.00	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00
10	.00	.00	3.00	3.00	1.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
11	.00	.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
12	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
13	.00	.00	1.00	1.00	.33	3.00	3.00	1.00	.33	1.00	.33	1.00	1.00	1.00	.33	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
14	.00	.00	1.00	1.00	.33	1.00	1.00	.33	.33	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
15	.00	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
16	.00	.00	3.00	3.00	1.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00
17	.00	.00	3.00	3.00	1.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00
18	.00	.00	3.00	3.00	1.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00
19	.00	.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	.33	.33
20	.00	.00	1.00	1.00	.33	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	1.00	.33	.33
21	.00	.00	1.00	1.00	.33	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	1.00	.33	.33
22	.00	.00	1.00	1.00	.33	1.00	1.00	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	.33	.33	1.00	1.00	1.00	1.00	.33	.33
23	.00	.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
24	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	5.00	5.00	1.00	1.00
25	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	5.00	5.00	1.00	1.00
26	.00	.00	1.00	1.00	.33	.33	.33	1.00	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.33	.33	1.00	.33	.33	.33	.33	.33
27	.00	.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.20	.20	3.00	1.00	.33	.33	.33	.33
28	.00	.00	1.00	1.00	.33	.33	.33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.20	.20	3.00	3.00	1.00	1.00	1.00	1.00
29	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00
30	.00	.00	3.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	3.00	1.00	1.00	1.00

To simplify matters, only two options are offered: a 1 if the variables are rated as equally important or a 3 if the variable is rated as more important (or $0.\bar{3}$ if it is rated less relevant). The table was evaluated by the five interview partners, and the most frequent mention was used in each case (cf. 5.2). Before weights are determined, the quality of the AHP matrix must be assessed by means of a consistency check (cf. 4.8).

Table 21 - Quality AHP

Quality criterion	Value
λ_{max}	26.60
Consistency Index	-.12
Random Index	1.68
Consistency Ratio	-.07

The consistency ratio is < 0.1 , which means that the matrix can be considered consistent and the following weighting can be used.

Table 22 - *Weights using the AHP method*

X_i	Weight	X_i	Weight
3	.021	17	.048
4	.021	18	.048
5	.051	19	.025
6	.040	20	.024
7	.040	21	.024
8	.031	22	.024
9	.048	23	.030
10	.040	24	.050
11	.036	25	.050
12	.035	26	.024
13	.032	27	.026
14	.030	28	.031
15	.032	29	.045
16	.048	30	.045

5.9.4. Weighting number of parameters

To consider in the index calculation that few secondary data are available for some states, which can be seen as a disadvantage of the model and the chosen parameters, a country-specific weighting factor (α) is introduced in addition to the parameter-specific weightings (γ_i). Since the application of imputation procedures leads to a distortion of the results, this weighting factor takes into account that states with few raw data are devalued. The devaluation of less available parameters can be done by reducing the individual weights (Greco, et al., 2019, p. 66). For simplification, three country groups are formed: all parameters are fully weighted if more than 20 items of raw data are available for a state, there is a weighting of 0.5 if less than 10 data items are available, or a three-quarters weighting is applied. Thus,

$$\begin{aligned}
 \alpha &= 1, |X_i| \geq 20, \\
 \alpha &= \frac{3}{4}, 20 > |X_i| \geq 10, \\
 \alpha &= \frac{1}{2}, |X_i| < 10.
 \end{aligned} \tag{55}$$

5.10. Aggregation

As explained in Section 4.9, the previously collected and analysed data can be aggregated in different ways. For the present case, a linear combination of additive and geometric aggregation is suitable, although the most common way of indexing is additive (Greco, et al., 2019, p. 66). Since the parameters are largely independent and identically normalised (cf. 5.6.2), the composition of a suitable aggregation function results from possible compensation effects.

The parameters X_i with $i = \{3, \dots, 30\}$ (i.e. those of the subindices Δ_2 and Δ_3 , which describe KETs and technological sovereignty) are combined by means of an additive

aggregation procedure. This can lead to possible undesirable compensation effects and thus to a loss of information. In the present case, however, compensations are desirable. For example, a state can achieve particular strength regarding digital sovereignty by strengthening some of the 28 parameters while others are weaker; thus, compensation is not a disadvantage here but part of the concept. Moreover, the parameters for KETs and technological sovereignty are so numerous that the compensatory effect of one parameter would generally be relatively small. Thus, the composite sub-index for the above two sub-indices is given by the following formula:

$$\Delta_2 + \Delta_3 = \sum_{i=3}^{30} \gamma_i X_i \quad (56)$$

The two parameters of the sub-index Δ_1 are aggregated multiplicatively, as the aim is to avoid possible compensation of relatively wealthy states that exhibit low levels of ‘good governance’.

$$\Delta_1 = \prod_{i=1}^2 X_i^{\gamma_i} \quad (57)$$

The factors included in the indexes Δ_1 and $\Delta_2 + \Delta_3$ are equally weighted and multiplicatively aggregated. Thus, all digitalisation parameters for KETs and technological sovereignty are given the same weight as the factor for state sovereignty with low compensation effects. This ensures that states with low levels of state sovereignty and prosperity do not receive a significantly high ranking due to high levels of the digitalisation technology parameters. The multiplier α additionally devalues the states for which less data are available. This results in an index composed of additive and geometric procedures, such as follows:

$$DSI = \alpha \Delta_1 (\Delta_2 + \Delta_3) \quad (58)$$

Factorisation satisfies the model principle that the definition of digital sovereignty used here is based on European values. This approach is strengthened by considering the prosperity of Europe (Δ_1). In contrast, a state whose technology focus is low will not receive a high ranking despite pronounced state sovereignty. For this reason, the geometric approach of index formation is suitable for avoiding compensatory effects. With the individual parameters, the formation of the digital sovereignty index is then as follows:

$$DSI = \alpha \prod_{i=1}^2 X_i^{\gamma_i} \sum_{i=3}^{30} \gamma_i X_i = \alpha \sqrt{(\Theta + \Xi) \text{GDP}_{\text{per capita}}} \left[\sum_{i=3}^{18} \gamma_i X_i + \sum_{i=19}^{30} \gamma_i X_i \right] = \alpha \Delta_1 [\Delta_2 + \Delta_3] \quad (59)$$

Since there is no ‘golden rule’ for index formation, the approach used here should be understood as a possible one (OECD, 2020d, pp. 102–105). Alternative aggregation methods,

for example by weighting the factors or purely additive, geometric or non-linear methods, represent further variants. The aggregation approach used here is a possible compromise, the validity of which must be demonstrated, and it fulfils Fisher's circularity requirement (Drechsler, 1973, p. 21).

5.11. Index formation DSI

A digital sovereignty index can thus be formed as follows:

$$DSI = \alpha(X_1^{\gamma_1} X_2^{\gamma_2}) \sum_{i=3}^{30} \gamma_i X_i,$$

$$\sum_{i=3}^{30} \gamma_i = 1, 0 \leq \gamma_i \leq 1 \text{ and } \gamma_{1,2} = \frac{1}{2}$$

$$X = X_n = \begin{pmatrix} x_{n1} \\ x_{n2} \\ \dots \\ x_{30n} \end{pmatrix}; x_{ni} \in \mathbb{R}; n, m \in \mathbb{N}; \alpha \in \left\{ \frac{1}{2}, \frac{3}{4}, \frac{1}{1} \right\}$$
(60)

Here, the x_{nm} are the parameters proposed in Section 5.4 and the γ_i and α are the possible weightings calculated in Section 5.9. The result of the index is determined by the selection of the data based on X_n , and the weighting procedure for γ_i is determined. The data basis is all permutations from the following:

- Normalisation procedure
 - without normalisation (raw data)
 - MinMax method
 - Mean = 0
 - Mean = 0 and standard deviation = 1
- Imputation procedure
 - Without imputation
 - kNN imputation
 - Mean value imputation.

With equal weighting, PCA and AHP are used as weighting procedures. From this, up to 36 index values can be calculated per state and the selection of the best permutation must be made.

5.12. Index validation, uncertainty and sensitivity analysis

By permuting the previously defined weighting, normalisation and imputation procedures, up to a single index value can be formed per state and a ranking of states can be derived from

this. The ranks, the dispersion and variance of which can be considerable, are used for evaluation, as can be seen in Figure 48.

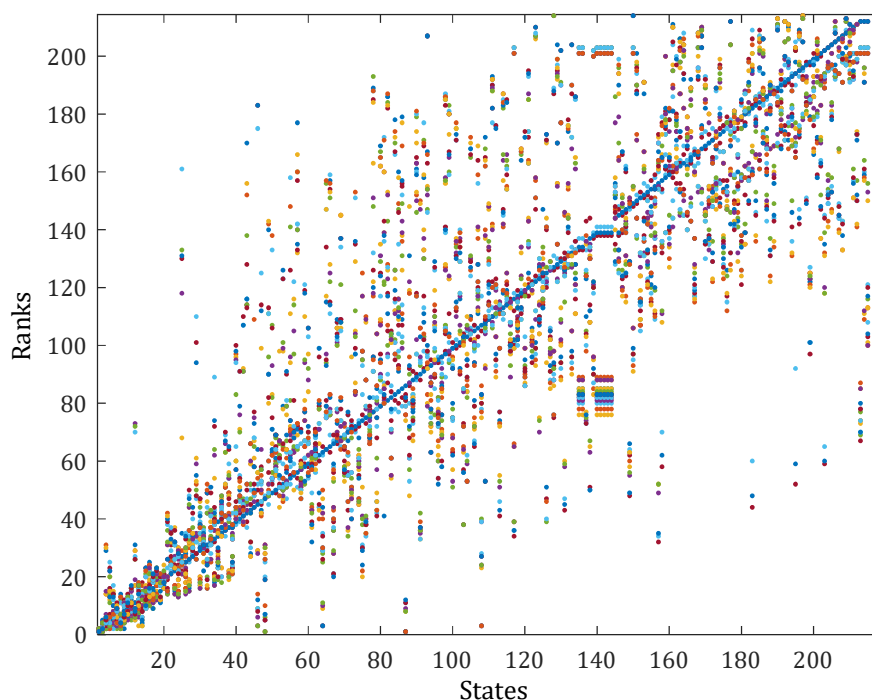


Figure 48 - All states ranks; sorted by rank, raw data, mean imputation, equal weighting, and supplementary ranks of other permutations

A suitable combination of normalisation, imputation and weighting methods must be found. This can be done using methods of uncertainty and sensitivity analysis. A combination is sought that has the smallest distance to all permutations. For this purpose, a variance analysis method is used that calculates the variances for the weighting and normalisation methods over all imputation methods. This is used to calculate how large the dispersion of the ranks is depending on the imputation procedure. The combination of smallest dispersion on average or in total is searched.

Table 23 - Extract from variance calculation ranks, MN = Mean, StD = Standard Deviation

	Variance with and without imputation											
	Raw			MinMax			MN=0			MN=0, StD=1		
	PCA	AHP	equal	PCA	AHP	equal	PCA	AHP	equal	PCA	AHP	equal
Aruba	72.25	100.00	56.25	4.00	2.25	4.00	81.00	110.25	132.25	196.00	225.00	784.00
Andorra	49.00	42.25	36.00	42.25	25.00	30.25	90.25	49.00	56.25	42.25	12.25	16.00
Afghanistan	64.00	72.25	81.00	2.25	2.25	4.00	20.25	49.00	25.00	30.25	72.25	49.00
...												
Cyprus	2.25	9.00	9.00	1.00	2.25	4.00	9.00	1.00	1.00	25.00	2.25	9.00
Czech Reput	9.00	6.25	9.00	2.25	2.25	4.00	20.25	16.00	20.25	36.00	49.00	16.00
Germany	1.00	1.00	0.25	16.00	12.25	12.25	4.00	4.00	2.25	1.00	1.00	1.00
...												
Congo, Dem.	992.25	930.25	930.25	1,444.00	1,482.25	1,406.25	90.25	64.00	56.25	2,809.00	3,192.25	3,249.00
Zambia	56.25	56.25	72.25	240.25	240.25	210.25	6.25	4.00	9.00	210.25	169.00	156.25
Zimbabwe	90.25	90.25	90.25	256.00	240.25	256.00	0.25	-	0.25	4.00	9.00	9.00

As can be seen in Table 24, the individual values are summed and the mean value is calculated to determine those with the smallest difference.

Table 24 - Sum and mean variance calculation

	Raw			MinMax			MN=0			MN=0, StD=1		
	PCA	AHP	equal	PCA	AHP	equal	PCA	AHP	equal	PCA	AHP	equal
Sum	25,059.50	25,907.50	24,969.25	62,177.00	63,138.50	62,821.00	40,583.50	49,113.50	47,897.00	97,503.00	110,100.00	100,745.00
Mean	117.10	121.06	116.68	290.55	295.04	293.56	189.64	229.50	223.82	455.62	514.49	470.77
Ranks sum	2	3	1	7	9	8	4	6	5	10	12	11
Rank mean	2	3	1	7	9	8	4	6	5	10	12	11

Using the methods from Section 4.10, it becomes apparent that a particularly suitable combination is created by means of equal weighting and the use of raw data. The question remains as to the best imputation method. Therefore, the difference is formed from the existing values without imputation, and this difference is then summed. The lower sum in each case corresponds to the best imputation procedure.

Table 25 - Mean value and kNN imputation as difference to index without imputation

	Mean	kNN
Sum	7	9
Rank	1	2

As a result, the combination of raw data without normalisation, equal weighting and mean imputation yields the best result. The kNN method, however, does not produce any significant deviations. In this case, the best result means that this method generates the smallest deviation when added to the other permutations. This applies to the population of all states.

States	Best	Second best	Imputation
All	equal Raw	PCA Raw	Mean
Europe + X	equal Raw	AHP Raw	Mean
Top 50	equal Raw	PCA Raw	Mean

Figure 49 - Best permutation, all states, EU + X, TOP 50

If the set of all states is reduced to the states of the EU and supplemented by additionally selected ones (Switzerland, Hong Kong, China, Taiwan, the US and Russia) or reduced to the TOP 50 ranks, the best combination of procedures does not change. Therefore, it is proposed to use equally weighted raw data for the index calculation, with missing data replaced by mean values.

Non-linear, non-compensatory ranking methods have not been included in the sensitivity analysers so far (cf. 4.9). To validate that the linear aggregation procedure is suitable, it is compared with the method for ranking using Copeland's procedure. The standard deviation for all normalisation and imputation procedures is clearly worse with Copeland's procedure than as shown in Table 24: in the following graph (Figure 50, left), the states are sorted according to ascending rank of index formation using the proposed linear procedure (equal weighting, mean imputation).

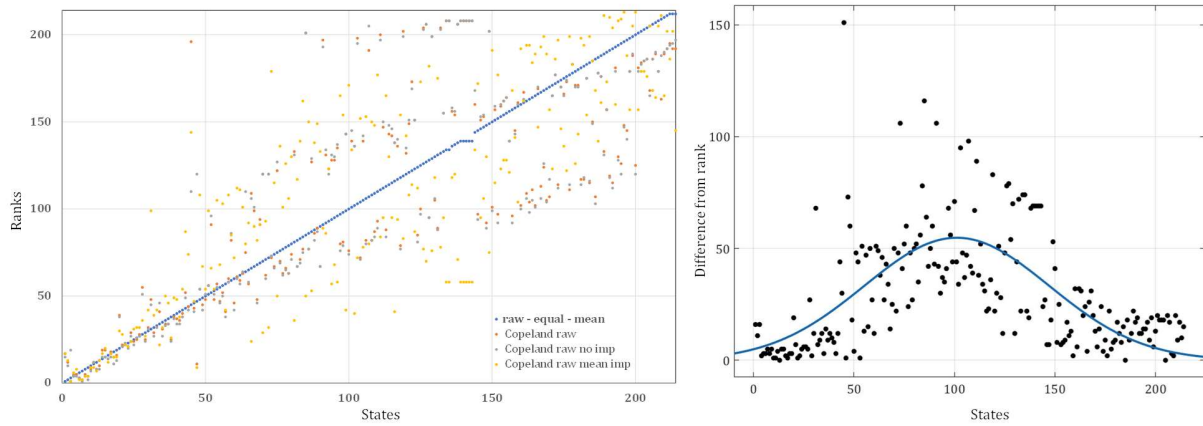


Figure 50 - Left: all states, sorted by rank, mean value imputation, supplementary rank according to Copeland's procedure, right: rank difference between Copeland method and mean imputation, supplementary: Gaussian fit function¹⁶⁷

The ranks are mapped according to Copeland's procedure based on different imputation procedures. It can be seen that the deviations are not only significant to the proposed linear procedure but also within the non-linear procedures based on different basic data. The figure on the right shows the largest differences between the non-linear and the proposed linear procedures. These essentially follow Gaussian normal distributions, and it is clear that, as described in the literature, the deviations are smaller in the highest and lowest ranks but larger in the middle ranks (Greco, et al., 2019, p. 78).

Therefore, the following analyses are conducted based on the proposed linear procedure.

5.13. Calculation of the DSI

Using the linear procedure described above, the DSI is calculated and evaluated. An overview of the states' populations shows that European and North American states, as well as Australia and New Zealand, top the rankings. The states of Africa occupy the lower ranks, which is confirmed by case studies (Fritzsche, Spoiala, 2022, pp. 20–23).

¹⁶⁷ $y = ae^{\left[-\left(\frac{x-b}{c}\right)^2\right]}$, $a = 54.79$, $b = 101.1$, $c = 65.16$ and $R^2 = 0.42$

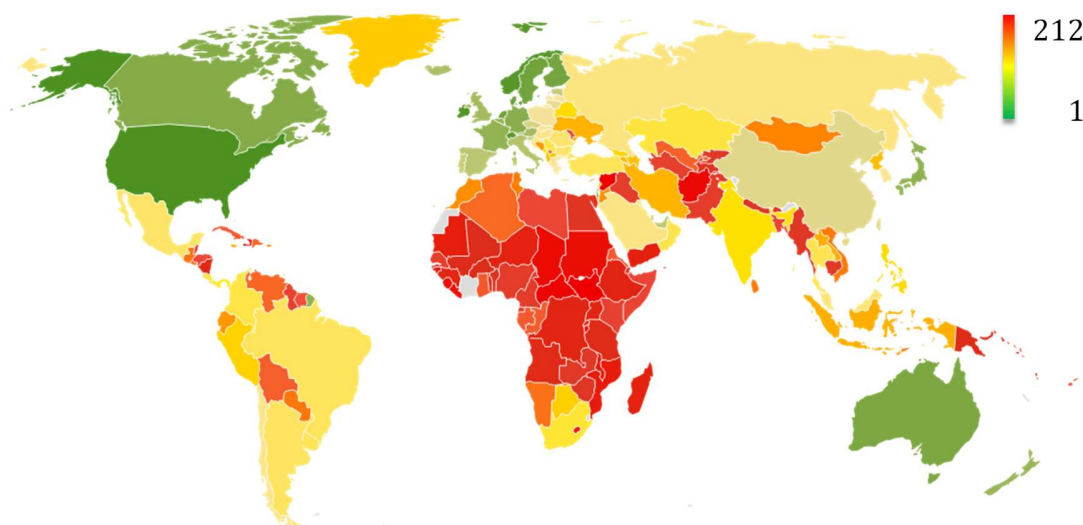


Figure 51 - Ranks DSI¹⁶⁸, colour-coded¹⁶⁹

The TOP 50 of the DSI also includes states that have relatively high DSI values despite not having a high sovereignty value. Examples include Hong Kong (rank 34), China (rank 31) and Saudi Arabia (rank 42).

Table 26 - TOP 50 DSI Ranks

Rank	Country	Rank	Country	Rank	Country
1	USA	18	UK	35	Poland
2	Ireland	19	Austria	36	Latvia
3	Norway	20	Iceland	37	Greece
4	Luxembourg	21	Israel	38	Korea Rep.
5	Switzerland	22	Italy	39	Slovak Rep.
6	Denmark	23	Spain	40	Hungary
7	Netherlands	24	Arab Emirates	41	Chile
8	Sweden	25	Estonia	42	Saudi Arabia
9	Finland	26	Portugal	43	Russia
10	Australia	27	Slovenia	44	Croatia
11	Singapore	28	Qatar	45	Monaco
12	Canada	29	Czech Rep.	46	Malaysia
13	Germany	30	Cyprus	47	Liechtenstein
14	Japan	31	China	48	Kuwait
15	France	32	Malta	49	Romania
16	New Zealand	33	Lithuania	50	Bulgaria
17	Belgium	34	Hong Kong		

¹⁶⁸ See online version at www.digital-sovereignty-index.org; complete list in Appendix

¹⁶⁹ Tool: <https://app.datawrapper.de/>

There are good reasons for this positioning, which essentially results from the strong expression of individual digitalisation factors. Authoritarian states, however, also use digitalisation to enforce their state interests, in contradiction to the European understanding of digital sovereignty. The analysis will not be examined in depth here, as the focus of the observations is on the EU states and a comparison of their positioning.

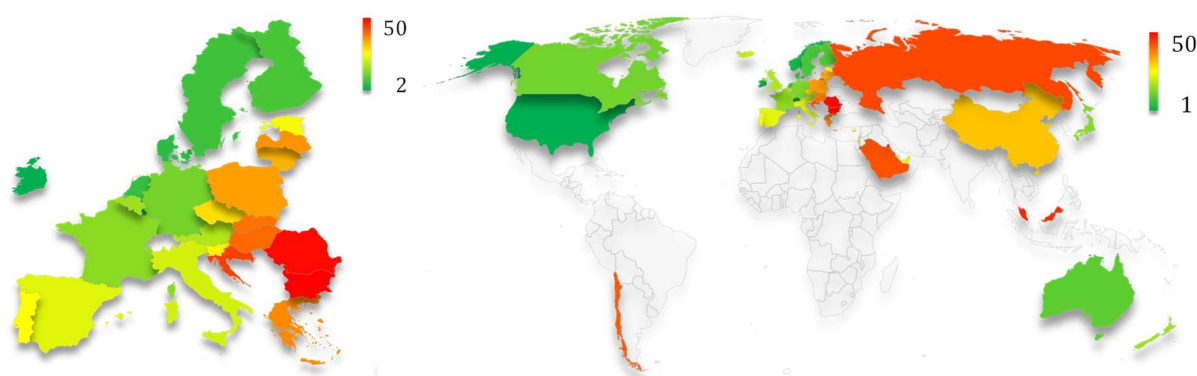


Figure 52 - Left: DSI rank EU-27, right: Top 50 DSI ranks worldwide, colour-coded¹⁷⁰

The EU-27 countries are entirely in the TOP-50 ranking, but there is a wide dispersion here, which is discussed in the following sections.

Table 27 - EU states' rankings

Rank EU	Country (world rank)	Rank EU	Country (world rank)
1	Ireland (2)	15	Slovenia (27)
2	Luxembourg (4)	16	Czech Republic (29)
3	Denmark (6)	17	Cyprus (30)
4	Netherlands (7)	18	Malta (32)
5	Sweden (8)	19	Lithuania (33)
6	Finland (9)	20	Poland (35)
7	Germany (13)	21	Latvia (36)
8	France (15)	22	Greece (37)
9	Belgium (17)	23	Slovak Republic (39)
10	Austria (19)	24	Hungary (40)
11	Italy (22)	25	Croatia (44)
12	Spain (23)	26	Romania (49)
13	Estonia (25)	27	Bulgaria (50)
14	Portugal (26)		

¹⁷⁰ Tool: Excel Mapdsigner

5.14. Verification through simulation

5.14.1. 'Stress test' DSI

To check robustness, the DSI is calculated, and each parameter is stochastically perturbed by $\pm 20\%$ in one iteration. The resulting new DSI value and the resulting ranking are stored and visualised. The programme logic is shown in the following function diagram (Figure 53).

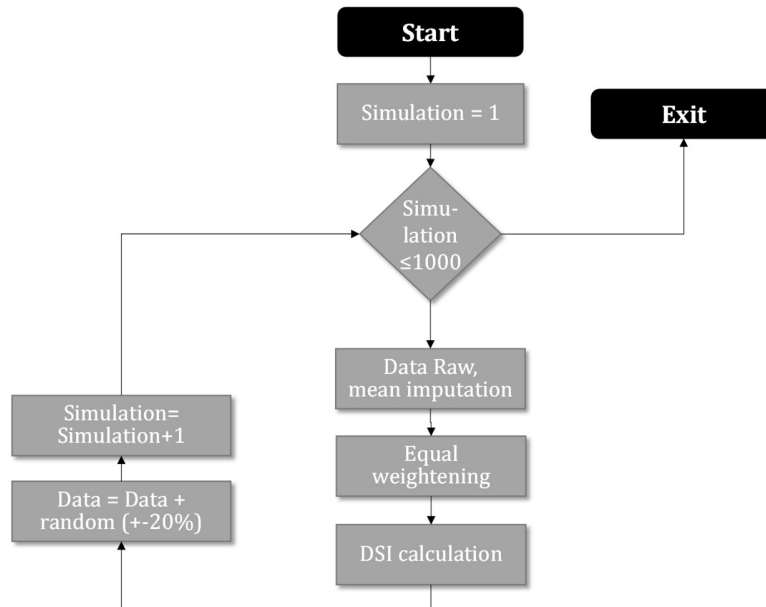


Figure 53 - Function diagram¹⁷¹, stress test of DSI calculations

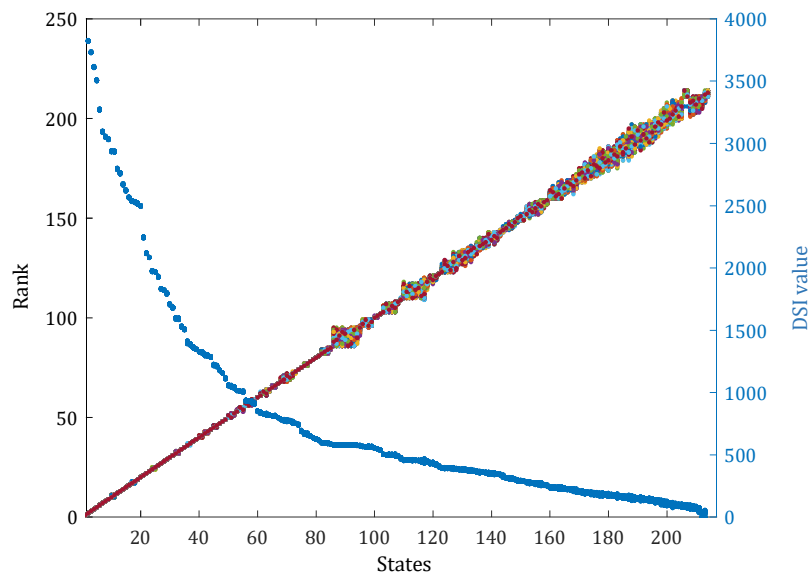


Figure 54 – Result stress test of DSI, DSI ranks and values, states sorted in ascending order by rank

¹⁷¹ Coded in MATLAB R2022a

With a repetition of 1,000 simulations, the dispersion of the index values and ranks increases as the DSI (-rank) decreases, but it remains essentially stable even when the individual parameters are significantly perturbed¹⁷². Thus, the parameters of the DSI, the weighting and the selected aggregation procedures can be assessed as robust. This is especially true for the countries in the upper ranks and thus for the EU countries.

5.14.2. Simulation in 150 periods

As further verification, the index development is simulated over 150 periods. Here, it is assumed that the parameters identified in the later analysis, which are determined to effectively increase the DSI for the states of the EU (cf. 5.16.3), increase by 5% in each period, meaning measures are taken so that the relevant levers improve. All parameters are additionally stochastically $\pm 5\%$.

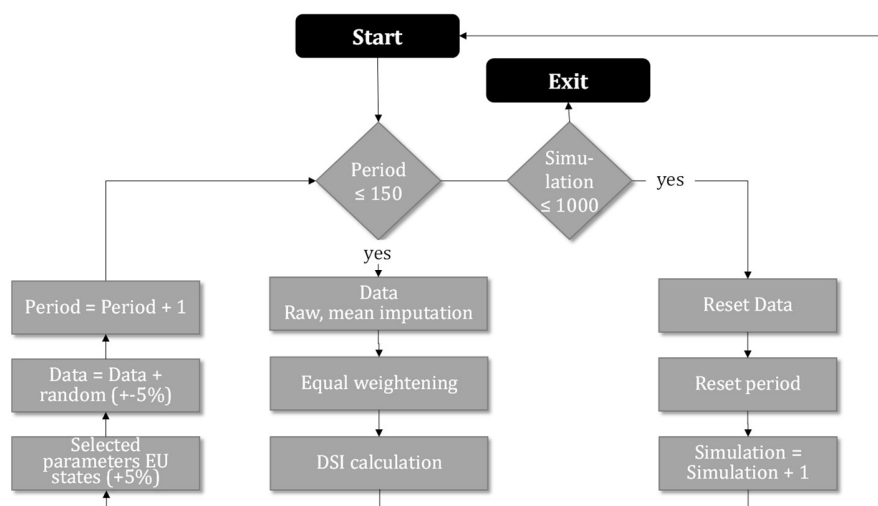


Figure 55 - Function diagram¹⁷³, simulation DSI

The rankings of the EU states per period in 1,000 simulations and the mean value per period are evaluated.

At the beginning of the simulation, the ranks of the EU states vary between 2–50 with a mean of 25.08 and a standard deviation of 13.69; the mean converges (only!) from period 43 onwards to 14 with a standard deviation of 7.94, placing the EU states in the top ranks of the DSI.

¹⁷² Comparable behaviour also observable with stochastic disturbances of $\pm 50\%$

¹⁷³ Coded in MATLAB R2022a

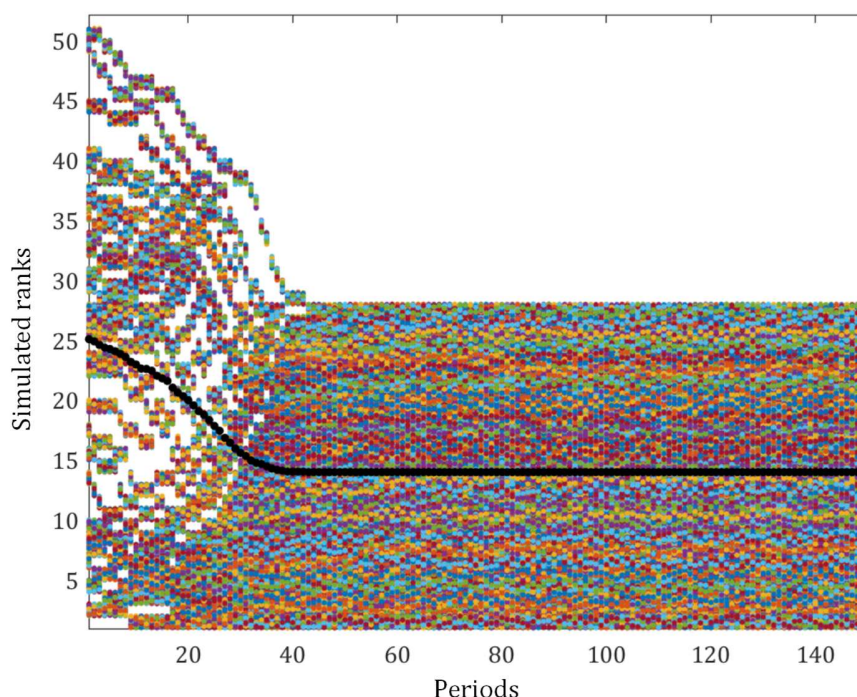


Figure 56 - Simulation EU – states over 150 periods: development of rank (coloured dots) and mean (black)

5.15. Verification and validation through existing country indices

In order to verify and validate the DSI, it is compared with other existing indices that are not directly related to digital sovereignty, the parameters of which have not been included identically in the calculation of the DSI. In addition to an Economic Freedom Index¹⁷⁴ (EFI) (Miller, et al., 2021), sustainability indices in the sense of economically, socially and ecologically relevant indices are also considered as comparative indices (Hampel, et al., 2016), for example the Food Safety Index¹⁷⁵ (FSI), the Better Life Index¹⁷⁶ (BLI), the Happy Planet Index¹⁷⁷ (HPI) and the Environmental Performance Index¹⁷⁸ (EPI). In addition, in Section 5.15.2 with the Sustainable Development Goals Index¹⁷⁹ (SDG), a dedicated, aggregated sustainability index is compared with the DSI.

In Sections 5.15.3 and 5.15.4, indices are used for validation that describe similar issues compared to the DSI. These include the Digital Dependence Index (DDI), which assesses the digital dependence of countries (Mayer, Lu, 2022) and the European Sovereignty Index (ESI), which measures the sovereignty of the EU-27 states in selected categories.

¹⁷⁴ Source: Global Footprint Network (2022)

¹⁷⁵ Source: Global Food Safety Initiative (2022)

¹⁷⁶ Source: OECD (2022a)

¹⁷⁷ Source: Wellbeing Economy Alliance (2022)

¹⁷⁸ Source: Wolf, et al. (2022)

¹⁷⁹ Source: Sachs, et al. (2022)

5.15.1. DSI Correlation to FSI, HPI, EPI, BLI and EFI

Economic parameters (e.g. GDP in PPS) can be supplemented or replaced by parameters that describe sustainability (Hampel, et al., 2016, p. 56). Some of these parameters, as well as other selected ones, will be used to investigate linear correlations with the newly formed DSI. Before analysing the correlations of the indices BLI, EFI, HPI and EPI with the DSI, a graphical overview of the correlations of the indices' ranks across all countries is provided.

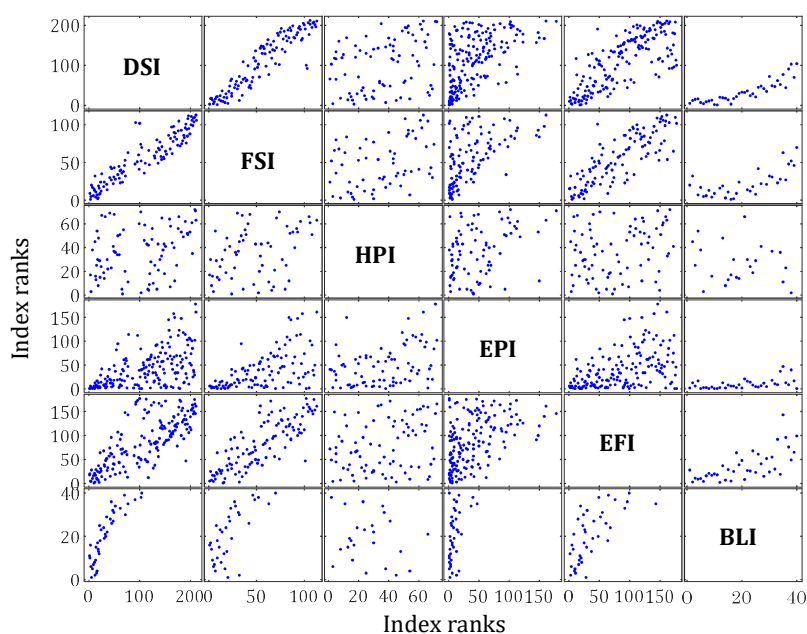


Figure 57 - Scatter plot index comparison ranks, all states, 2022

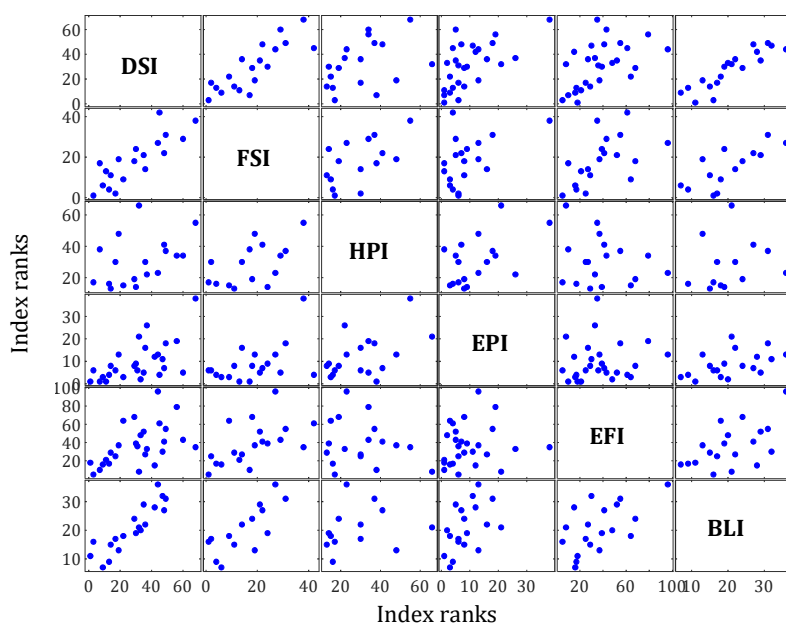


Figure 58 - Scatter plot index comparison ranks, EU states, 2022

Focusing on the DSI, it becomes graphically clear that linear relationships exist with the FSI, EFI and BLI indices. With the focus on the EU states, the linear correlations are reduced to the FSI and BLI.

The coefficients of determination show pronounced linear correlations of the DSI, with 86% to the FSI, 77% to the BLI and 57% to the EFI ranks. In contrast, the coefficient of determination for the HPI and EPI is very low.

Table 28 - left: Measures of determination, right: p-values, indices, all states, 2022

R²	DSI	FSI	HPI	EPI	EFI	BLI	p	DSI	FSI	HPI	EPI	EFI	BLI
DSI	1.00	.86	.10	.29	.57	.77	DSI	.00	.00	.00	.00	.00	.00
FSI	.86	1.00	.14	.32	.66	.48	FSI	.00	.00	.00	.00	.00	.00
HPI	.10	.14	1.00	.13	.07	.03	HPI	.00	.00	.00	.00	.02	.89
EPI	.29	.32	.13	1.00	.22	.24	EPI	.00	.00	.00	.00	.00	.00
EFI	.57	.66	.07	.22	1.00	.47	EFI	.00	.00	.02	.00	.00	.00
BLI	.77	.48	.03	.24	.47	1.00	BLI	.00	.00	.89	.00	.00	.00

The associated p-values are above the significance level of 5%, which confirms the validity of the model used. Thus, the regression results can be considered useful. The relevance of the usability of the EFI, FSI and BLI as validation or verification indices for the DSI are described below.

The EFI calculation focuses on the economic independence of a state's citizens. The state itself determines the degrees of freedom of labour, capital, goods and its use. In this respect, the index describes economic freedom as the fundamental right of citizens to be able to determine these things themselves. It is measured in four sub-indices with three sub-components each for 184 states (Miller, et al., 2021, 11ff). The index has linear correlations with other economic indices (e.g. Economic Freedom of the World) or logarithmic ones (e.g. GDP).

Table 29 - Regression analysis DSI Rank, EFI Rank

Results regression analysis, all		Results regression analysis, EU	
R ²	.57	R ²	.27
Number of observations	164	Number of observations	27
Error degrees of freedom	162	Error degrees of freedom	25
p-value	1.48 e ⁻³¹	p-value	5.05 e ⁻⁰²

When considering all states, a statistically significant correlation can be found with a coefficient of determination of 57%, which drops to 27% when reduced to the EU states. No better results can be achieved with other regression methods¹⁸⁰. Therefore, the EFI is not analysed further in connection with the DSI.

The Food Safety Index describes the affordability, availability and quality of food security for 113 countries based on 58 individual indices. The individual indices of the FSI

¹⁸⁰ With power regression $R^2 = 0.31$

largely do not overlap with the parameters of the DSI, and both indices show a high linear correlation.

Table 30 - Regression analysis DSI Rank, FSI Rank

Results regression analysis, all		Results regression analysis, EU	
R ²	.86	R ²	.70
Number of observations	112	Number of observations	19
Error degrees of freedom	110	Error degrees of freedom	17
p-value	1.89 e ⁻⁰⁶	p-value	1.4 e ⁻⁴⁸

At first glance, the FSI does not appear to be a validation index for the DSI. However, constructively comparable approaches are available in the index calculation (e.g. assessment of the political stability of states, FSI sub-index value 2.5 ‘Political and social barriers to access’), even if no explicit overlaps in the parameters are identified.

With the BLI, the OECD assesses elements of quality of life, such as household income, living conditions, education, environment, governance and security, within the framework of the Better Life Initiative (OECD, 2020b). Extensive overlaps with the DSI can be identified, which are also shown in the regression analysis.

Table 31 - Regression analysis DSI Rank, BLI Rank

Results regression analysis¹⁸¹, all		Results regression analysis, EU	
R ²	.77	R ²	.79
Number of observations	40	Number of observations	19
Error degrees of freedom	38	Error degrees of freedom	17
p-value	1.59 e ⁻¹³	p-value	3.49 e ⁻⁰⁷

When focusing on the EU states, the degree of determination remains largely identical. This confirms a linear dependence of the DSI on the BLI.

Due to comparable evaluation bases, the BLI is suitable in the sense of a verification of the selection of the correct parameters for the DSI; the FSI is suitable as a comparative index for a validation of the correct representation of reality by the DSI.

5.15.2. DSI correlation to sustainability indices

The validation by the FSI, which already contains elementary sustainability parameters, is strengthened by correlating the DSI with relatively new sustainability indices. If no strong correlations can be found with sustainability risk indices (e.g. ESG risk score¹⁸²), correlations with the SDG Index¹⁸³ can be identified. The SDG Index contains 122 individual parameters

¹⁸¹ Better coefficients of determination can be achieved by non-linear regression, e.g. by power regression with $f(x) = a \cdot x^b$ and $a = 2.8$ and $b = 0.6$: $R^2 = 0.79$.

¹⁸² Source: Sachs, et al. (2022) (Risk Score)

¹⁸³ Source: Sachs, et al. (2022) (Ranking)

from numerous social, ecological and structural, responsibility-oriented economic areas and is compiled for the UN states (Sachs, et al., 2022, pp. 63–71).

At the level of all UN states, there is a strong correlation of over 70% between the SDG and DSI ranks, which is significantly lower when focusing on the EU level, which is just under 50% but is still present. The associated p-values are above the significance level of 5%; thus, both correlation analyses are considered useful.

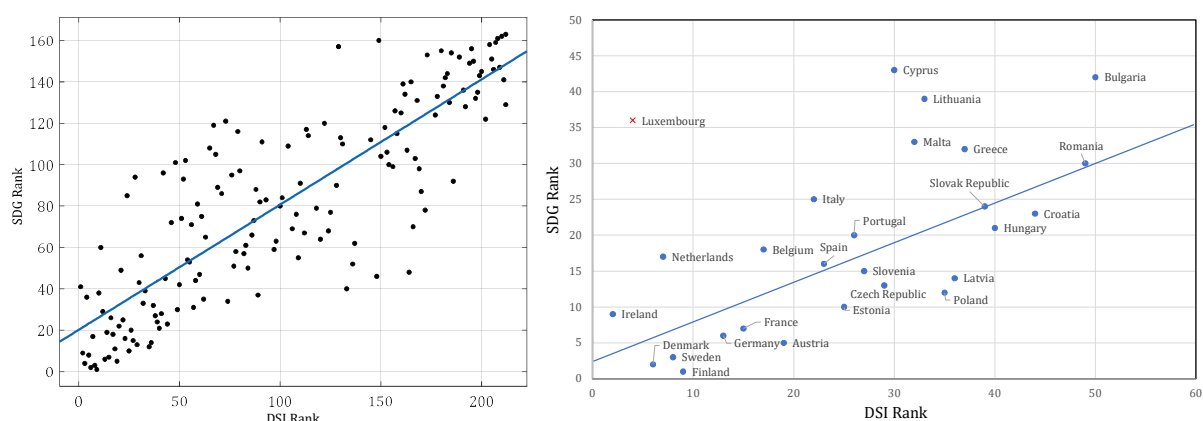


Figure 59 - Comparison of DSI and SDG ranks (left: all states, right: EU)

Table 32 - Regression analysis DSI and SDG rank, left: all states, right: EU states without Luxembourg (own calculation)

Results regression analysis		Results regression analysis	
R ²	.71	R ²	.48
Number of observations	160	Number of observations	26
Error degrees of freedom	158	Error degrees of freedom	24
p-value	4.88 e ⁻⁴⁴	p-value	0.24 e ⁻²

The results can be considered useful here, as the associated p-values are above the significance level of 5%. Thus, the SDG can be used as a validation for the DSI.

5.15.3. DSI Correlation to the Digital Dependence Index

In the development of the DSI, the Digital Dependence Index (DDI) takes on particular relevance, as it measures the extent to which states act in digital dependence on other states based on the G20 states as well as Kenya, Israel, Estonia and Singapore. This corresponds in part to the theoretical framework of the DSI in subindices Δ_2 (KET) and Δ_3 (technological sovereignty), although the latter refers more to digital sovereignty in the European value context, whereas the DDI looks exclusively at technological dependencies.

For the DDI, three sub-indices are formed—hardware, software and intellectual property—and secondary data is used to estimate a country's dependence. Secondary data include trade in ICT goods and services, ICT infrastructure and patents. Dependence in the sub-indices is calculated with values between 0 (low dependence) and 1 (high dependence). By weighting the sub-indices, differently centred DDI variants are formed alongside the

equally weighted standard DDI for trade, infrastructure and dependence on hardware, software and intellectual property. Missing data are replaced by mean imputation. Aggregation yields a DDI value per state that lies between 0 and 1 (Mayer, Lu, 2022).

The index is not validated or verified. Because of partly similar sub-indices or sub-components, it is recommended to analyse them. If there are significant correlations, a substantial part of the DSI can be validated. Since the DDI focuses on digital dependencies and does not measure digital sovereignty as a whole, it is recommended not to compare the indices as a whole, but only the sub-indices of the DSI.

Table 33 - Measures of determination – DDI and DSI (sub-indices)

	Δ_2	Δ_3	$\Delta_2 + \Delta_3$	DSI
DDI	.56	.12	.39	.21

Special attention is paid to the sub-index Δ_2 (KET) with a coefficient of determination of 56.4%, which has a disproportionate effect on the sum of sub-indices Δ_2 and Δ_3 . The linear regression analysis is significant.

Table 34 - Regression analysis of DDI and Δ_2

Source of Variation	SumSq	DF	MeanSq	F	pValue
Regression	725.75	1	725.75	27.13	3.67×10^{-5}
Error	561.72	21	26.74	1	0.5
Total	1287.50	22	58.52		
R²	.56				

Results regression analysis

R ²	.56
Number of observations	23
Error degrees of freedom	21
p-value	3.67×10^{-5}

Sub-index Δ_2 (KETs) has a coefficient of determination of 74.1% after elimination of outliers (Korea and Kenya)¹⁸⁴.

¹⁸⁴ At this point, it should be noted that the apparently contradictory relationship between high ICT goods import volumes and strong technological independence is reflected in this correlation analysis with a $R^2 = 0.66$ (ICT import goods and DDI-values).

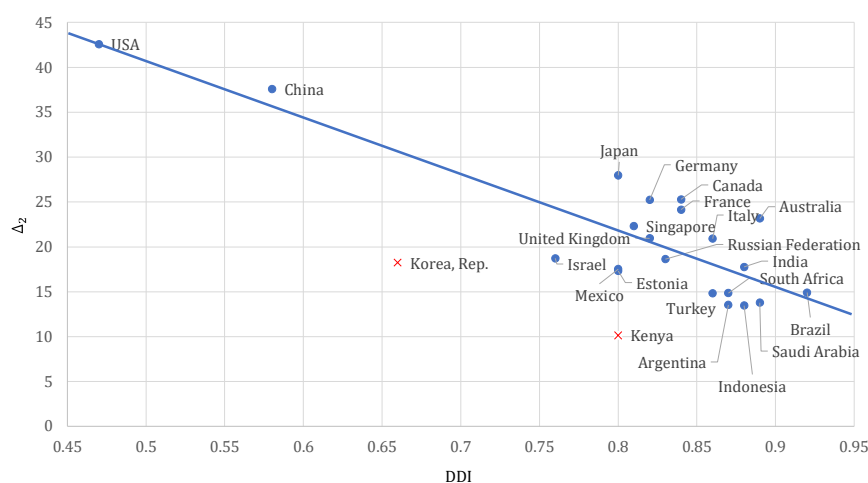


Figure 60 - Regression analysis of DDI and Δ_2 without outliers (red)

The coefficient of determination of the correlation of the DDI and $\Delta_2 + \Delta_3$ can be increased to 61.4% if outliers are excluded. If China and Australia are also eliminated, the coefficient of determination of the correlation of DSI and DDI increases from 21.6% to 57.3%. Therefore, sub-index Δ_2 can be explicitly validated with the help of the DDI, which then implicitly also applies to the DSI.

5.15.4. DSI Correlation to the European Sovereignty Index

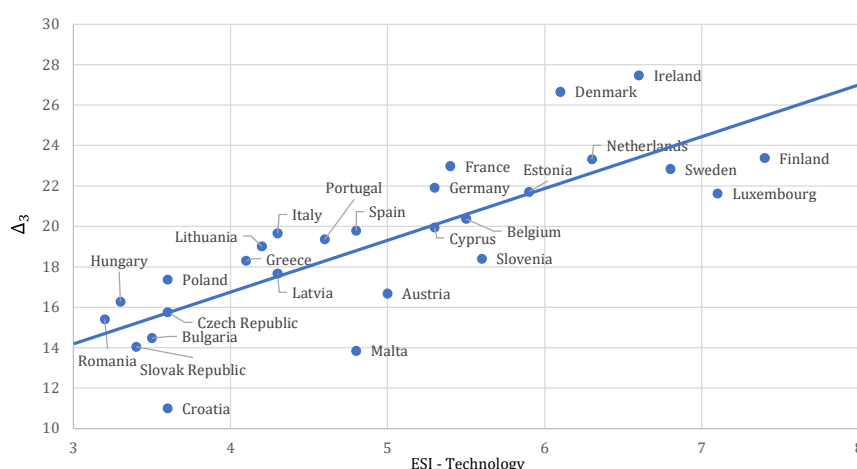
The European Council on Foreign Relations (ECFR) has developed a European Sovereignty Index¹⁸⁵ (ESI), which measures the sovereignty of the EU-27 states in the categories of climate, defence, economy, health, migration and technology on the basis of select parameters. The index is being calculated but has not yet been validated or verified. For each category, the ESI measures two key topics; for technology, these are AI (including big data) and digital infrastructure (including cloud computing, semiconductors, robotics, IoT, high-performance computing, telecommunications and cybersecurity). Therefore, some of the parameters of the ESI in the technology category, as well as from other categories (e.g. economy, here FDI), are comparable to those of the DSI, even if other data sources are used¹⁸⁶. Therefore, the ESI is considered a validation element for the present analysis. The data of the ESI is based on primary and secondary data as well as a normalised rating by the survey team from 0 (poor) to 10 (excellent). The parameters are weighted equally and aggregated additively. The index has not yet been discussed with regard to alternative methods of collection, aggregation, weighting and normalisation and can therefore only be used as a further validation method. The correlation matrix shows strong correlations in select sub-indices of the DSI and ESI, provided that the respective DSI and ESI scores are correlated.

¹⁸⁵ Puglierin, Zerka (2023)

¹⁸⁶ Ibid.

Table 35 - Measures of determination points ESI and DSI in EU-27 countries (own calculation)

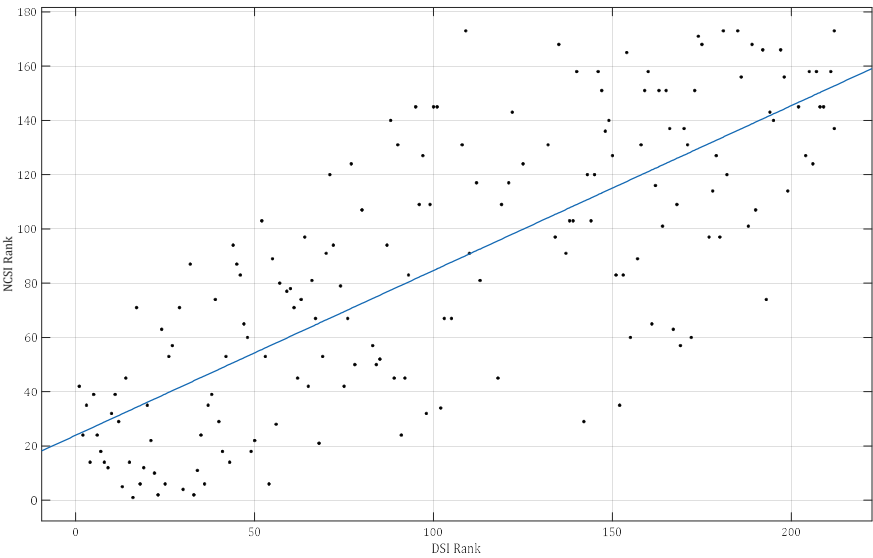
Sub-index ESI	Δ_1	Δ_2	Δ_3	DSI
Overall	0.49	0.52	0.55	0.63
Climate	0.42	0.32	0.40	0.50
Defence	0.01	0.29	0.12	0.07
Economy	0.62	0.44	0.57	0.72
Health	0.65	0.50	0.51	0.74
Migration	0.12	0.00	0.12	0.10
Technology	0.71	0.23	0.65	0.73

Figure 61 - Regression analysis points ESI – technology and Δ_3 , EU-27 countries

The correlation of the ranks shows even stronger correlations. Therefore, the ESI with a coefficient of determination of 63%, especially the categories of technology (73%), health (74%) and economy (72%), can be considered as a validation of the DSI and its sub-indices.

5.15.5. DSI Correlation to the NCSI

As it was recommended in advance not to integrate existing cybersecurity measures into the DSI, which is designed with a democratic 'DNA' due to too strong biases in selected states, but cybersecurity is conceptually linked to digital sovereignty, a regression analysis between the DSI and the NCSI is performed at this point (cf. 2.2.6.10). This can be seen as a confirmation of the conceptual interdependence between cybersecurity and digital sovereignty. It shows that there is a large deviation between NCSI/DSI ranks for dedicated states (e.g. USA 42/1, Saudi Arabia 14/42, Dominican Republic 29/124, Malaysia 22/46). This confirms the decision made in the conception not to include the NCSI in the calculation of the DSI, as otherwise the result would be distorted for specific countries. However, with a coefficient of determination of 59%, the correlation between the populations of all countries is strong.



Results regression analysis

R ²	.59
Number of observations	170
Error degrees of freedom	168
p-value	1.29 e ⁻³⁴

Figure 62 - Regression analysis of DSI and NCSI

With this strong correlation, the conceptual interdependence of digital sovereignty and state cyber resilience can be demonstrated, and can provide an approach for in-depth research (see discussion).

5.16. Analysis of individual states and clustering

As defined in Section 5.10, the parameters of sub-index Δ_1 (sovereignty, also referred to as sovereignty parameters), are given special priority through their exposed valuation as an equally weighted factor in the index formation. As a result, states with high values for these parameters are ‘preferred’ in the calculation of the DSI¹⁸⁷. Due to the high prioritisation of the sovereignty parameters, in the following analysis, the parameters X_1 and X_2 are evaluated separately.

The parameters of sub-indices Δ_2 (Digitalisation/KETs) and Δ_3 (Digitalisation/technological sovereignty) are weighted, as shown in Section 5.4, and used for the index formation. The parameters of both sub-indices are referred to in the following as digitalisation parameters and evaluated together.

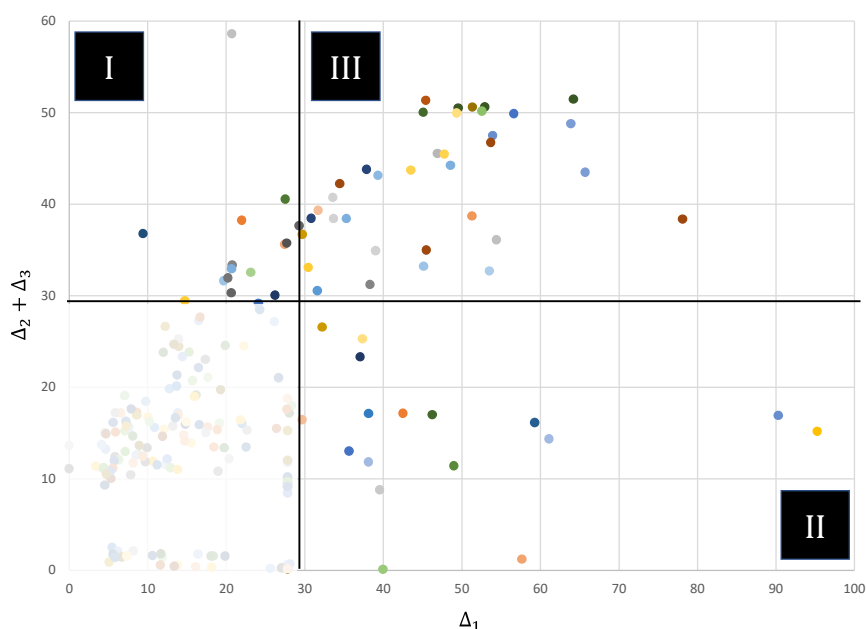
¹⁸⁷ The weighting generates a value-oriented, European, liberal-democratic view of the index. States whose democratically oriented sovereignty or prosperity is less pronounced are devalued in the index formation.

In order to assign states to clusters, the categories are derived from the values of the sovereignty parameters (Δ_1) and the digitalisation parameters ($\Delta_2 + \Delta_3$).

Table 36 - Statistical values of Δ_1 and $\Delta_2 + \Delta_3$

	Sovereignty parameters (Δ_1)	Digitalisation parameters ($\Delta_2 + \Delta_3$)
Mean value	23.27	22.21
1 st quartile	10.53	13.03
Median	19.05	16.95
3rd quartile	29.39	29.25

The countries can be classified by a categorisation cut, which is shown graphically in Figure 63. The 3rd quartile is chosen as the categorisation cut line in each case to identify a group of top performers and to delineate others in this regard (cf. 4.10).



Category/3 rd quartile	Δ_1	$\Delta_2 + \Delta_3$	Description
I	smaller	bigger	Potential challengers
II	bigger	smaller	Observing differentiations
III	bigger	bigger	Top Performer

Figure 63 - Categorisation cut, all states, sub-index points for Δ_1 and $\Delta_2 + \Delta_3$

Countries with a below-3rd quartile score of Δ_1 and $\Delta_2 + \Delta_3$ are excluded. The remaining set can be divided into three categories.

Since the DSI is largely a product of Δ_1 and $\Delta_2 + \Delta_3$, the analysis is based on the states per category with reference to the reasons why they are found in their respective categories.

From this, suggestions for improvements of the categories can be developed to enable the EU states to strengthen their digital sovereignty. For each category, the five main FoA that lead to deviations are identified in order to propose them for strengthening digital sovereignty.

5.16.1. Category I states – potential challengers

This category contains states with a relatively low level of prosperity or lower state sovereignty and a relatively high level of digitalisation.

Table 37 - Overview of states, Category I (EU states in **bold**)

DSI Rank	State	DSI Rank	State
31	China	50	Bulgaria
35	Poland	51	Mexico
36	Latvia	54	Argentina
40	Hungary	55	Brazil
41	Chile	56	Turkey
43	Russian Federation	62	Kazakhstan
44	Croatia	73	India
46	Malaysia	77	Georgia

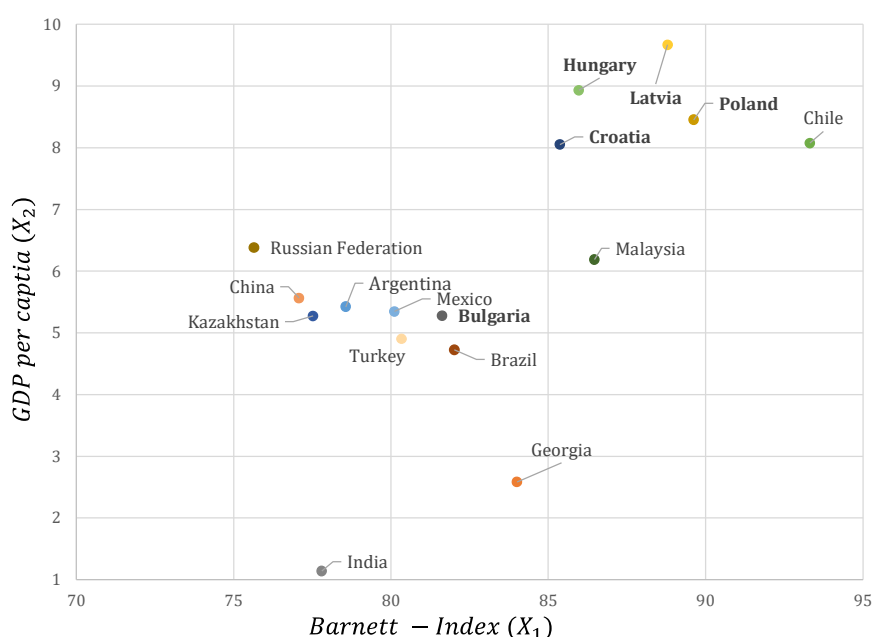


Figure 64 - Analysis of sovereignty parameters, Category I: X_1, X_2 (EU countries in **bold**)

Category I includes countries that have a below-average product of the Barnett Index (X_1) and GDP per capita (X_2), thus low sovereignty parameters (Δ_1). The EU states in this category

have relatively high values, whereas they have relatively low values in the digitalisation parameters.

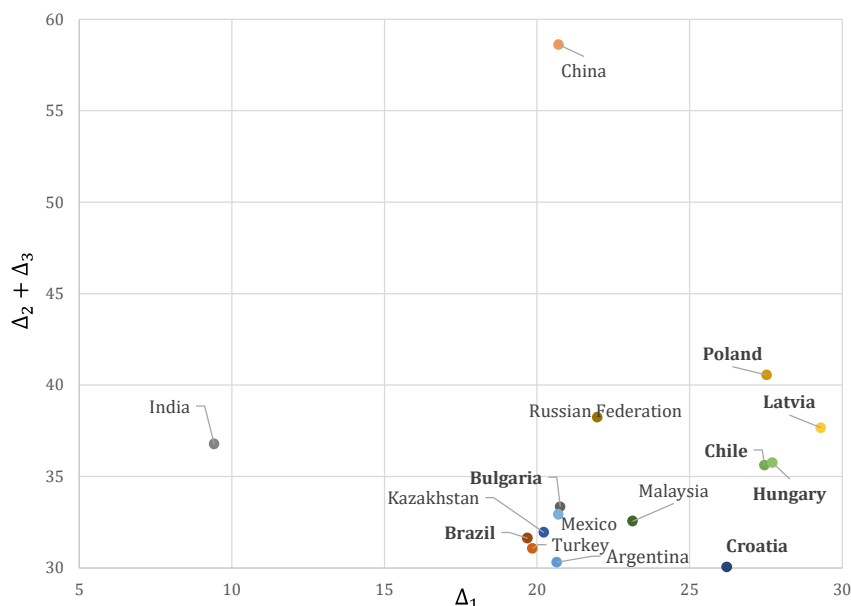


Figure 65 – Analysis of DSI sub-indices, Category I: $\Delta_1, \Delta_2 + \Delta_3$ (EU countries in **bold**)

The countries in Category I receive a high DSI rank due to their high scores on the digitalisation parameters. China is an outlier in terms of the digitalisation parameters; it compensates for lower scores on the sovereignty factor with disproportionately high digital parameters, resulting in a relatively high DSI rank (31)¹⁸⁸.

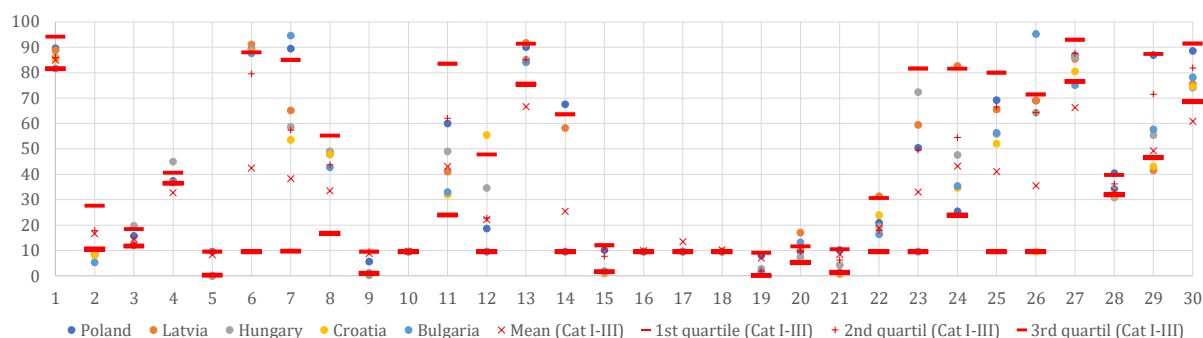


Figure 66 - Raw data, mean imputation, $X_i, i = 1, \dots, 30$, EU states of Category I, percentiles and mean value Categories I-III

Category I includes numerous EU states of the former 'Eastern Bloc': Russia, former Soviet republics, Hungary, Poland and Bulgaria, and states of the former Yugoslavia (Croatia)¹⁸⁹.

¹⁸⁸ Here it should be pointed out again that China or other autocratic states could be ranked higher in the definition of a 'digital autarky' for digital sovereignty, which was not used in this dissertation.

¹⁸⁹ Romania is not part in any of the three categories.

The heat map for the EU states in Category I (referred to as Group C in the following) determined from the deviation analysis shows the most important FoA on which to focus. FoA are developed by identifying the difference between the parameter value per state and the best quartile of states from category I-III. This results in a fields-of-action sequence of focus. The five parameters with the largest mean deviation are shown in Figure 67.

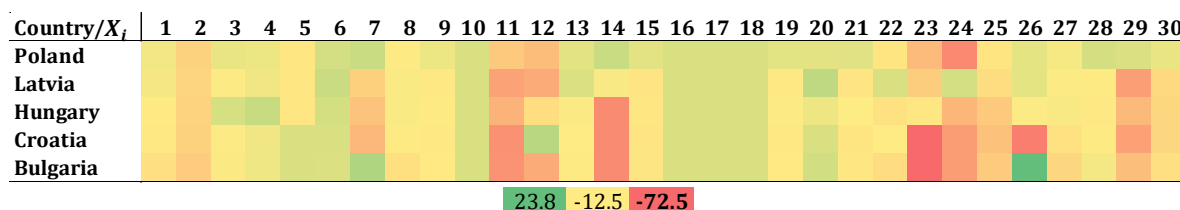


Figure 67 - Heatmap Group C, deviation from 1st quartile, Category I

5.16.2. Category II states – observing differentiations

This category includes countries with smaller populations, lower GDPs¹⁹⁰ and underrepresented digitalisation parameters. These are states that sometimes have high sovereignty parameters but do not play a significant role in the digitalisation sub-indices. This category does not include any EU states.

Table 38 - Overview of states, Category II

DSI Rank	State	DSI Rank	State
45	Monaco	72	Andorra
47	Liechtenstein	75	Bahamas
48	Kuwait	81	Greenland
52	Brunei Darussalam	94	Saint Kitts and Nevis
53	Bahrain	96	Aruba
63	Macau	99	Puerto Rico
66	Cayman Islands	102	Virgin Islands (U.S.)
70	San Marino	116	Guam

Many of the parameters for these states are not available and have been replaced. In addition, the Barnett Index is replaced by mean imputation¹⁹¹, which does not seem unsuitable for the evaluation of the DSI but is conspicuous in the evaluation, for example, by identical X_1 .

Apart from the wealthy states of Liechtenstein and Monaco, there are no notable outliers, and the states in this category currently play a subordinate role in the discussion on digital sovereignty.

¹⁹⁰ The exception is Puerto Rico, with 3.2 million inhabitants and USD 105 billion GDP.

¹⁹¹ The mean of the Barnett Index values is 81.52.

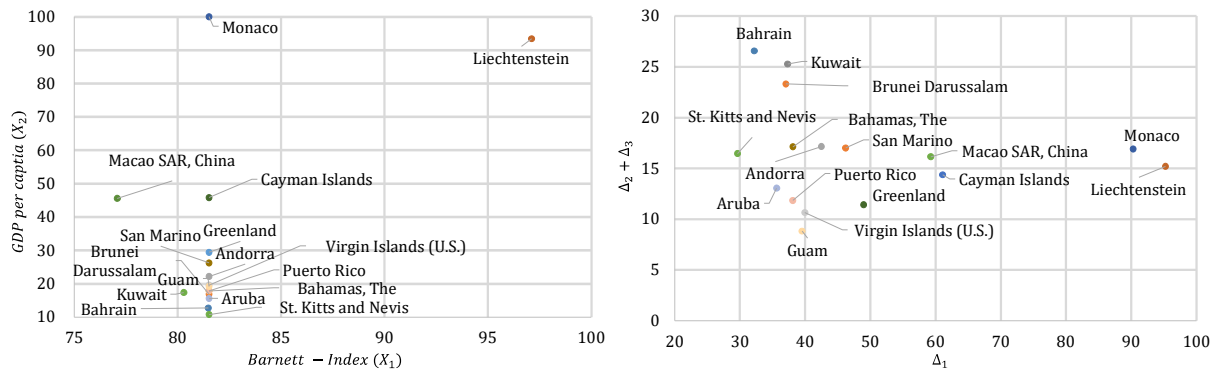


Figure 68 - Analysis of DSI parameters and sub-indices, Category II: X_1, X_2 , and $\Delta_1, \Delta_2 + \Delta_3$

The heat map is not presented in this category, as optimisation approaches cannot be derived to a large extent due to low data availability and are irrelevant for strengthening the digital sovereignty of the EU states. At a minimum, the states with high prosperity in this category should continue to be monitored.

5.16.3. Category III states – top performer

Most (21) of the EU states are in Category III. The G7 countries with 10% of the world's population and 45% of global GDP are also entirely included in category III and could be used as a joint optimisation initiative. Since the focus of the analysis is on the EU states, this is not done in this dissertation.

Compared to Category I, it is noticeable that the Baltic states of Estonia and Latvia, as former republics of the USSR, have a higher DSI than the states of the former 'Eastern Bloc'. This is due to a focussed digitalisation policy over the last 25 years, which can be seen in high values for the digitalisation parameters (especially Estonia, cf. 2.2.5.4). Category III states occupy all of the first 30 rankings in the DSI (China, as a Category I element, is ranked 31st, followed only by Malta, another Category III state).

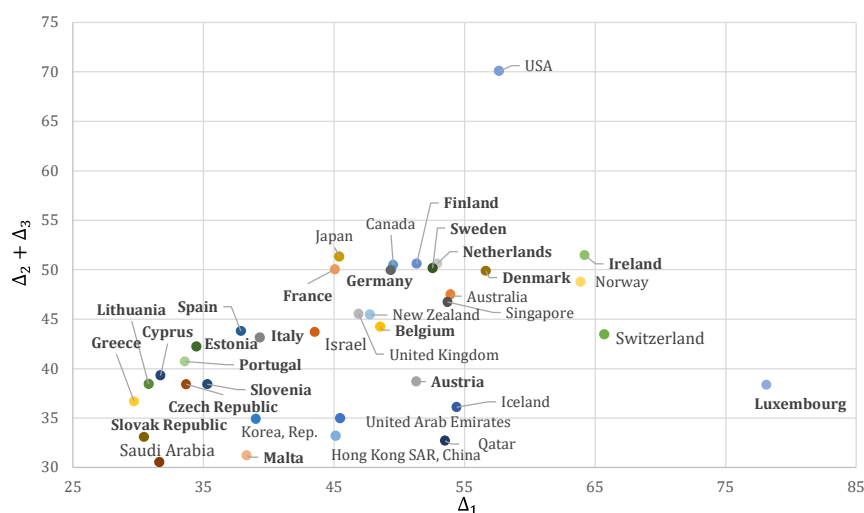
In addition to the EU states, the top 20 rankings include the US, Norway, Switzerland, Australia, Singapore, Canada, Japan, New Zealand, the UK and Iceland.

The US, due to its high digitalisation factor, and especially Luxembourg, due to its high sovereignty factor, can be considered outliers in the top DSI rankings in Category III. Among the EU states, Ireland has disproportionately high values in the sovereignty and digitalisation parameters, placing it second. Denmark, the Netherlands, Sweden, Finland, Germany and France form another group in Category III, followed by the other EU states, which have somewhat low parameter values.

The autocratic Saudi Arabia can be considered particularly critical within the assessment as a top performer. The state has high values for the digitalisation parameters due to its consistent digitalisation policy, which leads to its positioning in this category. In addition, a relatively high level of prosperity with a relatively low Barnett value results in a relatively high sovereignty sub-index overall.

Table 39 - Overview of states, Category III (EU states in **bold**)

DSI Rank	Country	DSI Rank	Country
1	USA	20	Iceland
2	Ireland	21	Israel
3	Norway	22	Italy
4	Luxembourg	23	Spain
5	Switzerland	24	United Arab Emirates
6	Denmark	25	Estonia
7	Netherlands	26	Portugal
8	Sweden	27	Slovenia
9	Finland	28	Qatar
10	Australia	29	Czech Republic
11	Singapore	30	Cyprus
12	Canada	32	Malta
13	Germany	33	Lithuania
14	Japan	34	Hong Kong SAR, China
15	France	37	Greece
16	New Zealand	38	Korea, Rep.
17	Belgium	39	Slovak Republic
18	United Kingdom	42	Saudi Arabia
19	Austria		

Figure 69 - DSI parameters, Category III: Δ_1 , $\Delta_2 + \Delta_3$, (EU states in **bold**)

Luxembourg, Norway, Ireland, Switzerland, the US and Denmark achieve a high sovereignty sub-index through high per capita GDP and Barnett Index values (Δ_1). Coupled with high digitalisation rankings, these countries thus top the digital sovereignty rankings.

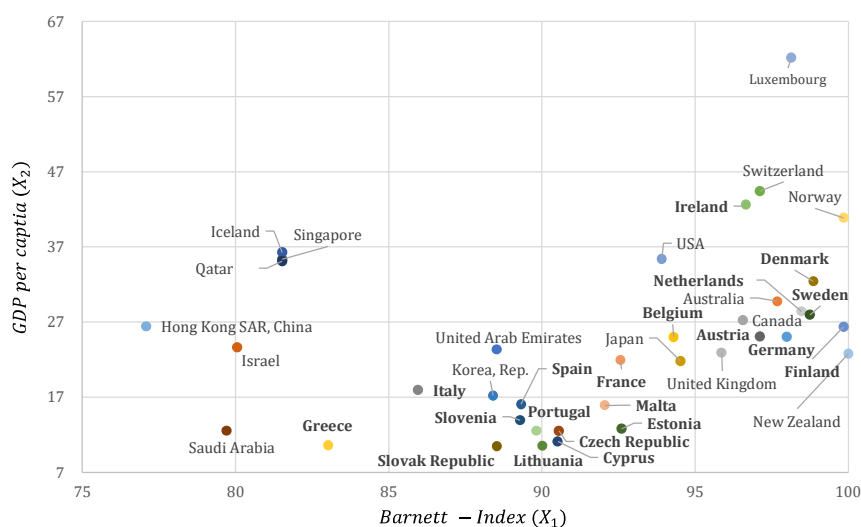


Figure 70 - DSI parameters, Category III: X_1, X_2 , (EU states in **bold**)

Ireland, Switzerland, Luxembourg and Norway dominate, with particularly high per capita GDP values. The US achieves first place in the ranking due to a disproportionately strong digitalisation factor.

If the previous analysis is reduced to the EU states in Category III, correlations between the sovereignty and digitalisation parameters become apparent. If the outlier Luxembourg is not included, the EU states show a significant linear correlation.

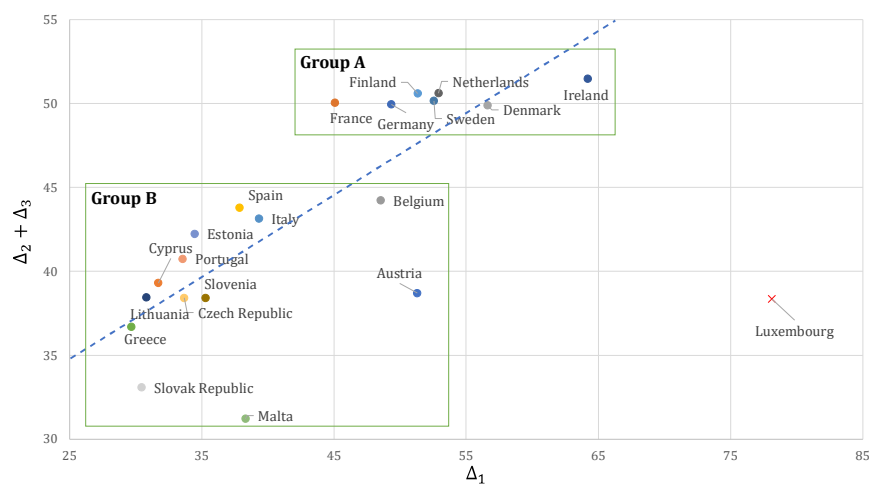


Figure 71 - Analysis of sub-indices for DSI, Category III: $\Delta_1, \Delta_2 + \Delta_3$, EU states without Luxembourg¹⁹²

¹⁹² $R^2 = 25.41$ with Luxembourg

Table 40 - Regression analysis

Results of the regression analysis	
R^2	.61
Number of observations	20
Error degrees of freedom	18
p-value	4.05 e^{-05}

Due to its high per capita GDP, Luxembourg is an outlier and is excluded from the regression. With a coefficient of determination of 61% and a p-value <0.05 , the regression can be assessed as useful. This means that in Category III, the EU states show a linear relationship between the sovereignty and digitalisation parameters.

In the correlation shown, the EU states can be divided into two groups, as outlined above. Group A includes the states that have high rankings and above-average sovereignty and digitalisation parameters, such as Ireland, the Netherlands, Germany, France, Sweden, Finland and Denmark; the remaining states are in Group B.

The objective regarding recommendations for strengthening digital sovereignty can therefore be to develop the states of Group B at least to the level of the states of Group A to generate homogeneity within the EU states.

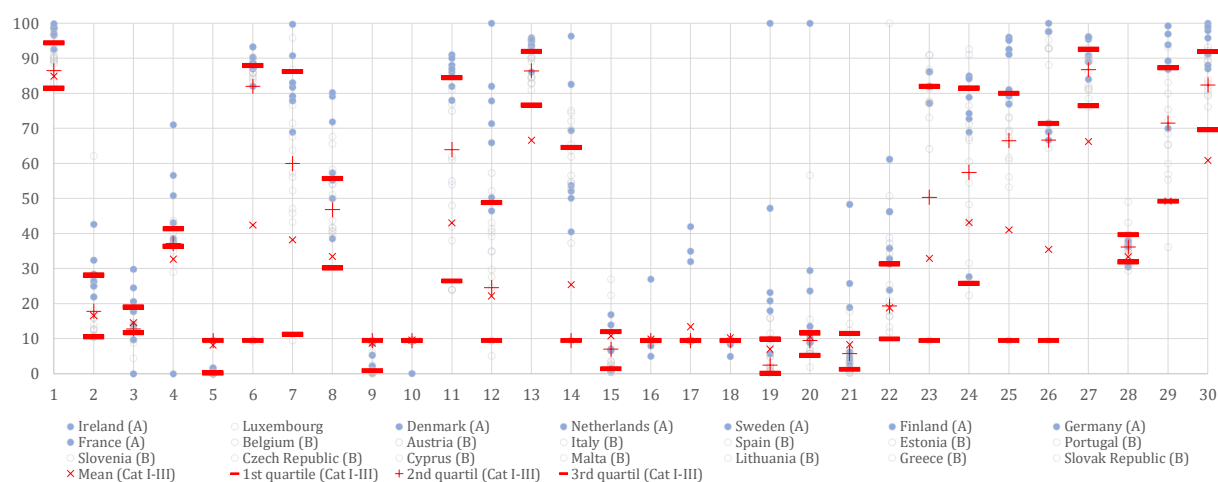


Figure 72 - Raw data X_i , $i = 1, \dots, 30$, EU countries in Category III, mean and percentiles categories I–III, Groups A and B

The analysis of the individual parameters shows that, in most EU countries, there is a need for action in X_6 , X_7 , X_{11} , X_{12} , X_{14} , X_{23} , X_{24} , X_{25} and X_{29} (red colouring on the heat map). This means that there is little need for action regarding the sovereignty parameters, but rather for the digitalisation parameters.

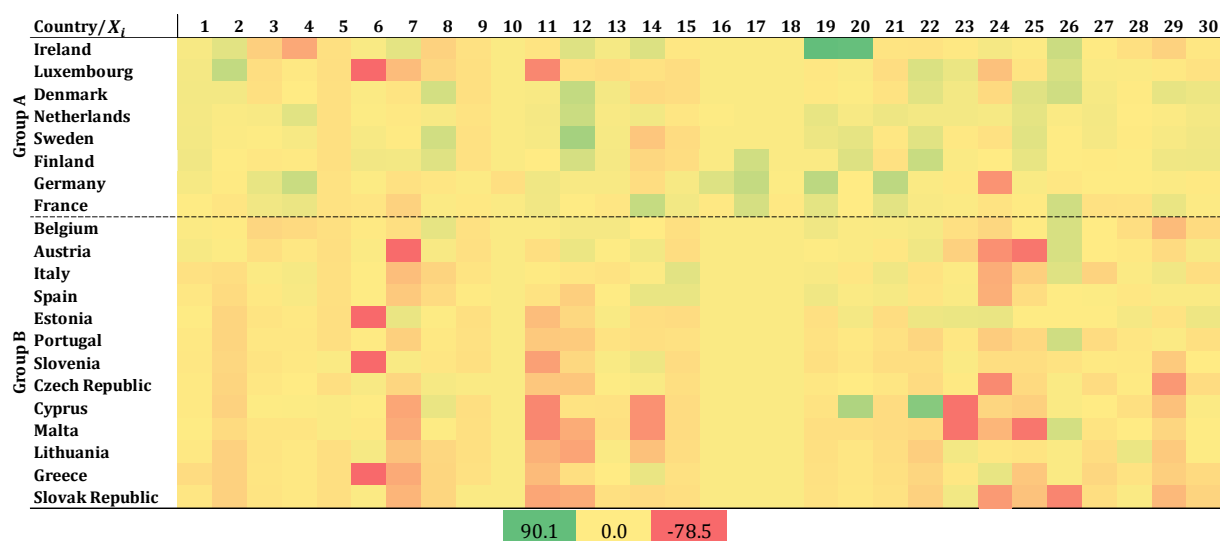


Figure 73 – Heatmap of the EU states, deviation 1st quartile, Category III

It is recommended for the states in Group A to aim for the 1st quartile of all states per parameter as a target value in order to strengthen their digital sovereignty. Assuming that all the EU states in Group A reach the best quartile, the FoAs, shown in Table 41, can be identified. The smaller the rank, the higher the averaged quotient of the 1st quartile and corresponding parameter.

Table 41 - Sequence of FoA, Group A

FoA	Parameter	FoA	Parameter	FoA	Parameter
1	Start-up investments	11	Internet access	21	WDC
2	ICT employees	12	Digital STRI	22	Education
3	R&D expenditure	13	TIMMS	23	Data protection
4	Patents	14	Human capital	24	Quantum funding
5	EDBI	15	TOP university	25	Import ICT services
6	FDI inflow	16	Governments online	26	Import ICT goods/ t
7	SIM cards	17	E-Government rank	27	Leiden Index
8	Blockchain Start-ups	18	FDI outflow	28	Cambrian Index
9	Asses. reading, maths	19	Barnett-Sovereignty	29	Exp. ICT services/ ft ¹⁹³
10	OpenData Index	20	GDP per capita	30	Exp. ICT services/ t

The states in Group B should, at a minimum, reach the average of the Group A states to strengthen their digital sovereignty. In this way, the EU states achieve a largely balanced increase in the level of digital sovereignty. The lower the rank, the higher the average quotient of the mean value of Group A states to the corresponding parameters.

¹⁹³ 't' stands for total and 'ft' for from total.

Table 42 - Sequence of FoA, Group B

FoA	Parameter	FoA	Parameter	FoA	Parameter
1	Export ICT services/ t	11	TIMMS	21	FDI outflow
2	Import ICT goods total	12	FDI inflow	22	E-Government Ran
3	Export ICT services/ ft	13	Import ICT services	23	Data protection EU
4	Cambrian Index	14	Governments online	24	Barnett-Sovereignty
5	Leiden Index	15	EDBI	25	Internet access
6	TOP university	16	Education	26	Human capital
7	GDP per capita	17	ICT employees	27	SIM cards
8	WDC	18	OpenData Index	28	Blockchain Start-ups
9	Asses. reading, maths	19	Quantum funding	29	Patents
10	R&D expenditure	20	Digital STRI	30	Start-up investments

5.17. Overarching view

To show differences between the states of the EU and selected others, the states of Group A (Group B is only outlined) as well as following additional states are discussed.

- The USA, which is first in the ranking; their software and hardware products dominate the market in EU countries and worldwide.
- China, due to high hardware dominance, especially in the telecommunications market, and the production of hardware for other countries. China is of particular interest due to its autocratic political system, state intervention in the economy and focus on technology leadership (Hannas, Tatlow, 2021).
- (South) Korea, which, along with Japan, has developed into a state with a strong digital industry in Asia (and does not produce in China). Korea's proximity to the EU can also be seen, for example, in the fact that it was the only non-EU state to be included in the German GAIA-X cloud campaign. In addition, Korea plays a prominent role in the DDI (Mayer, Lu, 2022, 22ff),
- India, because despite low sovereignty parameters, there are relatively high digitalisation parameters due to many ICT service providers and the significant offshoring location,
- Russia, as an autocratic state with a longstanding ambition to become technologically self-sufficient and, especially in the recent past, to cooperate more closely with China and India. This also applies to digital technologies (Hsiung, 2021) (DeNardis, et al., 2020) (Musiani, 2022),
- The EU, assumed here as a fictitious state in the sense of a common economic area regulated in digital aspects, of which the 30 parameters are simulated as if the EU were a state pursuing a common, integrated digital policy. The parameter values of the EU from the individual EU states are used either as a sum or as an arithmetic mean, depending on suitability.

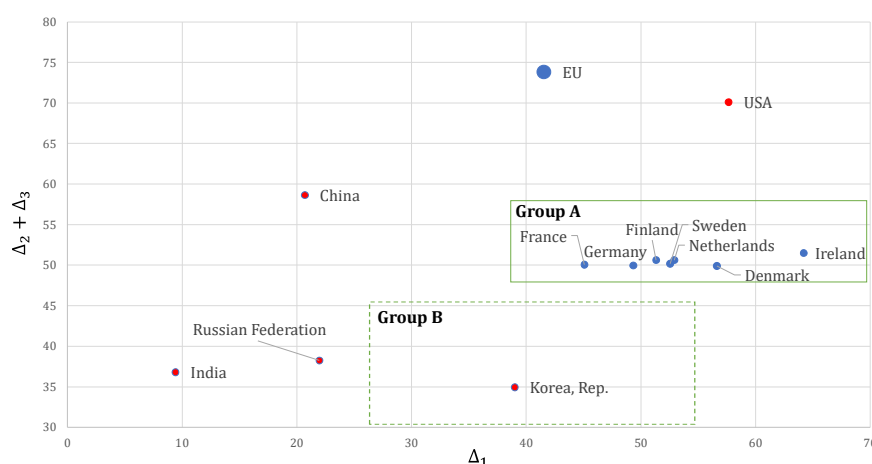


Figure 74 - Analysis of sub-indices for DSI, Category III, Group A and China, the USA, Russia, India, Korea and the EU: $\Delta_1, \Delta_2 + \Delta_3$

As a ‘fictitious state’, the EU scores high on the sovereignty and (especially) the digitalisation parameters. The EU as a single state would be ranked 4th in the DSI; a common economic digital alliance with a shared understanding of digital sovereignty would strengthen the EU. The individual 30 parameters were calculated either individually or by summing up values from individual EU countries (e.g. GDP per capita or ICT service exports) or as an average value (e.g. Barnett Index or Digital STRI).

With the development of the EU states of Group B to the level of Group A, a ‘fictitious EU single state’ could even top the rankings.

The analysis confirms that (South) Korea plays a special role among the Asian countries and lies in the corridor of EU countries in Group B (framed by a dashed line in Figure 74). This also applies when comparing the digitalisation parameters of sub-indices Δ_2 and Δ_3 . In addition, India and Russia have a lot of catching up to do in order to achieve higher rankings than the EU countries with regard to the sovereignty and digitalisation parameters.

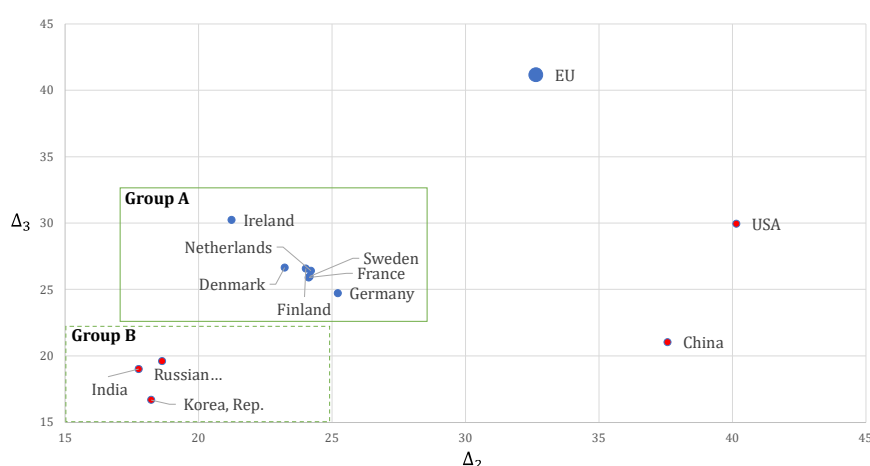


Figure 75 - Analysis of digitalisation parameters for DSI, Category III, Group A and China, the USA, Russia, India, Korea and the EU: Δ_2, Δ_3

China is strong in the components for KETs but relatively weak in technological sovereignty regarding the parameters of the index. The USA, on the other hand, is strong in both aspects. The ‘fictitious EU state’ stands out significantly from the states of Group A in both digitalisation parameters.

The US dominates; in particular, the parameters of the sub-index regarding KETs determine its positioning. This is also shown by a relative deviation analysis of the EU countries in Group A to the parameter values of the US. The parameters of the sub-indices for state sovereignty (Δ_1) and technological sovereignty (Δ_3) are, with a few exceptions, balanced, while the US achieves significantly higher values for KETs (Δ_2).

China, on the other hand, shows significant deviations compared to the EU states in Group A. Where the EU states are still clearly stronger in the sovereignty parameter (Δ_1), this is not true for the digitalisation parameters (Δ_2 and Δ_3). Significantly better values can be found for the EU countries (e.g. for OpenData or Digital STRI), as can significantly worse values (e.g. for the number of ICT experts, number of patents, number of Start-up investments or research and development expenditure).

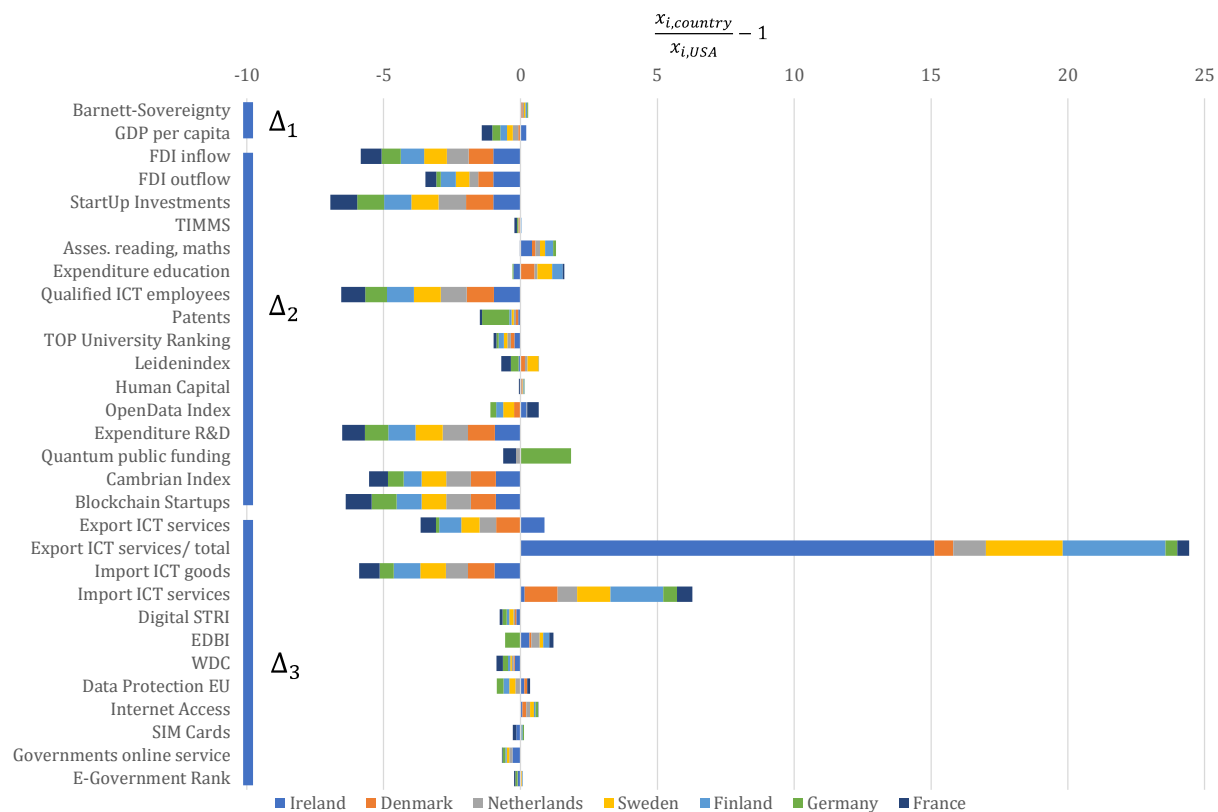


Figure 76 - Deviation EU countries, Group A to USA

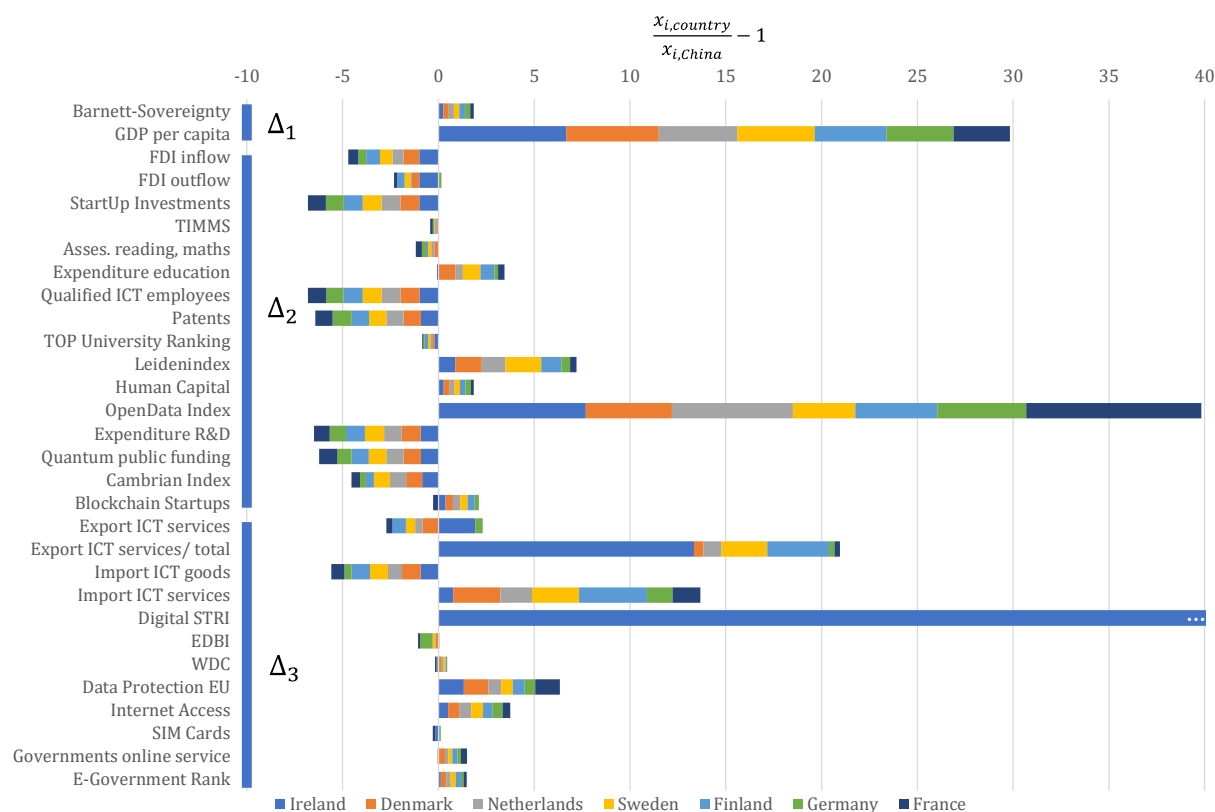


Figure 77 - Deviation EU countries, Group A to China

It is recommended that all EU states initiate a common digital market and industrial policy, including beyond country-specific individual interests, in order to confront the two digital superpowers, the USA and China, with significantly higher performances and purchasing powers.

Through these recommendations, the states of the EU can strengthen their individual and joint digital sovereignty, thus counterbalancing the autarky and autonomy of other major powers and allowing the EU to once again act as a recognised shaper and negotiating partner in the ICT context. In this way, the EU can implement the idea of global, strategic autonomy, producing technology in select fields resiliently and independently, obtaining technology sovereignly from other states and developing it through market power, also with regard to standardisation and transparency, so that it remains controllable.

6. Recommendations

The calculation of the DSI and the analysis of its parameters make it possible to identify FoA that strengthen the digital sovereignty of states. However, the recommendations can only be derived on the basis of the closed-loop nature of the model. This means that there may be additional levers that cannot be recommended here due to the nature of the model.

6.1. Fields of Actions at individual state-level/individual analysis

With the model developed in this dissertation, individual strengths and weaknesses of states in the 30 parameters can be identified at the state-level through comparative analyses and measures for optimisation can be derived from them. Identifying the FoA at individual state-level is only a first step. It will then be important to identify ‘how large a step’ needs to be taken, i.e. a quantitative statement on the improvement of a field of action. This can be done, for example, by comparing countries, as outlined below. The procedure of indexing allows for country-oriented comparisons with a focus on the EU-27 countries to identify FoA that can increase the index rank for digital sovereignty. For better orientation, the minima and maxima of Group A states are shown parameter by parameter, along with the state that holds the maximum value.

Table 43 - Group A states compared with Categories I and III states, 3rd quartile, maximum

X_i	3 rd quart. Group A	3 rd quart. Cat I-III	Max Group A	Max State Cat I-III ¹⁹⁴	Max EU Group A	Min EU Group A
1	96.66	94.41	99.85	New Zealand	Finland	France
2	25.07	28.20	42.64	Monaco	Ireland	France
3	9.74	19.06	29.84	USA	Germany	Ireland
4	37.69	41.34	71.04	Japan	Germany	Ireland
5	.24	9.50	1.71	USA	France	Ireland
6	87.06	87.98	93.28	Singapore	Finland	France
7	77.92	86.26	99.72	China	Ireland	France
8	50.00	55.73	80.21	Norway	Sweden	Ireland
9	.91	9.50	8.67	China	Germany	Denmark
10	9.50	9.50	9.50	China	Ireland	Germany
11	82.00	84.50	91.00	USA	Germany	Ireland
12	50.31	48.81	100.00	Sweden	Sweden	France
13	93.49	91.95	95.88	Australia	Denmark	France
14	50.09	64.59	96.31	Korea, Rep.	France	Sweden
15	1.44	12.08	16.90	USA	France	Sweden
16	8.00	9.50	27.00	China	Germany	France
17	9.50	9.50	42.00	USA	Germany	Ireland
18	8.40	9.50	9.50	USA	Ireland	France
19	9.93	9.92	100.00	Ireland	Ireland	Denmark
20	8.91	11.63	100.00	Ireland	Ireland	Germany
21	3.99	11.56	48.35	USA	Germany	Finland
22	31.34	31.34	61.19	Cyprus	Finland	Ireland
23	77.23	81.96	86.15	Norway	Denmark	Ireland
24	68.94	81.53	84.97	New Zealand	Ireland	Germany
25	79.23	79.98	96.01	USA	Denmark	France
26	69.05	71.43	100.00	Ireland	Ireland	Germany

¹⁹⁴ Maximum of states from category I-III = 100

X_i	3 rd quart. Group A	3 rd quart. Cat I–III	Max Group A	Max State Cat I–III ¹⁹⁴	Max EU Group A	Min EU Group A
27	89.04	92.60	96.20	Kuwait	Netherlands	France
28	31.54	39.71	38.03	Macau	Finland	Ireland
29	86.92	87.31	99.23	Hong Kong	Denmark	Ireland
30	88.05	91.93	100.00	Denmark	Denmark	Ireland

From this table, countries can identify other approaches by comparison, adapting them if necessary to generate higher parameter values. Another good overview and detailed profiles of the level of digitalisation in the EU countries is provided by the DESI overview outlined in Section 2.2.6.5, where country profiles¹⁹⁵ of the EU states can be viewed.

It is therefore recommended that the EU states use the model developed here to implement individual state FoA. This will be illustrated by means of an example. The degree of deviation of the field of action at individual state-level can be derived from the model and quantified. Table 43 shows that the US is the frontrunner for investments in Start-ups. The investment in the USA amounts to 135,649 million USD according to the OECD Entrepreneur Financing Database. That of all countries in Group A totals USD 6,319 million, with Germany and France being the most strongly represented in the group with a good USD 2,143 million and USD 2,320 million respectively. To reach the level of the USA, Germany and France would have to increase their Start-up investments by a factor of 60. If the Start-up investment of all countries in Group A were evaluated together, these countries would have to increase their investment by a factor of 21. Even if the total investment of all EU states is valued at 8,146 million USD, the difference to the USA is still 17-fold.

In this way, as shown in Figures 75 and 76, a profile of strengths and weaknesses can be drawn up for each state, from which FoA can be derived and quantified, as in the previous example.

The addressees for the analysis of the individual states, the comparative analysis and the derivation of specific, measurable FoA have to be sought at the national level. In the meantime, responsibilities for dealing with the megatrend of digitalisation have largely been established at government level in the EU states. In Germany, responsibility is shared between the Ministries of Transport and Digital Affairs, the Ministry of Economics and the Federal Chancellery. The Digital Strategy for Germany¹⁹⁶, which is intended to embody the measures to strengthen digital sovereignty, was drawn up between these responsible government bodies to embed it in operational government action. In the Czech Republic, the Innovation Strategy¹⁹⁷ of the Czech Republic could be a suitable recipient organisation. In France, the Digital Directorate¹⁹⁸ would be a suitable recipient of the model developed here. This list could be continued for other EU countries and supplemented by policy-related

¹⁹⁵ <https://digital-strategy.ec.europa.eu/en/policies/countries-digitisation-performance>

¹⁹⁶ <https://digitalstrategie-deutschland.de/>

¹⁹⁷ <https://digital-skills-jobs.europa.eu/en/actions/national-initiatives/national-strategies/czech-republic-innovation-strategy-2019-2030>

¹⁹⁸ <https://www.numerique.gouv.fr/dinum/>

associations (e.g. D21¹⁹⁹ or Bitkom). These national organisations are thus responsible for implementing digitisation measures and strengthening digital sovereignty. To simplify the analysis, initial proposals for FoA for the individual EU states are recommended in the following Section. For these, too, the evaluation of the FoA and their implementation must be in the responsibilities of the national governments.

6.2. Fields of Actions at individual state-level/group analysis

The preceding analysis of the DSI proposed in this dissertation offers the possibility of categorising all states into three categories with the help of the measurement point of the third quartile. The EU member states can be divided into three groups. If a state does not derive any individual state FoAs as recommended in the previous Section, then the FoAs identified at group-level for the respective state of a group can be used from the preceding analysis and be implemented in each of the member states in the group.

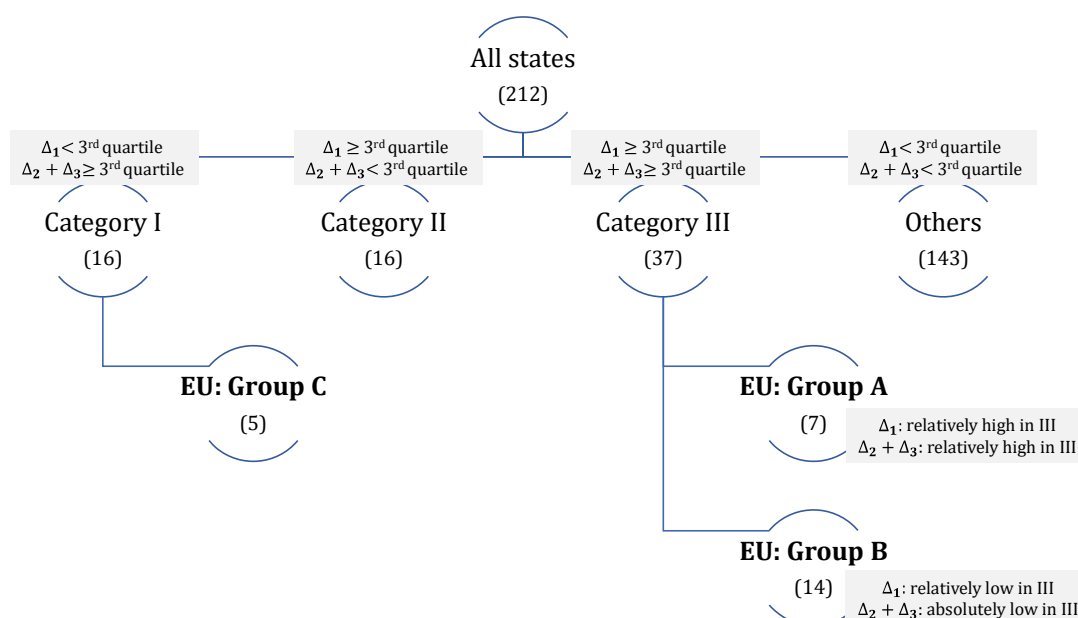


Figure 78 - Summary categorisation states, number of states²⁰⁰ in brackets

For these three groups, section 5.16 identifies FoA to strengthen digital sovereignty in each country. The top 5 FoA for each group are as follows.

¹⁹⁹ www.initiative21.de

²⁰⁰ Romania is not part of categories I–III, i.e. ‘others’, and is not explicitly listed - cf. also footnote 189.

Table 44 - Top five FoA, Groups A, B and C

FoA	Group A	Group B	Group C
1	Start-up investments	Export ICT services total	Top University
2	Qualified ICT employees	Import ICT goods total	OpenData Index
3	R&D expenditure	Export ICT services/ total	digital STRI
4	Patents	Cambrian Index	EDBI
5	EDBI	Leiden Index	Governments online

Each field of action corresponds to one of the 30 parameters of the DSI and has the effect of strengthening digital sovereignty according to the model approach used in this dissertation, as shown in Figure 56.

The (statistical) differentiation of the EU states into three groups shows that Group A exclusively contains states that have historically (since the end of the Second World War) operated with a high degree of political and economic independence and prosperity.

With the exception of Belgium and Austria, the states in group B are at a medium level of prosperity or have been historically influenced by other states like the soviet union. The latter applies in particular to the states of the former 'Eastern Bloc' and ex Yugoslavia.

The remaining countries—i.e. group C—are exclusively former 'Eastern Bloc' countries and Yugoslavia and were influenced by socialist and communist regimes until the end of the 1990s.

In order to formulate the main FoA, the recommendations are formulated as follows on the basis of the three groups and the main FoA shown in Table 44.

6.2.1. States with historically high independence

Group A states are historically economically and politically independent. After the end of the Second World War, the policies of the Western victorious powers focused on an independent economic and political identity combined with the development of economic ecosystems (e.g. EEC). For these countries in Category III, Group A—Ireland, Denmark, the Netherlands, Sweden, Finland, Germany and France—the following FoA are recommended: increase investment in Start-ups, increase the number of qualified people in the ICT labour market, increase the research and development budget and the number of ICT patents (X_{10}) and lower the barriers to setting up businesses. This is equivalent to the parameters: Start-up investments (X_5), Qualified ICT employees(X_9), R&D expenditure (X_{15}), ICT-Patents(X_{10}), and EDBI (X_{24}).

For these wealthy EU states, the first three FoA are particularly relevant: a political and economic initiative is needed to enable a much stronger Start-up ecosystem. The quality of education in this group of countries is already excellent, but know-how needs to be retained more strongly in these countries, to which a thriving technological Start-up landscape contributes significantly.

Excellence in teaching—at school, in job-training and at university-level—must be developed much more strongly in ICT as a further field of action, since the number of ICT

specialists is far too low in international comparison. This means that ICT could be introduced as a compulsory subject in schools, following the example of Estonia, that industry must create more training investment and that universities must set up significantly more ICT chairs. So the solution to this field of action is indeed a quantitative approach. To strengthen ICT innovation as the third, essential field of action, R&D budgets must be increased. ICT research does not have to take place exclusively in universities, but can also take place through increased cooperation between universities and industry.

6.2.2. States with historically dependency, medium level of prosperity

Group B countries have historically been politically or economically dependent and, with some exceptions, are at the EU's average level of prosperity.

Dependent states are states that have historically been politically or economically dependent on at least one other state, with varying degrees of dependence. This includes, for example, the dependency of Greece, which has historically been heavily dependent on other European powers: in the 19th century it was part of the Ottoman Empire and later under the influence of various European powers, in particular Britain, France and Germany. More recently, Greece's debt crisis has made it heavily dependent on financial support from the EU and the International Monetary Fund (IMF). Another example is the Baltic States: they became independent at the end of the First World War, but throughout history they have had to defend their independence against other great powers. They were part of the Soviet Union and experienced strong political and economic dependence during the Soviet occupation. Another example is the 'Eastern Bloc': in this group of countries in Eastern Europe, the Soviet Union gained great influence after the Second World War and established certain political systems and economic structures.

For the countries in Category III, Group B—Austria, Belgium, Italy, Spain, Malta, Estonia, Slovenia, Portugal, Czech Republic, Cyprus, Lithuania, Greece and the Slovak Republic—it is recommended that the intermediate level of Group A be achieved. To this end, the following FoA are of particular relevance: increasing exports of ICT services and imports of ICT goods; stronger focus on KETs, especially AI; and strengthening cooperation between universities and industry. This corresponds to the following parameters in Group B: Export ICT services (total) (X_{19}), Import ICT goods total (X_{21}), Export ICT services from total (X_{20}), AI Index (X_{17}), and Leiden Index (X_{12}).

Due to the high level of prosperity of Belgium and Austria in this group, these two countries play a special role. In principle, the group could be divided into two sub-groups, but this is not done here. For all countries, therefore, the first three FoA should be seen as essential principles for strengthening their digital sovereignty.

In addition to strengthening industrial locations through more intensive trade in ICT services and goods, capabilities in research and industrial implementation of KETs should be strengthened. This includes a well-founded training of specialists in edge technologies in order to understand the KETs identified today and those of the future and to transfer them into industrial applications, i.e. to make them producible. There is an urgent need to prevent

well-educated ICT specialists from migrating abroad in search of better research or career opportunities. This also includes strengthening cooperation between universities and industrial companies to make research relevant to practice. This can also help to keep trained ICT professionals in national or at least EU companies.

6.2.3. States with historically high dependency

The remaining states have historically been subject to a high degree of economic and political dependence, which may be one reason for the aggregation in this group.

As a result of the used DSI-model, it is recommended that the states in Category I, Group C—Poland, Latvia, Hungary, Croatia and Bulgaria—aim for the best quartile of the comparison group. To this end, the following FoA should be focused on: reducing barriers to digital trade, expanding work in university education, promoting business activity, using open data and strengthening digitally accessible public authorities. This corresponds to the following parameters to be strengthened in Group C: TOP University (X_{11}), OpenData Index (X_{14}), digital STRI (X_{23}), EDBI(X_{24}) and Governments online (X_{29}). In this way, the states in this group achieve a higher positioning among the Category I states.

With the exception of the former Yugoslav state of Croatia, this group consists exclusively of states of the former ‘Eastern Bloc’. Political and economic influence ended only about 25 years ago, so that the states in this group, as well as those in Group B to which this applies, can be described as ‘young’ democracies with ‘still upcoming’ economic systems. As former Soviet republics, the Baltic states also face a higher political risk, as the current threats from Russia make clear. Therefore, as Estonia’s digital successes show, the digital sovereignty of these states is particularly relevant, especially in the short term.

The first principle for strengthening their digital sovereignty should be to invest heavily in academic training at universities. There is an urgent need to keep ICT talent in the country and to avoid a brain drain to countries with more established university structures. In addition, openness should be strengthened in these countries to recognise data as the ‘new oil’. One way of doing this is to create an OpenData ecosystem that can be used by public administrations and industry, for example, as a basis for optimising processes. OpenData ecosystems make it attractive for entrepreneurs and ICT developers to set up Start-ups, and so the acceptance of such ecosystems goes hand in hand with the attractiveness of the business location. There is also a need to catch up in enabling digital trade, which will strengthen the ICT location in these countries.

6.3. Optimum: joint initiative at EU-level

Irrespective of optimisations at the level of individual countries identified through individual state FoAs or those from the group analysis, joint action at the EU level would be a valuable step forward. But a European initiative may take considerably longer to implement than the national approaches recommended before. It is recommended that all EU states initiate a common digital market and industrial policy, including beyond country-specific individual interests, in order to confront the two digital superpowers, the USA and China, with

significantly higher performances and purchasing powers. As a 'fictitious state', the EU would be highly ranked on digital sovereignty. The EU as 'one state' would come fourth in the DSI; a shared economic digital partnership with a common understanding of digital sovereignty would reinforce the EU. This recommendation could be the quickest and easiest to implement, as it is 'just a matter of politics' to strengthen cooperation. However, the political will of EU states to do so must be sustained and serious, and ICT must be recognised as a critical resource for defending the sovereignty of EU states.

If the focus of a European initiative to strengthen digital sovereignty is to unite the top three FoA of the three groups of EU states outlined above, then the following core principles of this policy can be applied in addition to a common digital policy.

- Strengthening the Start-up ecosystem, also with the help of freely available, i.e. open data, which can be the basis for new business models and thus strengthen the business location.
- Strengthen the education of ICT specialists to increase the number of ICT experts by providing more support to universities.
- Increasing research and development budgets at government level, e.g. funding for scientific institutes and industrial companies.
- Strengthen trade in ICT services and goods—both imports and exports—including by removing barriers to digital trade, thereby strengthening the business location.

Through the aforementioned recommendations, the states of the EU can strengthen their individual and joint digital sovereignty, thus counterbalancing the autarky and autonomy of other major powers and allowing the EU to once again act as a recognised shaper and negotiating partner in the ICT context.

Suitable addressees should be sought in the EU Commission ecosystem who can disseminate these recommendations and ensure their implementation. This can be done, for example, through the 'A Europe fit for the digital age' initiative, which is responsible for the development and adoption of the Digital Services Act, the Digital Markets Act, the development of the European Data Strategy, the EU-US Trade and Technology Council, or other initiatives, e.g. special interest groups on innovation, chaired by the EU Commissioner for Digital Affairs. Most of the parameters of the present model can be implemented through existing EU structures and initiatives - the need for a new initiative is not perceived. In this way, the EU can implement the idea of global, strategic autonomy, producing technology in select fields resiliently and independently, obtaining technology sovereignly from other states and developing it through market power, also regarding standardisation and transparency, so that it remains controllable.

7. Discussion

In analysing the terms *digitalisation* and *sovereignty*, the extensive literature research in Section 2 will not be repeated here. The possibly critical individual aspects of the index formation of the DSI were already discussed in Sections 4 and 5.3–5.14. Therefore, only selected aspects are referenced in the following.

7.1. Concept of digital sovereignty

Although there are different views on the impact of digitalisation (Brynjolfsson, et al., 2018) (Fend, Hofmann, 2022) (Schwab, 2018), there is a uniform understanding of the term in the literature. The intensive literature research in Section 2.2 is devoted to the historical derivation of digitalisation, the role of the state and the identification of possible key figures in the move to digitalisation.

A closer look at state and administrative functions reveals no significant differences between conceptual understandings of digitalisation, Industry 4.0 and State 4.0 (Schwab, 2018) (Rölle, 2020) (Rorissa, et al., 2011) (Yera, et al., 2020).

State sovereignty can be analysed on a purely historical basis or also taking into account digitalisation effects. It should be noted that these studies do not consider autocratic interpretations of sovereignty and digitalisation as a means of power.

Autocrats use sovereignty to strengthen their position and achieve political or personal goals. This autocratic view is at odds with Foucault's ideas about governmentality and the indeterminacy of power (Vasilache, 2014). In a system with a separation of powers, it is not possible to accept a single authority (Hillgruber, 2002). Sovereignty is therefore a complex concept and can be interpreted in various ways (August, 2021). This dissertation uses a simple view of state sovereignty, a decision that can be critically appraised.

A fundamental and relatively new link to be discussed critically is that between Foucault's ideas and digitalisation and digital sovereignty. Foucault's approaches are implicitly reflected in the digital world. Internet and internet governance (i.e. the influence of the internet on political decisions and the regulation of the internet by superordinate bodies such as ICANN²⁰¹) play an essential role in political discourse.

The very definition of internet governance is a political question (Abbate, 2017). Fundamental questions about what is centrally regulated, what is de-centrally permitted, which decisions are influenced and how, and which stakeholders play which roles are relevant to internet governance (DeNardis, et al., 2020).

In STS, tools exist that can describe the manifold connections between technological, social, political, technical and non-technical actors and analyse their interrelationships. Musiani describes the connections between STS and internet studies and the 'introduction

²⁰¹ Internet Corporation for Assigned Names and Numbers

of dispositifs and boundary objects' and refers to Foucault's definition of the 'dispositif'. (DeNardis, et al., 2020, p. 87).

Foucault repeatedly stayed in California for research purposes. Free thinking was important to him, and he established the first connections between 'governmentality' and information theory (August, 2021, p. 227). For example, in linguistic contexts, 'linguistics is linked to the social sciences in a new way, as the social can now be defined or described as a set of codes and information that characterise a particular group of senders and receivers' (Foucault, 2000, p. 826).

This new linkage and the diverse perspectives on state sovereignty have an impact on the concept of digital sovereignty. Depending on how aspects of digitalisation are defined, digital sovereignty can be understood as either the US understanding through platform capitalism or the Chinese cyberspace (Pohle, Thiel, 2021) (Pohle, 2020). The core question of whether states need to produce their own technologies in order to avoid digital colonisation (Haché, 2014) (Beltrán, 2016) or can also digitally sovereignly use those of other states in the sense of a strategic globalisation, is largely unanswered (van Veen, 2017). What is clear is that the digitally connoted concept of sovereignty is successively gaining relevance in the academic literature (Couture, Toupin, 2019).

The approach to digital sovereignty used in this dissertation, which is based on Foucault's notion of governmentality and forms the basis for, can therefore lead to too much focus and reduction of complexity.

It can also be controversially discussed that the concept of sovereignty is problematic in itself for describing technological interdependencies (Cottey, et al., 2022). This problem arises because the original concept of state sovereignty, as used in historical analysis, differs from how some authors have understood technological or data sovereignty. Sovereignty is essentially understood as the 'locus of political authority', whereas technological sovereignty is more likely to be understood as the capacity for 'strategic autonomy', i.e. the ability to act (Fiott, 2021).

The concept of strategic autonomy is older than that of European sovereignty, which is why this dissertation does not focus exclusively on strategic autonomy, i.e. the reduction of technological dependencies, but rather on the ability of states to maintain their own state sovereignty despite technological dependencies.

While strategic autonomy was initially understood as an element of decoupling from the US, in recent years the focus has increasingly shifted to the need for independence from China (Cottey, et al., 2022). However, a simple definition of digital sovereignty as the model for this dissertation will also be controversial, as the concepts of 'European sovereignty' and 'strategic autonomy' are controversial (Fiott, 2021).

On the one hand, an understandable definition is a necessary basis for the theoretical framework of the model, but on the other hand, simplification can lead to undesirable simplification.

7.2. Measurability of digital sovereignty

There are relatively few quantitative criteria in the literature to determine the measurability of state sovereignty, with very different views on the concept of sovereignty (Goertz, 2006, p. 120).

Although the explanations of *de jure* sovereignty are unambiguous (Dahlmann, 2009) further criteria for good governance (WGI) are proposed (The world bank, 1996) but are controversially discussed (Arndt, 2008). The WGI model has been criticised in the literature and subjected to scientific discourse by the authors of the WGI (Arndt, 2008) (Kaufmann, et al., 2007). It is not reproducible because many of the basic parameters and sources are not published; the number of sources is too complex, making it difficult to understand the index; and the selection of data is arbitrary because, for example, elements of surveys are not taken into account or are disregarded.

Moreover, there is no standard for whether the values determined for the individual categories are good or bad, data collection is not sufficiently balanced between business and household surveys, and comparability is not always available because the same data sources for the same countries are sometimes only available to a limited extent.

The proposal to measure sovereignty by aggregating the *de jure* and *de facto* WGI parameters can be criticised as it has a highly simplifying effect and differs too little in details. The same objections can be raised against the measurement of the digitalisation factors.

The choice of parameters is explained in this dissertation, but it cannot be ensured that the 30 variables chosen are complete and clearly precise in the quantitative measurement and thus represent a clear concept of digital sovereignty. The respective reference value can also be an aspect in the selection of parameters. In the present model, total values per country have been used in some cases as a basis for calculation. Alternatively, these could be standardised, e.g. per capita or to GDP or other reference values, which would change the result. The choice of factors and quantitative methods are always a matter of subjective choice (Bjerre, et al., 2019).

Although various validation and verification methods have been applied to check the results of the DSI and no indications have been found that the calculation approach is not correct, a critical evaluation cannot be ruled out as there is no fixed rule for the creation of an index (Bjerre, et al., 2019). Sensitivity analyses (Kennedy, 2010, 366ff) (Saisana, et al., 2005) and validation techniques of the index (Janová, et al., 2019) were also used in this dissertation.

Nevertheless, it is desirable that the individual processes of index generation are discussed and further research is considered, as it is common for indices to evolve (Worldbank, 2022).

7.3. Comparison with other existing indices

In Section 5.15.1 it is recommended to use the FSI as an index to validate the DSI, which does not seem conclusive at first due to the different contents of the two indices, as the FSI

measures the availability of food in states and the DSI measures digital sovereignty. Nevertheless, constructively comparable approaches are present in the index calculation (e.g. assessing the political stability of states, FSI sub-index values 2.5 Political and social barriers to access), even if no direct overlaps in the parameters could be identified (Global Food Safety Initiative, 2022).

Correlations between economic growth and FSI have already been shown (Jiménez, et al., 2014). Economic growth has a direct (X_2) and indirect (X_{30}) effect on the DSI, which is also confirmed by the correlation results. Therefore, it is recommended to use the FSI as a validation index.

On the other hand, the correlation analysed in Section 5.15.3 considers parameters that are fundamentally similar to those of the DSI, especially in the second sub-index. This may be an indication that the second sub-index of the DSI measures the effect of technology sovereignty and less the effect of KETs. However, the parameter evaluation of the DDI is done by expert assessment and not by standardised secondary data (Mayer, Lu, 2022).

Therefore, it is, in principle, suitable as a validation tool; however, it should be noted that the DDI has not yet been subjected to any scientific discourse or blind peer-review publication and was produced with the support of the Konrad Adenauer Foundation, a party political foundation affiliated with but independent of the Christian Democratic Union. The DDI is limited to the analysis of 20 states.

This dissertation does not examine how the DSI would shape up if complete values (i.e. without imputation) are available for the 30 DSI parameters. This could be a starting point for further research and an in-depth correlation analysis between DDI and DSI. Alternatively, both indices could also complement each other and be aggregated.

Supplementary, and in the context of the concepts of technology and sovereignty, the only index identified by the author, although published here without blind peer review, is the ESI (cf. 5.15.4), which is of great interest as a validation instrument for the DSI. The high coefficient of determination of 71% between the ESI technology sub-index and the DSI indicates the suitability and, above all, the validity of the DSI. The technology parameters of the ESI measure the sovereignty aspirations of KETs. These are strongly correlated with the DSI's third sub-index for technology sovereignty, which may indicate that the third sub-index measures KETs rather than the DSI's second sub-index.

Based on correlation analyses of the DSI with the DDI and ESI, it may therefore make sense to swap the designation of the second and third sub-indexes of the DSI. This could be the subject of more in-depth research.

A complementary form of validation could be country-specific case studies, on the basis of which the calculated index values are checked for plausibility. Especially for different types (e.g. strong regulation versus self-development versus diversification), it can be plausibilised in this way that the index sufficiently reflects different paths to sovereignty. (Sargent, 2013). This can also be the subject of further research.

7.4. Derivation of (political) recommendations

The fact that strengthening digital sovereignty—at least the strengthening of technology independence from other states—by taking governmental, political measures is unanimously assessed. Even in the persistently liberal economic policy of the USA, state, economic interventions are now no longer unusual (Holtmann, 2018).

The need for action is also noted for the states of the EU, although it is warned against reducing the FoA too much to ‘semantic debates’ such as digital sovereignty or strategic autonomy, but rather to strengthen the ability of EU citizens to act (Cottey, et al., 2022). Moreover, FoA should not be limited to technology, but should be broad. (Policy) recommendations should include elements from regulatory and competition economics (Budzinski, Eckert, 2018).

This dissertation makes a contribution to this and concretises the FoA that are to strengthen the ability to act. Due to the selection of the model, the FoA are limited exclusively to the 30 parameters identified. Further levers can therefore not be examined. Therefore, the list of recommendations should not be considered complete. Indirect FoA can also have an influence, such as the choice of a liberal economic policy with goals of reducing bureaucracy or lowering trade barriers. In order to find a compromise between general (e.g. implementation of a liberal economic policy) and specific (e.g. the state must increase the investment volume in Start-ups by a certain amount) recommendations, the proposed FoA should therefore be taken into account.

The author has decided against creating master plans for stronger digital sovereignty for each individual state, but rather suggested that groups of states be formed.

The grouping of EU states into A, B and C is intended to help identify recommendations for prioritising the 30 parameters. State-specific and state-group-specific FoA can be derived based on the prioritisation along with the ‘best state’ as shown in Table 43. The objective of the dissertation was not to elaborate on these at the state level but to develop a model that enables the identification of recommendations to strengthen digital sovereignty. This is inclusive of the policy recommendation to ‘reduce economic and technological dependence on China, especially in key sectors and technologies, while maintaining as much as possible an open trading relationship and the possibility of cooperation in other areas’ (Cottey, et al., 2022, p. 46).

In order to create a successful European model, it is essential that the values of Europe are promoted and positive digital developments are encouraged. Only when necessary should regulation be put in place in order to ensure that these advancements are not compromised. Furthermore, the focus should be on strategic autonomy, which means being aware of global interdependencies (Weber, 2022). It is also imperative to critically discuss the FoA propounded by the model of this dissertation. This will be discussed using the example of two FoA.

Concerning the EU countries in Group A, one of the recommendations entails increasing the number of ICT specialists in the labour market, i.e. investing in training and recruiting specialists. However, since a lack of skilled workers is a generally recognised problem and

ICT skilled workers are needed to ensure ICT competence, the recommendation seems uncritical, and intuitive. For this reason, it will not be discussed much in all likelihood. Rather, the question when implementing this recommendation for action, is how exactly and in what time frame the number of ICT specialists will be increased.

For group B, one of the recommendations is to increase imports of ICT goods, which may appear to be contradictory. The relative or absolute strength of the respective country's digital economy, and, accordingly, the need for imports of ICT goods can be discussed. In case the country does have a robust digital economy, imports and exports denote a sign of sovereignty. By contrast, if the digital economy is weak, a strong surplus of imports would (probably and) potentially be problematic. In addition, it is also necessary to elucidate whether imports from any state are interpreted identically positively or negatively. It is pertinent to note that the correlation analysis of the DDI and the DSI parameter for the import of ICT goods shows a high correlation. Thus, it can be implied, at least for the DDI, that states with high ICT imports are assessed as technologically independent.

Ultimately, policy recommendations need to be worked out individually for each country and adapted to not only the requirements of the respective political system but also the requirements of the EU, as a case in point. This can be a starting point for further research. Another recommendation is that in order to become economically stronger, especially vis-à-vis the USA and China, the EU states push for a common digital policy and bundle digital market power.

Another field of action, which has not been discussed in this paper so far, could be for EU states to make digital sovereignty a mandatory criterion in the procurement of ICT goods and services for public organisations, i.e. for civil servants, public employees and military personnel who are subject to EU procurement law. This could strengthen the EU's ICT industries and pool the EU's market power. As a result, the EU could be in a better negotiating position vis-à-vis strong hardware and software vendors and foster an alternative ICT industry in the EU.

This can be discussed controversially, as maintaining economic differentiation can also have strengths and forced homogenisation can only be achieved with difficulty. See also the discussion against the homogenisation of the 'strategic culture' of the EU states, especially in security policy issues, but for a 'common strategic mindset' and technological capabilities at the EU level (Cottey, et al., 2022).

7.5. Limitations of the theoretical model

7.5.1. Focusing coefficient of determination

In the analyses of this dissertation, a regression result is considered good if the coefficient of determination is $\geq 50\%$. This means that analyses for small coefficients of determination are omitted, as those correlations tend to have a greater dispersion. In such cases, interesting correlations may be present for the analysis of digital sovereignty, which may not have been investigated. The relevant literature indicates the risks of chasing the highest possible

coefficients of determination (Fahrmeir, et al., 2009, p. 100). On the one hand, the focus on coefficients of determination above 50% is a sensible one; on the other hand, it does not suppress potentially interesting dependencies. For this reason, this decision can be criticised for the fact that dependencies relevant to digital sovereignty are not sufficiently analysed in the dependency analysis, which can lead to recommendable FoA.

Therefore, a further field of research could be the analysis of dependencies with smaller measures of determination based on the model developed in this dissertation.

7.5.2. Model assumptions linear regression

In the context of the application of linear regression models, restrictive model assumptions such as linearity, homoscedasticity, no autocorrelation or normal distribution of the residuals are made (Fahrmeir, et al., 2009, 87ff).

This applies to the classical linear regression model, but in reality it usually does not correspond to the existing dependent and independent variables. The result of this, when applying statistical procedures, is that the covariance matrix, $Cov(\epsilon) = \delta^2 \cdot I$, cannot be used, as the variances and residuals are not correct; thus, the test statistics are not reliable. In this case, the general regression model can be used and the restrictive model assumptions can be neglected (Fahrmeir, et al., 2009, 124ff). The covariance matrix is replaced by using a modified matrix (W) as multiplier instead of the unit matrix: $Cov(\epsilon) = \delta^2 \cdot W$. This allows better statistical estimators to be achieved. In the present dissertation, the model assumptions were tested in selected cases, but not transformed and no alternative covariance matrices were determined, as the application of the simple linear model was sufficient for the practice-relevant statements.

This applied to the coefficient of determination in particular, since robust regression procedures in the econometrics software can be used to identify outliers and give them a lower weighting (Kennedy, 2010, 112ff). The advantage of the chosen approach is that the parameters used, which are normalised to 100 and missing data are filled in by means of mean imputation, are close to reality and have not been distorted by any statistical transformation.

Quantitative methods could be used to transform the 30 parameters used in this model to meet the above criteria in further scientific work. The resultant DSI values and rankings could then be the subject of further analysis.

7.5.3. Parameter selection

The parameters were selected after analyses of existing indices and validation from the interviews. The number of parameters, as well as their completeness and partial redundancy, may require discussion (Bjerre, et al., 2019). The same applies to the chosen weighting and aggregation procedures. In particular, the fact that very little data is available for a few of the parameters can invite criticism. For further development of the index, primary/secondary data for these parameters can be found for more states.

In addition, certain parameters appear unsuitable for index formation. Critically, GDP per capita could be discussed, because, among other effects, this contributes to the high index values of Switzerland, Luxembourg and Saudi Arabia, and the wealth of a state could be argued to be unsuitable for quantifying digital sovereignty. This can be examined in further developments of the index and replaced by alternative parameters, such as government share or lower weighting of prosperity. Some parameters are normalised (e.g. per capita) and others are used as total values (e.g. Start-up investment). This can also be critically discussed and a standardisation for all parameters can be demanded. The inclusion of parameters or aspects not currently used in the model, such as the NCSI for cybersecurity presented in a correlation analysis, can also be critically discussed. As emphasised several times, there is no 'golden rule' for index construction, and more in-depth research can be the basis for the index development.

However, through intensively discussed derivation of the parameters and the verification and validation investigations of this work, no indications could be found that the chosen parameters or the methods for index formation lead to unrealistic results.

7.5.4. Imputation procedure

Substituting non-existing secondary data can lead to problems (Rässler, 2000). Taking the Barnett Sovereignty Index and the Digital Competitiveness Index as an example, Figure 38 shows that the coefficient of determination used as a quality criterion decreases by almost 10%, i.e. the correlation of the data becomes worse through imputation: the regression analysis of the raw data with 57 paired values yields a coefficient of determination of 44.5%, after kNN imputation with 212 data sets, a coefficient of determination of 34.8% is obtained. Imputation could also have a positive effect, so that the application of an imputation procedure in the present case makes sense in order to calculate a DSI for all states, but states with many imputed parameters should be treated with caution with regard to their DSI evidence.

Therefore, although an imputation procedure can be used to replace missing values, parameters with fewer existing values could also be weighted less in the context of index formation. After analysing different imputation procedures, mean imputation is used for all parameters. Even better results could be achieved if the variables were imputed with different procedures. This applies in particular to states for which several data are missing and which were thus devalued in this dissertation by weighting with the factor α (OECD, 2020d). For the model developed in this dissertation, however, this is initially an auxiliary weighting factor that must be eliminated in the future. In the overall analysis, however, no state could be identified that was unrealistically evaluated due to an incorrect imputation method. As an additional research idea, the data imputed in this dissertation could be replaced by collecting primary data.

7.5.5. Democratic biased ‘DNA’ of the index and parameter standardisation

The index evaluates the European states’ understanding of sovereignty according to the definition on which this dissertation is based. The assumption could be that digital autarky, isolation and closure are understood as a successful model for reducing digital dependencies and strengthening national sovereignty.

This is the path that countries such as Russia and China are trying to implement in order to strengthen their digital sovereignty. The European model relies on value-based alliances, globalisation and democracy as a desirable political model and the basis for economic success. The basic assumption of this work therefore remains that the European model based on democracy, liberalism, globalisation and liberal value orientation will continue to exist in the future. This could be controversially discussed in the context of index formation, because the so formed DSI is biased as every used parameter evaluation is biased.

The parameters used in the index are equally standardised from 0 (very poor) to 100 (very good). This can be viewed critically for individual parameters, since, for example, a high value of the eGovernment Index is understood as positive for digital sovereignty, but a state such as Saudi Arabia uses digital services (including, for example, the population’s access to state services) to strengthen state authority.

Thus, individual parameters could be used by autocratic states to strengthen them in terms of digital sovereignty, albeit in contradiction to the European understanding of digital sovereignty (cf. also FDI in 5.4.2.1). It might be interesting to consider an upper limit for the weakest link in a chain. For example, if a state has a value of 100 in one dimension, but only 10 in another, then the index value is not the mean, but 10. The idea behind this is that for sovereignty, it is perhaps more important to have no gaps than high averages. For the relationship between economic relations and state sovereignty, for example, such an approach could make sense: without state options for political action, even the strongest digital economy is of limited use, because then only the dependencies shift. Alternatively, such sub-indices could also be weighted by each other and then included additively in the calculation. This, in turn, also depends on the basic design decision in the theoretical framework. To mitigate such potential issues, further individual political discussions would have to be held for each parameter regarding whether an expression is good or bad in the context of European digital sovereignty.

7.5.6. Evaluation according to territorial criteria

The digital sovereignty model developed in this dissertation is based on the assumption that a state’s power is limited to its own territorial borders, and a state’s cooperation with other states has an effect on foreign policy, but not in the sense of internal state sovereignty. Limiting states’ digital sovereignty to their territorial borders follows Bodin and the Peace of Westphalia. However, since Barlow’s declaration at the latest, the territorialisation of state action in the digital context has been the subject of critical debate, which is why the model’s current approach can be criticised (Mueller, 2020, 780 ff.). Treating states not as territorially limited but as virtual constructs would be an alternative approach.

It can be controversially discussed whether the understanding of the model changes when a state acts digitally beyond its own territorial borders, for example when a state acts as its own digital identity without a territorial border. This would require a reweighting of key parameters of the model developed in this dissertation. For example, prosperity and the import and export of goods and services would play a subordinate role to digital literacy, which would determine the strength of states through digital identities.

Estonia offers far-reaching approaches for digital identities (Tamppuu, Masso, 2019); states could, regardless of state sovereignty or wealth, provide all people in the state with a digital identity with only digital capabilities. For example, they could grant citizenship to foreign citizens or wage war exclusively digitally. The model used in this dissertation and the calculated parameters do not evaluate these aspects.

8. Conclusions

The digital sovereignty of states is a new concept that has been discussed in many ways, especially in the EU. It influences political and economic actions and provides a proxy for the discussion of whether dependence on the resources of other states influences state sovereignty. Global dependencies and the lack of resilient security mechanisms have affected EU states since the Covid-19 pandemic through interrupted logistics chains, especially for ICT goods. Dependencies have become particularly clear since Russia's war of aggression on Ukraine and the associated effects of fewer gas supplies from Russia and blocked transport routes for wheat. These are examples of how the actions of the EU states are directly dependent on the interventions of other states. Hedging mechanisms can be designed either by fortifying their own competence, raw materials and manufacturing industries or through bargaining power. Relying on the permanent functioning of global supply chains is not a suitable means for maintaining a state's ability to act. This applies equally to digital resources, both hardware and software products, and becomes increasingly important the more civil society, industry and state institutions depend on them with regard to ability to act. For this reason, the digital sovereignty of states has recently become the focus of political discussions.

However, it is still unclear what is meant by *digital sovereignty*—there is no uniform, scientifically accepted technological, political or social definition. Although the individual terms *digital* and *sovereignty*, as well as elements of digital sovereignty, are scientifically derived and have been described in many ways, a synthesis and a common understanding, which are the basis for common action, do not exist. The dissertation research question *How can states' digital sovereignty be scientifically defined?* can be concluded by proposing a definition based on the analysis and synthesis of the individual terms.

By closing the research gap, a contribution to theory is provided. The concept of digital sovereignty is defined scientifically and offers a proposal as a basis for a model that makes the digital sovereignty status of states understandable and measurable. A new definition is

based on the values of the European states and the Western world's understanding of democratic sovereignty and is a historical consequence of almost 500 years of the industrial and democratic development of the European states. It integrates the handling of KETs that are necessary for the maintenance of state action and names controllability with regard to technological freedom of choice as its basis. This definition propagates neither technological autonomy nor autarky but seeks a positioning between the digital protagonists—the USA and China—in the sense of a global, strategic autonomy.

An objective of this dissertation is to identify FoA with which the digital sovereignty of the EU states can be strengthened; thus, it is necessary, in addition to the qualitative descriptive definition, to better record the status quo of the states with regard to their digital sovereignty. To this end, a model must be developed to measure digital sovereignty. Here, the research question *How can the digital sovereignty of a state be measured quantitatively?* can be answered.

By closing this research gap, a contribution to theory is made. The development of a structured model based on the previously developed definition of digital sovereignty, the induction of measurable components, their quantitative analysis and the derivation of the relevant aspects for a comparative, country-based index form the basis for the theoretical contribution within the development of the methodical framework. Furthermore, the newly developed index provides a validation basis for other indices. From the definition of the term, a structured component model can be derived and quantified. The model consists of the three sub-indices—state sovereignty, KETs and technological sovereignty—and thus corresponds to the qualitatively descriptive definition of digital sovereignty. It can be substantiated with a further 10 sub-components and metrically backed by 30 parameters that are publicly available as secondary data. These can be unambiguously assigned to the subcomponents and are argued and weighted regarding their relevance for measuring the digital sovereignty of a state. Through equally weighted, additive and multiplicative aggregation, a DSI can be formed after replacing missing data and discussing weighting, aggregation and normalisation procedures.

Although political and economic measures on individual state-level and EU-level are already being taken to strengthen the digital sovereignty of a state, it remains unclear whether these measures will be effective and what effects they may have. With the help of the metric-valued component model, the parameter analysis and the DSI, these measures can be assessed in terms of their effect on the index, which answers the third research question: *What is the quantitative effect of (economic) policy measures on the digital sovereignty of a state?*

The identification of recommendations and FoA offers a practical contribution for decision-makers, from which politicians or decision-makers in business and academia can concretise measures to strengthen the digital sovereignty of a state. The measures have a special practical relevance because they do not correspond to general policy recommendations and are not formulated in too detailed a manner, but offer a practical, concrete orientation framework. With the definition and component model thus proposed,

it is shown that a DSI can be calculated, and from this each state can be ranked. The calculated index can be verified and described as valid; the parameter construction in the component model forms a transparent verification basis that can be mapped and checked in a computer model. The DSI can be further verified, as significant correlations can be found with the BLI, which has similar parameters. The same applies to the verification of the DSI with the ESI. Various methods are used to validate the model and the index formation. In addition to a visual plausibility check, it can be shown that it (or its sub-indices) indicate significant correlations with existing indices such as the FSI, the DDI, the ESI and the SDGI. The model describing digital sovereignty thus represents well the reality formulated by the definition of the new term with this dissertation as its basis. In addition, the index stands up to a stress test that stochastically perturbs the parameters, and effects of the recommendations to strengthen the digital sovereignty of the EU states can be simulated over several periods. The model behaves as constructed and analysed.

The top 50 rankings, led by the USA, Ireland and Norway, are predominantly states with democratic governments based on the Western model. The EU-27 states are entirely ranked in the top 50, with the emergence of three groups of EU states whose digital sovereignty can be strengthened through different packages of measures. The measures that need to be implemented at national state-level can be divided into three groups A, B and C. Group C countries are mainly former 'Eastern Bloc' or socialist countries with lower DSI ranks whose position results from relatively low sovereignty scores with high levels of digitalisation. The following FoA are recommended for these countries: reduce barriers to digital trade, focus on university education, promote business activity, use open data and strengthen digitally accessible authorities. This puts these countries in the first quartile of the reference group.

Group B comprises predominantly countries in Southern and Eastern Europe with higher (but no top) rankings compared to other EU countries. It is recommended that they focus on the following FoA in order to reach the level of Group A: increase exports of ICT services and imports of ICT goods; focus more on KETs, especially AI; and strengthen cooperation between universities and industry.

Group A countries are mostly located in Western and Northern Europe. They top the rankings but still need to take action internationally compared to the first quartile of leading countries. Thus, it is recommended that they focus on the following: investing more in Start-ups, increasing the number of qualified people in the ICT labour market, increasing the research and development budget and the number of ICT patents, and lowering the barriers to setting up businesses.

In addition to a nation-state approach by the respective national governmental organisations responsible, the states of the EU should pursue a more focused common digital policy. If successful, the investment and purchasing power of all EU states would be so strong that the EU could become a digital power. In this case, the EU would be between China and the USA in the sovereignty sub-index and superior to both states in the technological

sovereignty sub-index, though still inferior in the KETs sub-index. If the EU were to be evaluated as a common state, it would be ranked fourth after the USA, Ireland and Norway.

This shows that FoA at the national level that can strengthen state sovereignty can be identified. Alternatively, the EU can collaborate on a joint digital initiative and strengthen its own position and that of its member states in the short term.

The dissertation provides a pedagogical contribution, as one field for action for strengthening digital sovereignty has been elaborated that, on the one hand, more ICT professionals are needed to maintain freedom of choice in technologies and that there should be a higher interaction between teaching and industry. Therefore, the following applies to universities and other scientific institutes: Industry cooperation should be promoted, e.g. through cooperative publications, in order to establish practice-oriented ICT science, to bring students closer to practical topics and industrial companies at an early stage and thus to retain well-trained specialists in the companies of the EU states. Therefore, the results of this dissertation can be used by decision-makers in university management as well as industry to argue for the provision of funding for ICT education, lecturing and cooperation.

The previously mentioned FoAs for strengthening digital sovereignty are, like the secondary data on which they are based, relatively generic and must be specified when implemented in order to have an effect. The need for specification is outlined by the example of increasing the amount of ICT professionals as a field of action of group A: it will be essential to train or recruit the right ICT professionals, namely those who identify, understand, develop and thus master KETs and will be able to reduce technological dependencies. This measure gives the EU and its member states the opportunity to expand their digital sovereignty in the sense of strategic autonomy and freedom of choice in the medium to long term, and to position themselves as a third digital power alongside the USA and China.

The essential model to assess digital sovereignty and recommendations for FoA are developed in this dissertation. What remains important is a consistent and sustainable effort of the nation-states of the EU that is, above all, orchestrated at the EU level, to bundle the joint forces of the EU states regarding the digital sustainability of the European continent.

With a consistent and sustainable digital policy, the EU states can secure the digital and sovereign future of the continent and make up for the failures of the past decades, also with the support of the model developed in this dissertation.

9. References

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Appendix

Questionnaire, long form

Questions and answer patterns of the interviewees

Interview partners

Experts in industry and research who have been working on digital sovereignty for several decades were chosen as interviewees. The interview partners wished to remain anonymous, so only their industry, role and country are outlined.

Table 45 - Interview partners' surveys

Interview	Area	Role	Country
1	Teaching/Academia	Research and teaching, international affairs, politics	Germany
2	Consulting for the public sector	Technology development, expert level	Austria
3	Hardware producer	CEO	USA
4	Software producer	CEO	Germany
5	Security solution provider for the public sector	CEO	Germany

Depending on the level of expertise, not all interviewees answered all questions, as can be seen in the overview shown in Table 46.

Table 46 - Questions/interviewees - matrix

Question	Interview 1	Interview 2	Interview 3	Interview 4	Interview 5
1	X	X	X	X	X
2	X	X			
3	X	X		X	X
4	X	X			
5	X	X	X	X	X
6	X	X		X	X
7	X	X		X	X
8	X	X		X	X
9	X				X
10	X	X	X		

Answers in the structured interviews

Question 1: What (in your opinion) is the reason why European countries do not play in the digitalisation top league of China or the USA?

Interview 1:

Strength was and still is in industrial manufacturing and production processes, and the focus on exporting industrial products. Digitalisation/computerisation with a focus on industrial manufacturing, hence the term *Industry 4.0*, which is coined in Europe and Germany and not used in other parts of Asia and the USA.

Interview 2:

Investment in ICT technology fields is underrepresented in Europe. Dependence on others is interpreted in good faith as uncritical—see gas dependence on Russia. Therefore, no perceived need to develop ICT into another industry. In addition, no use is made of the EU economic area; the small states of the EU cannot stand up to the economic power of the USA and China.

Interview 3:

Dominance of European industry on larger and big companies. Little room for development of new ideas and Start-ups developing ICT technologies. Too slow, bureaucratic development in Europe ('dusty old Europe') and reliance on the post-WWII industrial success story being unstoppable.

Interview 4:

No clear political and economic focus on training, research and economic promotion of ICT technologies. Research is good, but there are no attractive working conditions for well-trained technologists. The best are lured away to non-EU countries. Too little strategic foresight on the part of the EU states for the next decades, opportunity/risk assessment and no programmatic approaches (e.g. through investment promotion) that are stringently implemented.

Interview 5:

According to EU public procurement law, state enterprises are obliged to put all services, including ICT, out to tender and to select the most economically advantageous provider. This means that state institutions cannot favour European companies and promote developments but must purchase 'better' and 'cheaper' non-EU technology.

Question 2: What is the reason European neighbouring countries, such as Finland, Estonia or Sweden, lead the digitalisation ranking in Europe?

Interview 1:

Sweden, Finland and Estonia are not NATO states and have always relied on a special, also military kind of strategic autonomy. This may be one reason why certain industries are promoted differently than those in the other EU states.

Interview 2:

The states are smaller than, for example, Germany or France and therefore easier to digitise, and citizens have a high level of trust in the state. Concepts for an electronic identity or an

electronic health card were implemented early on in these countries. Estonia and Sweden have a clear digitalisation strategy, also to strengthen their competitive position in order to lower barriers for their own citizens or for industries in other countries. Sweden is known for developing highly innovative products that scale, such as the payment service Klarna, Spotify, the game Minecraft or Candy Crush. With Ericsson, Sweden is also home to one of the largest, more traditional technology companies that benefited from the mobile phone boom in the 1990s.

Question 3: What factors should EU states particularly promote in order to strengthen digital sovereignty in Europe?

Interview 1:

State institutions must be role models for digital sovereignty. They therefore need a mandatory strategic orientation to identify technologies that are strategically relevant and influence state sovereignty. The technologies must then be demanded so that there is at least freedom of choice, i.e. dependencies on a technology that is or may become untrustworthy are reduced. In addition, the training of ICT experts must be increased, who on the one hand can develop technologies and on the other hand can also evaluate dependencies. Support for the migration of skilled workers from non-EU countries to the EU to strengthen digital competence is necessary.

Interview 2:

Strengthening citizens' trust in the performance of the state and its handling of personal data. Today, citizens are more likely to give data to technology companies such as Meta or Apple rather than to their state for the digital use of citizen services. To achieve this, data protection must be strengthened while at the same time reducing the complexity and hurdles that currently exist in the context of compliance with data protection requirements.

Interview 4:

Promoting industries that specialise in KETs of the future in order to reduce dependencies, such as those that exist today in cloud or mobile technologies, in the future. This includes state subsidies as well as the reduction of bureaucracy. For a transitional period, the promotion of foreign technologies by European states or companies is necessary, which secures the rights to the technologies in order to make them available in the EU states if necessary. At the same time, non-European investments in EU technologies must be increased.

Interview 5:

Promoting Start-ups and innovative technology industries, enabling the opening of markets of EU states also for public consumers. Technological dependency must be allowed to be a criterion for tenders or even be made obligatory in order to be able to give preference to those industries that offer trustworthy technology products.

Question 4: How relevant is the differentiation of political and economic systems (e.g. USA, China and EU) in the future for success in digitalisation?

Interview 1:

China pursues a focused investment policy that is controlled by the state. The promotion of technology sectors serves primarily to become independent of other states, essentially the USA. Technology development is thus a political instrument of power that China has consistently pursued successfully for decades. While China was still described as a 'copycat' a few years ago, a tipping point has already been reached and China is not only penetrating new technology sectors faster than the EU states, but also original industries such as the automotive industry. The goals can only be guessed at: independence or even power over other states. Therefore, China's autarky-seeking political system plays an important, if not the main, role in the success of Chinese technology companies. The ban on Chinese technology in some states has long since turned technology competition into a diplomatic instrument of power.

Interview 2:

The political and economically liberal values of the USA have given technology companies an optimal growth ecosystem in certain regions (West Coast Silicon Valley, East Coast Greater Washington, but now also in Austin/Texas). While in Europe new business ideas in traditional industries (e.g. taxis, automotive industry) are not promoted through regulation, digital business models in the US have good growth conditions (e.g. Uber, autonomous driving). Through experimental, incremental work (fault tolerance), deregulation and a focus on speed, this system attracts investors and entrepreneurs, as it offers good scaling opportunities for new business ideas. In addition, government institutions, especially the US military, promote the development of new technologies through investment and an attractive market potential.

Question 5: What are the three most important KETs on whose states the EU is particularly dependent and in which it needs to invest heavily?

Interview 1: Quantum technology because of the possibility to 'break' conventional cryptography faster, mobile communication with low latency and high fire widths in order to control critical infrastructures autonomously, digital identities/digital twin, which makes it possible to digitise data from citizens.

Interview 2:

automation of industrial processes, artificial intelligence, encryption algorithms.

Interview 3:

Secure storage of data, to protect the data of entrepreneurs and citizens of EU countries. Edge technologies to reduce bandwidth, including Artificial Intelligence at the edge.

Interview 4:

Cognitive recognition by machines using artificial intelligence, autonomous driving and robotics (human-like machine), personal digital assistants.

Interview 5:

Machine learning through AI algorithms, securing processes through distributed ledger, mobile and secure data communication.

Question 6: What are the three most important success factors to bring Europe to a higher level of digitalisation?

Interview 1:

Investment in key enabling technologies, long-term, technological planning and consistent (industrial) policy implementation

Interview 2:

Build Europe as a market, define more common standards, regulate and deregulate procurement processes: more focus on the use of controllable, trustworthy technologies, more political courage to consistently expand digital sovereignty.

Interview 4:

Create an investment ecosystem of private and public, European and non-European institutions to promote digital industries, reduce bureaucracies and encourage risk-taking for Start-ups. Use state organisation as a catalyst for digital industries and digital sovereignty.

Interview 5:

Stronger training and retention of skilled workers in the EU, more cooperation from industry and research. ICT and digitalisation must become a compulsory programme like teaching one's mother tongue in schools.

Question 7: Where should Europe be in 10 years' time in terms of digital sovereignty?

Interview 1: The states of the EU should build up their own digital competence for particularly relevant technologies in the sense of self-sufficiency in order to be able to act independently without dependence on third parties if necessary.

Interview 2:

The EU states should make existing technologies usable for themselves by imposing transparency requirements on non-European providers. EU markets should only be accessible to those providers who meet these requirements. The logic follows deglobalisation.

Interview 4:

The EU cannot develop without technology from China or the US. Digital sovereignty therefore means harnessing the technologies so as not to slow down progress. In the longer term, own technologies must be built for critical areas.

Interview 5:

The states of the EU should use the technologies of non-European states unchanged and only build up their own competence in technologies that will be relevant in the future. When using non-European products, a zero-trust approach should be followed, i.e. every data access is initially distrusted until it has been confirmed as secure.

Question 8: Where will Europe be in 10 years' time in terms of digital sovereignty?

Interview 1:

As the discussion on digital sovereignty is already 10 years old and has developed little in Europe, speed remains the main challenge. Probably in 10 years, the same dependency discussions will be held as today, only in other fields of technology. The best example is the dependence on Russia's oil—a problem that has been known for decades and was only solved when Russia launched a war of aggression on Ukraine.

Interview 2:

Since China and the USA are currently pushing the development of digital technologies more strongly than the EU countries and are also pushing for stricter mutual regulation, technology development in these countries will also progress faster than in the European countries in the next 10 years. In addition, the EU countries are not able to build up a common market power vis-à-vis the USA and China. Therefore, Europe will be even further behind in 10 years.

Interview 4:

The EU states will only develop technologically if they are triggered by threats. This will succeed in selected scarce resources (e.g. silicon production triggered by the chip crisis in 2020/2021 or the development of cybersecurity know-how due to attacks from Russia), but will remain reactive. Due to a lack of strategic foresight and a lack of political programmes, there will still be a strong dependence on other, non-European countries for key resources in 10 years' time.

Interview 5:

In 10 years, little will change because the timeframe is too short and the speed in the EU states is too slow. Therefore, the situation will be essentially identical to today. EU countries will still make situational and ad hoc decisions on a case-by-case basis, as policy decisions are not designed for the long term. There will probably be more investment in STEM education and there will be some Start-up ecosystems that produce good tech Start-ups. However, there will be no larger technology corporations that produce ICT technologies independent of other countries.

Question 9: What is your recommendation to Europe to strengthen digital sovereignty?

Interview 1:

At the EU level (e.g. Digital Commissioner), establishment of guidelines for EU states to implement digital sovereignty: clear definition of key competences and annual review process, creation of an investment ecosystem for Start-ups and established industries in the EU states, mandatory introduction of open source use in critical policy/industry areas to reduce technological dependencies.

Interview 5:

Stronger regulation through the creation of a new data protection framework by harnessing existing technologies with a zero-trust approach while deregulating markets and public procurement.

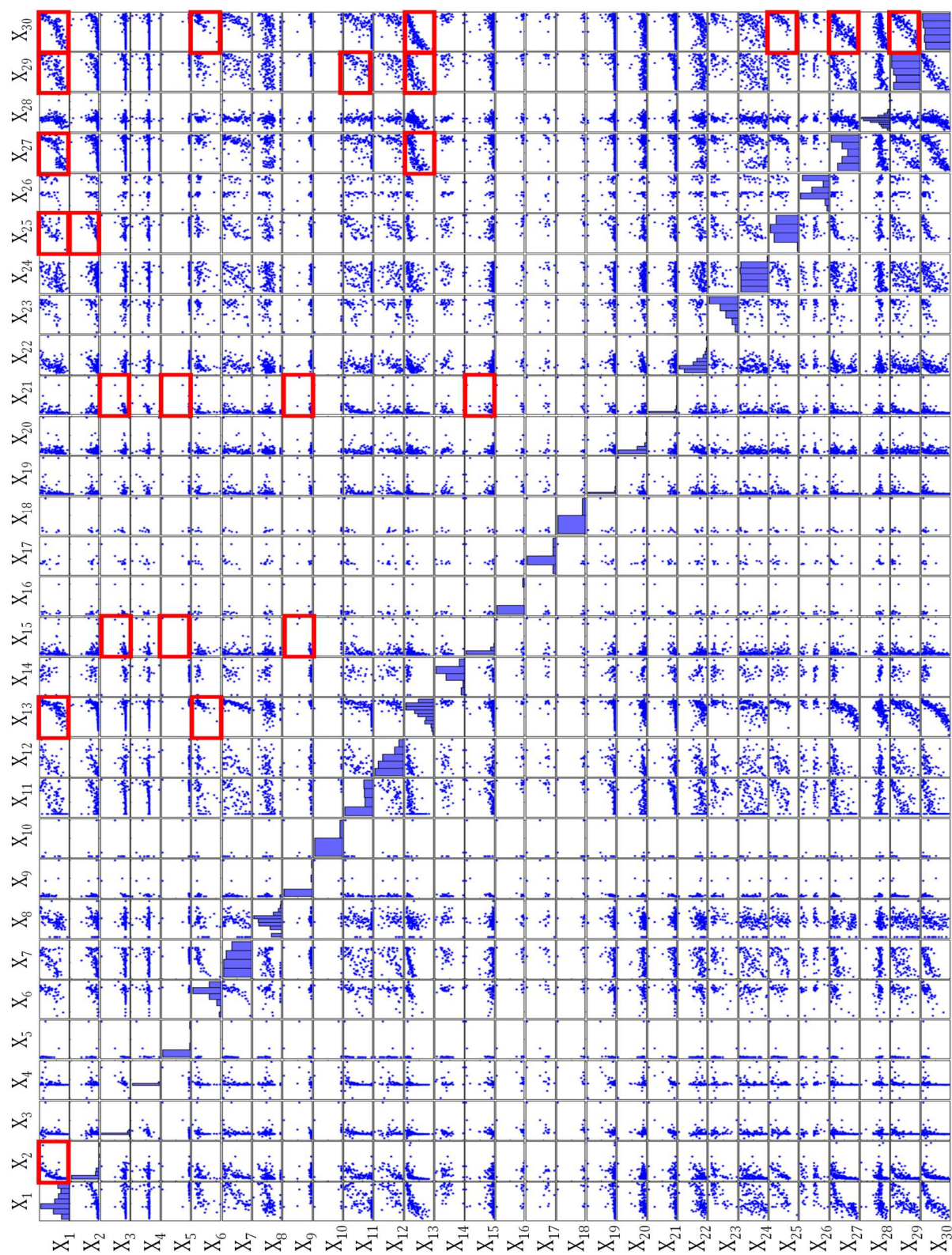
During the interviews, the author presented an AHP matrix to the interview partners as a proposal and discussed the assessment. The interview partners questioned the reasoning and suggested adjustments, which resulted in the AHP matrix used in this paper. In the case of differing assessments of the weighting parameters between the interview partners, the author decided which factor to choose.

Table 47 - Consolidated AHP matrix from interviews

[illegible]

DSI score and ranks

Country	DSI points	DSI Rank	Country	DSI points	DSI Rank	Country	DSI points	DSI Rank
Aruba	232.39	96	Gambia, The	34.52	202	Norway	3,117.30	3
Andorra	364.60	72	Guinea-Bissau	30.96	203	Nepal	84.96	169
Afghanistan	23.59	209	Equatorial Guinea	102.70	159	Nauru	124.27	146
Angola	63.61	185	Greece	1,089.32	37	New Zealand	2,171.53	16
Anguilla	132.09	134	Grenada	176.55	115	Oman	532.83	59
Albania	273.58	83	Greenland	279.58	81	Pakistan	101.25	160
Netherlands Antilles (former)	132.09	134	Guatemala	178.93	113	Panama	420.80	68
United Arab Emirates	1,591.30	24	French Guiana	127.37	139	Palau	204.60	105
Argentina	625.82	54	Guam	174.33	116	Peru	299.87	78
Armenia	256.45	86	Guyana	108.24	153	Philippines	326.13	76
American Samoa	151.82	123	Hong Kong SAR, China	1,124.36	34	Papua New Guinea	64.69	183
Antigua and Barbuda	241.22	95	Honduras	124.67	145	Poland	1,116.01	35
Australia	2,560.97	10	Croatia	787.99	44	Puerto Rico	225.71	99
Austria	1,985.55	19	Haiti	26.74	205	Korea, Dem. Rep.	261.40	85
Azerbaijan	264.26	84	Hungary	990.63	40	Portugal	1,367.54	26
Burundi	19.32	211	Indonesia	247.28	90	Paraguay	181.84	110
Belgium	2,147.87	17	India	346.44	73	Qatar	1,312.69	28
Benin	81.19	173	Ireland	3,304.83	2	Réunion	127.37	139
Burkina Faso	67.95	181	Iran, Islamic Rep.	252.64	88	Romania	704.24	49
Bangladesh	111.36	150	Iraq	104.06	158	Russian Federation	840.16	43
Bulgaria	692.20	50	Iceland	1,964.12	20	Rwanda	76.00	177
Bahrain	642.16	53	Israel	1,902.09	21	Samoa	76.09	175
Bahamas, The	326.75	75	Italy	1,696.85	22	Saudi Arabia	966.27	42
Bosnia and Herzegovina	232.05	97	Jamaica	244.02	93	Sudan	24.76	207
Belarus	337.51	74	Jordan	220.03	100	Senegal	104.52	157
Belize	108.05	154	Japan	2,331.17	14	Singapore	2,509.27	11
Bermuda	0.00	212	Kazakhstan	484.56	63	Solomon Islands	57.53	188
Bolivia	139.72	128	Kenya	109.76	152	Sierra Leone	24.82	206
Brazil	622.69	55	Kyrgyz Republic	88.51	164	El Salvador	169.08	118
Barbados	254.93	87	Cambodia	89.11	163	San Marino	393.31	70
Brunei Darussalam	648.12	52	Kiribati	53.83	193	Somalia	117.39	149
Bhutan	87.10	166	St. Kitts and Nevis	243.98	94	São Tomé and Príncipe	58.47	187
South Sudan	0.00	212	Korea, Rep.	1,021.70	38	Suriname	131.10	137
Botswana	285.39	79	Kuwait	708.17	48	Slovak Republic	1,007.78	39
Central African Republic	21.20	210	Lao PDR	245.59	91	Slovenia	1,356.14	27
Canada	2,501.03	12	Lebanon	279.68	80	Sweden	2,636.32	8
Switzerland	2,856.23	5	Liberia	28.38	204	Swaziland	127.41	138
Chile	977.81	41	Libya	124.10	147	Seychelles	244.04	92
China	1,213.74	31	St. Lucia	173.45	117	Syrian Arab Republic	0.00	212
Côte d'Ivoire	126.34	144	Liechtenstein	723.80	47	Chad	23.61	208
Cameroon	89.50	162	Sri Lanka	190.06	108	Togo	75.25	178
Congo, Rep.	55.80	190	Lesotho	46.37	198	Thailand	533.38	58
Cook Islands	141.88	127	Lithuania	1,184.60	33	Tajikistan	81.41	172
Colombia	511.00	61	Luxembourg	2,997.24	4	Turkmenistan	105.12	156
Comoros	40.97	201	Latvia	1,103.42	36	Timor-Leste	180.62	111
Cape Verde	212.49	103	Kosovo	167.12	119	Tonga	106.17	155
Costa Rica	518.04	60	Macao SAR, China	478.55	64	Trinidad and Tobago	420.82	67
Cuba	132.30	133	Morocco	214.23	101	Tunisia	202.37	106
Cayman Islands	438.85	66	Monaco	763.53	45	Turkey	616.57	56
Cyprus	1,246.71	30	Moldova	117.60	148	Tuvalu	84.12	171
Czech Republic	1,293.83	29	Madagascar	51.12	195	Taiwan, China	201.54	107
Germany	2,464.53	13	Maldives	179.47	112	Tanzania	64.23	184
Djibouti	70.86	180	Mexico	681.57	51	Uganda	55.56	191
Dominica	153.85	121	Marshall Islands	72.39	179	Ukraine	248.44	89
Denmark	2,824.72	6	North Macedonia	273.73	82	Uruguay	570.15	57
Dominican Republic	145.08	124	Mali	66.99	182	USA	4,179.97	1
Algeria	156.12	120	Malta	1,196.43	32	Uzbekistan	142.89	125
Ecuador	227.38	98	Myanmar	86.17	167	St. Vincent and the Grenadines	110.11	151
Egypt, Arab Rep.	84.91	170	Mongolia	206.43	104	Venezuela, RB	152.10	122
Eritrea	134.80	132	Montenegro	366.29	71	Virgin Islands (U.S.)	212.85	102
Spain	1,658.54	23	Mozambique	46.31	199	Vietnam	184.23	109
Estonia	1,455.69	25	Mauritania	47.78	197	Vanuatu	79.15	174
Ethiopia	54.30	192	Martinique	127.37	139	West Bank and Gaza	142.04	126
Finland	2,597.61	9	Mauritius	409.24	69	Jersey, Channel Islands	127.37	139
Fiji	131.80	136	Malawi	42.62	200	Yemen, Rep.	48.36	196
France	2,255.46	15	Malaysia	753.09	46	Serbia	509.21	62
Micronesia, Fed. Sts.	76.02	176	Namibia	177.29	114	South Africa	478.18	65
Gabon	138.32	130	Niger	52.71	194	Congo, Dem. Rep.	139.04	129
United Kingdom	2,134.76	18	Nigeria	99.69	161	Zambia	87.95	165
Georgia	325.40	77	Nicaragua	60.97	186	Zimbabwe	85.66	168
Ghana	135.16	131	Niue	127.37	139			
Guinea	57.43	189	Netherlands	2,677.92	7			

Scatter plot X_i 

Measures of determination X_i Mean and kNN imputation

X_i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
1	1.00	.30	.02	.03	.00	.08	.10	.12	.02	.01	.27	.12	.32	.00	.00	.01	.00	.00	.02	.05	.03	.17	.13	.10	.15	.03	.31	.05	.33	.40	
2	.30	1.00	.02	.01	.00	.03	.10	.03	.00	.00	.15	.10	.19	.00	.00	.01	.00	.01	.01	.04	.02	.10	.06	.06	.13	.02	.27	.09	.17	.27	
3	.02	.02	1.00	.57	.53	.06	.03	.02	.25	.02	.16	.01	.01	.00	.65	.17	.31	.26	.35	.00	.72	.00	.00	.03	.06	.00	.02	.00	.13	.07	
4	.03	.01	.57	1.00	.17	.05	.04	.01	.07	.01	.12	.04	.00	.01	.34	.11	.08	.06	.22	.05	.56	.00	.00	.01	.05	.01	.02	.00	.10	.05	
5	.00	.00	.53	.17	1.00	.00	.00	.00	.26	.04	.01	.01	.00	.00	.23	.04	.72	.77	.02	.00	.18	.01	.00	.00	.03	.00	.00	.00	.00	.00	
6	.08	.03	.06	.05	.00	.00	.17	.05	.01	.00	.06	.09	.04	.00	.01	.01	.01	.00	.02	.01	.04	.04	.02	.03	.05	.02	.02	.00	.04	.05	
7	.10	.10	.03	.04	.00	.17	.00	.10	.01	.01	.12	.15	.02	.01	.00	.01	.00	.00	.03	.03	.05	.04	.06	.09	.23	.00	.02	.00	.10	.04	
8	.12	.03	.02	.01	.00	.05	.10	.00	.00	.00	.07	.04	.02	.01	.01	.00	.00	.00	.03	.02	.02	.09	.03	.04	.02	.01	.01	.00	.06	.04	
9	.02	.00	.25	.07	.26	.01	.01	.00	.00	.53	.04	.00	.01	.00	.34	.49	.19	.02	.03	.00	.13	.02	.22	.00	.01	.15	.01	.00	.00	.00	
10	.01	.00	.02	.01	.04	.00	.01	.00	.53	.00	.00	.00	.00	.00	.05	.65	.07	.01	.00	.00	.02	.00	.16	.00	.00	.07	.00	.00	.00	.00	
11	.27	.15	.16	.12	.01	.06	.12	.07	.04	.00	.00	.11	.17	.01	.19	.00	.01	.00	.07	.02	.15	.04	.00	.09	.13	.00	.21	.03	.58	.23	
12	.12	.10	.01	.04	.01	.09	.15	.04	.00	.00	.11	.00	.03	.00	.00	.01	.02	.02	.00	.08	.00	.17	.13	.02	.20	.01	.02	.00	.03	.02	
13	.32	.19	.01	.00	.00	.04	.02	.02	.01	.00	.17	.03	.00	.00	.00	.01	.00	.00	.02	.00	.03	.01	.06	.02	.00	.53	.21	.23	.72	.02	
14	.00	.00	.00	.01	.00	.00	.01	.01	.00	.00	.01	.00	.00	.00	.00	.01	.00	.00	.01	.01	.01	.00	.01	.00	.00	.01	.00	.00	.00	.00	
15	.00	.00	.65	.34	.23	.01	.00	.01	.34	.05	.19	.00	.00	.00	.00	.31	.11	.06	.54	.00	.72	.01	.09	.01	.00	.01	.00	.01	.14	.03	
16	.01	.01	.17	.11	.04	.01	.01	.00	.49	.65	.00	.01	.01	.01	.31	.00	.04	.01	.16	.00	.31	.01	.18	.00	.00	.06	.01	.00	.00	.00	
17	.00	.00	.31	.08	.72	.01	.00	.00	.19	.07	.01	.02	.00	.00	.11	.04	.00	.66	.00	.01	.08	.00	.00	.00	.01	.00	.00	.00	.00	.00	
18	.00	.01	.26	.06	.77	.00	.00	.00	.02	.01	.00	.02	.00	.00	.06	.01	.66	.00	.00	.01	.05	.00	.03	.00	.02	.02	.00	.00	.00	.00	
19	.02	.01	.35	.22	.02	.02	.03	.03	.03	.00	.07	.00	.00	.00	.01	.54	.16	.00	.00	.00	.10	.74	.00	.00	.00	.02	.01	.03	.00	.02	.04
20	.05	.04	.00	.05	.00	.01	.03	.02	.00	.00	.02	.08	.02	.01	.00	.00	.01	.01	.10	.00	.00	.21	.01	.03	.01	.03	.02	.00	.03	.03	
21	.03	.02	.72	.56	.18	.04	.05	.02	.13	.02	.15	.00	.00	.01	.72	.31	.08	.05	.74	.00	.00	.00	.00	.03	.05	.00	.01	.00	.13	.07	
22	.17	.10	.00	.00	.01	.04	.09	.02	.00	.04	.17	.03	.00	.01	.01	.00	.00	.00	.21	.00	.00	.03	.03	.05	.00	.05	.00	.05	.00	.06	
23	.13	.06	.00	.00	.00	.02	.06	.03	.22	.16	.00	.13	.01	.01	.09	.18	.00	.03	.00	.01	.00	.03	.00	.02	.11	.14	.01	.00	.01	.01	
24	.10	.06	.03	.01	.00	.03	.09	.04	.00	.00	.09	.02	.06	.00	.01	.00	.00	.00	.02	.03	.03	.03	.02	.00	.13	.00	.10	.02	.19	.11	
25	.15	.13	.06	.05	.03	.05	.23	.02	.01	.00	.13	.20	.02	.00	.00	.00	.01	.02	.01	.01	.05	.05	.11	.13	.00	.00	.06	.02	.09	.04	
26	.03	.02	.00	.01	.00	.02	.00	.01	.15	.07	.00	.01	.00	.01	.01	.06	.00	.02	.03	.03	.00	.00	.14	.00	.00	.00	.00	.01	.00	.00	
27	.31	.27	.02	.02	.00	.02	.02	.01	.01	.00	.21	.02	.53	.00	.00	.01	.00	.00	.00	.02	.01	.05	.01	.10	.06	.00	.00	.27	.26	.66	
28	.05	.09	.00	.00	.00	.00	.00	.00	.00	.00	.03	.00	.21	.00	.01	.00	.00	.00	.02	.00	.00	.00	.00	.02	.02	.01	.27	.00	.06	.24	
29	.33	.17	.13	.10	.00	.04	.10	.06	.00	.00	.58	.03	.23	.00	.14	.00	.00	.00	.08	.03	.13	.05	.01	.19	.09	.00	.26	.06	.00	.38	
30	.40	.27	.07	.05	.00	.05	.04	.04	.00	.00	.23	.02	.72	.00	.03	.00	.00	.00	.04	.03	.07	.06	.01	.11	.04	.00	.66	.24	.38	.00	

X_i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1.00	.03	.00	.11	.00	.40	.16	.18	.00	.01	.23	.12	.57	.15	.00	.49	.17	.00	.01	.01	.01	.03	.18	.13	.29	.42	.32	.15	.42	.42
2	.03	1.00	.52	.27	.49	.23	.29	.05	.44	.43	.28	.07	.14	.06	.41	.00	.17	.53	.24	.59	.23	.53	.30	.18	.32	.22	.27	.18	.19	.27
3	.00	.52	1.00	.44	.74	.08	.13	.03	.65	.57	.16	.05	.04	.02	.72	.00	.21	.75	.25	.40	.51	.36	.11	.12	.13	.07	.06	.11	.09	.09
4	.11	.27	.44	1.00	.29	.24	.30	.07	.27	.23	.27	.10	.19	.07	.30	.02	.29	.30	.11	.14	.33	.21	.29	.12	.28	.22	.20	.23	.25	.25
5	.00	.49	.74	.29	1.00	.02	.04	.00	.77	.76	.08	.01	.01	.00	.62	.00	.16	.98	.15	.53	.29	.28	.04	.04	.05	.02	.02	.10	.03	.02
6	.40	.23	.08	.24	.02	1.00	.38	.23	.04	.02	.50	.15	.72	.20	.09	.11	.35	.03	.10	.10	.12	.13	.41	.33	.73	.97	.78	.35	.72	.96
7	.16	.29	.13	.30	.04	.38	1.00	.19	.06	.04	.30	.21	.30	.26	.07	.01	.17	.05	.15	.12	.18	.19	.75	.27	.54	.35	.30	.15	.31	.41
8	.18	.05	.03	.07	.00	.23	.19	1.00	.01	.00	.08	.71	.15	.71	.01	.05	.15	.00	.11	.03	.12	.13	.22	.32	.15	.24	.14	.12	.29	.24
9	.00	.44	.65	.27	.77	.04	.06	.01	1.00	.89	.12	.01	.01	.00	.71	.00	.13	.73	.24	.58	.31	.27	.02	.05	.07	.03	.03	.09	.05	.04
10	.01	.43	.57	.23	.76	.02	.04	.00	.89	1.00	.07	.00	.00	.00	.59	.00	.10	.69	.12	.53	.22	.27	.01	.04	.04	.01	.02	.10	.02	.02
11	.23	.28	.16	.27	.08	.50	.30	.08	.12	.07	1.00	.07	.42	.09	.25	.02	.23	.10	.16	.11	.21	.10	.30	.12	.58	.49	.54	.20	.64	.54
12	.12	.07	.05	.10	.01	.15	.21	.71	.01	.00	.07	1.00	.10	.76	.01	.01	.13	.01	.16	.05	.16	.14	.23	.27	.14	.15	.11	.11	.21	.16
13	.57	.14	.04	.19	.01	.72	.30	.15	.01	.00	.42	.10	1.00	.14	.03	.23	.30	.01	.05	.04	.06	.06	.33	.21	.57	.72	.61	.30	.56	.75
14	.15	.06	.02	.07	.00	.20	.26	.71	.00	.00	.09	.76	.14	1.00	.00	.01	.09	.00	.12	.03	.11	.09	.30	.30	.15	.21	.14	.10	.25	.21
15	.00	.41	.72	.30	.62	.09	.07	.01	.71	.59	.25	.01	.03	.00	1.00	.00	.18	.62	.33	.38	.47	.19	.04	.04	.13	.07	.06	.09	.10	.08
16	.49	.00	.00	.02	.00	.11	.01	.05	.00	.00	.02	.01	.23	.01	.00	1.00	.09	.00	.01	.00	.01	.01	.01	.01	.01	.01	.01	.09	.12	.11
17	.17	.17	.21	.29	.16	.35	.17	.15	.13	.10	.23	.13	.30	.09	.18	.09	1.00	.16	.15	.08	.31	.07	.17	.14	.32	.33	.28	.85	.30	.35
18	.00	.53	.75	.30	.98	.03	.05	.00	.73	.69	.10	.01	.01	.00	.62	.00	.16	1.00	.18	.54	.29	.30	.06	.04	.07	.03	.03	.10	.04	.04
19	.01	.24	.25	.11	.15	.10	.15	.11	.24	.12	.16	.16	.05	.12	.33	.01	.15	.18	1.00	.35	.55	.15	.11	.07	.13	.10	.05	.09	.09	.09
20	.01	.59	.40	.14	.53	.10	.12	.03	.58	.53	.11	.05	.04	.03	.38	.00	.08	.54	.35	1.00	.11	.51	.10	.10	.11	.10	.09	.10	.08	.10
21	.01	.23	.51	.33	.29	.12	.18	.12	.31	.22	.21	.16	.06	.11	.47	.01	.31	.29	.55	.11	1.00	.12	.12	.09	.17	.10	.08	.15	.11	.12
22	.03	.53	.36	.21	.28	.13	.19	.13	.27	.27	.10	.14	.06	.09	.19	.01	.07	.30	.15	.51	.12	1.00	.20	.17	.13	.13	.13	.08	.11	.15
23	.18	.30	.11	.29	.04	.41	.75	.22	.02	.01	.30	.23	.33	.30	.04	.01	.17	.06	.11	.10	.12	.20	1.00	.23	.50	.41	.33	.16	.34	.44
24	.13	.18	.12	.12	.04	.33	.27	.32	.05	.04	.12	.27	.21	.30	.04	.01	.14	.04	.07	.10	.09	.17	.23	1.00	.26	.31	.27	.15	.33	.34
25	.29	.32	.13	.28	.05	.73	.54	.15	.07	.04	.58	.14	.57	.15	.13	.05	.32	.07	.13	.11	.17	.13	.50	.26	1.00	.70	.60	.31	.60	.77
26	.42	.22	.07	.22	.02	.97	.35	.24	.03	.01	.49	.15	.72	.21	.07	.12	.33	.03	.10	.10	.10	.13	.41	.31	.70	1.00	.76	.35	.73	.95
27	.32	.27	.06	.20	.02	.78	.30	.14	.03	.02	.54	.11	.61	.14	.06	.10	.28	.03	.05	.09	.08	.13	.33	.27	.60	.76	1.00	.30	.66	.79
28	.15	.18	.11	.23	.10	.35	.15	.12	.09	.10	.20	.11	.30	.10	.09	.09	.85	.10	.09	.10	.15	.08	.16	.15	.31	.35	.30	1.00	.30	.36
29	.42	.19	.09	.25	.03	.72	.31	.29	.05	.02	.64	.21	.56	.25	.10	.12	.30	.04	.09	.08	.11	.11	.34	.33	.60	.73	.66	.30	1.00	.75
30	.42	.27	.09	.25	.02	.96	.41	.24	.04	.02	.54	.16	.75	.21	.08	.11	.35	.04	.09	.10	.12	.15	.44	.34	.77	.95	.79	.36	.75	1.00